

# **Custom Compiler™ analogLib Element Reference Manual**

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# About the analogLib Element Reference Manual

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The *analogLib Element Reference Manual* describes the analogLib library, which contains elements used for building testbenches and providing necessary stimulus for devices under test.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

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## New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the Custom Compiler Release Notes on the SolvNetPlus site.

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## Related Products, Publications, and Trademarks

For additional information about the Custom Compiler tool, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related Synopsys products:

- Synopsys® Custom Compiler™
- PrimeSim™ Continuum
- PrimeSim™
- PrimeSim™ Pro
- PrimeSim™ SPICE
- PrimeSim™ HSPICE®
- PrimeSim™ XA

- PrimeSim™ CCK
- PrimeSim™ MOSRA
- VCS® PrimeSim™ AMS
- Synopsys FineSim®
- Synopsys HSPICE®
- Synopsys VCS®
- Synopsys Verdi®
- Synopsys OptoCompiler™
- Synopsys StarRC™
- Mentor Graphics® Calibre®

## Conventions

The following conventions are used in Synopsys documentation.

| Convention            | Description                                                                                                                                                                                                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Courier               | Indicates syntax, such as <code>write_file</code> .                                                                                                                                                                                   |
| <i>Courier italic</i> | Indicates a user-defined value in syntax, such as <code>write_file design_list</code>                                                                                                                                                 |
| <b>Courier bold</b>   | Indicates user input—text you type verbatim—in examples, such as<br><code>prompt&gt; write_file top</code>                                                                                                                            |
| Purple                | <ul style="list-style-type: none"><li>• Within an example, indicates information of special interest.</li><li>• Within a command-syntax section, indicates a default, such as <code>include_enclosing = true   false</code></li></ul> |
| [ ]                   | Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>                                                                                                                                                   |
| ...                   | Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .                                                                                                                           |
|                       | Indicates a choice among alternatives, such as <code>low   medium   high</code>                                                                                                                                                       |
| \                     | Indicates a continuation of a command line.                                                                                                                                                                                           |

| Convention            | Description                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| /                     | Indicates levels of directory structure.                                                            |
| <b>Bold</b>           | Indicates a graphical user interface (GUI) element that has an action associated with it.           |
| <b>Edit &gt; Copy</b> | Indicates a path to a menu command, such as opening the <b>Edit</b> menu and choosing <b>Copy</b> . |
| Ctrl+C                | Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.                 |

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If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

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## Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working

environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

# 1

## analogLib Elements

---

*Describes cells inside analogLib elements and provides examples of how to use those devices.*

The analogLib library contains elements used for building testbenches and providing necessary stimulus for devices under test. These elements include resistors, capacitors, independent and behavioral voltage sources, and more.

Elements from analogLib are used in all testbenches. The simplest examples are power and ground supplies. A corresponding element from the analogLib is added so the simulator can understand which node is the ground node and which one is supply, as well as the supply value.

For information about how to copy symbols from the analogLib into your library or how to copy simulation views from the analogLib into your library, see the *Custom Compiler iPDK Developer Guide*.

You can find a sample model file for devices inside the analogLib library in `<product_installation_directory>/samples`.

The following sections describe the cells of the analogLib library. Each section includes an element description, a CDF parameter table, references to HSPICE documentation, and an example of the netlist with that device.

- [Passive Elements](#)
- [Active Elements](#)
- [Independent Sources](#)
- [Dependent Sources](#)
- [Global Nets](#)

---

## Passive Elements

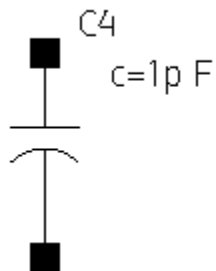
The following topics describe:

- [cap](#)
- [cap3](#)
- [cap4](#)
- [cccap](#)
- [core](#)
- [dcblock](#)
- [dcfeed](#)
- [delay](#)
- [ind](#)
- [Controlling Version Number for Legacy Designs](#)
- [iprobe, iprobe1](#)
- [ixfmr](#)
- [mind](#)
- [msline](#)
- [mtline](#)
- [phyres](#)
- [res](#)
- [res3](#)
- [res4](#)
- [switch](#)
- [tline](#)
- [u1wire - u5wire](#)
- [varactor](#)
- [varactor3](#)
- [vccap](#)

- [vcres](#)
- [winding](#)
- [xfmr](#)

## cap

Two-terminal capacitor device.



## CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Capacitance               | c             | 1p            | X       | X       | X     |
| Model Name                | model         |               | X       | X       | -     |
| Initial Condition         | ic            |               | X       | X       | -     |
| Temperature Coefficient 1 | tc1           |               | X       | X       | -     |
| Temperature Coefficient 2 | tc2           |               | X       | X       | -     |
| Width                     | w             |               | X       | X       | X     |
| Length                    | l             |               | X       | X       | X     |
| Device Area               | area          |               | -       | X       | -     |
| Multiplier                | m             |               | X       | X       | X     |
| Temp Difference           | dtemp         |               | X       | -       | -     |
| Scale Factor              | scale         |               | X       | X       | -     |



| CDF Parameter Prompt          | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-------------------------------|---------------|---------------|---------|---------|-------|
| Temperature rise from ambient | trise         |               | -       | X       | -     |
| Capacitor perimeter           | perim         |               | -       | X       | -     |
| Number of Polynomial Coeffs   | polyCoef      | 0             | -       | X       | -     |
| c1-20                         | c1-20         |               | -       | X       | -     |

Example

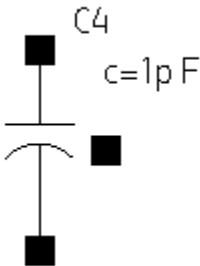
```
c4 plus minus c=1p
```

See Also

- [Passive Elements](#) in the *HSPICE® User Guide: Elements*

cap3

Three-terminal custom macro model capacitor device.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Capacitance          | c             | 1p            | X       | -       | X     |
| Model Name           | model         | cap3          | X       | -       | -     |
| Initial Condition    | ic            |               | X       | -       | -     |

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Temperature Coefficient 1 | tc1           |               | X       | -       | -     |
| Temperature Coefficient 2 | tc2           |               | X       | -       | -     |
| Width                     | w             |               | X       | -       | X     |
| Length                    | l             |               | X       | -       | X     |
| Multiplier                | m             |               | X       | -       | X     |
| Temp Difference           | dtemp         |               | X       | -       | -     |
| Scale Factor              | scale         |               | X       | -       | -     |

Example

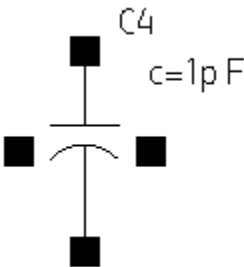
xc5 plus minus b cap3 c=1p

Model Statement

```
.subckt cap3 p m b c=1p
cmain p m c=c
.ends
```

cap4

Four-terminal custom macro model capacitor device.



## CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Capacitance               | c             | 1p            | X       | -       | X     |
| Model Name                | model         | cap4          | X       | -       | X     |
| Initial Condition         | ic            |               | X       | -       | X     |
| Temperature Coefficeint 1 | tc1           |               | X       | -       | X     |
| Temperature Coefficient 2 | tc2           |               | X       | -       | X     |
| Width                     | w             |               | X       | -       | X     |
| Length                    | l             |               | X       | -       | X     |
| Multiplier                | m             |               | X       | -       | X     |
| Temp Difference           | dtemp         |               | X       | -       | X     |
| Scale Factor              | scale         |               | X       | -       | X     |

## Example

```
xc6 plus minus b t cap4 c=1p
```

## Model Statement

```
.subckt cap4 p m b t c=1p
cmain p m c=c
.ends
```

## cccap

Two-terminal charge-conserved capacitor. Capacitance is calculated based on provided charge equation.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Charge Equation      | q             |               | X       | -       | X     |

Example

```
c7 n1 n2 q='(1e-9*v(10))'
```

See Also

- [Charge-conserved Capacitors](#) in the *HSPICE® User Guide: Elements*

core

Magnetic core element.

CORE4



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Model Name           | model         | core          | X       | X       | -     |
| Magnetization        | mag           |               | X       | -       | -     |
| Number of windings   | numOfL        | 1             | X       | -       | -     |
| Winding 1            | l1            |               | X       | -       | -     |
| Winding 2            | l2            |               | X       | -       | -     |
| Winding 3            | l3            |               | X       | -       | -     |
| Winding 4            | l4            |               | X       | -       | -     |
| Winding 5            | l5            |               | X       | -       | -     |
| Winding 6            | l6            |               | X       | -       | -     |
| Winding 7            | l7            |               | X       | -       | -     |
| Winding 8            | l8            |               | X       | -       | -     |
| Winding 9            | l9            |               | X       | -       | -     |
| Winding 10           | l10           |               | X       | -       | -     |
| Winding 11           | l11           |               | X       | -       | -     |
| Winding 12           | l12           |               | X       | -       | -     |
| Winding 13           | l13           |               | X       | -       | -     |
| Winding 14           | l14           |               | X       | -       | -     |
| Winding 15           | l15           |               | X       | -       | -     |
| Winding 16           | l16           |               | X       | -       | -     |
| Winding 17           | l17           |               | X       | -       | -     |
| Winding 18           | l18           |               | X       | -       | -     |
| Winding 19           | l19           |               | X       | -       | -     |
| Winding 20           | l20           |               | X       | -       | -     |
| Multiplier           | m             |               | -       | X       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Device area          | area          |               | -       | X       | -     |
| Physical length      | len           |               | -       | X       | -     |
| Inner diameter       | idiam         |               | -       | X       | -     |
| Outer diameter       | od            |               | -       | X       | -     |
| Gap length           | gap           |               | -       | X       | -     |

Example

```
11 2 0 nt=300
12 3 0 nt=90
13 0 4 nt=90
kcore1 11 12 13 core mag=0
```

Model Statement

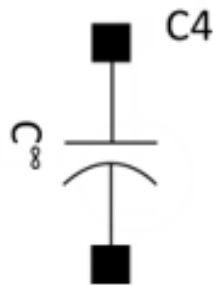
```
.model core l(ac=1e4 lc=100 hc=.1 tc=1u br=6.4k
+ bs=6.75k hs=.6 hcr=0)
```

See Also

- [Inductor Device Model and Equations](#) in the *HSPICE® Reference Manual: Device Models*. You can also check the HSPICE installation directory for a demo netlist:  
<HSPICE\_INSTALL\_DIR>/hspice/demo/hspice/mag/magcore.sp

dcblock

Two-terminal capacitor device, the capacitance is infinite.



## CDF Parameters

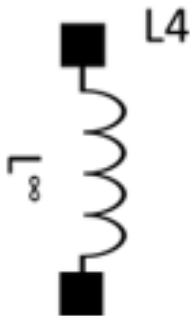
| CDF Parameter Prompt     | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------|---------------|---------------|---------|---------|-------|
| Initial Condition        | ic            |               | X       | X       | -     |
| Model name               | model         |               | -       | X       | -     |
| Capacitance used in tran | c             | 1u            | -       | X       | -     |

## Example

c4 plus minus infinity

## dcfeed

Two-terminal inductor device, the inductance is infinite.



## CDF Parameters

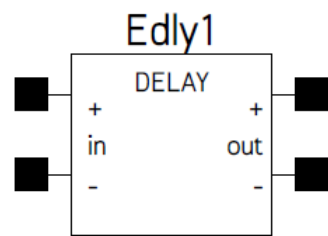
| CDF Parameter Prompt    | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-------------------------|---------------|---------------|---------|---------|-------|
| Initial Condition       | ic            |               | X       | X       | -     |
| Model name              | model         |               | -       | X       | -     |
| Inductance used in tran | l             | 1u            | -       | X       | -     |

### Example

l4 plus minus infinity

### delay

This element adjusts propagation delay of E-element which is used as a behavioral voltage source where the voltage source can be bias and/or frequency dependent.



### CDF Parameters

| CDF Parameter Prompt  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------|---------------|---------------|---------|---------|-------|
| Delay Time            | td            |               | X       | X       | -     |
| Gain                  | gain          | 1.0           | X       | X       | -     |
| Number of data points | npdelay       |               | X       | -       | -     |
| Multiplier            | m             |               | -       | X       | -     |

### Example

```
edly1 p n vcvs delay ps ns td=1n scale=1.0 npdelay=2
```

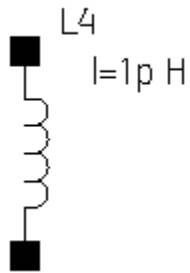
### See Also

- [Voltage-Dependent Voltage Sources – E-Elements](#) in the *HSPICE® User Guide: Elements*

### ind

Inductor element.





## CDF Parameters

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Inductance                     | l             | 1n            | X       | X       | X     |
| Model Name                     | model         |               | X       | X       | -     |
| Initial Condition              | ic            |               | X       | X       | -     |
| Temperature Coefficient 1      | tc1           |               | X       | -       | -     |
| Temperature Coefficient 2      | tc2           |               | X       | -       | -     |
| Resistance                     | r             |               | X       | X       | -     |
| Multiplier                     | m             |               | X       | X       | X     |
| Temp Difference                | dtemp         |               | -       | X       | -     |
| Scale Factor                   | scale         |               | X       | -       | -     |
| Maximum number of coefficients | polyCoef      | 0             | -       | X       | -     |
| Poly coefficient               | c1 – c20      | 0             | -       | X       | -     |
| Generate noise?                | isnoisy       |               | -       | X       | -     |
| Temperature rise from ambient  | trise         |               | -       | X       | -     |

## Example

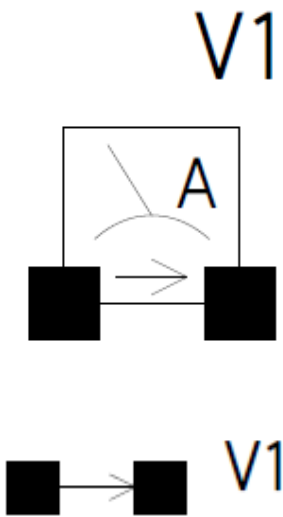
```
l12 plus minus l=1n
```

## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the `dbAnalogLibIndVersion` Tcl preference before the analogLib library is loaded.

### iprobe, iprobe1

Current Probe. In the HSPICE netlist, this creates a voltage source with 0 DC voltage. It computes the current flowing from the PLUS to the MINUS terminal.



### CDF Parameters

| CDF Parameter Prompt | CDF Parameter   | Default Value              | hspiceD | spectre | auCdl |
|----------------------|-----------------|----------------------------|---------|---------|-------|
| Source Type          | srcType         | dc                         | X       | -       | -     |
| Zero Voltage         | vdummy          | 0                          | -       | -       | -     |
| Ignore               | lvIgnore        | 1                          | -       | -       | -     |
| Remove lvs Device    | lvsRemoveDevice | (short(P<br>LUS<br>MINUS)) | -       | -       | -     |
| Remove lx Device     | lxRemoveDevice  | (short(P<br>LUS<br>MINUS)) | -       | -       | -     |

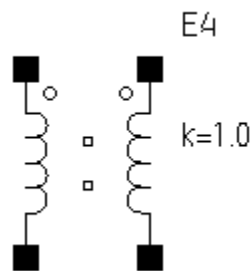
| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Action               | nlAction      | stop          | -       | -       | -     |

Example

```
v1 p n dc=0
```

ixfmr

Ideal transformer element.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Coupling Coefficient | k             | 1             | X       | -       | -     |

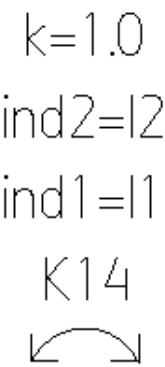
Example

```
e13 n+ n- transformer in+ in- 0.85
```

See Also

- [Ideal Transformer](#) in the *HSPICE® User Guide: Elements*

mind



CDF Parameters

| CDF Parameter Prompt    | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-------------------------|---------------|---------------|---------|---------|-------|
| First Coupled Inductor  | ind1          |               | X       | X       | -     |
| Second Coupled Inductor | ind2          |               | X       | X       | -     |
| Coupling Coefficient    | k             | 1             | X       | X       | -     |

Example

```
k14 l1 l2 k=1.0
```

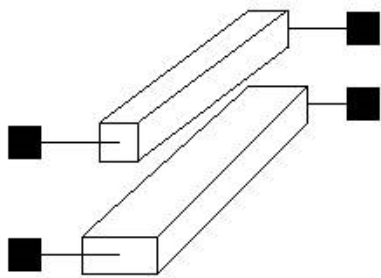
See Also

- [Inductor Device Model and Equations](#) in the *HSPICE® Reference Manual: Device Models*

msline

Microstrip line model.

The following figure illustrates the planar transmission line at the IC-circuits:



CDF Parameters

| CDF Parameter Prompt  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------|---------------|---------------|---------|---------|-------|
| Length                | l             | -             | -       | X       | -     |
| Width                 | w             | -             | -       | X       | -     |
| Substrate height      | h             | -             | -       | X       | -     |
| Conductor thickness   | t             | -             | -       | X       | -     |
| Relative permittivity | eps           | -             | -       | X       | -     |
| Max signal frequency  | fmax          | -             | -       | X       | -     |
| Multiplicity          | m             | -             | -       | X       | -     |

mtline

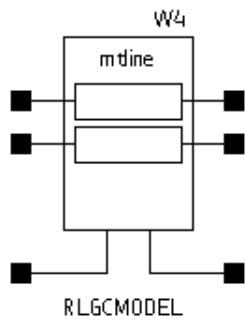
This device models distributed transmission lines using HSPICE W-element.

The W-element supports five formats to specify the transmission line properties:

- Model 1: RLGC-Model specification
  - Internally specified in a .model statement
  - Externally specified in a different file
- Model 2: U-Model specification
  - RLGC input for up to five coupled conductors
  - Geometric input (planer, coax, twin-lead)

- Measured-parameter input
- Skin effect
- Model 3: Built-in field solver model
  - Standard format (using geometric data with the W-element)
  - Tabular format
- Model 4: Frequency-dependent tabular model
- Model 5: S-parameter Model

The following figure illustrates the mtline element with two conductors:



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Number of Conductors | n             |               | X       | X       | -     |
| Model Type           | mtype         | RLGCMODEL     | X       | X       | -     |
| Model Name           | model         |               | -       | X       | -     |
| FS Model             | fsmodel       |               | X       | -       | -     |
| RLGC Model           | rlgcmode      |               | X       | -       | -     |
| RLGC file            | rlgcfile      |               | X       | -       | -     |
| RLGC data file       | file          |               | -       | X       | -     |
| U Model              | umodel        |               | X       | -       | -     |

| CDF Parameter Prompt     | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------|---------------|---------------|---------|---------|-------|
| TABLE Model              | tablemodel    |               | X       | -       | -     |
| S Model                  | smodel        |               | X       | -       | -     |
| Layer Stack              | layerstack    |               | X       | -       | -     |
| FS Options               | fsoptions     |               | X       | -       | -     |
| Coordinate Type          | coord         |               | X       | -       | -     |
| Output Format            | outputformat  | RLGCF ILE     | X       | -       | -     |
| Conductor 1 - 256        | conductor1    |               | X       | -       | -     |
| Imaginary Skin Effect?   | includersimag |               | X       | -       | -     |
| Complex Dielectric Loss? | includegdimag |               | X       | -       | -     |
| Cut-off Frequency        | fgd           |               | X       | -       | -     |
| Delay Optimization       | delayopt      |               | X       | -       | -     |
| DC Accuracy              | dcacc         |               | X       | -       | -     |
| Generate Noise?          | noise         |               | X       | -       | -     |
| Temperature Difference   | dtemp         |               | X       | -       | -     |
| LMG subcircuit file      | subcktfile    |               | -       | X       | -     |
| Maximum signal frequency | fmax          |               | -       | X       | -     |
| Multiplicity factor      | mf            | 1             | -       | X       | -     |
| Carrier frequency        | fc            |               | -       | X       | -     |
| Specify mtline choice?   | mtlineChoice  | 0             | -       | X       | -     |
| S-parameter File         | file1         |               | -       | X       | -     |
| R matrix per unit length | R             |               | -       | X       | -     |

| CDF Parameter Prompt                        | CDF Parameter  | Default Value | hspiceD | spectre | auCdl |
|---------------------------------------------|----------------|---------------|---------|---------|-------|
| L matrix per unit length                    | L              |               | -       | X       | -     |
| G matrix per unit length                    | G              |               | -       | X       | -     |
| C matrix per unit length                    | C              |               | -       | X       | -     |
| Skin effect res matrix per unit length      | rskin          |               | -       | X       | -     |
| Dielectric loss cond matrix per unit length | gdloss         |               | -       | X       | -     |
| Frequency scale factor                      | freqscale      |               | -       | X       | -     |
| Transmission line type                      | linetype       | microstrip    | -       | X       | -     |
| Number of dielectric layers                 | numlayer       | 1             | -       | X       | -     |
| Number of Ground Planes                     | numgnd         | 1             | -       | X       | -     |
| Rel dielectric const of layers(er)          | er             |               | -       | X       | -     |
| Dielectric layer thickness (d)              | layerthickness |               | -       | X       | -     |
| Dielectric loss type                        | dlosstype      |               | -       | X       | -     |
| Dielectric layer loss                       | dloss          |               | -       | X       | -     |
| Signal line width                           | linewidth      |               | -       | X       | -     |
| Signal line thickness                       | linethickness  |               | -       | X       | -     |
| Signal line height (h)                      | lineheight     |               | -       | X       | -     |
| Signal line spacing                         | linespace      |               | -       | X       | -     |
| Signal line conductivity                    | linesigma      |               | -       | X       | -     |
| Ground plane thickness                      | gndthickness   |               | -       | X       | -     |



| CDF Parameter Prompt                 | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------------|---------------|---------------|---------|---------|-------|
| Ground plane conductivity            | gndsigma      |               | -       | X       | -     |
| Characteristic impedance             | z0            |               | -       | X       | -     |
| Delay time                           | tdmt          |               | -       | X       | -     |
| Frequency                            | fmt           |               | -       | X       | -     |
| Normalized length                    | nlmt          |               | -       | X       | -     |
| Propagation velocity normalized to c | velmt         |               | -       | X       | -     |
| Corner frequency                     | corner        |               | -       | X       | -     |
| DC series rest/Length                | dcr           |               | -       | X       | -     |
| Loss resistance per unit length      | seriesr       |               | -       | X       | -     |
| Conductor loss at fc                 | alphac        |               | -       | X       | -     |
| Conductor loss quality factor        | qc            |               | -       | X       | -     |
| Dielectric loss frequency            | fd            |               | -       | X       | -     |
| Loss conductance per unit length     | shuntg        |               | -       | X       | -     |
| Dielectric loss                      | alphad        |               | -       | X       | -     |
| Dielectric loss quality factor       | gd            |               | -       | X       | -     |
| Use LMG subckt                       | uselmg        | 0             | -       | X       | -     |
| firelmg                              | firelmg       | false         | -       | X       | -     |
| Enter RLGC etc. matrices             | entermatrices | 0             | -       | X       | -     |
| Type of Input                        | modelType     |               | -       | X       | -     |
| Display Cross-section                | firedisplay   | nil           | -       | X       | -     |

| CDF Parameter Prompt     | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------|---------------|---------------|---------|---------|-------|
| Conductor loss frequency | fcmt          |               | -       | X       | -     |
| Physical length          | len           |               | X       | X       | -     |

Example

The W1 lossy transmission line connects the in node to the out node:

```
W1 in gnd out gnd RLGCfile=cable.rlgc N=1 L=5
```

See Also

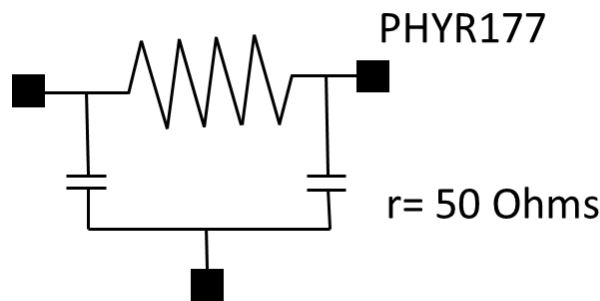
- [W-element](#) in the *HSPICE® User Guide: Elements*

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibMTlineVersion](#) Tcl preference before the analogLib library is loaded.

phyres

Physical resistor.



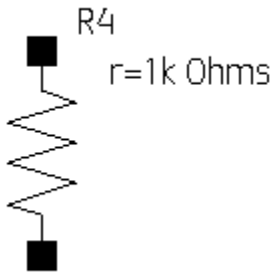
CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Model Name           | model         | phyres        | -       | X       | -     |
| Temp coefficient 1   | tc1           |               | -       | X       | -     |

| CDF Parameter Prompt              | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------------------|---------------|---------------|---------|---------|-------|
| Temp coefficient 2                | Tc2           |               | -       | X       | -     |
| Temperature rise from ambient     | trise         |               | -       | X       | -     |
| Multiplier                        | m             |               | -       | X       | -     |
| Resistance                        | r             | 50            | -       | X       | -     |
| Capacitance                       | c             |               | -       | X       | -     |
| Length                            | l             |               | -       | X       | -     |
| Width                             | w             |               | -       | X       | -     |
| Linear temp coefficient of cap    | tc1c          |               | -       | X       | -     |
| Quadratic temp coefficient of cap | tc2c          |               | -       | X       | -     |
| Estimated operating region        | region        |               | -       | X       | -     |

res

Two-terminal resistor element.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Resistance           | r             | 1k            | X       | X       | X     |

| CDF Parameter Prompt                            | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-------------------------------------------------|---------------|---------------|---------|---------|-------|
| Temperature Coefficient 1                       | tc1           |               | X       | X       | -     |
| Temperature Coefficient 2                       | tc2           |               | X       | X       | -     |
| Multiplier                                      | m             |               | X       | X       | -     |
| Model Name                                      | model         |               | X       | X       | -     |
| AC Resistance                                   | ac            |               | X       | X       | -     |
| Temp Difference                                 | dtemp         |               | X       | X       | -     |
| Width                                           | w             |               | X       | X       | X     |
| Length                                          | l             |               | X       | X       | X     |
| Scale Factor                                    | scale         |               | X       | X       | -     |
| Capacitance                                     | c             |               | X       | X       | -     |
| Generate Noise?                                 | isnoisy       |               | X       | X       | -     |
| Skin Effect Coefficient                         | rs            |               | X       | -       | -     |
| Temperature rise from ambient                   | trise         |               | -       | X       | -     |
| Resistance form                                 | resform       |               | -       | X       | -     |
| Alias for linear temperature coefficient        | tc1r          |               | -       | X       | -     |
| Alias for quadratic temperature coefficient     | tc2r          |               | -       | X       | -     |
| Linear temperature coefficient of linear cap    | tc1c          |               | -       | X       | -     |
| Quadratic temperature coefficient of linear cap | tc2c          |               | -       | X       | -     |

| CDF Parameter Prompt       | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------------|---------------|---------------|---------|---------|-------|
| Resistance scaling factor  | scaler        |               | -       | X       | -     |
| Capacitance scaling factor | scalec        |               | -       | X       | -     |

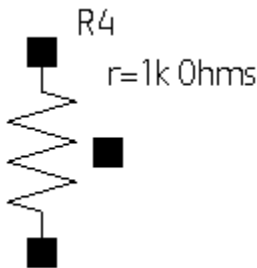
Example

r18 plus minus r=1k

See Also

- [Resistors](#) in the *HSPICE® User Guide: Elements*

res3



CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Resistance                | r             | 1k            | X       | -       | X     |
| Temperature Coefficient 1 | tc1           |               | X       | -       | -     |
| Temperature Coefficient 2 | tc2           |               | X       | -       | -     |
| Multiplier                | m             |               | X       | -       | -     |
| Model Name                | model         | res3          | X       | -       | -     |
| AC Resistance             | ac            |               | X       | -       | -     |

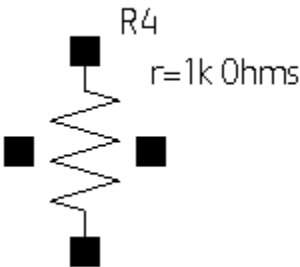
| CDF Parameter Prompt    | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-------------------------|---------------|---------------|---------|---------|-------|
| Temp Difference         | dtemp         |               | X       | -       | -     |
| Width                   | w             |               | X       | -       | X     |
| Length                  | l             |               | X       | -       | X     |
| Scale Factor            | scale         |               | X       | -       | -     |
| Capacitance             | c             |               | X       | -       | -     |
| Generate Noise?         | isnoisy       |               | X       | -       | -     |
| Skin Effect Coefficient | rs            |               | X       | -       | -     |

Example

```
xr19 plus minus b res3 r=1k
```

res4

Custom four-terminal resistor macro model.



CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Resistance                | r             | 1k            | X       | -       | X     |
| Temperature Coefficeint 1 | tc1           |               | X       | -       | -     |

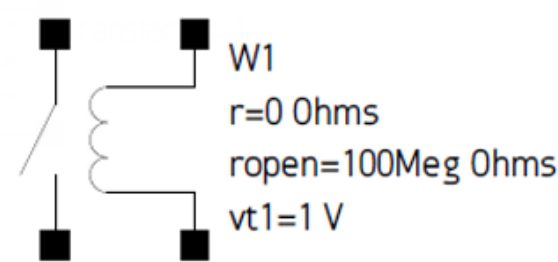
| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Temperature Coefficient 2 | tc2           |               | X       | -       | -     |
| Multiplier                | m             |               | X       | -       | -     |
| Model Name                | model         | res3          | X       | -       | -     |
| AC Resistance             | ac            |               | X       | -       | -     |
| Temp Difference           | dtemp         |               | X       | -       | -     |
| Width                     | w             |               | X       | -       | X     |
| Length                    | l             |               | X       | -       | X     |
| Scale Factor              | scale         |               | X       | -       | -     |
| Capacitance               | c             |               | X       | -       | -     |
| Generate Noise?           | isnoisy       |               | X       | -       | -     |
| Skin Effect Coefficient   | rs            |               | X       | -       | -     |

Example

```
xr24 plus minus b t res4 r=1k
```

switch

Voltage-controlled resistor.



## CDF Parameters

| CDF Parameter Prompt       | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------------|---------------|---------------|---------|---------|-------|
| Type                       | swType        | linear        | -       | -       | -     |
| Closed Resistance          | r             | 0             | x       | -       | -     |
| Open Resistance            | ropen         | 100Meg        | X       | -       | -     |
| Open Voltage               | vt1           |               | X       | X       | -     |
| Closed Voltage             | vt2           |               | X       | X       | -     |
| Open switch resistance     | ro            | 1T            | -       | X       | -     |
| Close switch resistance    | rc            | 1             | -       | X       | -     |
| Multiplier                 | m             |               | -       | X       | -     |
| Estimated operating region | region        |               | -       | X       | -     |
| Version                    | version       | 3             | -       | -       | -     |

### Note:

- For version 3, the switch element has empty values for the default of vt1 and vt2 parameters. For legacy designs with version 1 and 2, the switch element has old default values of vt1=0.1 and vt2=0.

## Example

```
gw0 n+ n- vcr relay nc+ nc- vt1 vt2 ropen r
```

### See Also

- [Voltage-Controlled Resistor](#) in the *HSPICE® User Guide: Elements*

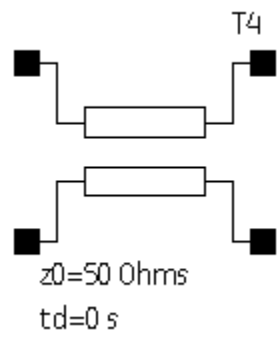
## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibSwitchVersion](#) Tcl preference before the analogLib library is loaded.



tline

Ideal transmission line model.



CDF Parameters

| CDF Parameter Prompt                 | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                           | model         |               | X       | X       | -     |
| Characteristic Impedance             | z0            | 50            | X       | -       | -     |
| Delay Time                           | td            |               | X       | X       | -     |
| Length                               | l             |               | X       | -       | -     |
| Frequency                            | freq          |               | X       | X       | -     |
| Normalized Length                    | nl            | 0.25          | X       | X       | -     |
| Voltage 1                            | v1            | 0             | X       | -       | -     |
| Current 1                            | i1            | 0             | X       | -       | -     |
| Voltage 2                            | v2            | 0             | X       | -       | -     |
| Current 2                            | i2            | 0             | X       | -       | -     |
| Multiplier                           | m             |               | -       | X       | -     |
| Propagation velocity normalized to c | vel           |               | -       | X       | -     |
| Resistance                           | r             | 1K            | -       | X       | -     |
| Loss resistance per unit length      | rs            | 1K            | -       | X       | -     |

| CDF Parameter Prompt             | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------------------|---------------|---------------|---------|---------|-------|
| Loss conductance per unit length | g             |               | -       | X       | -     |
| Physical Length                  | len           |               | -       | X       | -     |
| Characteristic impedance         | zo            | 50            | -       | X       | -     |

## Example

```
t25 in+ in- out+ out- z0=50 td=0 l=5u f=5G ic=v1,i1,v2,i2
```

## See Also

- [T-element](#) in the *HSPICE® User Guide: Elements*

## u1wire - u5wire

Lumped Transmission Lines model.

This can be used for modeling lossy non-ideal transmission lines in terms of lumped-element equivalent circuits.

The U-elements can model effects such as the following:

- Time delay
- Phase shift
- Coupling and Crosstalk
- Distortion

There are several cells which differ by input node counts from one to five.

## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Model Name           | model         |               | X       | -       | -     |
| Length               | l             |               | X       | -       | -     |
| Element Lumps        | lumps         |               | X       | -       | -     |

Example

```
u1 In10 In20 In30 Vss Out1 Out2 Out3 Vss IcWire L=200um
.model lcWire U LEVEL=3 Dlev=0 NI=3 Nlay=2 Plev=1 Elev=1
+ Llev=0 Ht=2u Wd=5u Sp=15u Th=1u Rho=2.8e-8 Kd=3.9
```

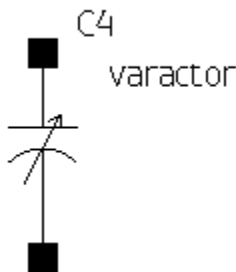
See Also

- [U-element Examples, Models, and Applications](#) in the *HSPICE® User Guide: Elements*

varactor

This macro-model device is intended to model voltage dependent capacitor, and has parameters for model name, width, length and multiplier.

In order to simulate this device, you need to provide model definition using .subckt.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Model Name           | model         | varactor      | X       | -       | X     |
| Width                | w             |               | X       | -       | X     |
| Length               | l             |               | X       | -       | X     |
| Multiplier           | m             |               | X       | -       | X     |

Example

```
xc27 n1 n2 varactor w=10u l=3u m=4
```

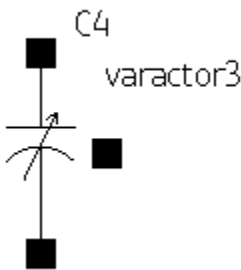
## Model Statement

```
.subckt varactor ng nds w=1u l=1u m=1
.param area='l*w'
.param Cmain='20E-11*2*w+1.15E-10*2*l+1.6E-3*area'
.param dC='-1.0E-11*2*w+4E-12*2*l+5.87E-03*area'
.param dVgs='-0.075-7E-05*(temper-27)'
.param Vgn='0.43+4E-04*(temper-27)'
cg ng nds 'm*(Cmain*(1.0+tanh((v(ng,nds)-dVgs)/Vgn)))'
.ends varactor
```

## varactor3

This macro-model device is intended to model voltage dependent capacitor, and has parameters for model name, width, length and multiplier.

In order to simulate this device, you need to provide model definition using .subckt.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Model Name           | model         | varactor      | X       | -       | X     |
| Width                | w             |               | X       | -       | X     |
| Length               | l             |               | X       | -       | X     |
| Multiplier           | m             |               | X       | -       | X     |

## Example

```
xc27 n1 n2 sub varactor3 w=10u l=3u m=4
```

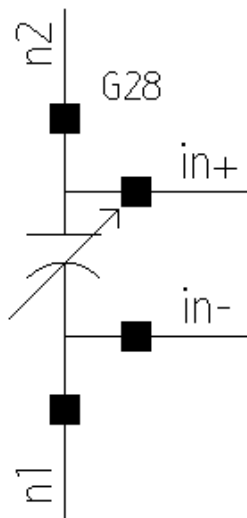
Model Statement

```
.subckt varactor3 ng nds nsub w=1u l=1u m=1
.param area='l*w'
.param Cmain='20E-11*2*w+1.15E-10*2*l+1.6E-3*area'
.param dC='-1.0E-11*2*w+4E-12*2*l+5.87E-03*area'
.param dVgs='-0.075-7E-05*(temper-27)'
.param Vgn='0.43+4E-04*(temper-27)'
cg ng nds 'm*(Cmain*(1.0+tanh((v(ng,nds)-v(nsub,0)-dVgs)/Vgn)))'
.ends varactor3
```

vccap

Voltage-controlled capacitor.

The capacitance value between nodes n1 and n2 depends on the voltage specified within nodes in+ and in-.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Delta                | delta         |               | X       | X       | -     |
| Controlling Pairs    | xypairs       | 2             | X       | X       | -     |
| Control Volt 1–50    | x1-x50        |               | X       | X       | -     |
| Corresp Element1–50  | y1-y50        |               | X       | X       | -     |

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Scale Factor                 | scale         |               | X       | X       | -     |
| Temperature Coefficient 1    | tc1           |               | X       | X       | -     |
| Temperature Coefficient 2    | tc2           |               | X       | X       | -     |
| Initial Condition            | hic           |               | X       | -       | -     |
| Multiplier                   | hm            | 1             | X       | X       | -     |
| Pwl Type                     | pwlType       | pwl           | X       | -       | -     |
| Smoothing Method             | smooth        | 2             | X       | -       | -     |
| Capacitance Calculation Mode | ctype         | 0             | X       | -       | -     |
| Type                         | csType        | linear        | X       | X       | -     |

Example

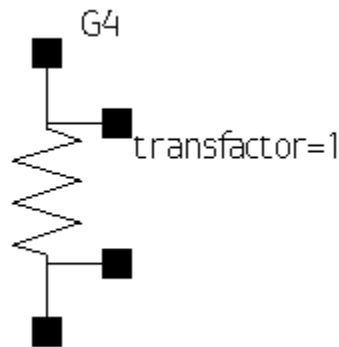
```
g28 n2 n1 vccap pwl(1) in+ in- 0.05, 20f 0.1, 35f 0.35, 85f smooth=2
ctype=0
```

See Also

- [Voltage-Controlled Capacitor](#) in the *HSPICE® User Guide: Elements*

vcres

Voltage-controlled resistor.



## CDF Parameters

| CDF Parameter Prompt       | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------------|---------------|---------------|---------|---------|-------|
| Volt/res Conversion factor | transfactor   |               | X       | X       | -     |
| Type                       | csType        | linear        | X       | X       | -     |
| Controlling Pairs          | xypairs       | 2             | X       | X       | -     |
| Delta                      | delta         |               | X       | -       | -     |
| Control Current 1–50       | x1-50         |               | X       | X       | -     |
| Corresp Element 1-50       | y1-50         |               | X       | X       |       |
| Maximum Resistance         | maxr          |               | X       | -       | -     |
| Minimum Resistance         | minr          |               | X       | -       | -     |
| Scale Factor               | scale         |               | X       | -       | -     |
| Temperature Coefficient 1  | tc1           |               | X       | -       | -     |
| Temperature Coefficient 2  | tc2           |               | X       | -       | -     |
| Initial Condition          | hic           |               | X       | -       | -     |
| Multiplier                 | hm            |               | X       | -       | -     |
| Pwl Type                   | pwlType       | pwl           | X       | -       | -     |
| Smoothing Method           | smooth        |               | X       | -       | -     |
| Resistance 1–50            | y1-y50        |               | X       | -       | -     |

## Example

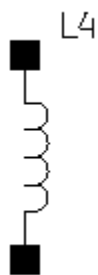
```
g28 n2 n1 vcr in+ in- 1 max=100k min=5k
g28 n2 n1 vcr pwl(1) in+ in- 0.05, 5k 0.1, 7k 0.35, 10k
```

## See Also

- [Voltage-Controlled Resistor](#) in the *HSPICE® User Guide: Elements*

## winding

Inductor with magnetic winding.



### CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Num of Turns              | turn          |               | X       | X       | -     |
| Model Name                | model         |               | X       | -       | -     |
| Initial Condition         | ic            |               | X       | X       | -     |
| Temperature Coefficient 1 | tc1           |               | X       | -       | -     |
| Temperature Coefficient 2 | tc2           |               | X       | -       | -     |
| Resistance                | r             |               | X       | X       | -     |
| Multiplier                | m             |               | X       | X       | -     |
| Temp Difference           | dtemp         |               | X       | -       | -     |
| Scale Factor              | scale         |               | X       | -       | -     |
| Name of core              | core          |               | -       | X       | -     |
| Res of the winding        | resis         |               |         | X       |       |

### Example

```
l4 n2 n1 nt=100 r=10
```

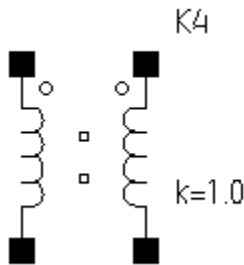


See Also

- [Inductors](#) in the *HSPICE® User Guide: Elements*

xfmr

Mutual inductors with coupling coefficient k.



CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Primary Inductance           | pi            |               | X       | -       | -     |
| Secondary Inductance         | si            |               | X       | -       | -     |
| Coupling Coefficient         | k             | 1             | X       | -       | -     |
| Multiplier                   | m             |               | -       | X       | -     |
| Number of turns on primary   | n1            |               | -       | X       | -     |
| Number of turns on secondary | n2            |               | -       | X       | -     |

Example

```
xk31 p+ p- s+ s- top_xk31
.subckt top_xk31 p1 p2 s1 s2
  l1 p1 p2 l=0.5n
  l2 s1 s2 l=0.75n
  k1 l1 l2 k=0.85
.ends top_xk31
```

### See Also

- [Mutual Inductors](#) in the *HSPICE® User Guide: Elements*

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## Active Elements

The following topics describe:

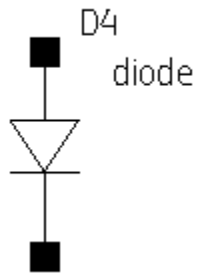
- [diode](#)
- [diode3](#)
- [diode4](#)
- [ibis\\_buffer](#)
- [mport](#)
- [nbsim](#)
- [pbsim](#)
- [nbsim4](#)
- [pbsim4](#)
- [njfet](#)
- [pjfet](#)
- [nmes](#)
- [pmes](#)
- [nmes4](#)
- [pmes4](#)
- [nmos](#)
- [pmos](#)
- [nmos4](#)
- [pmos4](#)
- [nmos4hv](#)
- [pmos4hv](#)
- [nmoshv](#)

- [pmoshv](#)
- [npn](#)
- [pnp](#)
- [npn4](#)
- [nnp4a](#)
- [pnp4](#)
- [pnp4a](#)
- [nsoi](#)
- [psoi](#)
- [nsoi4](#)
- [psoi4](#)
- [nsoifd](#)
- [psoidf](#)
- [nsoifd4](#)
- [psoidf4](#)
- [schottky](#)
- [sp1tswitch](#)
- [sp2tswitch](#)
- [sp3tswitch](#)
- [sp4tswitch](#)
- [u1wire](#), [u2wire](#), [u3wire](#), [u4wire](#), [u5wire](#)

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## diode

Two-terminal p-n diode.



## CDF Parameters

| CDF Parameter Prompt   | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------|---------------|---------------|---------|---------|-------|
| Model Name             | model         | diode         | X       | X       | X     |
| Device Area            | area          |               | X       | X       | X     |
| Initial Diode Voltage  | Vd            |               | X       | -       | -     |
| Periphery of Junction  | pj            |               | X       | X       | X     |
| Width of Polycap       | wp            |               | X       | X       | -     |
| Length of Polycap      | lp            |               | X       | X       | -     |
| Width of Metalcap      | wm            |               | X       | X       | -     |
| Length of Metalcap     | lm            |               | X       | X       | -     |
| Width                  | w             |               | X       | X       | -     |
| Length                 | l             |               | X       | X       | -     |
| Initially Off          | off           | 0             | -       | X       | -     |
| Multiplier             | m             |               | X       | X       | X     |
| Device Temperature     | dtemp         |               | X       | X       | -     |
| Amplitude scale factor | scale         |               | -       | X       | -     |

## Example

```
d32 n2 n1 diode
```

Model Statement

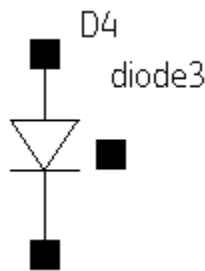
```
.model core 1(ac=1e4 lc=100 hc=.1 tc=1u br=6.4k  
+ bs=6.75k hs=.6 hcr=0)
```

See Also

- [Diode Element](#) in the *HSPICE® User Guide: Elements*

diode3

Three-terminal diode.



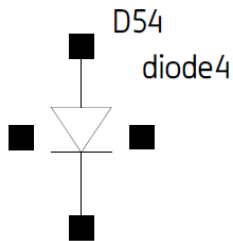
CDF Parameters

| CDF Parameter Prompt  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------|---------------|---------------|---------|---------|-------|
| Model Name            | model         | diode3        | X       | -       | X     |
| Device Area           | area          |               | X       | -       | X     |
| Initial Diode Voltage | Vd            |               | X       | -       | -     |
| Periphery of Junction | pj            |               | X       | -       | X     |
| Width of Polycap      | wp            |               | X       | -       | -     |
| Length of Polycap     | lp            |               | X       | -       | -     |
| Width of Metalcap     | wm            |               | X       | -       | -     |
| Length of Metalcap    | lm            |               | X       | -       | -     |
| Width                 | w             |               | X       | -       | -     |
| Length                | l             |               | X       | -       | -     |
| Initially Off         | off           | 0             | -       | -       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Multiplier           | m             |               | X       | -       | X     |
| Device Temperature   | dtemp         |               | X       | -       | -     |

## diode4

Four-terminal diode.



## CDF Parameters

| CDF Parameter Prompt  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------|---------------|---------------|---------|---------|-------|
| Model Name            | Model         | diode4        | -       | -       | -     |
| Device Area           | area          |               | X       | -       | X     |
| Initial Diode Voltage | Vd            |               | X       | -       | X     |
| Periphery of Junction | pj            |               | X       | -       | X     |
| Width of Polycap      | wp            |               | X       | -       | X     |
| Length of Polycap     | lp            |               | X       | -       | X     |
| Width of Metalcap     | wm            |               | X       | -       | X     |
| Length of Metalcap    | lm            |               | X       | -       | X     |
| Width                 | w             |               | X       | -       | X     |
| Length                | l             |               | X       | -       | X     |
| Initially Off         | off           | 0             | -       | -       | -     |
| Multiplier            | m             |               | X       | -       | X     |
| Device Temperature    | dtemp         |               | X       | -       | X     |

Example

```
d1 plus minus b t diode4
```

See Also

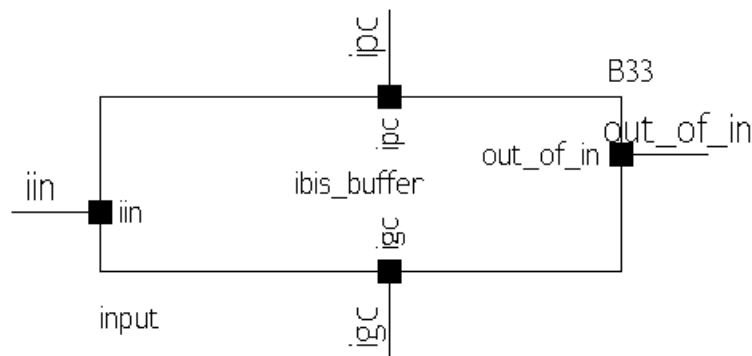
- [Diode Models](#) in the *HSPICE® Reference Manual: Device Models*

ibis\_buffer

A behavioral method of modeling input/output buffers based on V/I curve data derived from measurement or full circuit simulation.

There are different IBIS buffer types, including:

- Input buffer
- Output buffer
- Input ECL buffer
- Output ECL buffer
- Tri-state buffer
- Input/Output buffer



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Buffer Type          | buffer        | input         | X       | X       | X     |
| Power Ground Clamp?  | pgclamp       | 0             | X       | X       | X     |

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| IBIS File                      | file          |               | X       | -       | -     |
| Model Name                     | model         |               | X       | X       | X     |
| Type                           | typ           |               | X       | X       | -     |
| HSPICE Version                 | hsp_ver       |               | X       | X       | -     |
| Power                          | power         |               | X       | X       | -     |
| Interpolation                  | interpol      |               | X       | -       | -     |
| XV_PU State Node               | xv_pu         |               | X       | -       | -     |
| XV_PD State Node               | xv_pd         |               | X       | -       | -     |
| Ramp Falling Waveform          | ramp_fwf      |               | X       | -       | -     |
| Ramp Rising Waveform           | ramp_rwf      |               | X       | -       | -     |
| Falling Waveform Tune          | fwf_tune      |               | X       | -       | -     |
| Rising Waveform Tune           | rwf_tune      |               | X       | -       | -     |
| Falling Waveform Pullup Delay  | fwf_pu_dly    |               | X       | -       | -     |
| Rising Waveform Pulldown Delay | rwf_pd_dly    |               | X       | -       | -     |
| Pulldown Scale Factor          | pd_scal       |               | X       | -       | -     |
| Pullup Scale Factor            | pu_scal       |               | X       | -       | -     |
| Power Clamp Scale Factor       | pc_scal       |               | X       | -       | -     |
| Ground Clamp Scale Factor      | gc_scal       |               | X       | -       | -     |
| Falling Waveform Scale Factor  | fwf_scal      |               | X       | -       | -     |
| Rising Waveform Scale Factor   | rwf_scal      |               | X       | -       | -     |



| CDF Parameter Prompt         | CDF Parameter  | Default Value | hspiceD | spectre | auCdl |
|------------------------------|----------------|---------------|---------|---------|-------|
| Series Switch State          | ss_state       |               | X       | -       | -     |
| Remove Initial Delay Falling | rm_dly_fwf     |               | X       | -       | -     |
| Remove Initial Delay Rising  | rm_dly_rwf     |               | X       | -       | -     |
| Remove Flat Tails Falling    | rm_tail_fwf    |               | X       | -       | -     |
| Remove Flat Tails Rising     | rm_tail_rwf    |               | X       | -       | -     |
| Supress Warning Messages     | nowarn         | 9             | X       | -       | -     |
| C_comp Pullup                | c_com_pu       |               | X       | -       | -     |
| C_comp Pulldown              | c_com_pd       |               | X       | -       | -     |
| C_comp Power Clamp           | c_com_pc       |               | X       | -       | -     |
| C_comp Ground Clamp          | c_com_gc       |               | X       | -       | -     |
| Detect OTI MID               | detect_oti_mid |               | X       | -       | -     |
| Time Control                 | time_control   |               | X       | -       | -     |

## Example

```
b33 ipc igc iin out_of_in buffer=input
```

## See Also

- [Simulating Circuits with IBIS Models in HSPICE](#) in the *HSPICE® User Guide: Signal Integrity Modeling and Analysis*
- [Using IBIS Files with HSPICE](#) in the *HSPICE® User Guide: Signal Integrity Modeling and Analysis*
- [Modeling Input/Output Buffers Using IBIS Files](#) in the *HSPICE® User Guide: Elements*

## mport

To measure mixed mode S-parameters you can use a port element with three terminals. Except for the number of external terminals, the syntax of the port element remains the same. The LIN analysis function internally sets the necessary drive mode (common/differential) of these mixed mode port elements. This symbol specifies mixed mode Port element.

## Example

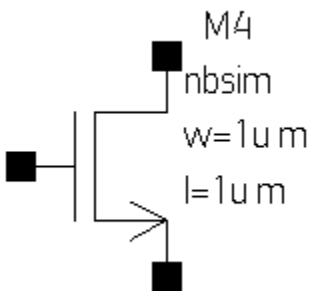
```
p1 plus minus ref dc=0 z0=50
```

### See Also

- [Using the Port Element for Mixed-Mode Measurement](#) in the *HSPICE® User Guide: Elements*

## nbsim

Three-terminal NMOS device with model name *nbsim*. By default, the bulk terminal of this device is connected to gnd!.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Model Name           | model         | nbsim         | X       | X       | X     |
| Width                | w             | 1u            | X       | X       | X     |
| Length               | l             | 1u            | X       | X       | X     |
| Drain Diffusion Area | ad            |               | X       | X       | -     |

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Source Diffusion Area        | as            |               | X       | X       | -     |
| Drain Junction Perimeter     | pd            |               | X       | X       | -     |
| Source Junction Perimeter    | ps            |               | X       | X       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | X       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | X       | -     |
| Additional Drain Resistance  | rdc           |               | X       | X       | -     |
| Additional Source Resistance | rsc           |               | X       | X       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | X       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | X       | X     |
| NQS flag                     | nqsmode       |               | -       | X       | -     |
| Drain diffusion length       | ld            |               | -       | X       | -     |
| Source diffusion length      | ls            |               | -       | X       | -     |
| Dist. OD and poly (one side) | sa            |               | -       | X       | -     |

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Dist. OD and poly (other side) | sb            |               | -       | X       | -     |
| Distance between fingers       | sd            |               | -       | X       | -     |
| Expected operating region      | region        |               | -       | X       | -     |
| Temperature rise from ambient  | trise         |               | -       | X       | -     |

Example

```
m34 d g s gnd! nbsim w=1u l=1u
```

Model Statement

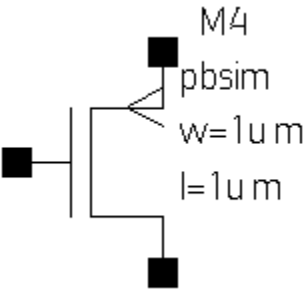
```
.model nbsim nmos level=54
```

See Also

- [MOSFETs](#) in the *HSPICE® User Guide: Elements*

pbsim

Three-terminal PMOS device with model name *pbsim*. By default, the bulk terminal of this device is connected to vdd!.



## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nbsim         | X       | X       | X     |
| Width                        | w             | 1u            | X       | X       | X     |
| Length                       | l             | 1u            | X       | X       | X     |
| Drain Diffusion Area         | ad            |               | X       | X       | -     |
| Source Diffusion Area        | as            |               | X       | X       | -     |
| Drain Junction Perimeter     | pd            |               | X       | X       | -     |
| Source Junction Perimeter    | ps            |               | X       | X       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | X       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | X       | -     |
| Additional Drain Resistance  | rdc           |               | X       | X       | -     |
| Additional Source Resistance | rsc           |               | X       | X       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | X       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | X       | X     |

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| NQS flag                       | nqsmod        |               | -       | X       | -     |
| Drain diffusion length         | ld            |               | -       | X       | -     |
| Source diffusion length        | ls            |               | -       | X       | -     |
| Dist. OD and poly (one side)   | sa            |               | -       | X       | -     |
| Dist. OD and poly (other side) | sb            |               | -       | X       | -     |
| Distance between fingers       | sd            |               | -       | X       | -     |
| Expected operating region      | region        |               | -       | X       | -     |
| Temperature rise from ambient  | trise         |               | -       | X       | -     |

Example

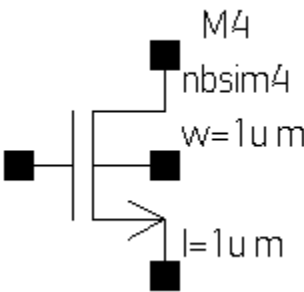
```
m1 s g d vdd! pbsim w=1u l=1u
```

Model Statement

```
.model pbsim pmos level=54
```

nbsim4

Four-terminal MOS device with model name *nbsim4*.



## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nbsim4        | X       | X       | X     |
| Width                        | w             | 1u            | X       | X       | X     |
| Length                       | l             | 1u            | X       | X       | X     |
| Drain Diffusion Area         | ad            |               | X       | X       | -     |
| Source Diffusion Area        | as            |               | X       | X       | -     |
| Drain Junction Perimeter     | pd            |               | X       | X       | -     |
| Source Junction Perimeter    | ps            |               | X       | X       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | X       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | X       | -     |
| Additional Drain Resistance  | rdc           |               | X       | X       | -     |
| Additional Source Resistance | rsc           |               | X       | X       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | X       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | X       | X     |

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Expected operating region      | region        |               | -       | X       | -     |
| Temperature rise from ambient  | trise         |               | -       | X       | -     |
| NQS flag                       | nqsmod        |               | -       | X       | -     |
| Drain diffusion length         | ld            |               | -       | X       | -     |
| Source diffusion length        | ls            |               | -       | X       | -     |
| Dist. OD and poly (one side)   | sa            |               | -       | X       | -     |
| Dist. OD and poly (other side) | sb            |               | -       | X       | -     |
| Distance between fingers       | sd            |               | -       | X       | -     |

Example

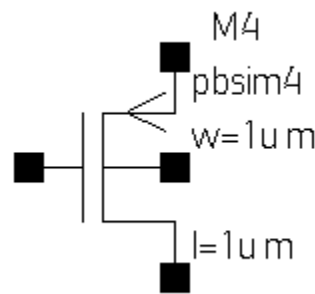
```
m34 d g s b nbsim4 w=1u l=1u
```

Model Statement

```
.model nbsim4 nmos level=54
```

pbsim4

Four-terminal PMOS device with model name *pbsim4*.





## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nbsim4        | X       | X       | X     |
| Width                        | w             | 1u            | X       | X       | X     |
| Length                       | l             | 1u            | X       | X       | X     |
| Drain Diffusion Area         | ad            |               | X       | X       | -     |
| Source Diffusion Area        | as            |               | X       | X       | -     |
| Drain Junction Perimeter     | pd            |               | X       | X       | -     |
| Source Junction Perimeter    | ps            |               | X       | X       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | X       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | X       | -     |
| Additional Drain Resistance  | rdc           |               | X       | X       | -     |
| Additional Source Resistance | rsc           |               | X       | X       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | X       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | X       | X     |

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Expected operating region      | region        |               | -       | X       | -     |
| Temperature rise from ambient  | trise         |               | -       | X       | -     |
| NQS flag                       | nqsmod        |               | -       | X       | -     |
| Drain diffusion length         | ld            |               | -       | X       | -     |
| Source diffusion length        | ls            |               | -       | X       | -     |
| Dist. OD and poly (one side)   | sa            |               | -       | X       | -     |
| Dist. OD and poly (other side) | sb            |               | -       | X       | -     |
| Distance between fingers       | sd            |               | -       | X       | -     |

Example

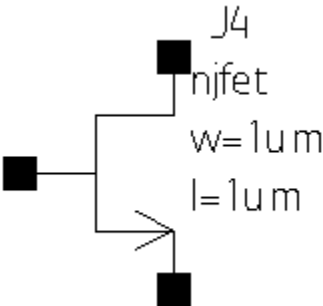
```
M4 s g d vdd! pbsim4 w=1u l=1u
```

Model Statement

```
.model pbsim4 pmos level=54
```

njfet

Three-terminal Junction Gate Field-Effect transistor.



## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | njfet         | X       | X       | X     |
| Device Area                  | area          |               | -       | X       | -     |
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Initially Off                | off           | 0             | X       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | X       | X     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Expected operating region    | region        |               | -       | X       | -     |

## Example

```
j35 d g s gnd! njfet w=1u l=1u
```

## Model Statement

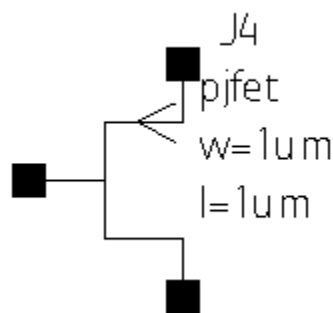
```
.model njfet njf vto=-1.2 beta=.179m lambda=2.2m rs=70 rd=70  
+ is=1.7e-14 cgs=14p cgd=14p
```

## See Also

- [JFETs and MESFETs](#) in the *HSPICE® User Guide: Elements*

## pjfet

Three-terminal p-type Junction Gate Field-Effect transistor. By default, bulk terminal is connected to global net vdd!.



CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | pjfet         | X       | X       | X     |
| Device Area                  | area          |               | -       | X       | -     |
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Initially Off                | off           | 0             | X       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | X       | X     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Expected operating region    | region        |               | -       | X       | -     |

Example

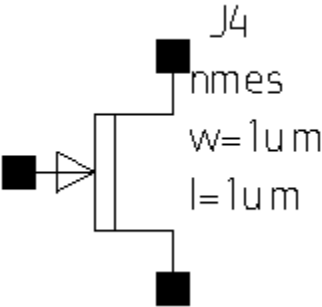
```
j4 s g d vdd! pjfet w=1u l=1u
```

Model Statement

```
.model pjfet pjf vto=-1.2 beta=.179m lambda=2.2m rs=70 rd=70  
+ is=1.7e-14 cgs=14p cgd=14p
```

nmes

Three-terminal Metal-Semiconductor Field-Effect transistor.



CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nmes          | X       | X       | 4     |
| Device Area                  | area          |               | -       | X       | -     |
| Width                        | w             | 1u            | X       | -       | 4     |
| Length                       | l             | 1u            | X       | -       | X     |
| Initially Off                | off           | 0             | X       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | X       | X     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Expected operating region    | region        |               | -       | X       | -     |

Example

```
j36 d g s gnd! nmes w=1u l=1u
```

Model Statement

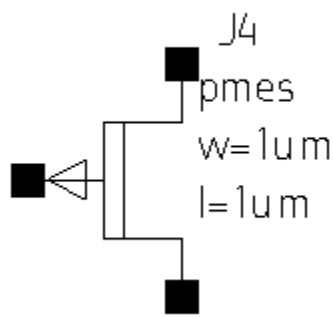
```
.model nmes njf vto=-1.2 beta=.179m lambda=2.2m rs=70 rd=70
+ is=1.7e-14 cgs=14p cgd=14p
```

See Also

- [JFETs and MESFETs](#) in the *HSPICE® User Guide: Elements*

pmes

Three-terminal Metal-Semiconductor Field-Effect transistor.



CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nmes          | X       | X       | X     |
| Device Area                  | area          |               | -       | X       | -     |
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Initially Off                | off           | 0             | X       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | X       | X     |
| Temp Difference              | dtemp         |               | X       | -       | -     |

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Expected operating region | region        |               | -       | X       | -     |

Example

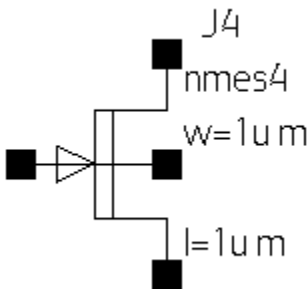
```
jm4 d g s gnd! pmes w=1u l=1u
```

Model Statement

```
.model pmes pjf vto=-1.2 beta=.179m lambda=2.2m rs=70 rd=70  
+ is=1.7e-14 cgs=14p cgd=14p
```

nmes4

Four-terminal Metal-Semiconductor Field-Effect transistor.



CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nmes4         | X       | -       | X     |
| Device Area                  | area          |               | -       | -       | -     |
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Initially Off                | off           | 0             | X       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |

| CDF Parameter Prompt        | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|---------------|---------------|---------|---------|-------|
| Gate-source Initial Voltage | Vgs           |               | X       | -       | -     |
| Multiplier                  | m             |               | X       | -       | X     |
| Temp Difference             | dtemp         |               | X       | -       | -     |

Example

```
j36 d g s b nmes4 w=1u l=1u
```

Model Statement

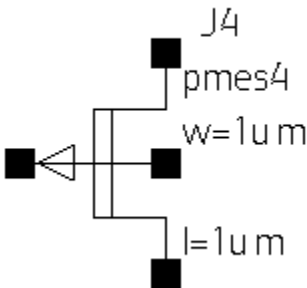
```
.model nmes4 njf vto=-1.2 beta=.179m lambda=2.2m rs=70 rd=70  
+ is=1.7e-14 cgs=14p cgd=14p
```

See Also

- [JFETs and MESFETs](#) in the *HSPICE® User Guide: Elements*

pmes4

Four-terminal Metal-Semiconductor Field-Effect transistor.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Model Name           | model         | nmes4         | X       | -       | X     |
| Device Area          | area          |               | -       | -       | -     |



| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Initially Off                | off           | 0             | X       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | -       | X     |
| Temp Difference              | dtemp         |               | X       | -       | -     |

## Example

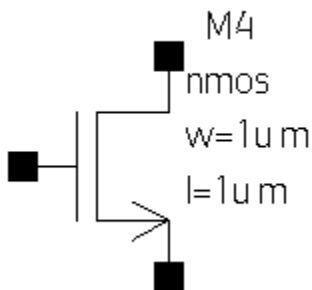
```
j m4 d g s b pmes4 w=1u l=1u
```

## Model Statement

```
.model pmes4 pjf vto=-1.2 beta=.179m lambda=2.2m rs=70 rd=70  
+ is=1.7e-14 cgs=14p cgd=14p
```

## nmos

Three-terminal N-MOS device. By default, bulk terminal is connected to gnd! net.



## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nmos          | X       | X       | X     |
| Width                        | w             | 1u            | X       | X       | X     |
| Length                       | l             | 1u            | X       | X       | X     |
| Drain Diffusion Area         | ad            |               | X       | X       | -     |
| Source Diffusion Area        | as            |               | X       | X       | -     |
| Drain Junction Perimeter     | pd            |               | X       | X       | -     |
| Source Junction Perimeter    | ps            |               | X       | X       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | X       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | X       | -     |
| Additional Drain Resistance  | rdc           |               | X       | X       | -     |
| Additional Source Resistance | rsc           |               | X       | X       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | X       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | X       | X     |

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Drain diffusion length         | ld            | -             | -       | X       | -     |
| Source diffusion length        | ls            | -             | -       | X       | -     |
| Dist. OD and poly (one side)   | sa            | -             | -       | X       | -     |
| Dist. OD and poly (other side) | sb            | -             | -       | X       | -     |
| Distance between fingers       | sd            | -             | -       | X       | -     |
| Hot-electron degradation       | degradation   | -             | -       | X       | -     |
| Expected operating region      | region        | -             | -       | X       | -     |
| Temperature rise from ambient  | trise         | -             | -       | X       | -     |

## Example

```
mj36 d g s gnd! nmos w=1u l=1u
```

## Model Statement

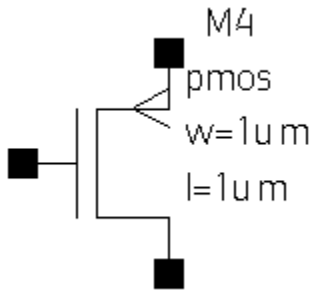
```
.model nmos nmos level=54
```

## See Also

- [MOSFETs](#) in the *HSPICE® User Guide: Elements*

## pmos

Three-terminal P-MOS device. By default, bulk terminal is connected to vdd! net.



## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nmos          | X       | X       | X     |
| Width                        | w             | 1u            | X       | X       | X     |
| Length                       | l             | 1u            | X       | X       | X     |
| Drain Diffusion Area         | ad            |               | X       | X       | -     |
| Source Diffusion Area        | as            |               | X       | X       | -     |
| Drain Junction Perimeter     | pd            |               | X       | X       | -     |
| Source Junction Perimeter    | ps            |               | X       | X       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | X       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | X       | -     |
| Additional Drain Resistance  | rdc           |               | X       | X       | -     |
| Additional Source Resistance | rsc           |               | X       | X       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Bulk-source Initial Voltage    | Vbs           |               | X       | -       | -     |
| Temp Difference                | dtemp         |               | X       | -       | -     |
| Source/Drain Selector          | geo           |               | X       | X       | -     |
| Threshold Voltage Shift        | delvto        |               | X       | -       | -     |
| Multiplier                     | m             |               | X       | X       | X     |
| Drain diffusion length         | ld            |               | -       | X       | -     |
| Source diffusion length        | ls            |               | -       | X       | -     |
| Dist. OD and poly (one side)   | sa            |               | -       | X       | -     |
| Dist. OD and poly (other side) | sb            |               | -       | X       | -     |
| Distance between fingers       | sd            |               | -       | X       | -     |
| Hot-electron degradation       | degradation   |               | -       | X       | -     |
| Expected operating region      | region        |               | -       | X       | -     |
| Temperature rise from ambient  | trise         |               | -       | X       | -     |

## Example

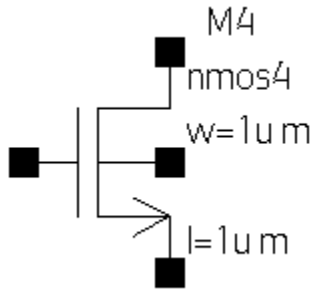
```
m4 d g s vdd! pmos w=1u l=1u
```

## Model Statement

```
.model pmos pmos level=54
```

## nmos4

Four-terminal N-MOS device.



## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nmos          | X       | X       | X     |
| Width                        | w             | 1u            | X       | X       | X     |
| Length                       | l             | 1u            | X       | X       | X     |
| Drain Diffusion Area         | ad            |               | X       | X       | -     |
| Source Diffusion Area        | as            |               | X       | X       | -     |
| Drain Junction Perimeter     | pd            |               | X       | X       | -     |
| Source Junction Perimeter    | ps            |               | X       | X       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | X       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | X       | -     |
| Additional Drain Resistance  | rdc           |               | X       | X       | -     |
| Additional Source Resistance | rsc           |               | X       | X       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Bulk-source Initial Voltage    | Vbs           |               | X       | -       | -     |
| Temp Difference                | dtemp         |               | X       | -       | -     |
| Source/Drain Selector          | geo           |               | X       | X       | -     |
| Threshold Voltage Shift        | delvto        |               | X       | -       | -     |
| Multiplier                     | m             |               | X       | X       | X     |
| Drain diffusion length         | ld            |               | -       | X       | -     |
| Source diffusion length        | ls            |               | -       | X       | -     |
| Dist. OD and poly (one side)   | sa            |               | -       | X       | -     |
| Dist. OD and poly (other side) | sb            |               | -       | X       | -     |
| Distance between fingers       | sd            |               | -       | X       | -     |
| Hot-electron degradation       | degradation   |               | -       | X       | -     |
| Expected operating region      | region        |               | -       | X       | -     |
| Temperature rise from ambient  | trise         |               | -       | X       | -     |

## Example

```
m4 d g s b nmos4 w=1u l=1u
```

## Model Statement

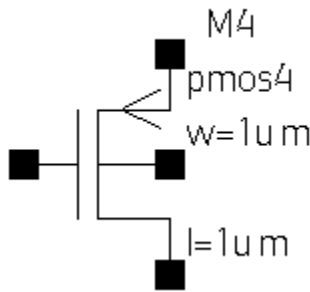
```
.model nmos4 nmos level=54
```

## See Also

- [MOSFETs](#) in the *HSPICE® User Guide: Elements*

pmos4

Four-terminal P-MOS device.



CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nmos          | X       | X       | X     |
| Width                        | w             | 1u            | X       | X       | X     |
| Length                       | l             | 1u            | X       | X       | X     |
| Drain Diffusion Area         | ad            |               | X       | X       | -     |
| Source Diffusion Area        | as            |               | X       | X       | -     |
| Drain Junction Perimeter     | pd            |               | X       | X       | -     |
| Source Junction Perimeter    | ps            |               | X       | X       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | X       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | X       | -     |
| Additional Drain Resistance  | rdc           |               | X       | X       | -     |
| Additional Source Resistance | rsc           |               | X       | X       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |



| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Drain-source Initial Voltage   | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage    | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage    | Vbs           |               | X       | -       | -     |
| Temp Difference                | dtemp         |               | X       | -       | -     |
| Source/Drain Selector          | geo           |               | X       | X       | -     |
| Threshold Voltage Shift        | delvto        |               | X       | -       | -     |
| Multiplier                     | m             |               | X       | X       | X     |
| Drain diffusion length         | ld            |               | -       | X       | -     |
| Source diffusion length        | ls            |               | -       | X       | -     |
| Dist. OD and poly (one side)   | sa            |               | -       | X       | -     |
| Dist. OD and poly (other side) | sb            |               | -       | X       | -     |
| Distance between fingers       | sd            |               | -       | X       | -     |
| Hot-electron degradation       | degradation   |               | -       | X       | -     |
| Expected operating region      | region        |               | -       | X       | -     |
| Temperature rise from ambient  | trise         |               | -       | X       | -     |

## Example

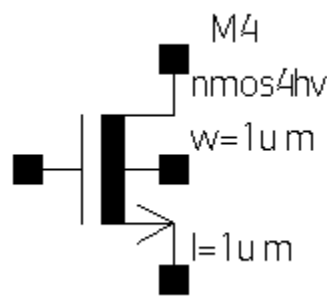
```
m4 d g s b pmos4 w=1u l=1u
```

## Model Statement

```
.model pmos4 pmos level=54
```

nmos4hv

Four-terminal high-voltage N-MOS device.



CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nmos4hv       | X       | -       | X     |
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Drain Diffusion Area         | ad            |               | X       | -       | -     |
| Source Diffusion Area        | as            |               | X       | -       | -     |
| Drain Junction Perimeter     | pd            |               | X       | -       | -     |
| Source Junction Perimeter    | ps            |               | X       | -       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | -       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | -       | -     |
| Additional Drain Resistance  | rdc           |               | X       | -       | -     |
| Additional Source Resistance | rsc           |               | X       | -       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | -       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | -       | X     |

Example

```
mj36 d g s b nmos4hv w=1u l=1u
```

Model Statement

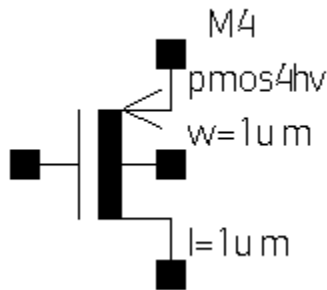
```
.model nmos4hv nmos level=54
```

See Also

- [MOSFETs](#) in the *HSPICE® User Guide: Elements*

pmos4hv

Four-terminal high-voltage P-MOS device.



## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nmos4hv       | X       | -       | X     |
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Drain Diffusion Area         | ad            |               | X       | -       | -     |
| Source Diffusion Area        | as            |               | X       | -       | -     |
| Drain Junction Perimeter     | pd            |               | X       | -       | -     |
| Source Junction Perimeter    | ps            |               | X       | -       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | -       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | -       | -     |
| Additional Drain Resistance  | rdc           |               | X       | -       | -     |
| Additional Source Resistance | rsc           |               | X       | -       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | -       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | -       | X     |

Example

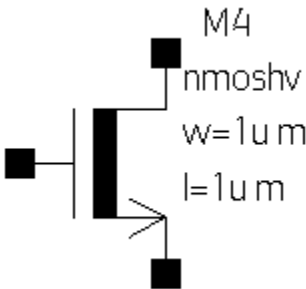
```
m4 d g s b pmos4hv w=1u l=1u
```

Model Statement

```
.model pmos4hv pmos level=54
```

nmoshv

Three-terminal high-voltage N-MOS device.



CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Model Name                | model         | nmos4hv       | X       | -       | X     |
| Width                     | w             | 1u            | X       | -       | X     |
| Length                    | l             | 1u            | X       | -       | X     |
| Drain Diffusion Area      | ad            |               | X       | -       | -     |
| Source Diffusion Area     | as            |               | X       | -       | -     |
| Drain Junction Perimeter  | pd            |               | X       | -       | -     |
| Source Junction Perimeter | ps            |               | X       | -       | -     |
| Drain Diff Resistor Sq    | nrd           |               | X       | -       | -     |
| Source Diff Resistor Sq   | nrs           |               | X       | -       | -     |

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Additional Drain Resistance  | rdc           |               | X       | -       | -     |
| Additional Source Resistance | rsc           |               | X       | -       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | -       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | -       | X     |

## Example

```
mj36 d g s gnd! nmoshv w=1u l=1u
```

## Model Statement

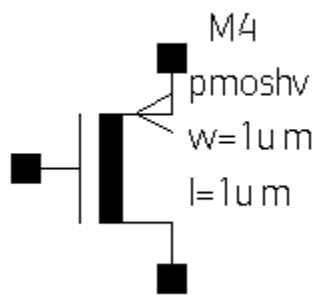
```
.model nmoshv nmos level=54
```

## See Also

- [MOSFETs](#) in the *HSPICE® User Guide: Elements*

## pmoshv

Three-terminal high-voltage P-MOS device. By default, bulk terminal is connected to vdd!.



CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nmos4hv       | X       | -       | X     |
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Drain Diffusion Area         | ad            |               | X       | -       | -     |
| Source Diffusion Area        | as            |               | X       | -       | -     |
| Drain Junction Perimeter     | pd            |               | X       | -       | -     |
| Source Junction Perimeter    | ps            |               | X       | -       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | -       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | -       | -     |
| Additional Drain Resistance  | rdc           |               | X       | -       | -     |
| Additional Source Resistance | rsc           |               | X       | -       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |

| CDF Parameter Prompt        | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|---------------|---------------|---------|---------|-------|
| Bulk-source Initial Voltage | Vbs           |               | X       | -       | -     |
| Temp Difference             | dtemp         |               | X       | -       | -     |
| Source/Drain Selector       | geo           |               | X       | -       | -     |
| Threshold Voltage Shift     | delvto        |               | X       | -       | -     |
| Multiplier                  | m             |               | X       | -       | X     |

Example

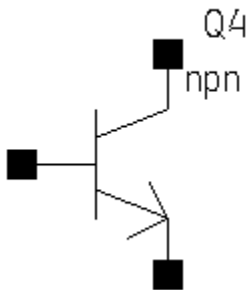
```
m4 d g s vdd! pmoshv w=1u l=1u
```

Model Statement

```
.model pmoshv pmos level=54
```

npn

Three-terminal N-P-N bipolar junction transistor.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Model Name           | model         | npn           | X       | X       | X     |
| Device Area          | area          |               | X       | X       | X     |



| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Base-emitter Voltage      | Vbe           |               | X       | -       | -     |
| Collector-emitter Voltage | Vce           |               | X       | -       | -     |
| Initially Off             | off           | 0             | -       | -       | -     |
| Multiplier                | m             |               | X       | X       | X     |
| Temp Difference           | dtemp         |               | X       | X       | -     |
| Base Area                 | areab         |               | X       | -       | -     |
| Collector Area            | areac         |               | X       | -       | -     |
| Temperature rise          | dtmp          |               | -       | -       | -     |
| Temperature rise          | dtempn        |               | -       | -       | -     |
| Expected operating region | region        |               | -       | X       | -     |

## Example

```
q37 c b e npn area=10u
```

## Model Statement

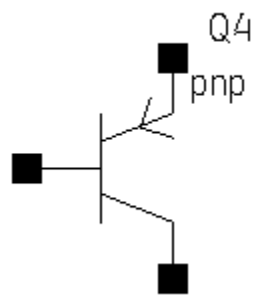
```
.model npn npn (bf=147 is=1.26e-17 br=0.7 ise=0 isc=0
+ ikf=2.7e-3 ikr=7.2e-3 ne=1.5 nc=1.5 tf=2.27e-10 tr=2.27e-9
+ cje=7.71e-14 cjc=5.74e-14 vje=0.88 vjc=0.74 mje=0.42 mjc=0.41
+ cjs=2.82e-13 vjs=0.67 mjs=0.44 vaf=17 var=4.25 rc=87.34
+ rb=1040 re=1.578 xtb=1.58936 eg=1.17 fc=0.5 xti=3.5)
```

## See Also

- [Bipolar Junction Transistor](#) in the *HSPICE® User Guide: Elements*

## pnp

Three-terminal P-N-P bipolar junction transistor.



CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Model Name                | model         | npn           | X       | X       | X     |
| Device Area               | area          |               | X       | X       | X     |
| Base-emitter Voltage      | Vbe           |               | X       | -       | -     |
| Collector-emitter Voltage | Vce           |               | X       | -       | -     |
| Initially Off             | off           | 0             | -       | -       | -     |
| Multiplier                | m             |               | X       | X       | X     |
| Temp Difference           | dtemp         |               | X       | X       | -     |
| Base Area                 | areab         |               | X       | -       | -     |
| Collector Area            | areac         |               | X       | -       | -     |
| Temperature rise          | dtmp          |               | -       | -       | -     |
| Temperature rise          | dtempn        |               | -       | -       | -     |
| Expected operating region | region        |               | -       | X       | -     |

Example

q4 c b e pnp area=10u

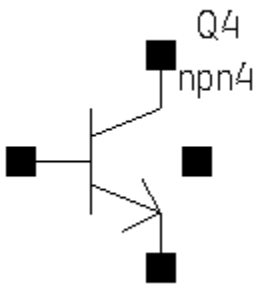
Model Statement

```
.model pnp pnp (bf=92 is=2.8e-16 br=0.5 ise=0 isc=0 ikf=.0003
+ ikr=9e-4 ne=1.5 nc=1.5 tf=3.54e-8 tr=1.06e-7
```

```
+ cje=1.14e-13 cjc=5.8e-13 vje=.74 vjc=.74
+ mje=.41 mjc=.41 cjs=0
+ vaf=16.1 var=5.38 rc=100 rb=50 re=5
+ xtb=1.3 eg=1.17 fc=.5 xti=3.5)
```

npn4

N-P-N bipolar junction transistor with substrate terminal.



CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Model Name                | model         | npn4          | X       | -       | X     |
| Device Area               | area          |               | X       | -       | X     |
| Base-emitter Voltage      | Vbe           |               | X       | -       | -     |
| Collector-emitter Voltage | Vce           |               | X       | -       | -     |
| Initially Off             | off           | 0             | -       | -       | -     |
| Multiplier                | m             |               | X       | -       | X     |
| Temp Difference           | dtemp         |               | X       | -       | -     |
| Base Area                 | areab         |               | X       | -       | -     |
| Collector Area            | areac         |               | X       | -       | -     |

Example

```
q37 c b e s npn area=10u
```

Model Statement

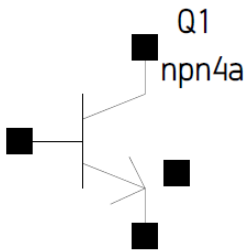
```
.model npn4 npn (bf=147 is=1.26e-17 br=0.7 ise=0 isc=0
+ cje=7.71e-14 cjc=5.74e-14 vje=0.88 vjc=0.74 mje=0.42 mjc=0.41
+ cjs=2.82e-13 vjs=0.67 mjs=0.44 vaf=17 var=4.25 rc=87.34
+ rb=1040 re=1.578 xtb=1.58936 eg=1.17 fc=0.5 xti=3.5)
+ ikf=2.7e-3 ikr=7.2e-3 ne=1.5 nc=1.5 tf=2.27e-10 tr=2.27e-9
```

See Also

- [Bipolar Junction Transistor](#) in the *HSPICE® User Guide: Elements*

npn4a

N-P-N bipolar junction transistor with substrate terminal.



CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Model Name                | model         | npn4a         | X       | -       | X     |
| Device Area               | area          |               | X       | -       | -     |
| Base-emitter Voltage      | Vbe           |               | X       | -       | -     |
| Collector-emitter Voltage | Vce           |               | X       | -       | -     |
| Initially Off             | off           | 0             | -       | -       | -     |
| Multiplier                | m             |               | X       | -       | X     |
| Temp Difference           | dtemp         |               | X       | -       | -     |
| Base Area                 | areab         |               | X       | -       | -     |
| Collector Area            | areac         |               | X       | -       | -     |

Example

```
q1 c b e s npn4a area=10u
```

Model Statement

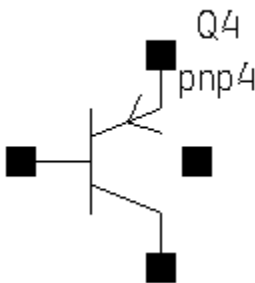
```
.model npn4a npn (bf=147 is=1.26e-17 br=0.7 ise=0 isc=0
+ ikf=2.7e-3 ikr=7.2e-3 ne=1.5 nc=1.5 tf=2.27e-10 tr=2.27e-9
+ cje=7.71e-14 cjc=5.74e-14 vje=0.88 vjc=0.74 mje=0.42 mjc=0.41
+ cjs=2.82e-13 vjs=0.67 mjs=0.44 vaf=17 var=4.25 rc=87.34
+ rb=1040 re=1.578 xtb=1.58936 eg=1.17 fc=0.5 xti=3.5)
```

See Also

- [BJT Model Equations: \(NPN and PNP\)](#) in the *HSPICE® Reference Manual: Device Models*

npn4

P-N-P bipolar junction transistor with substrate terminal.



CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Model Name                | model         | npn4          | X       | -       | X     |
| Device Area               | area          |               | X       | -       | X     |
| Base-emitter Voltage      | Vbe           |               | X       | -       | -     |
| Collector-emitter Voltage | Vce           |               | X       | -       | -     |
| Initially Off             | off           | 0             | -       | -       | -     |
| Multiplier                | m             |               | X       | -       | X     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Temp Difference      | dtemp         |               | X       | -       | -     |
| Base Area            | areab         |               | X       | -       | -     |
| Collector Area       | areac         |               | X       | -       | -     |

Example

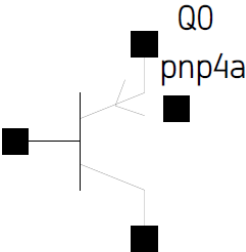
q4 c b e s pnp4 area=10u

Model Statement

```
.model pnp4 pnp (bf=92 is=2.8e-16 br=0.5 ise=0 isc=0 ikf=.0003
+ ikr=9e-4 ne=1.5 nc=1.5 tf=3.54e-8 tr=1.06e-7
+ cje=1.14e-13 cjc=5.8e-13 vje=.74 vjc=.74
+ mje=.41 mjc=.41 cjs=0
+ vaf=16.1 var=5.38 rc=100 rb=50 re=5
+ xtb=1.3 eg=1.17 fc=.5 xti=3.5)
```

pnp4a

P-N-P bipolar junction transistor with substrate terminal.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Model Name           | model         | pnp4a         | X       | -       | X     |
| Device Area          | area          |               | X       | -       | X     |
| Base-emitter Voltage | Vbe           |               | X       | -       | -     |

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Collector-emitter Voltage    | Vce           |               | X       | -       | -     |
| Initially Off                | off           | 0             | X       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | X       | X     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Expected operating region    | region        |               | -       | X       | -     |
| Base Area                    | areab         |               | -       | -       | -     |
| Collector Area               | areac         |               | -       | -       | -     |

## Example

q0 c b e s pnp4a

## Model Statement

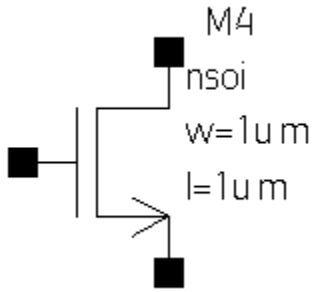
```
.model pnp4a pnp (bf=92 is=2.8e-16 br=0.5 ise=0 isc=0 ikf=.0003
+ ikr=9e-4 ne=1.5 nc=1.5 tf=3.54e-8 tr=1.06e-7
+ cje=1.14e-13 cjc=5.8e-13 vje=.74 vjc=.74
+ mje=.41 mjc=.41 cjs=0
+ vaf=16.1 var=5.38 rc=100 rb=50 re=5
+ xtb=1.3 eg=1.17 fc=.5 xti=3.5)
```

## See Also

- [BJT Model Equations: \(NPN and PNP\)](#) in the *HSPICE® Reference Manual: Device Models*

## nsoi

N-type three-terminal Silicon on Insulator (SOI) device. By default, bulk terminal is connected to gnd! node.



## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nsoi          | X       | -       | X     |
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Drain Diffusion Area         | ad            |               | X       | -       | -     |
| Source Diffusion Area        | as            |               | X       | -       | -     |
| Drain Junction Perimeter     | pd            |               | X       | -       | -     |
| Source Junction Perimeter    | ps            |               | X       | -       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | -       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | -       | -     |
| Additional Drain Resistance  | rdc           |               | X       | -       | -     |
| Additional Source Resistance | rsc           |               | X       | -       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |



| CDF Parameter Prompt        | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|---------------|---------------|---------|---------|-------|
| Bulk-source Initial Voltage | Vbs           |               | X       | -       | -     |
| Temp Difference             | dtemp         |               | X       | -       | -     |
| Source/Drain Selector       | geo           |               | X       | -       | -     |
| Threshold Voltage Shift     | delvto        |               | X       | -       | -     |
| Multiplier                  | m             |               | X       | -       | X     |

Example

```
m38 d g s gdn! nsoi w=1u l=1u
```

Model Statement

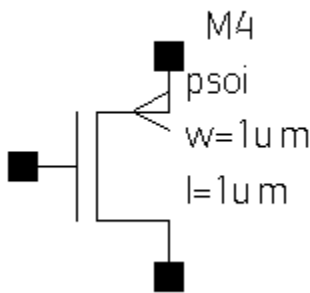
```
.model nsoi nmos level=57
```

See Also

- [LEVEL 57: BSIM3 SOI Model](#) in the *HSPICE® Reference Manual: MOSFET Models*

psoi

P-type three-terminal Silicon on Insulator (SOI) device. By default, bulk terminal is connected to vdd! node.



## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | psoi          | X       | -       | X     |
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Drain Diffusion Area         | ad            |               | X       | -       | -     |
| Source Diffusion Area        | as            |               | X       | -       | -     |
| Drain Junction Perimeter     | pd            |               | X       | -       | -     |
| Source Junction Perimeter    | ps            |               | X       | -       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | -       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | -       | -     |
| Additional Drain Resistance  | rdc           |               | X       | -       | -     |
| Additional Source Resistance | rsc           |               | X       | -       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | -       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | -       | X     |

Example

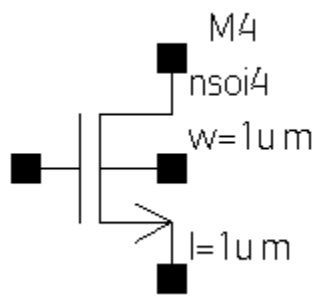
```
m38 d g s vdd! psoi w=1u l=1u
```

Model Statement

```
.model psoi pmos level=57
```

nsoi4

N-type four-terminal Silicon on Insulator (SOI) device.



CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Model Name                | model         | nsoi4         | X       | -       | X     |
| Width                     | w             | 1u            | X       | -       | X     |
| Length                    | l             | 1u            | X       | -       | X     |
| Drain Diffusion Area      | ad            |               | X       | -       | -     |
| Source Diffusion Area     | as            |               | X       | -       | -     |
| Drain Junction Perimeter  | pd            |               | X       | -       | -     |
| Source Junction Perimeter | ps            |               | X       | -       | -     |
| Drain Diff Resistor Sq    | nrd           |               | X       | -       | -     |
| Source Diff Resistor Sq   | nrs           |               | X       | -       | -     |

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Additional Drain Resistance  | rdc           |               | X       | -       | -     |
| Additional Source Resistance | rsc           |               | X       | -       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | -       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | -       | X     |

## Example

```
m38 d g s b nsoi4 w=1u l=1u
```

## Model Statement

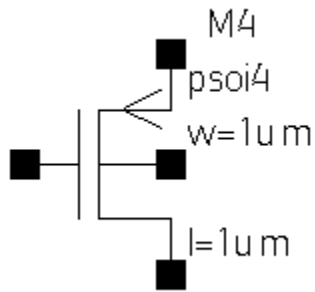
```
.model nsoi4 nmos level=57
```

## See Also

- [LEVEL 57: BSIM3 SOI Model](#) in the *HSPICE® Reference Manual: MOSFET Models*

## psoi4

N-type four-terminal Silicon on Insulator (SOI) device.



## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nsoi4         | X       | -       | X     |
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Drain Diffusion Area         | ad            |               | X       | -       | -     |
| Source Diffusion Area        | as            |               | X       | -       | -     |
| Drain Junction Perimeter     | pd            |               | X       | -       | -     |
| Source Junction Perimeter    | ps            |               | X       | -       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | -       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | -       | -     |
| Additional Drain Resistance  | rdc           |               | X       | -       | -     |
| Additional Source Resistance | rsc           |               | X       | -       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |

| CDF Parameter Prompt        | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|---------------|---------------|---------|---------|-------|
| Bulk-source Initial Voltage | Vbs           |               | X       | -       | -     |
| Temp Difference             | dtemp         |               | X       | -       | -     |
| Source/Drain Selector       | geo           |               | X       | -       | -     |
| Threshold Voltage Shift     | delvto        |               | X       | -       | -     |
| Multiplier                  | m             |               | X       | -       | X     |

Example

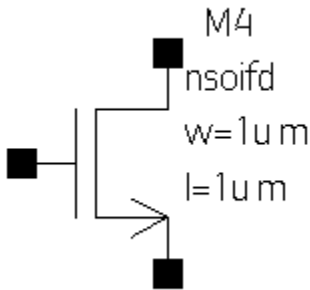
```
m38 d g s b psoi4 w=1u l=1u
```

Model Statement

```
.model psoi4 pmos level=57
```

nsoifd

Three-terminal Fully Depleted Silicon On Insulator, or FD-SOI device. By default, bulk terminal is connected to gnd!.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Model Name           | model         | nsoifd        | X       | -       | X     |

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Drain Diffusion Area         | ad            |               | X       | -       | -     |
| Source Diffusion Area        | as            |               | X       | -       | -     |
| Drain Junction Perimeter     | pd            |               | X       | -       | -     |
| Source Junction Perimeter    | ps            |               | X       | -       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | -       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | -       | -     |
| Additional Drain Resistance  | rdc           |               | X       | -       | -     |
| Additional Source Resistance | rsc           |               | X       | -       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | -       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | -       | X     |

Example

```
m38 d g s gnd! nsoifd w=1u l=1u
```

Model Statement

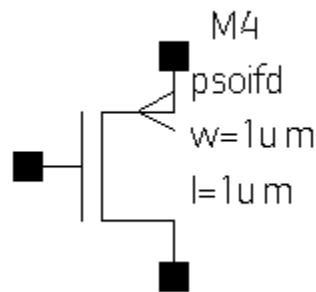
```
.model nsoifd nmos level=59
```

See Also

- [LEVEL 59: BSIM3 SOI FD Model](#) in the *HSPICE® Reference Manual: MOSFET Models*

psoifd

Three-terminal Fully Depleted Silicon On Insulator, or FD-SOI device. By default, bulk terminal is connected to vdd!.



CDF Parameters

| CDF Parameter Prompt     | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------|---------------|---------------|---------|---------|-------|
| Model Name               | model         | nsoifd        | X       | -       | X     |
| Width                    | w             | 1u            | X       | -       | X     |
| Length                   | l             | 1u            | X       | -       | X     |
| Drain Diffusion Area     | ad            |               | X       | -       | -     |
| Source Diffusion Area    | as            |               | X       | -       | -     |
| Drain Junction Perimeter | pd            |               | X       | -       | -     |



| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Source Junction Perimeter    | ps            |               | X       | -       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | -       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | -       | -     |
| Additional Drain Resistance  | rdc           |               | X       | -       | -     |
| Additional Source Resistance | rsc           |               | X       | -       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | -       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | -       | X     |

## Example

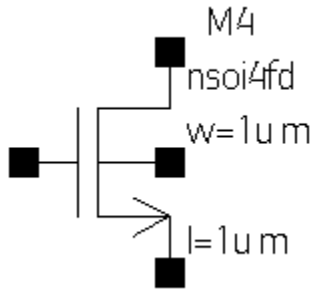
```
m38 d g s vdd! nsoifd w=1u l=1u
```

## Model Statement

```
.model psoifd pmos level=59
```

## nsoifd4

Four-terminal Fully Depleted Silicon On Insulator, or FD-SOI device.



## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nsoifd        | X       | -       | X     |
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Drain Diffusion Area         | ad            |               | X       | -       | -     |
| Source Diffusion Area        | as            |               | X       | -       | -     |
| Drain Junction Perimeter     | pd            |               | X       | -       | -     |
| Source Junction Perimeter    | ps            |               | X       | -       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | -       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | -       | -     |
| Additional Drain Resistance  | rdc           |               | X       | -       | -     |
| Additional Source Resistance | rsc           |               | X       | -       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |

| CDF Parameter Prompt        | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|---------------|---------------|---------|---------|-------|
| Bulk-source Initial Voltage | Vbs           |               | X       | -       | -     |
| Temp Difference             | dtemp         |               | X       | -       | -     |
| Source/Drain Selector       | geo           |               | X       | -       | -     |
| Threshold Voltage Shift     | delvto        |               | X       | -       | -     |
| Multiplier                  | m             |               | X       | -       | X     |

Example

```
m38 d g s b nsoi4fd w=1u l=1u
```

Model Statement

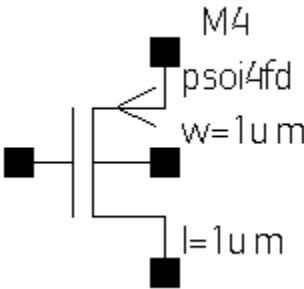
```
.model nsoi4fd nmos level=59
```

See Also

- [LEVEL 59: BSIM3 SOI FD Model](#) in the *HSPICE® Reference Manual: MOSFET Models*

psoifd4

Four-terminal Fully Depleted Silicon On Insulator, or FD-SOI device.



## CDF Parameters

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Model Name                   | model         | nsoifd        | X       | -       | X     |
| Width                        | w             | 1u            | X       | -       | X     |
| Length                       | l             | 1u            | X       | -       | X     |
| Drain Diffusion Area         | ad            |               | X       | -       | -     |
| Source Diffusion Area        | as            |               | X       | -       | -     |
| Drain Junction Perimeter     | pd            |               | X       | -       | -     |
| Source Junction Perimeter    | ps            |               | X       | -       | -     |
| Drain Diff Resistor Sq       | nrd           |               | X       | -       | -     |
| Source Diff Resistor Sq      | nrs           |               | X       | -       | -     |
| Additional Drain Resistance  | rdc           |               | X       | -       | -     |
| Additional Source Resistance | rsc           |               | X       | -       | -     |
| Initially Off                | off           | 0             | -       | -       | -     |
| Drain-source Initial Voltage | Vds           |               | X       | -       | -     |
| Gate-source Initial Voltage  | Vgs           |               | X       | -       | -     |
| Bulk-source Initial Voltage  | Vbs           |               | X       | -       | -     |
| Temp Difference              | dtemp         |               | X       | -       | -     |
| Source/Drain Selector        | geo           |               | X       | -       | -     |
| Threshold Voltage Shift      | delvto        |               | X       | -       | -     |
| Multiplier                   | m             |               | X       | -       | X     |

Example

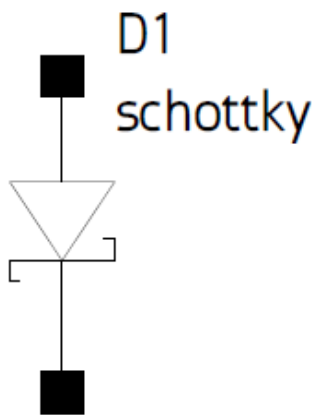
```
m38 d g s b psoi4fd w=1u l=1u
```

Model Statement

```
.model psoi4fd pmos level=59
```

schottky

Schottky barrier diode element.



CDF Parameters

| CDF Parameter Prompt  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------|---------------|---------------|---------|---------|-------|
| Model Name            | model         | schottky      | X       | X       | -     |
| Device Area           | area          |               | X       | X       | X     |
| Initial Diode Voltage | Vd            |               | X       | -       | -     |
| Periphery of Junction | pj            |               | X       | X       | X     |
| Width of Polycap      | wp            |               | X       | X       | -     |
| Length of Polycap     | lp            |               | X       | X       | -     |
| Width of Metalcap     | wm            |               | X       | X       | -     |
| Length of Metalcap    | lm            |               | X       | X       | -     |
| Width                 | w             |               | X       | X       | -     |

| CDF Parameter Prompt   | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------|---------------|---------------|---------|---------|-------|
| Length                 | l             | 0             | X       | X       | -     |
| Initially Off          | off           |               | -       | -       | -     |
| Multiplier             | m             |               | X       | X       | X     |
| Device Temperature     | dtemp         |               | X       | X       | -     |
| Amplitude scale factor | scale         |               | -       | X       | -     |

Example

```
dl p n schottky
```

Model Statement

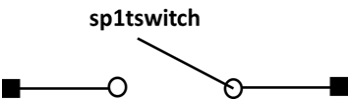
```
.MODEL schottky D
+ LEVEL=1 XP =0.0 EG =1.1
+ XOI =0.0 XOM =0.0 XM =0.0
+ WP =0.0 WM =0.0 LP =0.0
+ LM =0.0 AF =1.0 JSW =0.0
+ PB =0.65 PHP =0.8 M =0.2994
+ EG =0.69
+ FC =0.95 FCS =0.4 MJSW=0.5
+ TT =2.446e-9 BV =4.65 RS =19
+ IS =1.485e-11 CJO =1.09e-9 CJP =0.0
+ PJ =0.0 N =1.615 IK =0.0
+ IKR =1.100e-2 IBV =2.00e-2
```

See Also

- [Diode Types](#) in the *HSPICE® Reference Manual: Device Models*

sp1tswitch

Ideal switch with 1 and 2 positions.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter    | Default Value | hspiceD | spectre | auCdl |
|----------------------|------------------|---------------|---------|---------|-------|
| Offset voltage       | offset           |               | X       | X       | -     |
| Switch position      | position         | 0             | X       | X       | -     |
| dc_position          | dc_position      | -1            | X       | -       | -     |
| ac_position          | ac_position      | -1            | X       | -       | -     |
| ic_position          | ic_position      | -1            | X       | -       | -     |
| tran_position        | tran_position    | -1            | X       | -       | -     |
| DC position          | dcPosition       | 0             | -       | X       | -     |
| AC position          | acPosition       | 0             | -       | X       | -     |
| IC position          | icPosition       | 0             | -       | X       | -     |
| Tran position        | tranPosition     | 0             | -       | X       | -     |
| Switch position      | position_str     | 0             | -       | X       | -     |
| DC position          | dcPosition_str   | 0             | -       | X       | -     |
| AC position          | acPosition_str   | 0             | -       | X       | -     |
| IC position          | icPosition_str   | 0             | -       | X       | -     |
| Tran position        | tranPosition_str | 0             | -       | X       | -     |
| Multiplier           | m                |               | X       | X       | -     |
| Parameter Type       | paramType        | cyclic        | -       | X       | -     |

## Example

```

xsl t0 t1 t2 sp2tswitch offset=0
x2 t1 t0 spltswitch offset=0
.hdl
'<installation_dir>/auxx/samples/analogLib/sp2tswitch/veriloga/veriloga.
va'
.hdl
'<installation_dir>/auxx/samples/analogLib/spltswitch/veriloga/veriloga.
va'

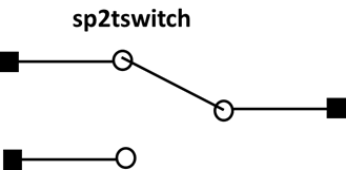
```

## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the `dbAnalogLibSpltswitchVersion` Tcl preference before the analogLib library is loaded.

### sp2tswitch

Ideal switch with 1 and 2 positions.



### CDF Parameters

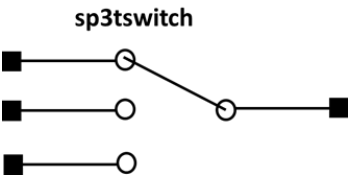
| CDF Parameter Prompt | CDF Parameter    | Default Value | hspiceD | spectre | auCdl |
|----------------------|------------------|---------------|---------|---------|-------|
| Offset voltage       | offset           |               | X       | X       | -     |
| Switch position      | position         | 0             | X       | X       | -     |
| dc_position          | dc_position      | -1            | X       | -       | -     |
| ac_position          | ac_position      | -1            | X       | -       | -     |
| ic_position          | ic_position      | -1            | X       | -       | -     |
| tran_position        | tran_position    | -1            | X       | -       | -     |
| DC position          | dcPosition       | 0             | -       | X       | -     |
| AC position          | acPosition       | 0             | -       | X       | -     |
| IC position          | icPosition       | 0             | -       | X       | -     |
| Tran position        | tranPosition     | 0             | -       | X       | -     |
| Switch position      | position_str     | 0             | -       | X       | -     |
| DC position          | dcPosition_str   | 0             | -       | X       | -     |
| AC position          | acPosition_str   | 0             | -       | X       | -     |
| IC position          | icPosition_str   | 0             | -       | X       | -     |
| Tran position        | tranPosition_str | 0             | -       | X       | -     |



|  |           |        |   |   |   |
|--|-----------|--------|---|---|---|
|  | m         |        | - | X | - |
|  | paramType | cyclic | - | X | - |

sp3tswitch

Voltage-controlled switch.



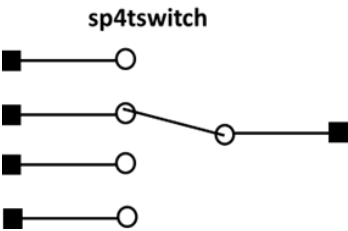
CDF Parameters

| CDF Parameter Prompt | CDF Parameter    | Default Value | hspiceD | spectre | auCdl |
|----------------------|------------------|---------------|---------|---------|-------|
| Offset voltage       | offset           |               | X       | X       | -     |
| Switch position      | position         | 0             | X       | X       | -     |
| dc_position          | dc_position      | -1            | X       | -       | -     |
| ac_position          | ac_position      | -1            | X       | -       | -     |
| ic_position          | ic_position      | -1            | X       | -       | -     |
| tran_position        | tran_position    | -1            | X       | -       | -     |
| DC position          | dcPosition       | 0             | -       | X       | -     |
| AC position          | acPosition       | 0             | -       | X       | -     |
| IC position          | icPosition       | 0             | -       | X       | -     |
| Tran position        | tranPosition     | 0             | -       | X       | -     |
| Switch position      | position_str     | 0             | -       | X       | -     |
| DC position          | dcPosition_str   | 0             | -       | X       | -     |
| AC position          | acPosition_str   | 0             | -       | X       | -     |
| IC position          | icPosition_str   | 0             | -       | X       | -     |
| Tran position        | tranPosition_str | 0             | -       | X       | -     |

|  |           |        |   |   |   |
|--|-----------|--------|---|---|---|
|  | m         |        | - | X | - |
|  | paramType | cyclic | - | X | - |

sp4tswitch

Voltage-controlled switch.



CDF Parameters

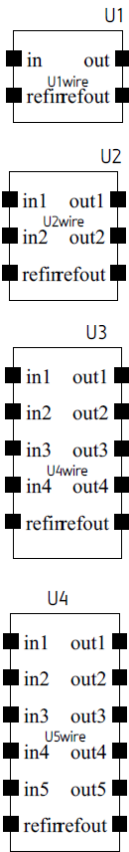
| CDF Parameter Prompt | CDF Parameter  | Default Value | hspiceD | spectre | auCdl |
|----------------------|----------------|---------------|---------|---------|-------|
| Offset voltage       | offset         |               | X       | X       | -     |
| Switch position      | position       | 0             | X       | X       | -     |
| dc_position          | dc_position    | -1            | X       | -       | -     |
| ac_position          | ac_position    | -1            | X       | -       | -     |
| ic_position          | ic_position    | -1            | X       | -       | -     |
| tran_position        | tran_position  | -1            | X       | -       | -     |
| DC position          | dcPosition     | 0             | -       | X       | -     |
| AC position          | acPosition     | 0             | -       | X       | -     |
| IC position          | icPosition     | 0             | -       | X       | -     |
| Tran position        | tranPosition   | 0             | -       | X       | -     |
| Switch position      | position_str   | 0             | -       | X       | -     |
| DC position          | dcPosition_str | 0             | -       | X       | -     |
| AC position          | acPosition_str | 0             | -       | X       | -     |
| IC position          | icPosition_str | 0             | -       | X       | -     |

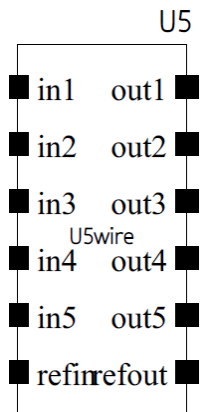
|               |                  |        |   |   |   |
|---------------|------------------|--------|---|---|---|
| Tran position | tranPosition_str | 0      | - | X | - |
|               | m                |        | - | X | - |
|               | paramType        | cyclic | - | X | - |

**u1wire, u2wire, u3wire, u4wire, u5wire**

U-element is useful for modeling lossy non-ideal transmission lines in terms of lumped-element equivalent circuits.

The U-element models single and coupled lossy transmission lines for various planar, coaxial, and twinlead structures. When a U-element is included in your netlist, the simulation creates an internal network of R, L, C, and G-elements to represent up to five lines and their coupling capacitances and inductances.





CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Model Name           | model         |               | -       | -       | -     |
| Length               | l             |               | X       | -       | -     |
| Element Lumps        | lumps         |               | X       | -       | -     |

Example

```
u1 in refin out refout ustrip
u2 in1 in2 refin out1 out2 refout ustrip
u3 in1 in2 in3 in4 refin out1 out2 out3 out4 refout ustrip
u4 in1 in2 in3 in4 in5 refin out1 out2 out3 out4 out5 refout ustrip
u5 in1 in2 in3 in4 in5 refin out1 out2 out3 out4 out5 refout usrip
l=0.178 lumps=3
```

Model Statement

```
.Model USTRIP U LEVEL=3 PLev=1 Elev=1 Dlev=2 Nl=3 Ht=381u
+ Wd=305u Th=25u Sp=102u Ts=838u Kd=4.7
.Probe v(13) v(7) v(8) v(6)
.End
```

See Also

- [Lumped Transmission Line -- UModel](#) in the *HSPICE® User Guide: Elements*

---

## Independent Sources

The following topics describe:

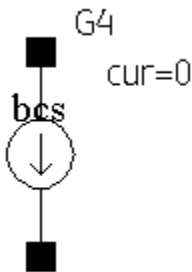
- [bcs](#)
- [bvs](#)
- [iam](#)
- [ibit](#)
- [idc](#)
- [iexp](#)
- [ilfsr](#)
- [ipat](#)
- [ipulse](#)
- [ipwl](#)
- [ipwlf](#)
- [ipwlz](#)
- [iprbs](#)
- [isffm](#)
- [isin](#)
- [isource](#)
- [ivmrf](#)
- [multibit](#)
- [n1port](#)
- [n2port](#)
- [n3port](#)
- [n4port](#)
- [ncs](#)
- [nport](#)
- [nport\\_ref](#)

- [nvs](#)
- [pam](#)
- [pdc](#)
- [pexp](#)
- [plfsr](#)
- [port](#)
- [port3t](#)
- [pprbs](#)
- [ppulse](#)
- [ppwl](#)
- [ppwlf](#)
- [psffm](#)
- [psin](#)
- [vam](#)
- [vbit](#)
- [vcos](#)
- [vdc](#)
- [vexp](#)
- [vgpulse](#)
- [vlfsr](#)
- [vpat](#)
- [vprbs](#)
- [vpulse](#)
- [vpwl](#)
- [vpwlf](#)
- [vpwlz](#)
- [vsffm](#)

- [vsin](#)
- [vsource](#)
- [vwmrf](#)

## bcS

Behavioral current source. Useful if you want to model current source, which can be dependent on bias and/or frequency.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Current Equation     | cur           | 0             | X       | X       | -     |
| Min Current          | min           |               | X       | X       | -     |
| Max Current          | max           |               | X       | X       | -     |
| Scale Factor         | scale         |               | X       | -       | -     |
| Multiplier           | m             |               | X       | X       | -     |

## Example

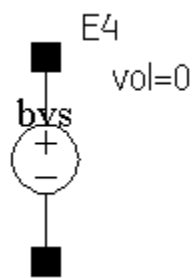
```
v1 1 0 1
r1 1 2 1k
g4 1 2 cur='v(1,2)*0.001'
rout 2 0 1meg
```

## See Also

- [Behavioral Current Source](#) in the *HSPICE® User Guide: Elements*

bvs

Behavioral voltage source. Useful if you want to model voltage source, which can be dependent on bias and/or frequency.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Voltage Equation     | vol           | 0             | X       | X       | -     |
| Min Voltage          | min           |               | X       | X       | -     |
| Max Voltage          | max           |               | X       | X       | -     |
| Scale Factor         | scale         |               | X       | -       | -     |

Example

```
r1 n1 gnd 1
r2 n2 gnd 1
e1 n3 gnd vol='v(n1)+v(n2)'
```

See Also

- [Behavioral Voltage Source](#) in the *HSPICE® User Guide: Elements*

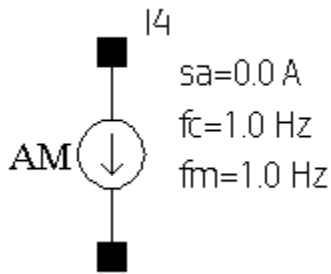
iam

Single-frequency AM current source.

The following expression defines the waveform shape:

*sourcevalue* = *sa* · {*oc* + SIN[2 · π · *fm* · (*Time* – *td*)]} · SIN[2 · π · *fc* · (*Time* – *td*)]





## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | idc           | 0             | X       | -       | -     |
| Source Type          | srcType       | am            | -       | -       | -     |
| Signal Amplitude     | sa            | 0             | X       | -       | -     |
| Carrier Frequency    | fc            | 1             | X       | -       | -     |
| Modulation Frequency | fm            | 1             | X       | -       | -     |
| Offset Constant      | oc            | 0             | X       | -       | -     |
| Delay Time           | td            | 0             | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | -       | -     |
| AC Phase             | acp           |               | X       | -       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1    |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1    |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | -       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |
| SNAC Magnitude       | snacm         |               | X       | -       | -     |
| SNAC Phase           | snacp         |               | X       | -       | -     |
| Multiplier           | m             |               | X       | -       | -     |
| Impedance            | z0            |               | X       | -       | -     |

Example

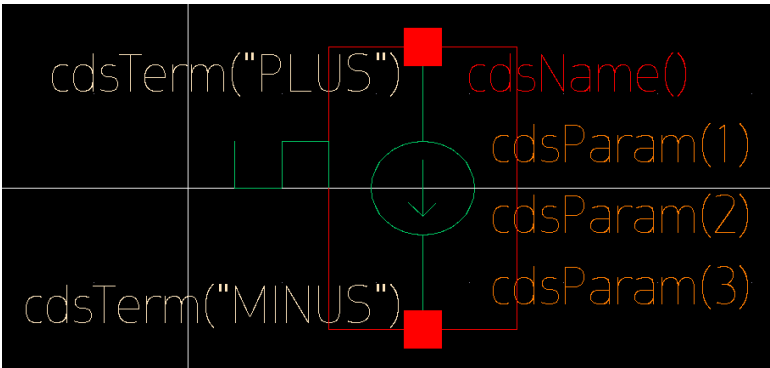
```
i44 plus minus dc=0 am ( 0.85 0.0 10k 1k 0 )
```

See Also

- [Single-frequency AM Source](#) in the *HSPICE® User Guide: Elements*

ibit

Current source that is a pattern source type with various parameters settings.

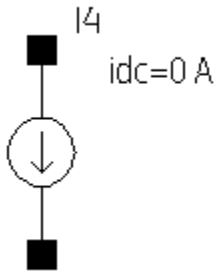


## CDF Parameters

| CDF Parameter Prompt               | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|------------------------------------|-----------------|---------------|---------|---------|-------|
| Pattern Parameter data             | data            |               | -       | X       | -     |
| Pattern Parameter data is variable | data_as_var     | 0             | -       | X       | -     |
| Pattern Parameter rptstart         | rptstart        | 1             | -       | X       | -     |
| Pattern Parameter rpttimes         | rpttimes        | 0             | -       | X       | -     |
| Trigger                            | trigger         |               | -       | X       | -     |
| RJ(rms)                            | rjrms           |               | -       | X       | -     |
| RJ(seed)                           | rjseed          |               | -       | X       | -     |
| Number of periodic jitters         | nPeriodicJitter | 0             | -       | X       | -     |
| PJ%d(amplitude)                    | pjump1-3        |               | -       | X       | -     |
| PJ%d(frequency)                    | pjfreq1-3       |               | -       | X       | -     |
| PJ%d(type)                         | pjtype1-3       | sine          | -       | X       | -     |
| One Value                          | val1            |               | -       | X       | -     |
| Zero Value                         | val0            |               | -       | X       | -     |
| Delay Time                         | td              |               | -       | X       | -     |
| Rise Time                          | tr              | 10f           | -       | X       | -     |
| Fall Time                          | tf              | 10f           | -       | X       | -     |
| Period                             | per             |               | -       | X       | -     |
| Source Type                        | srcType         | bit           | -       | X       | -     |

## idc

Single-frequency AM current source.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | idc           | 0             | X       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | -       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |

| CDF Parameter Prompt            | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|---------------------------------|-------------------|---------------|---------|---------|-------|
| SNAC Magnitude                  | snacm             |               | X       | -       | -     |
| SNAC Phase                      | snacp             |               | X       | -       | -     |
| Multiplier                      | m                 |               | X       | X       | -     |
| Source Type                     | srcType           | dc            | -       | X       | -     |
| PAC magnitude                   | pacm              |               | -       | X       | -     |
| PAC phase                       | pacp              |               | -       | X       | -     |
| Temp coefficient 1              | tc1               |               | -       | X       | -     |
| Temp coefficient 2              | tc2               |               | -       | X       | -     |
| XF magnitude                    | xfm               |               | -       | X       | -     |
| Number of noise/frequency pairs | FNpairs           | 0             | -       | X       | -     |
| Noise file name                 | noisefile         |               | -       | X       | -     |
| Noise file name is variable     | noise_file_as_var | 0             | X       | X       | -     |
| Frequency1-50                   | F1-50             |               | -       | X       | -     |
| Noise1-50                       | N1-50             |               | -       | X       | -     |
| Nominal temperature             | tnom              |               | -       | X       | -     |
| Impedance                       | z0                |               | X       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output | spectre Output |
|-------------------|----------------|----------------|
| noise_file_as_var |                |                |

| CDF Parameter   | hspiceD Output           | spectre Output        |
|-----------------|--------------------------|-----------------------|
| false (default) | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true            | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

Example

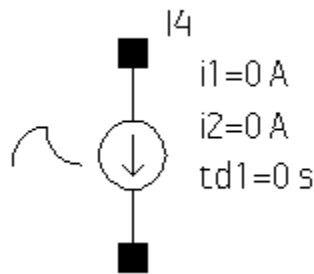
```
i4 plus minus dc=0
```

See Also

- [Independent Source Elements](#) in the *HSPICE® User Guide: Elements*

iexp

Exponential current source.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | idc           | 0             | X       | X       | -     |
| Source Type          | srcType       | exp           |         | X       | -     |
| Current1             | i1            | 0             | X       | X       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Current2             | i2            | 0             | X       | X       | -     |
| Rise Delay Time      | td1           | 0             | X       | X       | -     |
| Fall Delay Time      | td2           |               | X       | X       | -     |
| Rise Time Constant   | tau1          |               | X       | X       | -     |
| Fall Time Constant   | tau2          | 0             | X       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | -       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |
| SNAC Magnitude       | snacm         |               | X       | -       | -     |
| SNAC Phase           | snacp         |               | X       | -       | -     |
| Multiplier           | m             |               | X       | X       | -     |

| CDF Parameter Prompt            | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|---------------------------------|-------------------|---------------|---------|---------|-------|
| PAC magnitude                   | pacm              |               | -       | X       | -     |
| PAC phase                       | pacp              |               | -       | X       | -     |
| Temp coefficient 1              | tc1               |               | -       | X       | -     |
| Temp coefficient 2              | tc2               |               | -       | X       | -     |
| XF magnitude                    | xfm               | -             | -       | X       | -     |
| Number of noise/frequency pairs | FNpairs           | 0             | -       | X       | -     |
| Noise file name                 | noisefile         |               | -       | X       | -     |
| Noise file name is variable     | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                   | F1-50             |               | -       | X       | -     |
| Noise1-50                       | N1-50             |               | -       | X       | -     |
| Nominal temperature             | tnom              |               | -       | X       | -     |
| Delay time                      | td                |               |         | X       | -     |
| Impedance                       | z0                |               | X       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |



- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NoiseFile)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

## Example

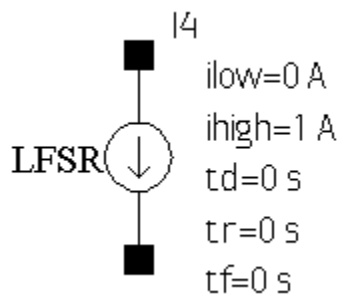
```
i4 n1 n2 dc=0 exp ( 0 50u 0 20n 0 30n )
```

## See Also

- [Exponential Source Function](#) in the *HSPICE® User Guide: Elements*

## ilfsr

Pseudo Random Bit Generator Source (PRBS) function as an independent current source. The PRBS source uses a Linear Feedback Shift Register (LFSR) to generate a pseudo random bit sequence. HSPICE supports eight-bit data byte conversion to a 10-bit transmission character-8B/10B encoding for the LFSR keyword.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | idc           | 0             | X       | X       | -     |
| Source Type          | srcType       | lfsr          |         | X       | -     |
| Current Low          | ilow          | 0             | X       | X       | -     |
| Current High         | ihigh         | 1             | X       | X       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Delay Time           | td            | 0             | X       | X       | -     |
| Rise Time            | tr            | 0             | X       | X       | -     |
| Fall Time            | tf            | 0             | X       | X       | -     |
| Bit Rate             | rate          | 1M            | X       | -       | -     |
| Seed                 | seed          | 1             | X       | -       | -     |
| Taps                 | taps          | [1,2]         | X       | -       | -     |
| Output Resistance    | rout          |               | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | -       | -     |
| AC Phase             | acp           |               | X       | -       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | -       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |
| SNAC Magnitude       | snacm         |               | X       | -       | -     |

| CDF Parameter Prompt | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|----------------------|-----------------|---------------|---------|---------|-------|
| SNAC Phase           | snacp           |               | X       | -       | -     |
| PAM4 modulation      | pam4_modulation | none          | X       | -       | -     |
| PAM4 mapping         | pam4_mapping    |               | X       | -       | -     |
| Impedance            | z0              |               | X       | -       | -     |

## Example

The following example shows the pattern source connected between nodes n1 and n2.

Example of LFSR, output is 1100011111001101:

```
i4 n1 n2 dc=0 lfsr ( 0 50u 0 1n 1n 10meg 3 [2,5] rout=10 )
```

Where:

- The output low voltage is 0, and the output high voltage is 1 v.
- The delay time is 0 ms.
- The rise and fall times are each 1 ns.
- The bit rate is 10 meg bits/s.
- The seed is 3 (bits: 00011).
- The taps are at the output of the 2nd and 5th shift registers and are xor'd together as the input to the first shift register.
- The output resistance is 10 ohms.

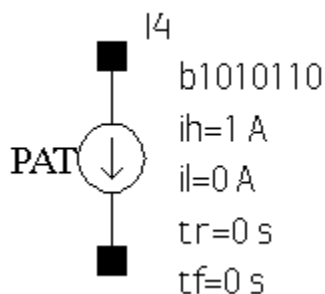
## See Also

- [Pseudo Random Bit Generator Source](#) in the *HSPICE® User Guide: Elements*

## ipat

Pattern source function. It uses four states, '1','0','m', and 'z', which represent the high, low, middle voltage, or current and high impedance state respectively. The series of these four states is a "b-string." HSPICE supports 8-bit data byte conversion to a 10-bit transmission character-8B/10B encoding, 64-bit data conversion to a 66-bit transmission

character-64B/66B encoding, 128-bit data byte conversion to a 130-bit or 132-bit transmission character-128B/130B or 128B/132B encoding, for the PAT sources.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Voltage           | vdc           | 0             | -       | -       | -     |
| Source Type          | srcType       | pat           | -       | -       | -     |
| Current High         | ih            | 1             | X       | -       | -     |
| Current Low          | il            | 0             | X       | -       | -     |
| Delay Time           | td            | 0             | X       | -       | -     |
| Rise Time            | tr            | 0             | X       | -       | -     |
| Fall Time            | tf            | 0             | X       | -       | -     |
| Sample Time          | tsample       | 1n            | X       | -       | -     |
| Data                 | data          | b1010<br>110  | X       | -       | -     |
| Repeat               | r             |               | X       | -       | -     |
| Repeat Bit           | rb            |               | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | -       | -     |
| AC Phase             | acp           |               | X       | -       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |

| CDF Parameter Prompt | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|----------------------|-----------------|---------------|---------|---------|-------|
| HB Harmonic1-7       | hbharm1-7       | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7       | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7    |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7    |               | X       | -       | -     |
| Open DC Connection   | dcopen          |               | -       | -       | -     |
| DC Resistance        | rdc             |               | X       | -       | -     |
| AC Resistance        | rac             |               | X       | -       | -     |
| HB Resistance        | rhb             |               | X       | -       | -     |
| HBAC Resistance      | rhbac           |               | X       | -       | -     |
| TRAN Resistance      | rtran           |               | X       | -       | -     |
| HBAC Magnitude       | hbacm           |               | X       | -       | -     |
| HBAC Phase           | hbacp           |               | X       | -       | -     |
| SNAC Magnitude       | snacm           |               | X       | -       | -     |
| SNAC Phase           | snacp           |               | X       | -       | -     |
| PAM4 modulation      | pam4_modulation | none          | X       | -       | -     |
| PAM4 mapping         | pam4_mapping    |               | X       | -       | -     |
| Impedance            | z0              |               | X       | -       | -     |

## Example

```
i4 n1 n2 pat ( 50u 10u 1n 1n 5n b1011 r=5 rb=2 )
```

In this example:

- High current is 50 uA
- Low current is 10 uA
- Time delay is 0 n
- Rise time is 1 n

- Fall time is 1 n
- Sample time is 5 n

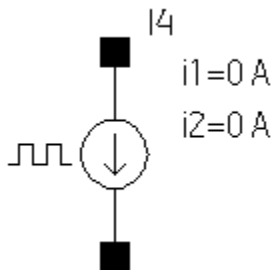
The first b-string is 1011, which repeats once and then repeats from the second bit, which is 0. The pattern is repeated 5 times.

See Also

- [Pattern Source](#) in the *HSPICE® User Guide: Elements*

ipulse

Trapezoidal pulse current source that starts with an initial delay from the beginning of the transient simulation interval to an onset ramp. During the onset ramp, the current changes linearly from its initial value to the pulse plateau value. After the pulse plateau, the current moves linearly along a recovery ramp back to its initial value. The entire pulse repeats, with a period named per, from onset to onset.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | idc           | 0             | X       | X       | -     |
| Source Type          | srcType       | pulse         | -       | X       | -     |
| Current1             | i1            | 0             | X       | X       | -     |
| Current2             | i2            | 0             | X       | X       | -     |
| Delay Time           | td            | 0             | X       | X       | -     |
| Rise Time            | tr            |               | X       | X       | -     |
| Fall Time            | tf            |               | X       | X       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Pulse Width          | pw            |               | X       | X       | -     |
| Period               | per           |               | X       | X       | -     |
| Period Jitter        | perjitter     |               | X       | -       | -     |
| Seed                 | seed          |               | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7      | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | X       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |
| SNAC Magnitude       | snacm         |               | X       | -       | -     |
| SNAC Phase           | snacp         |               | X       | -       | -     |
| Multiplier           | m             |               | X       | X       | -     |
| PAC magnitude        | pacm          |               | -       | X       | -     |

| CDF Parameter Prompt                     | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|-------------------|---------------|---------|---------|-------|
| PAC phase                                | pacp              |               | -       | X       | -     |
| Edge type                                | edgetype          | linear        | -       | X       | -     |
| Temp coefficient 1                       | tc1               |               | -       | X       | -     |
| Temp coefficient 2                       | tc2               |               | -       | X       | -     |
| XF magnitude                             | xfm               |               | -       | X       | -     |
| Number of noise/frequency pairs          | FNpairs           | 0             | -       | X       | -     |
| Noise file name                          | noisefile         |               | -       | X       | -     |
| Noise file name is variable              | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                            | F1-50             |               | -       | X       | -     |
| Noise1-50                                | N1-50             |               | -       | X       | -     |
| Frequency name for fundamental frequency | fundname          |               | -       | X       | -     |
| Nominal temperature                      | tnom              |               | -       | X       | -     |
| Impedance                                | z0                |               | X       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |



- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

Example

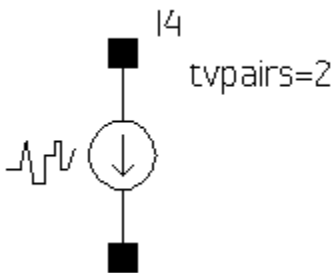
```
i4 plus gate dc=0 pulse ( 0 50u 0 1n 1n 10n 20n )
```

See Also

- [Trapezoidal Pulse Source](#) in the *HSPICE® User Guide: Elements*

ipwl

PWL independent current source.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | idc           | 0             | X       | X       | -     |
| Source Type          | srcType       | pwl           | -       | X       | -     |
| Time Value Pairs     | tvpairs       | 2             | X       | X       | -     |
| Display Pairs count  | dpcount       | 5             | X       | -       | -     |
| Time1-50             | t1-50         |               | X       | X       | -     |
| Current1-50          | i1-50         | 0             | X       | X       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Repeat               | r             |               | X       | -       | -     |
| Repeat Stop Time     | rstop         |               | X       | -       | -     |
| Repeat Stop Value    | stopvalue     |               | X       | -       | -     |
| Repeat Stop Slope    | stopslope     |               | X       | -       | -     |
| Delay Time           | td            | 0             | X       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | -       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |
| SNAC Magnitude       | snacm         |               | X       | -       | -     |
| SNAC Phase           | snacp         |               | X       | -       | -     |
| Multiplier           | m             |               | X       | X       | -     |

| CDF Parameter Prompt                     | CDF Parameter     | Default Value                 | hspiceD | spectre | auCdl |
|------------------------------------------|-------------------|-------------------------------|---------|---------|-------|
| Delay Time                               | delay             | 0                             | -       | X       | -     |
| Offset current                           | io                |                               | -       | X       | -     |
| Amplitude scale factor                   | scale             |                               | -       | X       | -     |
| Number of noise/frequency pairs          | FNpairs           | 0                             | -       | X       | -     |
| Noise file name                          | noisefile         |                               | -       | X       | -     |
| Noise file name is variable              | noise_file_as_var | 0                             | -       | X       | -     |
| Frequency1-50                            | F1-50             |                               | -       | X       | -     |
| Noise1-50                                | N1-50             |                               | -       | X       | -     |
| Frequency name for fundamental frequency | fundname          |                               | -       | X       | -     |
| Nominal temperature                      | tnom              |                               | -       | X       | -     |
| PAC magnitude                            | pacm              |                               | -       | X       | -     |
| PAC phase                                | pacp              |                               | -       | X       | -     |
| PWL period                               | pwlperiod         |                               | -       | X       | -     |
| Edge type                                | edgetype          | linear                        | -       | X       | -     |
| Transition width                         | twidht            |                               | -       | X       | -     |
| Time scaling factor                      | stretch           |                               | -       | X       | -     |
| Temp coefficient 1                       | tc1               |                               | -       | X       | -     |
| Temp coefficient 2                       | tc2               |                               | -       | X       | -     |
| XF magnitude                             | xfm               |                               | -       | X       | -     |
| Impedance                                | z0                |                               | X       | -       | -     |
| PWL                                      | pwl               | cmdPWL<br>Display("i<br>pwl") | -       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output            | spectre Output         |
|-------------------|---------------------------|------------------------|
| noise_file_as_var |                           |                        |
| false (default)   | noise file='NOISEFILE'    | noise file="NOISEFILE" |
| true              | noise file=str(NOISEFILE) | noise file=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

## Example

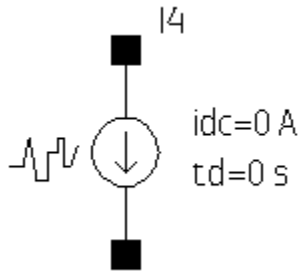
```
i4 n1 n2 dc=0 pwl ( 0 0.0 1n 0.5u 1.2n 0.5u 2n 1.2u )
```

## See Also

- [PWL Source](#) in the *HSPICE® User Guide: Elements*

## ipwlf

PWL source, that accepts a text file containing time and current pairs. The data in the text file can be either a table or a list.



## CDF Parameters

| CDF Parameter Prompt  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------|---------------|---------------|---------|---------|-------|
| PWL File              | fileName      |               | X       | X       |       |
| File name is variable | file_as_var   | 0             | X       | X       | -     |
| DC Current            | idc           | 0             | X       | X       | -     |
| Source Type           | srcType       | pwl           | -       | X       | -     |
| Repeat                | r             |               | X       | -       | -     |
| Repeat Stop Time      | rstop         |               | X       | -       | -     |
| Repeat Stop Value     | stopvalue     |               | X       | -       | -     |
| Repeat Stop Slope     | stopslope     |               | X       | -       | -     |
| Delay Time            | td            | 0             | X       | X       | -     |
| AC Magnitude          | acm           |               | X       | X       | -     |
| AC Phase              | acp           |               | X       | X       | -     |
| # HB Tones            | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7      | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7           | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7        | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7            | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7         | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7         | hbmodtone1-7  |               | X       | -       | -     |

| CDF Parameter Prompt                     | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|-------------------|---------------|---------|---------|-------|
| Open DC Connection                       | dcopen            |               | -       | -       | -     |
| DC Resistance                            | rdc               |               | X       | -       | -     |
| AC Resistance                            | rac               |               | X       | -       | -     |
| HB Resistance                            | rhb               |               | X       | -       | -     |
| HBAC Resistance                          | rhbac             |               | X       | -       | -     |
| TRAN Resistance                          | rtran             |               | X       | -       | -     |
| HBAC Magnitude                           | hbacm             |               | X       | -       | -     |
| HBAC Phase                               | hbacp             |               | X       | -       | -     |
| SNAC Magnitude                           | snacm             |               | X       | -       | -     |
| SNAC Phase                               | snacp             |               | X       | -       | -     |
| Multiplier                               | m                 |               | X       | X       | -     |
| Offset current                           | io                |               | -       | X       |       |
| Amplitude scale factor                   | scale             |               | -       | X       | -     |
| Number of noise/frequency pairs          | FNpairs           | 0             | -       | X       | -     |
| Noise file name                          | noisefile         |               | -       | X       | -     |
| Noise file name is variable              | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                            | F1-50             |               | -       | X       | -     |
| Noise1-50                                | N1-50             |               | -       | X       | -     |
| Frequency name for fundamental frequency | fundname          |               | -       | X       | -     |
| Nominal temperature                      | tnom              |               | -       | X       | -     |
| PAC magnitude                            | pacm              |               | -       | X       | -     |
| PAC phase                                | pacp              |               | -       | X       | -     |
| PWL period                               | pwlperiod         |               | -       | X       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Edge type            | edgetype      | linear        | -       | X       | -     |
| Transition width     | twidth        |               | -       | X       | -     |
| Desired rms value    | rms           |               | -       | x       | -     |
| Time scaling factor  | stretch       |               | -       | X       | -     |
| Temp coefficient 1   | tc1           |               | -       | X       | -     |
| Temp coefficient 2   | tc2           |               | -       | X       | -     |
| XF magnitude         | xfm           |               | -       | X       | -     |
| Impedance            | z0            |               | X       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

**Example**

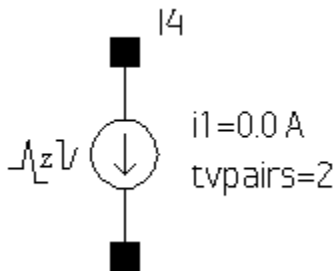
```
i4 n1 n2 dc=0 pw1 ( 0 0.0 1n 0.5u 1.2n 0.5u 2n 1.2u )
```

## See Also

- [PWL Source](#) in the *HSPICE® User Guide: Elements*

## ipwlz

The high Z state of the PWLZ source adds the capability to disconnect the PWL source for time periods marked with the keyword Z.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | idc           | 0             | X       | X       | -     |
| Source Type          | srcType       | pwlz          | -       | X       | -     |
| Time Value Pairs     | tvpairs       |               | X       | X       | -     |
| Time1-50             | t1-50         |               | X       | X       | -     |
| Current1-50          | i1-50         | 0             | X       | X       | -     |
| Z State1-50          | Z1-50         | 0             | X       | X       | -     |
| Repeat               | r             |               | X       | -       | -     |
| Delay Time           | td            | 0             | X       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |



| CDF Parameter Prompt            | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|---------------------------------|-------------------|---------------|---------|---------|-------|
| HB Harmonic1-7                  | hbharm1-7         | 1             | X       | -       | -     |
| HB Tone1-7                      | hbtone1-7         | 1-7           | X       | -       | -     |
| HB ModHarm1-7                   | hbmodharm1-7      |               | X       | -       | -     |
| HB ModTone1-7                   | hbmodtone1-7      |               | X       | -       | -     |
| Open DC Connection              | dcopen            |               | -       | -       | -     |
| DC Resistance                   | rdc               |               | X       | -       | -     |
| AC Resistance                   | rac               |               | X       | -       | -     |
| HB Resistance                   | rhb               |               | X       | -       | -     |
| HBAC Resistance                 | rhbac             |               | X       | -       | -     |
| TRAN Resistance                 | rtran             |               | X       | -       | -     |
| HBAC Magnitude                  | hbacm             |               | X       | -       | -     |
| HBAC Phase                      | hbacp             |               | X       | -       | -     |
| SNAC Magnitude                  | snacm             |               | X       | -       | -     |
| SNAC Phase                      | snacp             |               | X       | -       | -     |
| Multiplier                      | m                 |               | X       | X       | -     |
| Delay Time                      | delay             | 0             | -       | X       | -     |
| Offset current                  | io                |               | -       | X       | -     |
| Amplitude scale factor          | scale             |               | -       | X       | -     |
| Number of noise/frequency pairs | FNpairs           | 0             | -       | X       | -     |
| Noise file name                 | noisefile         |               | -       | X       | -     |
| Noise file name is variable     | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                   | F1-50             |               | -       | X       | -     |
| Noise1-50                       | N1-50             |               | -       | X       | -     |

| CDF Parameter Prompt                     | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|---------------|---------------|---------|---------|-------|
| Frequency name for fundamental frequency | fundname      |               | -       | X       | -     |
| Nominal temperature                      | tnom          |               | -       | X       | -     |
| PAC magnitude                            | pacm          |               | -       | X       | -     |
| PAC phase                                | pacp          |               | -       | X       | -     |
| PWL period                               | pwlperiod     |               | -       | X       | -     |
| Edge type                                | edgetype      | linear        | -       | X       | -     |
| Transition width                         | twidth        |               | -       | X       | -     |
| Time scaling factor                      | stretch       |               | -       | X       | -     |
| Temp coefficient 1                       | tc1           |               | -       | X       | -     |
| Temp coefficient 2                       | tc2           |               | -       | X       | -     |
| XF magnitude                             | xfm           |               | -       | X       | -     |
| Impedance                                | z0            |               | X       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NoiseFile)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

## Example

```
i4 n1 n2 dc=0 pwlz ( 0 0.0 1n z 1.2n 0.5u 2n 1.2u )
```

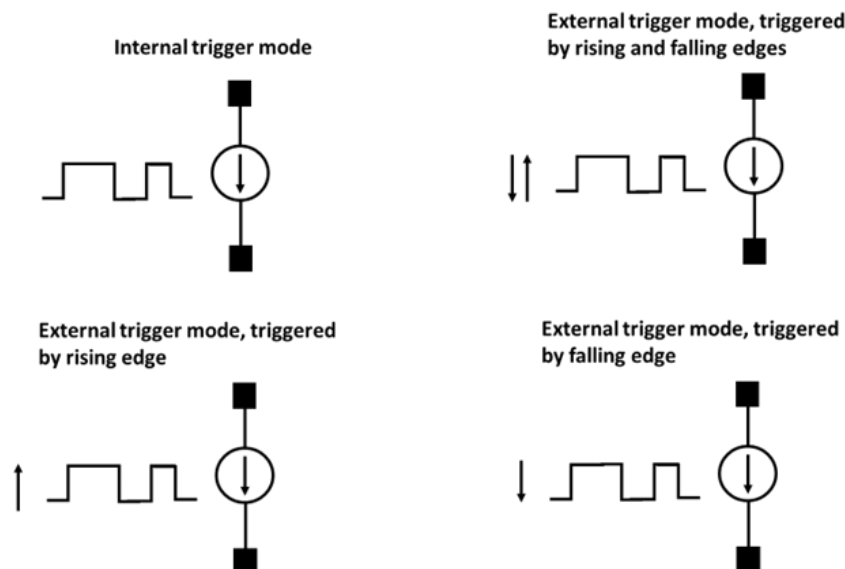
## See Also

- [High Impedance State PWL \(PWLZ\)](#) in the *HSPICE® User Guide: Elements*

## iprbs

Current source with pseudo random bit generator source transient function source.

The iprbs device is a PCell. The following figure shows four vprbs devices that have different trigger modes, which are Internal (on the upper-left), External by both edges (on the upper-right), External by falling edges (on the lower-left) and External by rising edges (on the lower-right). The internal trigger mode has 2 pins (PLUS and MINUS) and other three external trigger modes have 4 pins (P1, M1, P2, and M2). The netlisting procedure writes out their pins correspondingly.



## CDF Parameters

| CDF Parameter Prompt        | CDF Parameter       | Default Value                                          | hspiceD | spectre | auCdl |
|-----------------------------|---------------------|--------------------------------------------------------|---------|---------|-------|
| triggerValue                | triggerValue        | 1                                                      | -       | X       | -     |
| Delay time                  | tdPrbs              |                                                        | -       | X       | -     |
| Zero value                  | val0                |                                                        | -       | X       | -     |
| One value                   | val1                |                                                        | -       | X       | -     |
| Bit period                  | perPrbs             | 1n                                                     | -       | X       | -     |
| Rise time                   | tr                  |                                                        | -       | X       | -     |
| Fall time                   | tf                  |                                                        | -       | X       | -     |
| Transition reference        | transitionreference |                                                        | -       | X       | -     |
| Lower threshold(percent)    | lowerthreshold      |                                                        | -       | X       | -     |
| Upper threshold(percent)    | upperthreshold      |                                                        | -       | X       | -     |
| Edge type                   | edgetype            |                                                        | -       | X       | -     |
| Trigger                     | trigger             | Internal                                               | -       | X       | -     |
| Threshold                   | triggerthreshold    |                                                        | -       | X       | -     |
| LFSR Mode                   | lfsrmode            | PN32                                                   | -       | X       | -     |
| Browse and specify bit file | prbsFileBox         | 0                                                      | -       | X       | -     |
| Bit data file               | prbsFile            |                                                        | -       | X       | -     |
| Seed                        | lfsrseed            | 1 3 5 7 9<br>11 13 15<br>17 19 21<br>23 25 27<br>29 31 | -       | X       | -     |
| Taps                        | lfsrtaps            |                                                        | -       | X       | -     |
| RJ(rms)                     | rjrms               |                                                        | -       | X       | -     |
| RJ(seed)                    | rjseed              |                                                        | -       | X       | -     |

| CDF Parameter Prompt       | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|----------------------------|-----------------|---------------|---------|---------|-------|
| Number of periodic jitters | nPeriodicJitter | 0             | -       | X       | -     |
| PJ1(amplitude)             | pjump1          |               | -       | X       | -     |
| PJ1(frequency)             | pjfreq1         |               | -       | X       | -     |
| PJ1(type)                  | pjtype1         | sine          | -       | X       | -     |
| PJ2(amplitude)             | pjump2          |               | -       | X       | -     |
| PJ2(frequency)             | pjfreq2         |               | -       | X       | -     |
| PJ2(type)                  | pjtype2         | sine          | -       | X       | -     |
| PJ3(amplitude)             | pjump3          |               | -       | X       | -     |
| PJ3(frequency)             | pjfreq3         |               | -       | X       | -     |
| PJ3(type)                  | pjtype3         | sine          | -       | X       | -     |
| Pattern Parameter data     | data            |               | -       | X       | -     |
| Reference Value            | ref             |               | -       | X       | -     |
| Random delay time          | jitter          |               | -       | X       | -     |
| Seed                       | seed            |               | -       | X       | -     |
| Feedback Shift Register    | taps            |               | -       | X       | -     |
| Source type                | srcType         | prbs          | -       | X       | -     |
| Version                    | int             | 1             | -       | -       | -     |

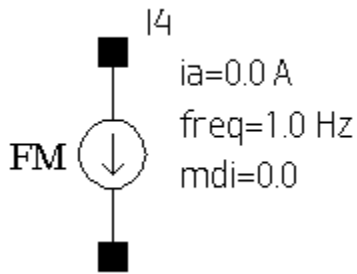
## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibIprbsVersion](#) Tcl preference before the analogLib library is loaded.

---

### isffm

Single-frequency FM current source.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | idc           | 0             | X       | -       | -     |
| Source Type          | srcType       | sffm          | -       | -       | -     |
| Output Amplitude     | ia            | 0             | X       | -       | -     |
| Offset Current       | io            | 0             | X       | -       | -     |
| Carrier Frequency    | freq          | 1             | X       | -       | -     |
| Modulation Index     | mdi           | 0             | X       | -       | -     |
| Signal Frequency     | fs            |               | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | -       | -     |
| AC Phase             | acp           |               | X       | -       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | -       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |
| SNAC Magnitude       | snacm         |               | X       | -       | -     |
| SNAC Phase           | snacp         |               | X       | -       | -     |
| Multiplier           | m             |               | X       | -       | -     |
| Impedance            | z0            |               | X       | -       | -     |

Example

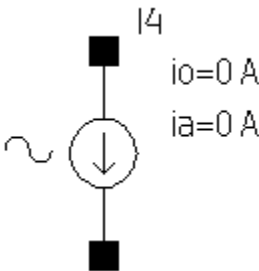
```
i4 n1 gnd! dc=25u sffm ( 0.0 30u 20k 10 5k )
```

See Also

- [Single-frequency FM Source](#) in the *HSPICE® User Guide: Elements*

isin

Sinusoidal current source that can be either a constant sinusoid or a damped, that produces a dying exponential with a sine wave.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | idc           | 0             | X       | X       | -     |
| Source Type          | srcType       | pwlz          | -       | X       | -     |
| Current Offset       | io            |               | X       | X       | -     |
| Current Amplitude    | ia            |               | X       | X       | -     |
| Frequency            | freq          |               | X       | X       | -     |
| Delay Time           | td            |               | X       | X       | -     |
| Damping Factor       | theta         |               | X       | X       | -     |
| Phase for sinusoid   | sinephase     |               | X       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | -       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |



| CDF Parameter Prompt            | CDF Parameter           | Default Value | hspiceD | spectre | auCdl |
|---------------------------------|-------------------------|---------------|---------|---------|-------|
| HBAC Phase                      | hbacp                   |               | X       | -       | -     |
| SNAC Magnitude                  | snacm                   |               | X       | -       | -     |
| SNAC Phase                      | snacp                   |               | X       | -       | -     |
| Multiplier                      | m                       |               | X       | X       | -     |
| Phase shift                     | phase                   |               | -       | X       |       |
| Amplitude 2                     | ia2                     |               | -       | X       | -     |
| Amp modulation frequency        | ammodfreq               |               | -       | X       | -     |
| Amp modulation index            | ammodindex              |               | -       | X       | -     |
| Amp modulation phase            | ammodphase              |               | -       | X       | -     |
| Number of FM Files              | filenums                |               | -       | X       | -     |
| Name of FM File1                | fmmodfile1              |               | -       | X       | -     |
| FM file1 name is variable       | fmmodfile1_as_0<br>_var |               | -       | X       | -     |
| Name of FM file 2               | fmmodfile2              |               | -       | X       | -     |
| FM file2 name is variable       | fmmodfile2_as_0<br>_var |               | -       | X       | -     |
| FM frequency                    | fmmodfreq               |               | -       | X       | -     |
| FM index                        | fmmodindex              |               | -       | X       | -     |
| Frequency 2                     | freq2                   |               | -       | X       | -     |
| Number of noise/frequency pairs | FNpairs                 | 0             | -       | X       | -     |
| Noise file name                 | noisefile               |               | -       | X       | -     |
| Noise file name is variable     | noise_file_as_0<br>var  |               | -       | X       | -     |
| Frequency1-50                   | F1-50                   |               | -       | X       | -     |
| Noise1-50                       | N1-50                   |               | -       | X       | -     |

| CDF Parameter Prompt                     | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|---------------|---------------|---------|---------|-------|
| Frequency name for fundamental frequency | fundname      | -             | -       | X       | -     |
| Second frequency name                    | fundname2     | -             | -       | X       | -     |
| Noise temperature                        | noisetemp     | -             | -       | X       | -     |
| Nominal temperature                      | tnom          | -             | -       | X       | -     |
| PAC magnitude                            | pacm          | -             | -       | X       | -     |
| PAC phase                                | pacp          | -             | -       | X       | -     |
| Phase for sinusoid 2                     | sinephase2    | -             | -       | X       | -     |
| Temp coefficient 1                       | tc1           | -             | -       | X       | -     |
| Temp coefficient 2                       | tc2           | -             | -       | X       | -     |
| XF magnitude                             | xfm           | -             | -       | X       | -     |
| Impedance                                | z0            | X             | -       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output            | spectre Output         |
|-------------------|---------------------------|------------------------|
| noise_file_as_var |                           |                        |
| false (default)   | noise file='NOISEFILE'    | noise file="NOISEFILE" |
| true              | noise file=str(NOISEFILE) | noise file=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

## Example

```
i4 n1 0 dc=0 sin ( 0 100u 100Meg 1n 1e+6 0 )
```

This damped sinusoidal source connects between nodes n1 and 0. In this waveform:

- Peak value is 100 uA
- Offset is 0 A
- Frequency is 100 MHz
- Time delay is 1 ns
- Damping factor is 1e+6
- Phase delay is 0 deg

## See Also

- [Sinusoidal Source Function](#) in the *HSPICE® User Guide: Elements*

## Controlling Version Number for Legacy Designs

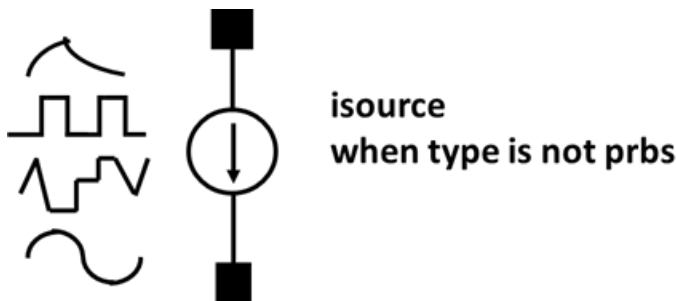
To control the version number for legacy designs, create and use the `dbAnalogLibIsinVersion` Tcl preference before the analogLib library is loaded.

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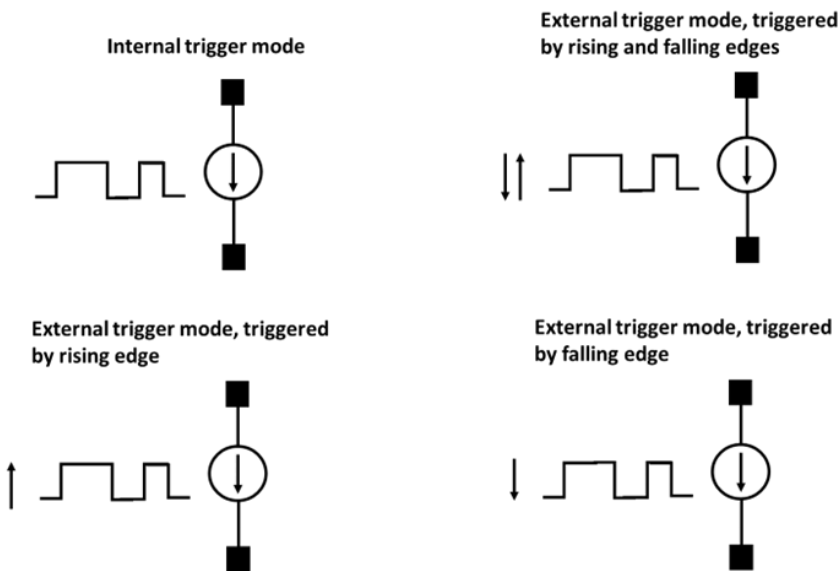
## isource

Current source with various parameters settings.

The isource device is a PCell. Its PCell behavior is controlled by its source type. For source type other than prbs, it is a 2-pin device and its symbol looks as follows:



For prbs source type, its PCell behavior is affected by the trigger parameter. The following figure shows four isource devices that have different trigger modes, which are Internal (on the upper-left), External by rising edges (on the upper-right), External by falling edges (on the lower-left) and External by both edges (on the lower-right). The internal trigger mode has 2 pins (PLUS and MINUS) and other three external trigger modes have 4 pins (P1, M1, P2, and M2). The netlisting procedure writes out their pins correspondingly.



The bit source type is mainly supported for Spectre netlisting.

### CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | idc           | 0             | X       | X       | -     |
| Source Type          | srcType       | dc            | X       | X       | -     |

| CDF Parameter Prompt  | CDF Parameter  | Default Value | hspiceD | spectre | auCdl |
|-----------------------|----------------|---------------|---------|---------|-------|
| Frequency name 1      | fundname       |               | -       | X       | -     |
| Waveform entry method | pwlEntryMethod | File          | X       | X       | -     |
| File name             | fileName       |               | X       | X       | -     |
| File name is variable | file_as_var    | 0             | X       | X       | -     |
| Time Value Pairs      | tvpairs        | 0             | X       | X       | -     |
| Time1                 | t1             |               | X       | X       | -     |
| Current1              | i1             |               | X       | X       | -     |
| Time2                 | t2             |               | X       | X       | -     |
| Current2              | i2             |               | X       | X       | -     |
| Zero value            | val0           |               | X       | X       | -     |
| One value             | val1           |               | X       | X       | -     |
| Rise Delay Time       | td1            | 0             | X       | X       | -     |
| Fall Delay Time       | td2            | 0             | X       | X       | -     |
| Rise Time Constant    | tau1           | 0             | X       | X       | -     |
| Fall Time Constant    | tau2           | 0             | X       | X       | -     |
| Exp period            | expperiod      |               | -       | x       | -     |
| Rise Time             | tr             |               | X       | X       | -     |
| Fall Time             | tf             |               | X       | X       | -     |
| Pulse Width           | pw             |               | X       | X       | -     |
| Period                | per            |               | X       | X       | -     |
| Time3-50              | t3-50          |               | -       | X       | -     |
| Current3-50           | i3-50          |               | -       | X       | -     |
| Repeat                | r              |               | X       | -       | -     |
| Repeat Stop Time      | rstop          |               | X       | -       | -     |
| Repeat Stop Value     | stopvalue      |               | X       | -       | -     |

| CDF Parameter Prompt    | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-------------------------|---------------|---------------|---------|---------|-------|
| Repeat Stop Slope       | stopslope     |               | X       | -       | -     |
| Current Offset          | io            |               | X       | -       | -     |
| Frequency               | freq          |               | X       | X       | -     |
| Current Amplitude       | ia            |               | X       | X       | -     |
| Phase for sinusoid      | sinephase     | 0             | X       | X       | -     |
| Sine DC level           | sinedc        |               | -       | X       | -     |
| Delay Time              | td            | 0             | X       | X       | -     |
| Damping Factor          | theta         | 0             | X       | X       | -     |
| # HB Tones              | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7        | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7             | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7          | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7              | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7           | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7           | hbmodtone1-7  |               | X       | -       | -     |
| Power Switch            | power         | V             | X       | -       | -     |
| Impedance               | z0            |               | X       | -       | -     |
| Open DC Connection      | dcopen        |               | -       | -       | -     |
| Display second sinusoid | numofsines    | 0             | -       | X       | -     |
| Frequency name 2        | fundname2     |               | -       | X       | -     |
| Frequency 2             | freq2         |               | -       | X       | -     |
| Amplitude 2 (lpk)       | ia2           |               | -       | X       | -     |
| Phase for sinusoid 2    | sinephase2    |               | -       | X       | -     |
| Display multi sinusoid  | vecofsines    | 0             | -       | X       | -     |

| CDF Parameter Prompt       | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|----------------------------|-------------------|---------------|---------|---------|-------|
| Number of frequencies      | freqvec           | 0             | X       | X       | -     |
| Sinusoid frequency 1-9     | freqv1-9          |               | -       | X       |       |
| Sinusoid amplitude 1-9     | vav1-9            |               | -       | -       | -     |
| Sinusoid phase 1-9         | phasev1-9         |               | -       | X       | -     |
| Sinusoid max harmonic 1-9  | harmv1-9          |               | -       | X       | -     |
| Display modulation params  | modulation        | 0             | -       | X       | -     |
| Amp modulation frequency 1 | ammodfreq         |               | -       | X       | -     |
| Amp modulation index 1     | ammodindex        |               | -       | X       | -     |
| Amp modulation phase 1     | ammodphase        |               | -       | X       | -     |
| Number of FM Files         | filenums          | none          | -       | X       | -     |
| Name of FM File1           | fmmodfile1        |               | -       | X       | -     |
| FM file1 name is variable  | fmmodfile1_as_var | 0             | -       | X       | -     |
| Name of FM file 2          | fmmodfile2        |               | -       | X       | -     |
| FM file2 name is variable  | fmmodfile2_as_var | 0             | -       | X       | -     |
| FM index 1                 | fmmodindex        |               | -       | X       | -     |
| FM frequency 1             | fmmodfreq         |               | -       | X       | -     |
| IQ modulation I File       | imodfile          |               | -       | X       | -     |
| I File name is variable    | imodfile_as_var   | 0             | -       | X       | -     |
| IQ modulation Q File       | qmodfile          |               | -       | X       | -     |

| CDF Parameter Prompt               | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|------------------------------------|-----------------|---------------|---------|---------|-------|
| Q File name is variable            | qmodfile_as_var | 0             | -       | X       | -     |
| DC offset                          | offset          |               | -       | X       | -     |
| Pattern Parameter data             | data            |               | -       | X       | -     |
| Pattern Parameter data is variable | data_as_var     |               | -       | X       | -     |
| Pattern Parameter rptstart         | rptstart        | 1             | -       | X       | -     |
| Pattern Parameter rpttimes         | rpttimes        | 0             | -       | X       | -     |
| Amplitude scale factor             | scale           |               | -       | X       | -     |
| Desired rms value                  | rms             |               | -       | x       | -     |
| Time scaling factor                | stretch         |               | -       | X       | -     |
| Filter for pwl source              | pwlfilter       |               | -       | X       | -     |
| Rolloff factor                     | rolloff         |               | -       | X       | -     |
| Bandwidth of the pwl signal        | pwlbandwidth    |               | -       | X       | -     |
| PWL period                         | pwlperiod       |               | -       | X       | -     |
| Edge type                          | edgetype        |               | X       | X       | -     |
| RMS Jitter                         | perjitter       |               | X       | -       | -     |
| Period Jitter                      | perjitter       |               | X       | -       | -     |
| Width of transition                | twidh           |               | -       | X       | -     |
| Breakpoints                        | allbrkpts       | no            | -       | X       | -     |
| Period start time                  | pwlperiodstart  |               | -       | X       | -     |
| Delay time                         | tdPrbs          |               | -       | X       | -     |
| Bit period                         | perPrbs         |               | -       | X       | -     |



| CDF Parameter Prompt           | CDF Parameter       | Default Value                                          | hspiceD | spectre | auCdl |
|--------------------------------|---------------------|--------------------------------------------------------|---------|---------|-------|
| Transition reference           | transitionreference |                                                        | -       | X       | -     |
| Lower threshold(percent)       | lowerthreshold      |                                                        | -       | X       | -     |
| Upper threshold(percent)       | upperthreshold      |                                                        | -       | X       | -     |
| Edge type                      | edgetypePrbs        |                                                        | -       | X       | -     |
| Rise delay                     | td01                |                                                        | -       | X       | -     |
| Fall delay                     | td10                |                                                        | -       | X       | -     |
| PAM4 modulation                | pam4_modulation     | none                                                   | -       | X       | -     |
| PAM4 mapping                   | pam4_mapping        |                                                        | -       | X       | -     |
| triggerValue                   | triggerValue        | 1                                                      | -       | X       | -     |
| Trigger                        | trigger             | Internal                                               | -       | X       | -     |
| Threshold                      | triggerthreshold    |                                                        | -       | X       | -     |
| LFSR Mode                      | lfsrmode            |                                                        | -       | X       | -     |
| Browse and specify bit file    | prbsFileBox         | 0                                                      | -       | -       | -     |
| Bit data file                  | prbsFile            |                                                        | -       | X       | -     |
| Bit data file name is variable | prbsFile_as_var     | 0                                                      | -       | X       | -     |
| Seed                           | lfsrseed            | 1 3 5 7 9<br>11 13 15<br>17 19 21<br>23 25 27<br>29 31 | -       | X       | -     |
| Taps                           | lfsrtaps            |                                                        | -       | X       | -     |
| RJ(rms)                        | rjrms               |                                                        | -       | X       | -     |
| RJ(seed)                       | rjseed              |                                                        | -       | X       | -     |

| CDF Parameter Prompt          | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|-------------------------------|-----------------|---------------|---------|---------|-------|
| Number of periodic jitters    | nPeriodicJitter | 0             | -       | X       | -     |
| PJ1(amplitude)                | pjump1          |               | -       | X       | -     |
| PJ1(frequency)                | pjfreq1         |               | -       | X       | -     |
| PJ1(type)                     | pjtype1         | sine          | -       | X       | -     |
| PJ2(amplitude)                | pjump2          |               | -       | X       | -     |
| PJ2(frequency)                | pjfreq2         |               | -       | X       | -     |
| PJ2(type)                     | pjtype2         | sine          | -       | X       | -     |
| PJ3(amplitude)                | pjump3          |               | -       | X       | -     |
| PJ3(frequency)                | pjfreq3         |               | -       | X       | -     |
| PJ3(type)                     | pjtype3         | sine          | -       | X       | -     |
| Display small signal params   | smallSig        | 0             | X       | X       | -     |
| AC Magnitude                  | acm             |               | X       | X       | -     |
| AC Phase                      | acp             |               | X       | X       | -     |
| PAC magnitude                 | pacm            |               | -       | X       | -     |
| PAC phase                     | pacp            |               | -       | X       | -     |
| XF magnitude                  | xfm             |               | -       | X       | -     |
| HBAC Magnitude                | hbacm           |               | X       | -       | -     |
| HBAC Phase                    | hbacp           |               | X       | -       | -     |
| SNAC Magnitude                | snacm           |               | X       | -       | -     |
| SNAC Phase                    | snacp           |               | X       | -       | -     |
| Display resistance parameters | resistanceParam | 0             | X       | -       | -     |
| DC Resistance                 | rdc             |               | X       | -       | -     |
| TRAN Resistance               | rtran           |               | X       | -       | -     |
| AC Resistance                 | rac             |               | X       | -       | -     |

| CDF Parameter Prompt                                         | CDF Parameter         | Default Value              | hspiceD | spectre | auCdl |
|--------------------------------------------------------------|-----------------------|----------------------------|---------|---------|-------|
| HB Resistance                                                | rhb                   |                            | X       | -       | -     |
| HBAC Resistance                                              | rhbac                 |                            | X       | -       | -     |
| Display temperature params                                   | tempParam             | 0                          | -       | X       | -     |
| Nominal temperature                                          | tnom                  |                            | -       | X       | -     |
| Linear temperature coefficient                               | tc1                   |                            | -       | X       | -     |
| Quadratic temperature coefficient                            | tc2                   |                            | -       | X       | -     |
| Display noise parameters                                     | noiseParam            | 0                          | -       | X       | -     |
| Generate noise?                                              | isnoisyNoiseModel     |                            | -       | X       | -     |
| Noise entry method                                           | noiseEntryMethod      | File                       | -       | X       | -     |
| Noise file name                                              | noisefile             |                            | -       | X       | -     |
| Noise file name is variable                                  | noise_file_as_var     | 0                          | -       | X       | -     |
| Noise type                                                   | noiseModelType        | elsource                   | -       | X       | -     |
| Interpolation method                                         | noiseModelInterMethod | source                     | -       | X       | -     |
| Number of noise/frequency pairs                              | FNpairs               | 0                          | -       | X       | -     |
| Frequency1-50                                                | F1-50                 |                            | -       | X       | -     |
| Noise1-50                                                    | N1-50                 |                            | -       | X       | -     |
| Noise temperature                                            | noisetemp             |                            | -       | X       | -     |
| This dummy parameter exists only for using its default value | paramForDefault       | evalDefaultZeroOrEmpty("") | -       | -       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Multiplier           | m             |               | -       | X       | -     |
| Version              | version       | 3             | -       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noiseFile parameter is set as *NOISEFILE*, the netlisting behavior for noiseFile and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noiseFile='NOISEFILE'    | noiseFile="NOISEFILE" |
| true              | noiseFile=str(NOISEFILE) | noiseFile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noiseFile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.
- In Spectre-format netlisting, you can write out frequency/noise pairs (FNpairs) or noiseFile parameters even they are hidden by noiseParam. To skip frequency/noise pairs (FNpairs) or noiseFile parameters for legacy designs, create and use the dbAnalogLibIsSourceVersion Tcl preference.

**Example**

```
i4 n1 n2 dc=10u power=0
i4 n1 n2 dc=0 exp ( 0 100u 0 20n 0 30n )
i4 n1 n2 dc=10u pulse ( 0 100u 1n 1n 1n 10n 20n )
```

This damped sinusoidal source connects between nodes n1 and 0. In this waveform:

- Peak value is 100 uA
- Offset is 0 A
- Frequency is 100 MHz
- Time delay is 1 ns
- Damping factor is 1e+6
- Phase delay is 0 deg

### See Also

- [Independent Sources and Stimuli](#) in the *PrimeSim Continuum User Guide: Elements*

## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibIsourceVersion](#) Tcl preference before the analogLib library is loaded.

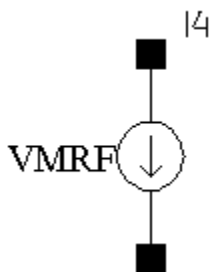
## Using isource to Set up Harmonic Balance or Shooting PSS Simulations

isource can be used as stimulus for one-, two-, and N-tone Harmonic Balance and Shooting Newton simulations. Its use is analogous to vsource and similar to port. Refer to [vsource](#) and [port](#) documentation for details of isource setup.

---

### ivmrf

Vector Modulated RF current source. It allows digital modulation of an RF carrier using in-phase and quadrature components created from a binary data stream.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | idc           | 0             | X       | -       | -     |
| Source Type          | srcType       | dc            | -       | -       | -     |
| Current Amplitude    | amp           |               | X       | -       | -     |
| Carrier Frequency    | freq          |               | X       | -       | -     |
| Carrier Phase        | phase         |               | X       | -       | -     |
| Modulation Method    | mod           |               | X       | -       | -     |
| Filter Method        | filter        |               | X       | -       | -     |
| Filter Parameter     | filcoef       |               | X       | -       | -     |
| Bit Rate             | rate          |               | X       | -       | -     |
| Bitstream            | bitstream     |               | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | -       | -     |
| AC Phase             | acp           |               | X       | -       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | -       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |
| SNAC Magnitude       | snacm         |               | X       | -       | -     |
| SNAC Phase           | snacp         |               | X       | -       | -     |
| Multiplier           | m             |               | X       | -       | -     |
| Impedance            | z0            |               | X       | -       | -     |

## Example

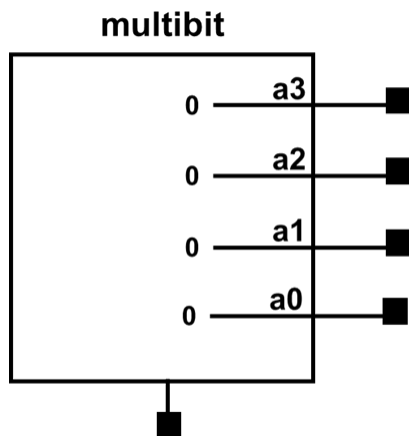
```
i55 n1 n2 dc=0 vmrf amp=100u freq=10k phase=0 rate=100Meg
+ mod=bpsk filter=rect bitstream=01010010011100b
```

## See Also

- [Vector-modulated RF \(VMRF\) Source](#) in the *HSPICE® User Guide: Elements*

## multibit

Generate a subcircuit of vdc sources, set the dc value according to the pattern specified.



## CDF Parameters

| CDF Parameter Prompt   | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------|---------------|---------------|---------|---------|-------|
| Input Mode             | mode1         | Binary        | X       | X       | -     |
| Number of Bits         | numbits       | 4             | X       | X       | -     |
| Expand Bus             | expand        | Yes           | X       | X       | -     |
| Bit Pattern(MSB...LSB) | pattern       | 0000          | X       | X       | -     |
| Decimal                | dec           | 0             | X       | X       | -     |
| Bit 1 voltage level    | vbit1         | 1             | X       | X       | -     |
| Bit 0 voltage level    | vbit0         | 0             | X       | X       | -     |

## Examples

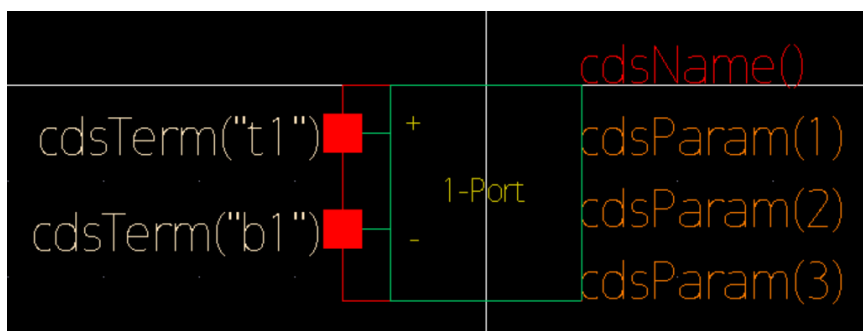
```
subckt multibit_1 ( a0 a1 a2 a3 ref)
parameters a0=fmod(11, 2) a1=fmod(5, 2) a2=fmod(2, 2) a3=fmod(1, 2)
vbit0=0 \
vbit1=1
V3 (a3 ref) vsource type=dc dc=( (a3>0) ? vbit1 : vbit0 )
V2 (a2 ref) vsource type=dc dc=( (a2>0) ? vbit1 : vbit0 )
V1 (a1 ref) vsource type=dc dc=( (a1>0) ? vbit1 : vbit0 )
V0 (a0 ref) vsource type=dc dc=( (a0>0) ? vbit1 : vbit0 )
ends multibit_1

I125 (net296 net295 net294 net293 net297) multibit_1 vbit1=1 vbit0=0
a0=fmod(int((11/1)), 2) \
a1=fmod(int((11/2)), 2) a2=fmod(int((11/4)), 2) a3=fmod(int((11/8)), 2)
```

## n1port

The n1port element is a special case of the generic nport element with specific port number.





**Note:**

The **nport** element has certain behavior which depends on S, Y or Z-parameter files.

## CDF Parameters

| CDF Parameter Prompt           | CDF Parameter       | Default Value  | hspiceD | spectre | auCdl |
|--------------------------------|---------------------|----------------|---------|---------|-------|
| Browse and select s-data file  | nportFileB          | 0              | X       | X       | -     |
| Model Type                     | modelFileType       | TSTONE<br>FILE | X       | -       | -     |
| S-parameter data file          | dataFile            |                | X       | X       | -     |
| Interpolation method           | interp              | linear         | -       | X       | -     |
| Tran convolution parameters    | tranParaLabel       | 1              | -       | X       | -     |
| Accuracy                       | accuracyMode        | default        | -       | X       | -     |
| Advanced transient parameters  | tranAdvanPara Label | 0              | X       | X       | -     |
| Max sampling points            | maxn                |                | -       | X       | -     |
| Max frequency of interest      | fmax                |                | X       | X       | -     |
| Impulse response truncation    | imptrunc            |                | -       | X       | -     |
| ROM data file                  | romdatfile          |                | -       | X       | -     |
| ROM data file name is variable | romdatfile_as_var   | 0              | -       | X       | -     |

| CDF Parameter Prompt           | CDF Parameter  | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|----------------|---------------|---------|---------|-------|
| Rational order                 | ratorder       |               | -       | X       | -     |
| Relative error                 | relerr         |               | -       | X       | -     |
| Absolute error                 | abserr         |               | -       | X       | -     |
| Noise parameters               | noiseParaLabel | 0             | X       | X       | -     |
| Thermal noise                  | thermalnoise   | Yes           | X       | X       | -     |
| Thermal noise model            | noisemodel     |               |         | X       | -     |
| Rarely used parameters         | otherParaLabel | 0             | X       | X       | -     |
| S-parameter data format        | datafmt        |               | -       | -       | -     |
| Frequency sampling interval    | fdelta         |               | X       | X       | -     |
| Use smooth data windowing      | usewindow      |               | -       | X       | -     |
| Multiplier                     | m              |               | X       | X       | -     |
| Scale factor                   | scale          |               | -       | X       | -     |
| Model name                     | hmname         | MDN           | X       | -       | -     |
| Enable mixed mode              | mixedmode      | 0             | X       | -       | -     |
| The order of indices           | datatype       |               | X       | -       | -     |
| Characteristic impedance       | z0             | 50            | X       | -       | -     |
| Hspice S-parameter data format | datafmtHspice  | touchstone    | X       | X       | -     |
| Enable passive checker         | passive        | 0             | X       | -       | -     |
| Hspice Interpolation method    | interpolation  |               | X       | -       | -     |
| Delay frequency                | delayfreq      |               | X       | -       | -     |

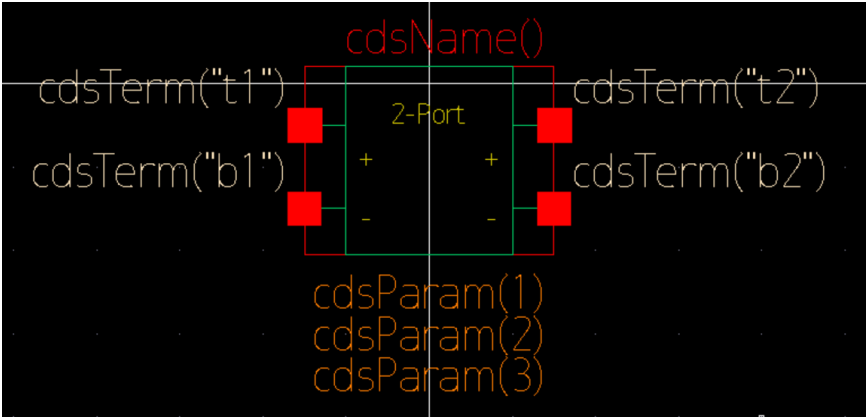
| CDF Parameter Prompt           | CDF Parameter       | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------------|---------------|---------|---------|-------|
| Extracts a system delay        | delayhandle         | 0             | X       | -       | -     |
| Temperature difference         | dtemp               |               | X       | -       | -     |
| High freq extrapolate method   | highpass            | 3             | X       | -       | -     |
| Linear interpolation data type | intdattyp           | MA            | X       | -       | -     |
| Low freq extrapolate method    | lowpass             | 1             | X       | -       | -     |
| Enable noise passive checker   | noipassivechk       | 1             | X       | -       | -     |
| Precondition factor keyword    | precfac             | 0.75          | X       | -       | -     |
| Enable rational function       | rational_func       | 0             | X       | -       | -     |
| Reuse rational function data   | rational_func_reuse | 1             | X       | -       | -     |
| Method of smooth               | smooth              | 0             | X       | -       | -     |
| Width of the smoothing window  | smoothpts           |               | X       | -       | -     |
| Stamping method                | stamp               |               | X       | -       | -     |

## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibNxPortVersion](#) Tcl preference before the analogLib library is loaded.

### n2port

The n2port element is a special case of the generic nport element with specific port number.



**Note:**

The [nport](#) element has certain behavior which depends on S, Y or Z-parameter files.

CDF Parameters

| CDF Parameter Prompt          | CDF Parameter       | Default Value  | hspiceD | spectre | auCdl |
|-------------------------------|---------------------|----------------|---------|---------|-------|
| Browse and select s-data file | nportFileB          | 0              | X       | X       | -     |
| Model Type                    | modelFileType       | TSTONE<br>FILE | X       | -       | -     |
| S-parameter data file         | dataFile            |                | X       | X       | -     |
| Interpolation method          | interp              | linear         | -       | X       | -     |
| Tran convolution parameters   | tranParaLabel       | 1              | -       | X       | -     |
| Accuracy                      | accuracyMode        | default        | -       | X       | -     |
| Advanced transient parameters | tranAdvanPara Label | 0              | X       | X       | -     |
| Max sampling points           | maxn                |                | -       | X       | -     |
| Max frequency of interest     | fmax                |                | X       | X       | -     |
| Impulse response truncation   | imptrunc            |                | -       | X       | -     |
| ROM data file                 | romdatfile          |                | -       | X       | -     |

| CDF Parameter Prompt           | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|-------------------|---------------|---------|---------|-------|
| ROM data file name is variable | romdatfile_as_var | 0             | -       | X       | -     |
| Rational order                 | ratorder          |               | -       | X       | -     |
| Relative error                 | relerr            |               | -       | X       | -     |
| Absolute error                 | abserr            |               | -       | X       | -     |
| Noise parameters               | noiseParaLabel    | 0             | X       | X       | -     |
| Thermal noise                  | thermalnoise      | Yes           | X       | X       | -     |
| Thermal noise model            | noisemodel        |               | -       | X       | -     |
| Rarely used parameters         | otherParaLabel    | 0             | X       | X       | -     |
| S-parameter data format        | datafmt           |               | -       | -       | -     |
| Frequency sampling interval    | fdelta            |               | X       | X       | -     |
| Use smooth data windowing      | usewindow         |               | -       | X       | -     |
| Multiplier                     | m                 |               | X       | X       | -     |
| Scale factor                   | scale             |               |         | X       | -     |
| Model name                     | hmname            | MDN           | X       | -       | -     |
| Enable mixed mode              | mixedmode         | 0             | X       | -       | -     |
| The order of indices           | datatype          |               | X       | -       | -     |
| Characteristic impedance       | z0                | 50            | X       | -       | -     |
| Hspice S-parameter data format | datafmtHspice     | touchstone    | X       | X       | -     |
| Enable passive checker         | passive           | 0             | X       | -       | -     |
| Hspice Interpolation method    | interpolation     |               | X       | -       | -     |

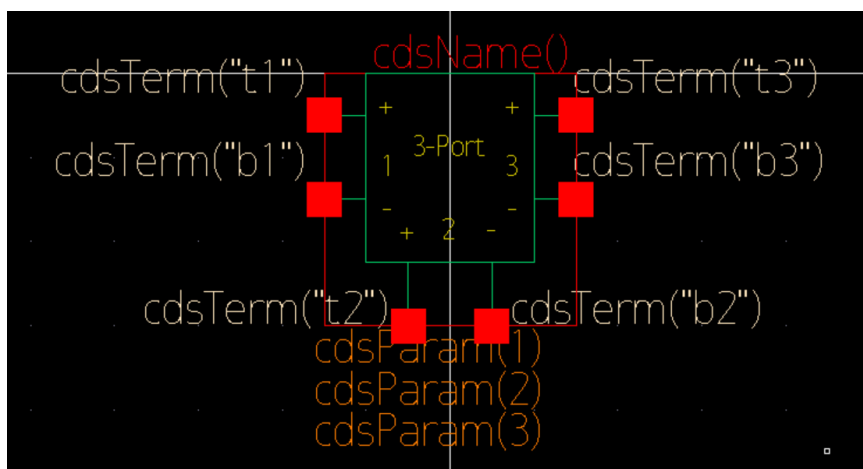
| CDF Parameter Prompt           | CDF Parameter       | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------------|---------------|---------|---------|-------|
| Delay frequency                | delayfreq           |               | X       | -       | -     |
| Extracts a system delay        | delayhandle         | 0             | X       | -       | -     |
| Temperature difference         | dtemp               |               | X       | -       | -     |
| High freq extrapolate method   | highpass            | 3             | X       | -       | -     |
| Linear interpolation data type | intdattyp           | MA            | X       | -       | -     |
| Low freq extrapolate method    | lowpass             | 1             | X       | -       | -     |
| Enable noise passive checker   | noipassivechk       | 1             | X       |         | -     |
| Precondition factor keyword    | precfac             | 0.75          | X       | -       | -     |
| Enable rational function       | rational_func       | 0             | X       | -       | -     |
| Reuse rational function data   | rational_func_reuse | 1             | X       | -       | -     |
| Method of smooth               | smooth              | 0             | X       | -       | -     |
| Width of the smoothing window  | smoothpts           |               | X       | -       | -     |
| Stamping method                | stamp               |               | X       | -       | -     |

## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibNxPortVersion](#) Tcl preference before the analogLib library is loaded.

## n3port

The n3port element is a special case of the generic nport element with specific port number.



**Note:**

The **nport** element has certain behavior which depends on S, Y, or Z-parameter files.

## CDF Parameters

| CDF Parameter Prompt          | CDF Parameter       | Default Value | hspiceD | spectre | auCdl |
|-------------------------------|---------------------|---------------|---------|---------|-------|
| Browse and select s-data file | nportFileB          | 0             | X       | X       | -     |
| S-parameter data file         | dataFile            |               | X       | X       | -     |
| Interpolation method          | interp              | linear        | -       | X       | -     |
| Tran convolution parameters   | tranParaLabel       | 1             | -       | X       | -     |
| Accuracy                      | accuracyMode        | default       | -       | X       | -     |
| Advanced transient parameters | tranAdvanPara Label | 0             | X       | X       | -     |
| Max sampling points           | maxn                |               | -       | X       | -     |
| Max frequency of interest     | fmax                |               | X       | X       | -     |
| Impulse response truncation   | imptrunc            |               | -       | X       | -     |
| ROM data file                 | romdatfile          |               | -       | X       | -     |

| CDF Parameter Prompt           | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|-------------------|---------------|---------|---------|-------|
| ROM data file name is variable | romdatfile_as_var | 0             | -       | X       | -     |
| Rational order                 | ratorder          |               | -       | X       | -     |
| Relative error                 | relerr            |               | -       | X       | -     |
| Absolute error                 | abserr            |               | -       | X       | -     |
| Noise parameters               | noiseParaLabel    | 0             | X       | X       | -     |
| Thermal noise                  | thermalnoise      | Yes           | X       | X       | -     |
| Thermal noise model            | noisemodel        |               | -       | X       | -     |
| Rarely used parameters         | otherParaLabel    | 0             | X       | X       | -     |
| S-parameter data format        | datafmt           |               | -       | -       | -     |
| Frequency sampling interval    | fdelta            |               | X       | X       | -     |
| Use smooth data windowing      | usewindow         |               | -       | X       | -     |
| Multiplier                     | m                 |               | X       | X       | -     |
| Scale factor                   | scale             |               | -       | X       | -     |
| Model name                     | hmname            | MDN           | X       | -       | -     |
| Enable mixed mode              | mixedmode         | 0             | X       | -       | -     |
| The order of indices           | datatype          |               | X       | -       | -     |
| Characteristic impedance       | z0                | 50            | X       | -       | -     |
| Hspice S-parameter data format | datafmtHspice     | touchstone    | X       | X       | -     |
| Enable passive checker         | passive           | 0             | X       | -       | -     |
| Hspice Interpolation method    | interpolation     |               | X       | -       | -     |



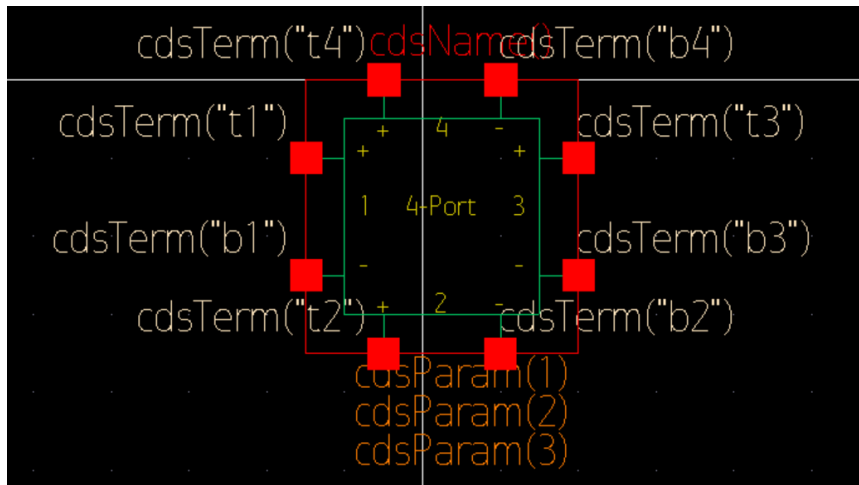
| CDF Parameter Prompt           | CDF Parameter       | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------------|---------------|---------|---------|-------|
| Delay frequency                | delayfreq           |               | X       | -       | -     |
| Extracts a system delay        | delayhandle         | 0             | X       | -       | -     |
| Temperature difference         | dtemp               |               | X       | -       | -     |
| High freq extrapolate method   | highpass            | 3             | X       | -       | -     |
| Linear interpolation data type | intdattyp           | MA            | X       | -       | -     |
| Low freq extrapolate method    | lowpass             | 1             | X       | -       | -     |
| Enable noise passive checker   | noipassivechk       | 1             | X       | -       | -     |
| Precondition factor keyword    | precfac             | 0.75          | X       | -       | -     |
| Enable rational function       | rational_func       | 0             | X       | -       | -     |
| Reuse rational function data   | rational_func_reuse | 1             | X       | -       | -     |
| Method of smooth               | smooth              | 0             | X       | -       | -     |
| Width of the smoothing window  | smoothpts           |               | X       | -       | -     |
| Stamping method                | stamp               |               | X       | -       | -     |

## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibNxPortVersion](#) Tcl preference before the analogLib library is loaded.

### n4port

The n4port element is a special case of the generic nport element with specific port number.



The `nport` element has certain behavior which depends on S, Y or Z-parameter files.

| CDF Parameter Prompt          | CDF Parameter       | Default Value | hspiceD | spectre | auCdl |
|-------------------------------|---------------------|---------------|---------|---------|-------|
| Browse and select s-data file | nportFileB          | 0             | X       | X       | -     |
| S-parameter data file         | dataFile            |               | X       | X       | -     |
| Interpolation method          | interp              | linear        | -       | X       | -     |
| Tran convolution parameters   | tranParaLabel       | 1             | -       | X       | -     |
| Accuracy                      | accuracyMode        | default       | -       | X       | -     |
| Advanced transient parameters | tranAdvanPara Label | 0             | X       | X       | -     |
| Max sampling points           | maxn                |               | -       | X       | -     |
| Max frequency of interest     | fmax                |               | X       | X       | -     |
| Impulse response truncation   | imptrunc            |               | -       | X       | -     |
| ROM data file                 | romdatfile          |               | -       | X       | -     |

| CDF Parameter Prompt           | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|-------------------|---------------|---------|---------|-------|
| ROM data file name is variable | romdatfile_as_var | 0             | -       | X       | -     |
| Rational order                 | ratorder          |               | -       | X       | -     |
| Relative error                 | relerr            |               | -       | X       | -     |
| Absolute error                 | abserr            |               | -       | X       | -     |
| Noise parameters               | noiseParaLabel    | 0             | X       | X       | -     |
| Thermal noise                  | thermalnoise      | Yes           | X       | X       | -     |
| Thermal noise model            | noisemodel        |               | -       | X       | -     |
| Rarely used parameters         | otherParaLabel    | 0             | X       | X       | -     |
| S-parameter data format        | datafmt           |               | -       | -       | -     |
| Frequency sampling interval    | fdelta            |               | X       | X       | -     |
| Use smooth data windowing      | usewindow         |               | -       | X       | -     |
| Multiplier                     | m                 |               | X       | X       | -     |
| Scale factor                   | scale             |               |         | X       | -     |
| Model name                     | hmname            | MDN           | X       | -       | -     |
| Enable mixed mode              | mixedmode         | 0             | X       | -       | -     |
| The order of indices           | datatype          |               | X       | -       | -     |
| Characteristic impedance       | z0                | 50            | X       | -       | -     |
| Hspice S-parameter data format | datafmtHspice     | touchstone    | X       | X       | -     |
| Enable passive checker         | passive           | 0             | X       | -       | -     |
| Hspice Interpolation method    | interpolation     |               | X       | -       | -     |

| CDF Parameter Prompt           | CDF Parameter       | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------------|---------------|---------|---------|-------|
| Delay frequency                | delayfreq           |               | X       | -       | -     |
| Extracts a system delay        | delayhandle         | 0             | X       | -       | -     |
| Temperature difference         | dtemp               |               | X       | -       | -     |
| High freq extrapolate method   | highpass            | 3             | X       | -       | -     |
| Linear interpolation data type | intdattyp           | MA            | X       | -       | -     |
| Low freq extrapolate method    | lowpass             | 1             | X       | -       | -     |
| Enable noise passive checker   | noipassivechk       | 1             | X       | -       | -     |
| Precondition factor keyword    | precfac             | 0.75          | X       | -       | -     |
| Enable rational function       | rational_func       | 0             | X       | -       | -     |
| Reuse rational function data   | rational_func_reuse | 1             | X       | -       | -     |
| Method of smooth               | smooth              | 0             | X       | -       | -     |
| Width of the smoothing window  | smoothpts           |               | X       | -       | -     |
| Stamping method                | stamp               |               | X       | -       | -     |

## Controlling Version Number for Legacy Designs

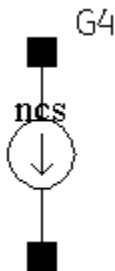
To control the version number for legacy designs, create and use the [dbAnalogLibNxPortVersion](#) Tcl preference before the analogLib library is loaded.

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### ncs

Behavioral noise source.

In HSPICE advanced analog analyses, you can use the G-element to specify noise sources. Frequency domain noise analyses (.NOISE, .HBNOISE, and .PHASENOISE) take these noise sources into account.



CDF Parameters

| CDF Parameter Prompt        | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|-------------------|---------------|---------|---------|-------|
| Noise Expression            | noise             |               | X       | -       | -     |
| Noise File                  | noisefile         |               | X       | -       | -     |
| Noise file name is variable | noise_file_as_var | 0             | X       | -       | -     |
| Noise entry method          | noiseEntryMethod  | file          | X       | -       | -     |

Note:

- The default of noise\_file\_as\_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, str(NOISEFILE) is used as the value format.

- The `noiseEntryMethod` parameter controls both Hspice-format and Spectre-format netlisting behavior. This `noiseEntryMethod` parameter has two choices: `file` and `expression`. Based on its value, the Property Editor shows corresponding parameters such as `noise`, `noisefile`, and `noise_file_as_var`. With the `noiseEntryMethod` parameter, either `noise` or `noisefile` is netlisted in the Hspice-format and Spectre-format netlisting. The netlisting results are mutually exclusive to choose either `noise` or `noisefile`.

To hide parameters such as `noise`, `noisefile`, and `noise_file_as_var` for legacy designs, create and use the [dbAnalogLibNcsVersion](#) Tcl preference.

## Example

Noise expression form

```
g4 n1 n2 noise='4*1.3806266e-23*(TEMPER+273.15)*0.001'
```

Noise file form

```
g4 n1 n2 noisefile='out_noise.txt'
```

## See Also

- [Behavioral Current Noise Source](#) in the *HSPICE® User Guide: Elements*
- [Behavioral Noise Sources](#) in the *HSPICE® User Guide: Advanced Analog Simulation and Analysis*

## Controlling Version Number for Legacy Designs

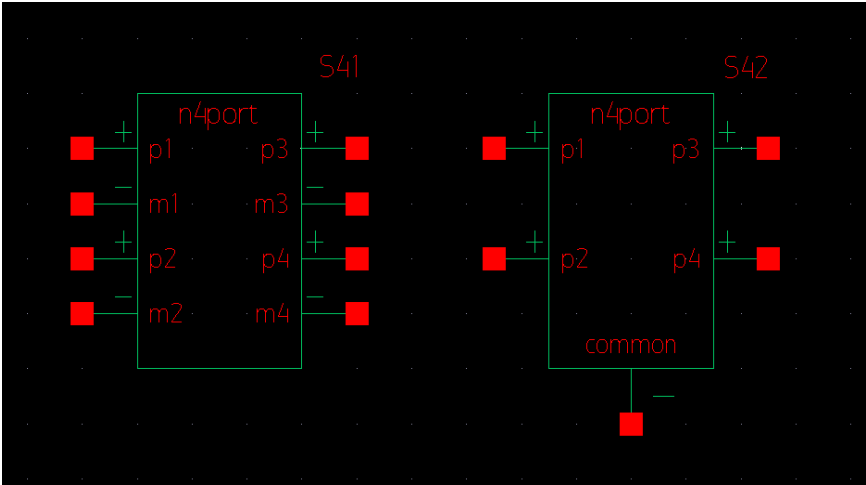
To control the version number for legacy designs, create and use the [dbAnalogLibNcsVersion](#) Tcl preference before the `analogLib` library is loaded.

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## nport

The `nport` cell is a PCell device that is controlled by version parameter.

The following figure shows two `nport` devices that have `numports == 4` (`p == 4`). The one on the left is with common mode off and another one on the right is with common mode on.



CDF Parameters

| CDF Parameter Prompt               | CDF Parameter       | Default Value | hspiceD | spectre | auCdl |
|------------------------------------|---------------------|---------------|---------|---------|-------|
| Common reference                   | nmode               | 0             | -       | X       | -     |
| numports                           | numports            | 1             | X       | -       | -     |
| Model Name                         | mname               |               | X       | -       | -     |
| Model File                         | modelFile           |               | X       | -       | -     |
| Data Source                        | fileOrCV            | file          | -       | X       | -     |
| S-parameter data file              | dataFile            |               | -       | X       | -     |
| S-parameter file as Design Var?    | sparam_file_as _var | 0             | -       | X       | -     |
| Model Type                         | modelFileType       | TSTONE FILE   | X       | -       | -     |
| S-parameter Lib                    | sparamLib           |               | -       | X       | -     |
| S-parameter Cell                   | sparamCell          |               | -       | X       | -     |
| S-parameter View                   | sparamView          |               | -       | X       | -     |
| Browse and specify S-parameter LCV | sparamlcvB          | 0             | -       | -       | -     |
| Resistance                         | r                   | 50            | -       | -       | -     |
| Rational Func Approx               | rational_func       |               | X       | -       | -     |

| CDF Parameter Prompt                               | CDF Parameter        | Default Value | hspiceD | spectre | auCdl |
|----------------------------------------------------|----------------------|---------------|---------|---------|-------|
| checkFlag                                          | checkFlag            | 1             | -       | X       | -     |
| Number of ports                                    | p                    | 1             | -       | X       | X     |
| Browse and select s-data file                      | nportFileB           | 0             | -       | X       | -     |
| Interpolation method                               | interp               | linear        | -       | X       | -     |
| Passivity                                          | passivity            | check         | -       | X       | -     |
| Passivity                                          | passivity_bbspace    | fit_enforce   | -       | X       | -     |
| Tran convolution parameters                        | tranParaLabel        | 1             | -       | X       | -     |
| Accuracy                                           | accuracyMode         | default       | -       | X       | -     |
| Advanced transient parameters                      | tranAdvanParaLabel   | 0             | -       | X       | -     |
| Max sampling points                                | maxn                 |               | -       | X       | -     |
| Base Frequency                                     | fbase                |               | X       | -       | -     |
| Max Frequency                                      | fmax                 |               | X       | -       | -     |
| Impulse response truncation                        | imptrunc             |               | -       | X       | -     |
| Causality correction                               | causality            | fmax          | -       | X       | -     |
| Enforce causality                                  | s_causality          |               | X       | -       | -     |
| Max frequency multiplier for causality enforcement | causality_fmax_scale |               | X       | -       | -     |
| DC extrapolation                                   | dcextrap             | constant      | -       | X       | -     |
| ROM data file                                      | romdatfile           |               | -       | X       | -     |
| ROM data file name is variable                     | romdatfile_as_var    | 0             | -       | X       | -     |
| Rational order                                     | ratorder             |               | -       | X       | -     |
| Relative error                                     | relerr               |               | -       | X       | -     |



| CDF Parameter Prompt            | CDF Parameter             | Default Value | hspiceD | spectre | auCdl |
|---------------------------------|---------------------------|---------------|---------|---------|-------|
| Absolute error                  | abserr                    |               | -       | X       | -     |
| Noise parameters                | noiseParaLabel            | 0             | -       | X       | -     |
| Generate Noise?                 | noise                     | 1             | X       | -       | -     |
| Noise Passive Check             | noipassivechk             |               | X       | -       | -     |
| Temp Difference                 | dtemp                     |               | X       | -       | -     |
| Thermal noise                   | thermalnoise              | yes           | -       | X       | -     |
| Thermal noise model             | noisemodel                |               | -       | X       | -     |
| Noise correlation               | noisecorr                 |               | -       | X       | -     |
| Rarely used parameters          | otherParaLabel            | 0             | -       | X       | -     |
| Interpolation Method            | interpolation             |               | X       | -       | -     |
| Model Name                      | mname                     |               | X       |         | -     |
| Modeling method (rational_func) | rational_func_for_spectre |               | -       | X       | -     |
| Parameter Type                  | type                      | S             | X       | -       | -     |
| Frequency Behavior              | fqmodel                   |               | X       | -       | -     |
| Characteristic Impedance        | z0                        | 50            | X       | -       | -     |
| Precondition Factor             | precfac                   | 0.75          | X       | -       | -     |
| Delay Handler                   | delayhandle               |               | X       | -       | -     |
| Delay Frequency                 | delayfreq                 | iPar("fmax")  | X       | -       | -     |
| Interpolation Data Type         | intdattyp                 | MA            | X       | -       | -     |
| High Extrapolation Method       | highpass                  | 3             | X       | -       | -     |
| Low Extrapolation Method        | lowpass                   | 2             | X       | -       | -     |

| CDF Parameter Prompt                    | CDF Parameter                 | Default Value | hspiceD | spectre | auCdl |
|-----------------------------------------|-------------------------------|---------------|---------|---------|-------|
| Stamp                                   | stamp                         |               | X       | -       | -     |
| Mixed Mode                              | mixedmode                     | 0             | X       | -       | -     |
| Data Type String                        | datatype                      |               | X       | -       | -     |
| Write causal S-param?                   | write_causality_enforced_file |               | X       | -       | -     |
| Causal s-param output file              | outFile                       |               | -       | X       | -     |
| Rational Func Reuse                     | rational_func_reuse           |               | X       | -       | -     |
| Use smooth data windowing               | usewindow                     |               | -       | X       | -     |
| S-parameter data format                 | datafmt                       |               | -       | X       | -     |
| Passivity tolerance                     | pabstol                       |               | -       | X       | -     |
| Data truncation threshold               | datatrunc                     |               | -       | X       | -     |
| Frequency sampling interval             | fdelta                        |               | -       | X       | -     |
| Multiplier                              | m                             |               | X       | X       | -     |
| Scale factor                            | scale                         |               | -       | X       | -     |
| High freq extrapolation                 | hfextrap                      |               | -       | X       | -     |
| Flag for matrix form input              | matrixform                    |               | -       | X       | -     |
| Matrix entry data file                  | matrixfile                    |               | -       | X       | -     |
| Matrix entry data file name is variable | matrixfile_as_var             | 0             | -       | X       | -     |
| Type of port 1-20                       | porttype1-20                  |               | -       | X       | -     |
| Quantity of Port 1-20                   | portquantity1-20              |               | -       | X       | -     |

| CDF Parameter Prompt      | CDF Parameter           | Default Value | hspiceD | spectre | auCdl |
|---------------------------|-------------------------|---------------|---------|---------|-------|
| Prioritize Accuracy Range | prioritizeAccuracyRange | 0             | -       | X       | -     |
| Start Frequency           | startFrequency          |               | -       | X       | -     |
| End Frequency             | endFrequency            |               | -       | X       | -     |
| Additional parameter list | additionalParam         |               | -       | X       | -     |
| Passive Mode              | passive                 | 0             | X       | -       | -     |
| Enforce Passivity         | enforce_passive         |               | X       | -       | -     |
| Additional parameter list | additionalSpiceParam    |               | X       | -       | -     |
| AC Model                  | acModel                 | freqdomain    | -       | X       | -     |
| RLGC data file            | file                    |               | -       | -       | -     |
| check box status          | checkBoxValue           | 1             | -       | X       | -     |
| Version                   | version                 | 2             | -       | -       | -     |

## Example

```
s7 p1 m1 p2 m2 mname=mymodel type='S' z0=50 fbase=5G
+ fmax=15G precfac=0.75 delayfreq=15G
+ interpolation='LINEAR' intdattyp='MA' highpass=3
+ lowpass=2 mixedmode=0 noise=1 passive=0
.model mymodel s TSTONEFILE=<path_to_file>
```

## See Also

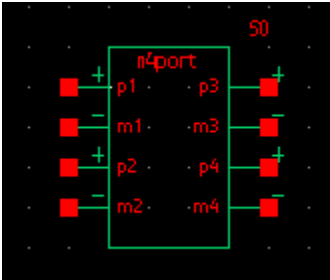
- [https://solvnet.synopsys.com/dow\\_retrieve/latest/hspice/hspice\\_olh/index.htm#context/cidSynopsysCustom/s-parameter\\_modeling](https://solvnet.synopsys.com/dow_retrieve/latest/hspice/hspice_olh/index.htm#context/cidSynopsysCustom/s-parameter_modeling)

## Using nport to Run PrimeSim Analyses

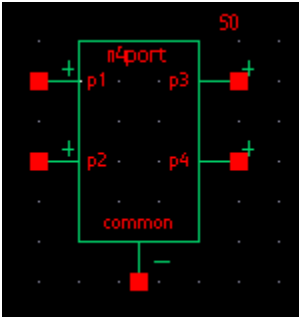
Use the nport element to specify S-parameters. In many cases, the number of ports and the file name containing S-parameters (usually in Touchstone format) is the only required setup in the Property Editor.

| Prompt                          | Value                    |
|---------------------------------|--------------------------|
| Common reference                | <input type="checkbox"/> |
| numports                        | 4                        |
| Model File                      |                          |
| Model Type                      | TSTONEFILE               |
| Modeling method (rational_func) |                          |
| Advanced transient parameters   | <input type="checkbox"/> |
| Noise parameters                | <input type="checkbox"/> |
| Other Parameters                | <input type="checkbox"/> |

In its default form, the nport symbol has a return terminal per port, so that the total number of terminals is 2N, as shown in the following figure.



Check the **Common reference** option in the Property Editor to create a symbol with a shared return terminal, as a way of simplifying wiring of nports with many terminals, as shown in the following figure.



In earlier versions of Custom Compiler, nport required you to specify the name of a model netlisted as an HSPICE `.model` statement. This specification is no longer required, and the netlister automatically handles requirements such as the insertion of model statements.

| Prompt                          | Value                               |
|---------------------------------|-------------------------------------|
| Common reference                | <input checked="" type="checkbox"/> |
| numports                        | 4                                   |
| Model File                      | test.s4p                            |
| Model Type                      | TSTONEFILE                          |
| Modeling method (rational_func) |                                     |

The **Modeling method** option controls the behavior of S-parameters in transient, frequency-domain, and Shooting Newton simulations:

|                                            | rational_func=0 | rational_func=1 or unspecified | rational_func=2 |
|--------------------------------------------|-----------------|--------------------------------|-----------------|
| Transient (including HB initial transient) | IFFT            | Rational                       | Rational        |
| Any frequency dependent analyses           | Raw data        | Raw data                       | Rational        |
| Shooting Newton                            | Rational        | Rational                       | Rational        |

Set **Modeling method** to 0 to use IFFT (convolution) method in transient. Raw data from the S-parameter file is used for frequency domain methods, except when overridden by setting **Modeling method=2** to use rational fitting. Shooting Newton always uses rational modeling.

In recent versions of Custom Compiler and PrimeSim, the nport interface and S-element technology were simplified and improved to require, in most cases, specifying only **Model File**, **numports** and **Modeling method** options. The remaining S-element parameters are exposed by checking the **Advanced transient parameters**, **Noise parameters**, and **Other Parameters** options.

| Prompt                          |                                     |
|---------------------------------|-------------------------------------|
| Common reference                | <input checked="" type="checkbox"/> |
| numports                        | 4                                   |
| Model File                      |                                     |
| Model Type                      | TSTONEFILE                          |
| Modeling method (rational_func) |                                     |
| Advanced transient parameters   | <input type="checkbox"/>            |
| Noise parameters                | <input type="checkbox"/>            |
| Other Parameters                | <input type="checkbox"/>            |

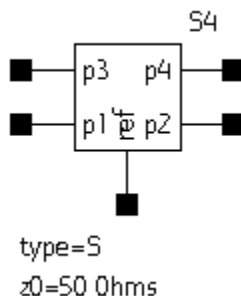
Refer to S-element (Scattering Parameter Data) in the *PrimeSim Continuum User Guide: Elements* for details of S-element options. In most cases, you should leave them at default values and allow simulation engines to determine optimal settings automatically. The following list describes a few noteworthy parameters in **Advanced transient parameters** and **Other Parameters** categories:

- **Enforce causality:** Newer versions of PrimeSim SPICE support IFFT causality enforcement. Causality enforcement is on by default and recommended to leave unspecified (preferred) or set to 1.
- **Max frequency multiplier for causality enforcement:** When causality is enforced, PrimeSim extrapolates raw data to an automatically selected frequency larger than the maximum data file frequency. If you observe log file messages warning about differences between causal and raw data, increase this parameter to between 2 and 10. The same suggestion applies if you run into convergence problems or suspected accuracy issues in IFFT.
- **Max frequency of interest:** It is recommended to leave this setting unspecified.
- **Interpolation Method:** In recent versions of PrimeSim SPICE, **Interpolation Method** controls raw data interpolation in frequency domain analyses. **Modeling method (rational\_func)** is used to control rational fitting vs. convolution (IFFT) methods. Spline and Linear methods work best. The behavior of **Interpolation Method=rational** is not defined and should not be used.
- **Additional parameter list:** Specify a list of parameter-value pairs in the form p1=val1 p2=val2... to send to the simulator any simulation parameters not exposed in the Property Editor. An additional parameter list is appended to the end of the S-element instance line.

---

## nport\_ref

You can use the nport\_ref element to describe a multi-terminal network circuit analyses within most HSPICE and RF analyses.



## CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value    | hspiceD | spectre | auCdl |
|---------------------------|---------------|------------------|---------|---------|-------|
| Model Name                | mname         |                  | X       | -       | -     |
| Model File                | modelFile     |                  | X       | -       | -     |
| Model Type                | modelFileType | TSTONE<br>FILE   | X       | -       | -     |
| Frequency Behavior        | mqmodel       |                  | X       | -       | -     |
| Parameter Type            | type          | S                | X       | -       | -     |
| Characteristic Impedance  | z0            | 50               | X       | -       | -     |
| Resistance                | r             | 50               | -       | -       | -     |
| Base Frequency            | fbase         |                  | X       | -       | -     |
| Max Frequency             | fmax          |                  | X       | -       | -     |
| Precondition Factor       | precfac       | 0.75             | X       | -       | -     |
| Delay Handler             | delayhandle   |                  | X       | -       | -     |
| Delay Frequency           | delayfreq     | iPar("fma<br>x") | X       | -       | -     |
| Interpolation Method      | interpolation | LINEAR           | X       | -       | -     |
| Interpolation Data Type   | intdattyp     | MA               | X       | -       | -     |
| High Extrapolation Method | highpass      | 3                | X       | -       | -     |
| Low Extrapolation Method  | lowpass       | 2                | X       | -       | -     |
| Mixed Mode                | mixedmode     | 0                | X       | -       | -     |
| Data Type String          | datatype      |                  | X       | -       | -     |
| Generate Noise?           | noise         | 1                | X       | -       | -     |
| Noise Passive Check       | noipassivechk |                  | X       | -       | -     |
| Temp Difference           | dtemp         |                  | X       | -       | -     |

| CDF Parameter Prompt            | CDF Parameter       | Default Value | hspiceD | spectre | auCdl |
|---------------------------------|---------------------|---------------|---------|---------|-------|
| Passive Mode                    | passive             | 0             | X       | -       | -     |
| Enforce Passive Mode            | enforce_passive     |               | X       | -       | -     |
| Rational Func Approx            | rational_func       |               | X       | -       | -     |
| checkFlag                       | checkFlag           | 1             | -       | X       | -     |
| S-parameter file as Design Var? | sparam_file_as_var  | 0             | -       | X       | -     |
| Rational Func Reuse             | rational_func_reuse |               | X       | -       | -     |
| Stamp                           | stamp               |               | X       | -       | -     |
| Multiplier                      | m                   |               | X       | -       | -     |

## Example

```
s7 m1 m2 p1 p2 ref mname=mymodel type='S' z0=50 fbase=5G
+ fmax=15G precfac=0.75 delayfreq=15G interpolation='LINEAR'
+ intdattyp='MA' highpass=3 lowpass=2 mixedmode=1 noise=1
+ passive=0
.model mymodel s TSTONEFILE=<path_to_file>
```

## See Also

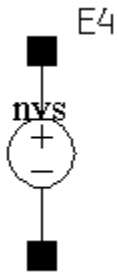
- [Mixed-Mode S-parameters](#) in the *HSPICE® User Guide: Signal Integrity Modeling and Analysis*

## nvs

Behavioral noise voltage source.

In HSPICE advanced analog analyses, you can use the E-element to specify noise voltage sources.





## CDF Parameters

| CDF Parameter Prompt        | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|-------------------|---------------|---------|---------|-------|
| Noise Expression            | noise             |               | X       | -       | -     |
| Noise File                  | noisefile         |               | X       | -       | -     |
| Noise file name is variable | noise_file_as_var | 0             | X       | -       | -     |
| Noise entry method          | noiseEntryMethod  | file          | X       | -       | -     |

### Note:

- The default of noise\_file\_as\_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- The noiseEntryMethod parameter controls both Hspice-format and Spectre-format netlisting behavior. This noiseEntryMethod parameter has two choices: *file* and *expression*. Based on its value, the Property Editor shows corresponding parameters such as noise, noisefile, and

noise\_file\_as\_var. With the noiseEntryMethod parameter, either noise or noisefile is netlisted in the Hspice-format and Spectre-format netlisting. The netlisting results are mutually exclusive to choose either noise or noisefile.

To hide parameters such as noise, noisefile, and noise\_file\_as\_var for legacy designs, create and use the [dbAnalogLibNvsVersion](#) Tcl preference.

Example

```
e4 n1 n2 noisefile='noise_v.txt'  
e4 n1 n2 noise='4*1.3806266e-23*(TEMPER+273.15)*0.001'
```

See Also

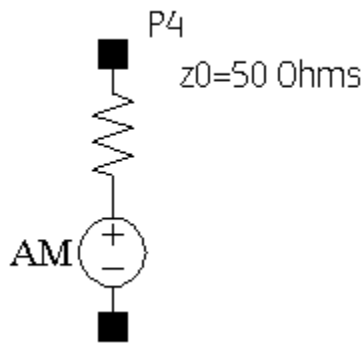
- [Behavioral Voltage Noise Source](#) in the *HSPICE® User Guide: Elements*

Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibNvsVersion](#) Tcl preference before the analogLib library is loaded.

pam

Amplitude modulated port element.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Source Type          | sourceType    | port          | -       | -       | -     |
| Source Type          | srcType       | am            | -       | -       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Port Number          | num           |               | X       | -       | -     |
| Impedance            | z0            | 50            | X       | -       | -     |
| Resistance           | r             | 50            | -       | -       | -     |
| Power Switch         | power         |               | X       | -       | -     |
| DC Value             | dc            |               | X       | -       | -     |
| DC Voltage           | vdc           |               | -       | -       | -     |
| Signal Amplitude     | sa            | 0             | X       | -       | -     |
| Carrier Frequency    | fc            | 1             | X       | -       | -     |
| Modulation Frequency | fm            | 0             | X       | -       | -     |
| Offset Constant      | oc            | 0             | X       | -       | -     |
| Delay Time           | td            | 0             | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | -       | -     |
| AC Phase             | acp           |               | X       | -       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | X       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | -       | -       | -     |
| HBAC Resistance      | rhbac         |               | -       | -       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |
| SNAC Magnitude       | snacm         |               | X       | -       | -     |
| SNAC Phase           | snacp         |               | X       | -       | -     |

Example

```
p4 n1 gnd! port=1 am ( 0.0 0.0 10k 1k 0 ) z0=50
```

See Also

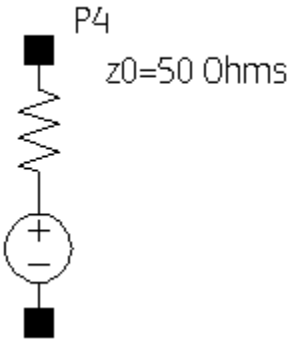
- [Port Element](#) in the *HSPICE® User Guide: Elements*

Updating Port Number Automatically

You can create and use the [dbAnalogLibAutoUpdatePortNum](#) Tcl preference to control whether to automatically update the port number of port family devices.

pdc

DC port element.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Source Type          | sourceType    | port          | -       | -       | -     |
| Source Type          | srcType       | dc            | -       | X       | -     |
| Port Number          | num           |               | X       | X       | -     |
| Impedance            | z0            | 50            | X       | -       | -     |
| Resistance           | r             | 50            | -       | X       | -     |
| Power Switch         | power         |               | X       | -       | -     |
| DC Value             | dc            |               | X       | -       | -     |
| DC Voltage           | vdc           |               | -       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | X       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |

| CDF Parameter Prompt            | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|---------------------------------|-------------------|---------------|---------|---------|-------|
| HBAC Phase                      | hbacp             |               | X       | -       | -     |
| SNAC Magnitude                  | snacm             |               | X       | -       | -     |
| SNAC Phase                      | snacp             |               | X       | -       | -     |
| Number of noise/frequency pairs | FNpairs           | 0             | -       | X       | -     |
| Noise file name                 | noisefile         |               | -       | X       | -     |
| Noise file name is variable     | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                   | F1-50             |               | -       | X       | -     |
| Noise1-50                       | N1-50             |               | -       | X       | -     |
| Multiplier                      | m                 |               |         | X       |       |
| Nominal temperature             | tnom              |               | -       | X       | -     |
| PAC magnitude                   | pacm              |               | -       | X       | -     |
| PAC phase                       | pacp              |               | -       | X       | -     |
| Temp coefficient 1              | tc1               |               | -       | X       | -     |
| Temp coefficient 2              | tc2               |               | -       | X       | -     |
| XF magnitude                    | xfm               |               | -       | X       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output        | spectre Output        |
|-------------------|-----------------------|-----------------------|
| noise_file_as_var |                       |                       |
| false (default)   | noisefile='NOISEFILE' | noisefile="NOISEFILE" |

| CDF Parameter | hspiceD Output           | spectre Output      |
|---------------|--------------------------|---------------------|
| true          | noisefile=str(NOISEFILE) | noisefile=NOISEFILE |

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

Example

```
p4 n1 gnd! port=1 dc=3.3 z0=50
```

See Also

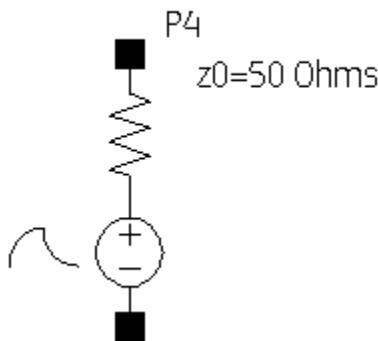
- [Port Element](#) in the *PrimeSim Continuum User Guide: Elements*

Updating Port Number Automatically

You can create and use the [dbAnalogLibAutoUpdatePortNum](#) Tcl preference to control whether to automatically update the port number of port family devices.

pexp

Exponential port element.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Source Type          | sourceType    | port          | -       | -       | -     |
| Source Type          | srcType       | exp           | -       | X       | -     |
| Port Number          | num           |               | X       | X       | -     |
| Impedance            | z0            | 50            | X       | -       | -     |
| Resistance           | r             | 50            | -       | X       | -     |
| Power Switch         | power         |               | X       | -       | -     |
| DC Value             | dc            |               | X       | -       | -     |
| DC Voltage           | vdc           |               | -       | X       | -     |
| Value 1              | v1            | 0             | X       | X       | -     |
| Value 2              | v2            | 0             | X       | X       | -     |
| Rise Delay Time      | td1           | 0             | X       | X       | -     |
| Fall Delay Time      | td2           |               | X       | X       | -     |
| Rise Time Constant   | tau1          |               | X       | X       | -     |
| Fall Time Constant   | tau2          | 0             | X       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | X       | -       | -     |



| CDF Parameter Prompt            | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|---------------------------------|-------------------|---------------|---------|---------|-------|
| DC Resistance                   | rdc               |               | X       | -       | -     |
| AC Resistance                   | rac               |               | X       | -       | -     |
| HB Resistance                   | rhb               |               | X       | -       | -     |
| HBAC Resistance                 | rhbac             |               | X       | -       | -     |
| TRAN Resistance                 | rtran             |               | X       | -       | -     |
| HBAC Magnitude                  | hbacm             |               | X       | -       | -     |
| HBAC Phase                      | hbacp             |               | X       | -       | -     |
| SNAC Magnitude                  | snacm             |               | X       | -       | -     |
| SNAC Phase                      | snacp             |               | X       | -       | -     |
| Number of noise/frequency pairs | FNpairs           | 0             | -       | X       | -     |
| Noise file name                 | noisefile         |               | -       | X       | -     |
| Noise file name is variable     | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                   | F1-50             |               | -       | X       | -     |
| Noise1-50                       | N1-50             |               | -       | X       | -     |
| Multiplier                      | m                 |               |         | X       |       |
| Nominal temperature             | tnom              |               | -       | X       | -     |
| PAC magnitude                   | pacm              |               | -       | X       | -     |
| PAC phase                       | pacp              |               | -       | X       | -     |
| Temp coefficient 1              | tc1               |               | -       | X       | -     |
| Temp coefficient 2              | tc2               |               | -       | X       | -     |
| XF magnitude                    | xfm               |               | -       | X       | -     |
| Delay time                      | td                |               | -       | X       |       |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

**Example**

```
p4 n1 gnd! port=1 dc=3.3 exp ( 0 2.5 5n 1n 15n 2n ) z0=50
```

**See Also**

- [Port Element](#) in the *HSPICE® User Guide: Elements*

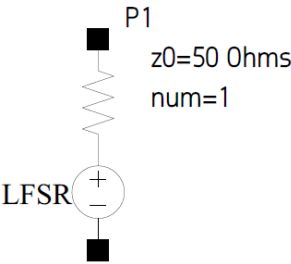
**Updating Port Number Automatically**

You can create and use the `dbAnalogLibAutoUpdatePortNum` Tcl preference to control whether to automatically update the port number of port family devices.

---

**plfsr**

The plfsr element specifies lfsr waveform type and analysis settings for P-element.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Source Type          | sourceType    | port          | -       | -       | -     |
| Source Type          | srcType       | lfsr          | -       | -       | -     |
| Port Number          | num           |               |         |         |       |
| Resistance           | r             | 50            | X       | -       | -     |
| Power Switch         | power         |               | X       | -       | -     |
| DC Value             | dc            |               | X       | -       | -     |
| DCVoltage            | vdc           |               | X       | -       | -     |
| Low Value            | vl            | 0             | X       | -       | -     |
| High Value           | vh            | 1             | X       | -       | -     |
| Delay Time           | td            | 0             | X       | -       | -     |
| Rise Time            | tr            | 0             | X       | -       | -     |
| Fall Time            | tf            | 0             | X       | -       | -     |
| Bit Rate             | rate          | 1M            | X       | -       | -     |
| Seed                 | seed          | 1             | X       | -       | -     |
| Taps                 | taps          | [1,2]         | X       | -       | -     |
| Output Resistance    | rout          |               | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | -       | -     |
| AC Phase             | acp           |               | X       | -       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |

| CDF Parameter Prompt | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|----------------------|-----------------|---------------|---------|---------|-------|
| HB Magnitude 1-7     | hbmag1-7        |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7      | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7       | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7       | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7    |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7    |               | X       | -       | -     |
| Open DC Connection   | dcopen          |               | X       | -       | -     |
| DC Resistance        | rdc             |               | X       | -       | -     |
| AC Resistance        | rac             |               | X       | -       | -     |
| HB Resistance        | rhb             |               | X       | -       | -     |
| HBAC Resistance      | rhbac           |               | X       | -       | -     |
| TRAN Resistance      | rtran           |               | X       | -       | -     |
| HBAC Magnitude       | hbacm           |               | X       | -       | -     |
| HBAC Phase           | hbacp           |               | X       | -       | -     |
| SNAC Magnitude       | snacm           |               | X       | -       | -     |
| SNAC Phase           | snacp           |               | X       | -       | -     |
| PAM4 modulation      | pam4_modulation | none          | X       | -       | -     |
| PAM4 mapping         | pam4_mapping    |               | X       | -       | -     |
| Impedance            | z0              |               | X       | -       | -     |

## Example

```
p1 p gnd! port=1 dc=0 lfsr ( 0 1 0 0 0 1meg 1 [1,2] ) z0=50
```

## See Also

- [Port Element](#) in the *HSPICE® User Guide: Elements*

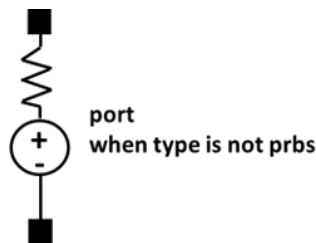
## Updating Port Number Automatically

You can create and use the `dbAnalogLibAutoUpdatePortNum` Tcl preference to control whether to automatically update the port number of port family devices.

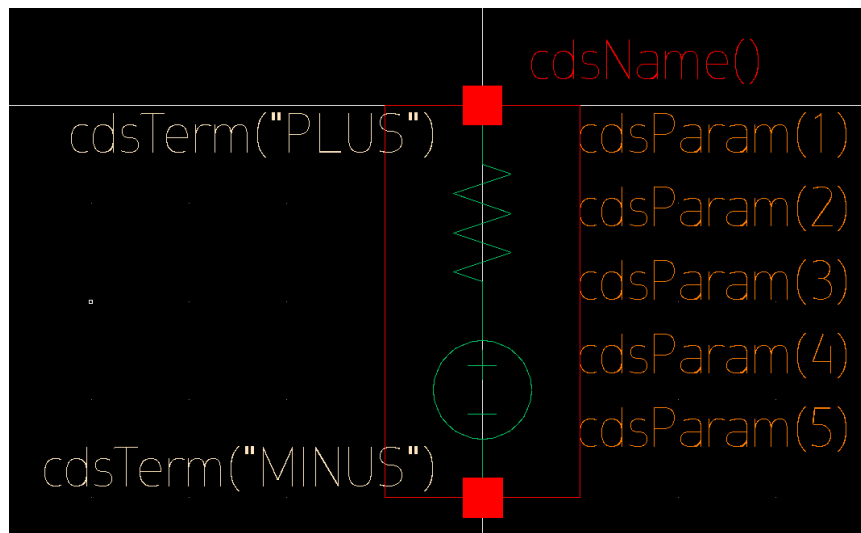
### port

Port element with selectable transient waveform type. Source type can be dc, exp, pulse, pwl, sine, and prbs.

This port element can also be used for Algorithmic Modeling Interface (AMI) to enable StatEye analysis of systems with Serializer-Deserializer (SERDES) devices for equalizers in both transmitter and receiver sides.

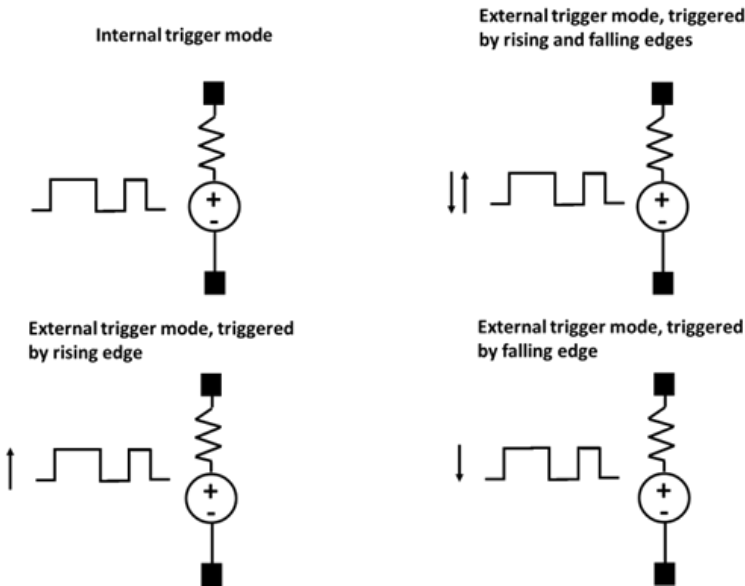


The port device is a PCell. Its PCell behavior is affected by its source type. For source type other than prbs, it is a 2-pin device and its symbol looks as follows:



For prbs source type, its PCell behavior is affected by its triggerValue parameter. The following figure shows four vprbs devices that have different trigger modes, which are Internal (on the upper-left), External by both edges (on the upper-right), External by falling edges (on the lower-left), and External by rising edges (on the lower-right) modes. The internal trigger mode has 2 pins (PLUS and MINUS) and other three external trigger

modes have 4 pins (P1, M1, P2, and M2). The netlisting procedure writes out their pins accordingly.



CDF Parameters

| CDF Parameter Prompt  | CDF Parameter  | Default Value | hspiceD | spectre | auCdl |
|-----------------------|----------------|---------------|---------|---------|-------|
| Source Type           | srcType        | dc            | X       | X       | -     |
| Source Type           | sourceType     | port          | -       | X       | -     |
| Port Number           | num            |               | X       | X       | -     |
| Impedance             | z0             |               | X       | -       | -     |
| Resistance            | r              | 50            | -       | X       | -     |
| Reactance             | x              |               | -       | X       | -     |
| Power Switch          | power          |               | X       | -       | -     |
| DC Value              | dc             |               | X       | -       | -     |
| DC Voltage            | vdc            |               | -       | X       | -     |
| Waveform entry method | pwlEntryMethod | File          | -       | X       | -     |
| File name             | fileName       |               | -       | X       | -     |

| CDF Parameter Prompt  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------|---------------|---------------|---------|---------|-------|
| File name is variable | file_as_var   | 0             | -       | X       | -     |
| Time Value Pairs      | tvpairs       | 0             | X       | X       | -     |
| Time1-50              | t1-50         |               | X       | X       | -     |
| Z State1-50           | Z1-50         | 0             | -       | -       | -     |
| Value1-50             | v1-50         |               | X       | X       | -     |
| Zero value            | val0          |               | -       | X       | -     |
| One value             | val1          |               | -       | X       | -     |
| Rise Delay Time       | td1           | 0             | X       | X       | -     |
| Fall Delay Time       | td2           | 0             | X       | X       | -     |
| Rise Time Constant    | tau1          | 0             | X       | X       | -     |
| Fall Time Constant    | tau2          | 0             | X       | X       | -     |
| Exp period            | expperiod     |               | -       | x       | -     |
| Rise Time             | tr            |               | X       | X       | -     |
| Fall Time             | tf            |               | X       | X       | -     |
| Pulse width           | pw            |               | X       | X       | -     |
| Period                | per           |               | X       | X       | -     |
| Offset Value          | vo            | 0             | X       | X       | -     |
| DC offest             | offset        |               | -       | X       | -     |
| Desired rms value     | rms           |               | -       | x       | -     |
| Time scaling factor   | stretch       |               | -       | X       | -     |
| Repeat                | rpt           |               | X       | -       | -     |
| Repeat Stop Time      | rstop         |               | X       | -       | -     |
| Repeat Stop Value     | stopvalue     |               | X       | -       | -     |
| Repeat Stop Slope     | stopslope     |               | X       | -       | -     |
| PWL period            | pwlperiod     |               | -       | X       | -     |

| CDF Parameter Prompt               | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------------|---------------|---------------|---------|---------|-------|
| Edge type                          | edgetype      |               | -       | X       | -     |
| Breakpoints                        | allbrkpts     | no            | -       | X       | -     |
| Width of transition                | twidht        |               | -       | X       | -     |
| Frequency name 1                   | fundname      |               | -       | X       | -     |
| Frequency                          | freq          |               | X       | X       | -     |
| Amplitude Value                    | va            | 0             | X       | X       | -     |
| Amplitude 1 (dBm)                  | vaDBm         |               | -       | X       | -     |
| Phase for sinusoid                 | sinephase     | 0             | X       | X       | -     |
| Voltage level                      | sinedc        |               | -       | X       | -     |
| Signal Amplitude                   | sa            | 0             | X       | -       | -     |
| Carrier Frequency                  | fc            | 1             | X       | -       | -     |
| Modulation Frequency               | fm            | 0             | X       | -       | -     |
| Offset Constant                    | oc            | 0             | X       | -       | -     |
| Voltage High                       | vh            | 1             | X       | -       | -     |
| Voltage Low                        | vl            | 0             | X       | -       | -     |
| Sample Time                        | tsample       | 1n            | X       | -       | -     |
| Data                               | data          | b1010<br>110  | X       | -       | -     |
| Pattern Parameter data is variable | data_as_var   |               | -       | X       | -     |
| Pattern Parameter rptstart         | rptstart      | 1             | X       | X       | -     |
| Pattern Parameter rpttimes         | rpttimes      | 0             | X       | X       | -     |
| Repeat Bit                         | rb            |               | X       | -       | -     |
| Modulation Index                   | mdi           | 0             | X       | -       | -     |
| Signal Frequency                   | fs            |               | X       | -       | -     |



| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Voltage Low               | vlow          | 0             | X       | -       | -     |
| Voltage High              | vhigh         | 1             | X       | -       | -     |
| Delay Time                | td            | 0             | X       | X       | -     |
| Bit Rate                  | rate          | 1M            | X       | -       | -     |
| Seed                      | seed          | 1             | X       | -       | -     |
| Taps                      | taps          | [1,2]         | X       | -       | -     |
| Output Resistance         | rout          |               | X       | -       | -     |
| Encode (DW8b10b)          | enc           | 0             | X       | -       | -     |
| Initial Running Disparity | ird           |               | X       | -       | -     |
| Damping Factor            | theta         | 0             | X       | X       | -     |
| # HB Tones                | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7          | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7               | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7            | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7                | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7             | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7             | hbmodtone1-7  |               | X       | -       | -     |
| SI Parameters             | siParameters  | 0             | X       | -       | -     |
| Emphasis Level            | emphasisLevel |               | X       | -       | -     |
| Emphasis Time             | emphasisTime  |               | X       | -       | -     |
| Tap Coef. Pairs           | dfepairs      | 0             | X       | -       | -     |
| DFE Tap 1-50              | tap1-50       |               | X       | -       | -     |
| Coefficient 1-50          | coef1-50      |               | X       | -       | -     |
| DFE Vth                   | dfe_vth       | 0             | X       | -       | -     |
| DFE Clk                   | dfe_clk       | 0             | X       | -       | -     |

| CDF Parameter Prompt                     | CDF Parameter       | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|---------------------|---------------|---------|---------|-------|
| Duty Cycle Distortion                    | dcd                 |               | X       | -       | -     |
| Duty Cycle Distortion Variation Type     | dcdType             |               | X       | -       | -     |
| Periodic Jitter Voltage                  | pj                  |               | X       | -       | -     |
| Periodic Jitter Type                     | pjType              |               | X       | -       | -     |
| Random Jitter                            | rj                  |               | X       | -       | -     |
| Voltage Noise                            | vn                  |               | X       | -       | -     |
| Algorithmic Modeling Options             | algorithmicModeling | n/a           | X       | -       | -     |
| Algorithmic Modeling Interface Object    | amiObj              |               | X       | -       | -     |
| Algorithmic Modeling Interface Parameter | amiParam            |               | X       | -       | -     |
| AMI Param XX Name                        | amiParamXxName      |               | X       | -       | -     |
| AMI Param XX Value                       | amiParamXxValue     |               | X       | -       | -     |
| Index number of AMI corner               | amiCorner           |               | X       | -       | -     |
| AMI Clock Data Recovery                  | amiCdr              |               | X       | -       | -     |
| Rise delay                               | td01                |               | -       | X       | -     |
| Fall delay                               | td10                |               | -       | X       | -     |
| PAM4 modulation                          | pam4_modulation     | none          | X       | X       | -     |
| PAM4 mapping                             | pam4_mapping        |               | X       | X       | -     |
| IBIS AMI file                            | ibisAmiFile         |               | X       | -       | -     |
| IBIS AMI model                           | ibisAmiModel        |               | X       | -       | -     |

| CDF Parameter Prompt                                       | CDF Parameter      | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------------------------|--------------------|---------------|---------|---------|-------|
| Use VRANGE value info in *.ibs file in IBIS-AMI simulation | ibisAmiVrange Mode |               | X       | -       | -     |
| Use RAMP info in *.ibs file                                | ibisAmiUseRamp     |               | X       | -       | -     |
| Use VRANGE info in *.ibs file                              | ibisAmiUseVrange   |               | X       | -       | -     |
| Use linearized pull-up or pull-down info in *.ibs file     | ibisAmiUseZsrc     |               | X       | -       | -     |
| Display second sinusoid                                    | numofsines         | 0             | -       | X       | -     |
| Frequency name 2                                           | fundname2          |               | -       | X       | -     |
| Frequency 2                                                | freq2              |               | -       | X       | -     |
| Amplitude 2 (Vpk)                                          | va2                |               | -       | X       | -     |
| Amplitude 2 (dBm)                                          | vaDBm2             |               | -       | X       | -     |
| Phase for sinusoid 2                                       | sinephase2         |               | -       | X       | -     |
| Display multi sinusoid                                     | vecofsines         | 0             | -       | X       | -     |
| Number of frequencies                                      | freqvec            | 0             | -       | X       | -     |
| Sinusoid frequency 1-9                                     | freqv1-9           |               | -       | X       | -     |
| Sinusoid amplitude 0-9                                     | vav0-9             |               | -       | X       | -     |
| Sinusoid amplitude 1-9 (DBm)                               | vaDBmv1-9          |               | -       | X       | -     |
| Sinusoid phase 1-9                                         | phasev1-9          |               | -       | X       | -     |
| Sinusoid max harmonic 1-9                                  | harmv1-9           |               | -       | X       | -     |
| Display modulation params                                  | modulation         | 0             | -       | X       | -     |

| CDF Parameter Prompt        | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|-------------------|---------------|---------|---------|-------|
| Amp modulation index 1      | ammodindex        |               | -       | X       | -     |
| Amp modulation frequency 1  | ammodfreq         |               | -       | X       | -     |
| Amp modulation phase 1      | ammodphase        |               | -       | X       | -     |
| FM index 1                  | fmmodindex        |               | -       | X       | -     |
| FM frequency 1              | fmmodfreq         |               | -       | X       | -     |
| IQ modulation I File        | imodfile          |               | -       | X       | -     |
| I File name is variable     | imodfile_as_var   | 0             | -       | X       | -     |
| IQ modulation Q File        | qmodfile          |               | -       | X       | -     |
| Q File name is variable     | qmodfile_as_var   | 0             | -       | X       | -     |
| Number of FM files          | filenums          | none          | -       | X       | -     |
| Name of FM file 1           | fmmodfile1        |               | -       | X       | -     |
| FM file1 name is variable   | fmmodfile1_as_var | 0             | -       | X       | -     |
| Name of FM file 2           | fmmodfile2        |               | -       | X       | -     |
| FM file2 name is variable   | fmmodfile2_as_var | 0             | -       | X       | -     |
| Amplitude scale factor      | scale             |               | -       | X       | -     |
| Power of PWL waveform       | pwldbm            |               | -       | X       | -     |
| Filter for pwl source       | pwlfilter         |               | -       | X       | -     |
| Rolloff factor              | rolloff           |               | -       | X       | -     |
| Bandwidth of the pwl signal | pwlbandwidth      |               | -       | X       | -     |
| Delay time                  | tdPrbs            |               | -       | X       | -     |

| CDF Parameter Prompt           | CDF Parameter       | Default Value                                          | hspiceD | spectre | auCdl |
|--------------------------------|---------------------|--------------------------------------------------------|---------|---------|-------|
| Bit period                     | perPrbs             |                                                        | -       | X       | -     |
| Transition reference           | transitionreference |                                                        | -       | X       | -     |
| Lower threshold(percent)       | lowerthreshold      |                                                        | -       | X       | -     |
| Upper threshold(percent)       | upperthreshold      |                                                        | -       | X       | -     |
| Edge type                      | edgetypePrbs        |                                                        | -       | X       | -     |
| triggerValue                   | triggerValue        | 1                                                      | -       | X       | -     |
| Trigger                        | trigger             | Internal                                               | -       | X       | -     |
| Threshold                      | triggerthreshold    |                                                        | -       | X       | -     |
| LFSR Mode                      | lfsrmode            |                                                        | -       | X       | -     |
| Browse and specify bit file    | prbsFileBox         | 0                                                      | -       | -       | -     |
| Bit data file                  | prbsFile            |                                                        | -       | X       | -     |
| Bit data file name is variable | prbsFile_as_var     | 0                                                      | -       | X       | -     |
| Seed                           | lfsrseed            | 1 3 5 7 9<br>11 13 15<br>17 19 21<br>23 25 27<br>29 31 | -       | X       | -     |
| Taps                           | lfsrtaps            |                                                        | -       | X       | -     |
| RJ(rms)                        | rjrms               |                                                        | -       | X       | -     |
| RJ(seed)                       | rjseed              |                                                        | -       | X       | -     |
| Number of periodic jitters     | nPeriodicJitter     | 0                                                      | -       | X       | -     |
| PJ1(amplitude)                 | pjump1              |                                                        | -       | X       | -     |
| PJ1(frequency)                 | pjfreq1             |                                                        | -       | X       | -     |
| PJ1(type)                      | pjtype1             | sine                                                   | -       | X       | -     |

| CDF Parameter Prompt          | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|-------------------------------|-----------------|---------------|---------|---------|-------|
| PJ2(amplitude)                | pjamp2          |               | -       | X       | -     |
| PJ2(frequency)                | pjfreq2         |               | -       | X       | -     |
| PJ2(type)                     | pjtype2         | sine          | -       | X       | -     |
| PJ3(amplitude)                | pjamp3          |               | -       | X       | -     |
| PJ3(frequency)                | pjfreq3         |               | -       | X       | -     |
| PJ3(type)                     | pjtype3         | sine          | -       | X       | -     |
| Display small signal params   | smallSig        | 0             | X       | X       | -     |
| HBAC Magnitude                | hbacm           |               | X       | -       | -     |
| HBAC Phase                    | hbacp           |               | X       | -       | -     |
| SNAC Magnitude                | snacm           |               | X       | -       | -     |
| SNAC Phase                    | snacp           |               | X       | -       | -     |
| PAC magnitude                 | pacm            |               | -       | X       | -     |
| PAC magnitude (dBm)           | pacDBm          |               | -       | X       | -     |
| PAC phase                     | pacp            |               | -       | X       | -     |
| AC Magnitude                  | acm             |               | X       | X       | -     |
| AC Phase                      | acp             |               | X       | X       | -     |
| XF magnitude                  | xfm             |               | -       | X       | -     |
| Display resistance parameters | resistanceParam | 0             | X       | -       | -     |
| Open DC Connection            | dcopen          | ""            | X       | -       | -     |
| DC Resistance                 | rdc             |               | X       | -       | -     |
| AC Resistance                 | rac             |               | X       | -       | -     |
| HB Resistance                 | rhb             |               | X       | -       | -     |
| HBAC Resistance               | rhbac           |               | X       | -       | -     |
| TRAN Resistance               | rtran           |               | X       | -       | -     |

| CDF Parameter Prompt              | CDF Parameter          | Default Value                     | hspiceD | spectre | auCdl |
|-----------------------------------|------------------------|-----------------------------------|---------|---------|-------|
| Display temperature params        | tempParam              | 0                                 | -       | X       | -     |
| Linear temperature coefficient    | tc1                    |                                   | -       | -       | -     |
| Quadratic temperature coefficient | tc2                    |                                   | -       | X       | -     |
| Nominal temperature               | tnom                   |                                   | -       | X       | -     |
| Display noise parameters          | noiseParam             | 0                                 | -       | X       | -     |
| Generate noise?                   | isnoisyNoiseModel      |                                   | -       | X       | -     |
| Noise temperature                 | noisetemp              |                                   | -       | -       | -     |
| Noise entry method                | noiseEntryMethod       | File                              | -       | X       | -     |
| Number of noise/frequency pairs   | FNpairs                | 0                                 | -       | X       | -     |
| Frequency1-50                     | F1-50                  |                                   | -       | X       | -     |
| Noise1-50                         | N1-50                  |                                   | -       | X       | -     |
| Noise file name                   | noisefile              |                                   | -       | X       | -     |
| Noise file name is variable       | noise_file_as_var      | 0                                 | X       | X       | -     |
| Noise type                        | noiseModelType         | Noise Voltage(V <sup>2</sup> /Hz) | -       | X       | -     |
| Interpolation method              | noiseModelInterMethod  |                                   | -       | X       | -     |
| High-Z impedance                  | highz                  |                                   | -       | X       | -     |
| Min high-Z transition width       | min_z_transition_width |                                   | -       | X       | -     |
| Multiplier                        | m                      |                                   | -       | X       | -     |
| Version                           | version                | 2                                 | -       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output            | spectre Output         |
|-------------------|---------------------------|------------------------|
| noise_file_as_var |                           |                        |
| false (default)   | noise file='NOISEFILE'    | noise file="NOISEFILE" |
| true              | noise file=str(NOISEFILE) | noise file=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the `dbAnalogLibPortVersion` Tcl preference before the analogLib library is loaded.

## Updating Port Number Automatically

You can create and use the `dbAnalogLibAutoUpdatePortNum` Tcl preference to control whether to automatically update the port number of port family devices.

## Using port to Set up Harmonic Balance or Shooting PSS simulations

The sine type is commonly used to set up single- and multi-tone Harmonic Balance (HB) and Shooting Newton simulations. The parameter **Impedance** sets the value of the port impedance (default 50 Ohms). DC Value is used to set the value to use in DC simulations, which defaults to the waveform's value at t=0.

**Note:**

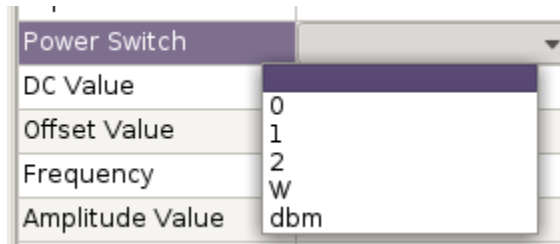
HB Tones parameters are supported only for legacy reasons in the PrimeSim HSPICE tool, but their ongoing use is not recommended.



| Parameters   |       |
|--------------|-------|
| Prompt       | Value |
| Source Type  | sine  |
| Port Number  | 1     |
| Impedance    |       |
| Power Switch |       |
| DC Value     |       |

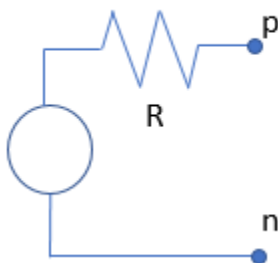
Use the **Power Switch** parameter to control how the port's amplitude is interpreted:

- 0 or unspecified if it specifies the open-circuit voltage
- 1 or W if it specifies available power in Watts
- 2 or dbm for available power in dbm



*A note about port's amplitude definition:*

The port's equivalent model is shown in the following figure.



When **Power Switch=0**, the port's amplitude is defined as the open-circuit voltage across (p, n). Starting with FineSim and PrimeSim SPICE, the option `.option port_element_voltage_matchload` can be used to tell the engines to treat the amplitude as the voltage across (p, n) when a matched resistor is connected across (p, n).

Single-sinusoid Specification

The single-sine source implements the function:

$$v(t) = v_0 + v_a e^{-q(t-t_d)} \sin(2\pi f(t - t_d) + \varphi)$$

|                    |   |
|--------------------|---|
| Offset Value       | 0 |
| Frequency          |   |
| Amplitude Value    |   |
| Phase for sinusoid |   |
| Delay Time         |   |
| Damping Factor     |   |

Simulation defaults are zero for each of the parameters. The mapping between the parameters and the functional expression is as shown in the following table.

| Parameter          | Maps to ... |
|--------------------|-------------|
| Offset Value       | $v_0$       |
| Frequency          | $f$         |
| Amplitude Value    | $v_a$       |
| Phase for sinusoid | $\varphi$   |
| Delay Time         | $t_d$       |
| Damping Factor     | $q$         |

Damping factor is used in transient simulations, but ignored in HB and Shooting Newton simulations.

Two-sinusoid specification

To set up a two-tone source, click **Display second sinusoid**.

|                    |                                     |                                                                                     |
|--------------------|-------------------------------------|-------------------------------------------------------------------------------------|
| Display second ... | <input checked="" type="checkbox"/> |  |
| Frequency 2        |                                     |                                                                                     |
| Amplitude 2 (Vpk)  |                                     |                                                                                     |
| Phase for sinus... |                                     |                                                                                     |

The meaning of the second sinusoid parameters is analogous to the first. It implements the function:

$$v(t) = v_0 + e^{-q(t-t_d)}[v_a \sin(2\pi f(t - t_d) + \varphi) + v_{a2} \sin(2\pi f_2(t - t_d) + \varphi_2)]$$

As before, **Power Switch** determines the meaning of  $v_{a2}$ , whether it is measured as open-circuit voltage or as available power in Watts or dbm.

**Multi-sinusoid specification**

You can add additional sinusoids using **Display multi sinusoid** and specifying the number of frequencies to add.

|                       |   |
|-----------------------|---|
| Number of frequencies | 3 |
| Sinusoid Frequency 1  |   |
| Sinusoid Ampl 1 (Vpk) |   |
| Sinusoid Ampl 1 (dbm) |   |
| Sinusoid Phase 1      |   |
| Sinusoid Maxharm 1    |   |
| Sinusoid Frequency 2  |   |
| Sinusoid Ampl 2 (Vpk) |   |
| Sinusoid Ampl 2 (dbm) |   |
| Sinusoid Phase 2      |   |
| Sinusoid Maxharm 2    |   |
| Sinusoid Frequency 3  |   |
| Sinusoid Ampl 3 (Vpk) |   |
| Sinusoid Ampl 3 (dbm) |   |
| Sinusoid Phase 3      |   |
| Sinusoid Maxharm 3    |   |

**Delay Time**, **Damping Factor**, and **Power Switch** do not apply to additional sinusoids. Instead of **Power Switch**, additional sinusoids are specified in Volts by setting **Sinusoid Ampl N (Vpk)**, or in dbm by setting **Sinusoid Ampl N (dbm)**. You can specify levels in Volts or dbm, but not both.

Maxharm settings are currently ignored.

Although you can treat multi-sinusoids as additions to single- and two-tone specifications (to specify tone 3 and higher), it is recommended to not to mix them together. Instead, use

one form or the other but not both. When using the multi-sinusoid form, the functional form of the waveform is:

$$v(t) = v_0 + \sum_{n=1}^N v_{ampl\ n} \sin (2\pi f_n + \varphi_n)$$

$v_{ampl\ n}$  corresponds to **Sinusoid Ampl n (Vpk)** or **Sinusoid Ampl n (dbm)**,  $f_n$  is **Sinusoid Frequency n**, and so forth.  $v_0$  is the **Offset Value** parameter.

Small signal parameters

Small signal parameters set the source amplitudes in AC, small-signal HB (HBAC), and small-signal Shooting Newton (SNAC) analysis. **Power Switch** determines whether the source is specified in open-circuit Volts, available Watts, or available dbm.

|                              |                                     |
|------------------------------|-------------------------------------|
| Display small signal para... | <input checked="" type="checkbox"/> |
| HBAC Magnitude               |                                     |
| HBAC Phase                   |                                     |
| SNAC Magnitude               |                                     |
| SNAC Phase                   |                                     |
| AC Magnitude                 |                                     |
| AC Phase                     |                                     |

Resistance parameters

The use of resistance parameters is not recommended.

|                             |                                     |
|-----------------------------|-------------------------------------|
| Display resistance param... | <input checked="" type="checkbox"/> |
| Open DC Connection          |                                     |
| DC Resistance               |                                     |
| AC Resistance               |                                     |
| HB Resistance               |                                     |
| HBAC Resistance             |                                     |
| TRAN Resistance             |                                     |

Noise parameters

|                          |                                     |
|--------------------------|-------------------------------------|
| Display noise parameters | <input checked="" type="checkbox"/> |
| Generate noise?          | no                                  |
| Noise temperature        |                                     |

By default, the port's noise voltage contribution is

$$\sqrt{4KTR},$$

where R is port **Impedance** and the default value of T is the circuit temperature. You can turn off noise contribution using **Generate noise**. You can set T to temperature different than circuit using **Noise temperature**.

In noise figure simulations using .LIN, .HBAC, or .SNAC analysis, the input port's temperature is 290 if unspecified, and the output port's contribution is ignored.

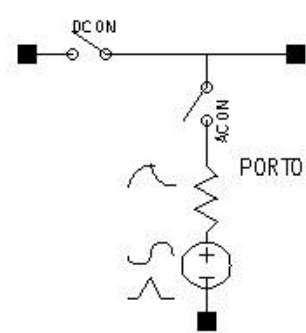
See Also

- [Port Element](#) in the *PrimeSim Continuum User Guide: Elements*

port3t

The port3t element is a port with T-shaped connection of DC (chock inductor) and AC (blocking capacitor) control.

The port element is a voltage source with internal resistance. See [port](#).



CDF Parameters

| CDF Parameter Prompt        | CDF Parameter        | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|----------------------|---------------|---------|---------|-------|
| Frequency name 1            | fundname             | -             | -       | x       | -     |
| Frequency name 2            | fundname2            | -             | -       | x       | -     |
| Noise file name             | noisefile            | -             | -       | x       | -     |
| Noise file name is variable | noise_file_as_ 0 var | -             | -       | X       | -     |
| File name                   | fileName             | -             | -       | x       | -     |

| CDF Parameter Prompt          | CDF Parameter    | Default Value | hspiceD | spectre | auCdl |
|-------------------------------|------------------|---------------|---------|---------|-------|
| Display second sinusoid       | numofsines       | 0             | -       | x       | -     |
| Display modulation params     | modulation       | 0             | -       | x       | -     |
| Display temperature params    | tempParam        | 0             | -       | x       | -     |
| Display small signal params   | smallSig         | 0             | -       | x       | -     |
| Display noise parameters      | noiseParam       | 0             | -       | x       | -     |
| Num. of noise/freq pairs      | FNpairs          | 0             | -       | x       | -     |
| Freq 1-10                     | F1-10            |               | -       | x       | -     |
| Noise 1-10                    | N1-10            |               | -       | x       | -     |
| Waveform Entry Method         | pwlEntryMethod   | File          | -       | x       | -     |
| Noise Entry Method            | noiseEntryMethod | File          | -       | x       | -     |
| Number of PWL/Time pairs      | tvpairs          | 0             | -       | x       | -     |
| Time 1-50                     | t1-50            |               | -       | x       | -     |
| Source type                   | sourceType       | port          | -       | x       | -     |
| Resistance                    | r                | 50            | -       | x       | -     |
| Chock ind for net analyser    | lchock           |               | -       | x       | -     |
| Blocking cap for net analyser | cblock           |               | -       | x       | -     |
| Port number                   | num              |               | -       | x       | -     |
| DC voltage                    | vdc              |               | -       | x       | -     |
| Delay time                    | td               |               | -       | x       | -     |
| Zero value                    | val0             |               | -       | x       | -     |

| CDF Parameter Prompt     | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------|---------------|---------------|---------|---------|-------|
| One value                | val1          |               | -       | X       | -     |
| Period of waveform       | per           |               | -       | X       | -     |
| Rise time                | tr            |               | -       | X       | -     |
| Fall time                | tf            |               | -       | X       | -     |
| Pulse width              | pw            |               | -       | X       | -     |
| Rise time start          | td1           |               | -       | X       | -     |
| Rise time constant       | tau1          |               | -       | X       | -     |
| Fall time start          | td2           |               | -       | X       | -     |
| Fall time constant       | tau2          |               | -       | X       | -     |
| Frequency 1              | freq          |               | -       | X       | -     |
| Amplitude 1 (Vpk)        | va            |               | -       | X       | -     |
| Amplitude 1 (dBm)        | vaDBm         |               | -       | X       | -     |
| Phase for Sinusoid 1     | sinephase     |               | -       | X       | -     |
| Sine DC level            | sinedc        |               | -       | X       | -     |
| Frequency 2              | freq2         |               | -       | X       | -     |
| Amplitude 2 (Vpk)        | va2           |               | -       | X       | -     |
| Amplitude 2 (dBm)        | vaDBm2        |               | -       | X       | -     |
| Phase for Sinusoid 2     | sinephase2    |               | -       | X       | -     |
| FM modulation index      | fmmodindex    |               | -       | X       | -     |
| FM modulation freq       | fmmodfreq     |               | -       | X       | -     |
| AM modulation index      | ammodindex    |               | -       | X       | -     |
| AM modulation freq       | ammodfreq     |               | -       | X       | -     |
| AM modulation phase      | ammodphase    |               | -       | X       | -     |
| Damping factor           | theta         |               | -       | X       | -     |
| Linear temp. coefficient | tc1           |               | -       | X       | -     |

| CDF Parameter Prompt    | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-------------------------|---------------|---------------|---------|---------|-------|
| Quadratic temp. coeff.  | tc2           |               | -       | x       | -     |
| Nominal temperature     | tnom          |               | -       | x       | -     |
| PAC Magnitude (Vpk)     | pacm          |               | -       | x       | -     |
| PAC Magnitude (dBm)     | pacDBm        |               | -       | x       | -     |
| PAC phase               | pacp          |               | -       | x       | -     |
| AC Magnitude (Vpk)      | acm           |               | -       | x       | -     |
| AC phase                | acp           |               | -       | x       | -     |
| XF Magnitude (Vpk)      | xfm           |               | -       | x       | -     |
| Noise temperature       | noisetemp     |               | -       | x       | -     |
| DC offset               | offset        |               | -       | x       | -     |
| Amplitude scale factor  | scale         |               | -       | x       | -     |
| Time scale factor       | stretch       |               | -       | x       | -     |
| Period                  | pwlperiod     |               | -       | x       | -     |
| Power of PWL waveform   | pwldbm        |               | -       | x       | -     |
| Transition width        | twidth        |               | -       | x       | -     |
| Multiplier              | m             |               | -       | x       | -     |
| Display multi sinusoid  | vecofsines    | 0             | -       | x       | -     |
| Number of Frequencies   | freqvec       | 0             | -       | x       | -     |
| Sinusoid Frequency 1-9  | freqv1-9      |               | -       | x       | -     |
| Sinusoid Ampl 1-9 (Vpk) | vav1-9        |               | -       | x       | -     |
| Sinusoid Ampl 1-9 (dbm) | vaDBmv1-9     |               | -       | x       | -     |



| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Sinusoid Phase 1-9   | phasev1-9     |               | -       | x       | -     |
| Sinusoid Maxharm 1-9 | harmv1-9      |               | -       | x       | -     |
| Number of FM Files   | filenums      | none          | -       | x       | -     |
| Name of FM File1     | fmmodfile1    |               | -       | x       | -     |
| Name of FM File2     | fmmodfile2    |               | -       | x       | -     |
| Reactance            | x             |               | -       | x       | -     |
| Source type          | srcType       | sine          | -       | x       | -     |
| Rise delay           | td01          |               | -       | x       | -     |
| Fall delay           | td10          |               | -       | x       | -     |
| Exp period           | expperiod     |               | -       | x       | -     |
| Desired rms value    | rms           |               | -       | x       | -     |
| Breakpoints          | allbrkpts     |               | -       | x       | -     |

**Note:**

- The default of noise\_file\_as\_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise\_file\_as\_var parameters is as follows:

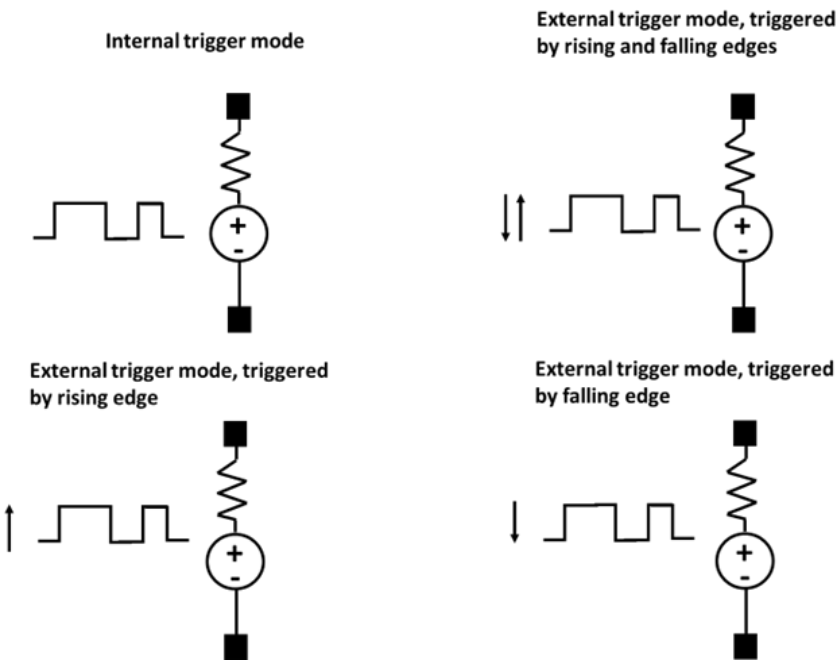
| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

pprbs

Port element with a pseudo random binary sequence function. This element is relevant only for Spectre netlisting and simulation at this moment.

The pprbs device is a PCell. Its PCell behavior is affected by the triggerValue parameter. The following figure shows four vprbs devices that have different trigger modes, which are Internal (on the upper-left), External by both edges (on the upper-right), External by falling edges (on the lower-left), and External by rising edges (on the lower-right) modes. The internal trigger mode has 2 pins (PLUS and MINUS) and other three external trigger modes have 4 pins (P1, M1, P2, and M2). The netlisting procedure writes out their pins accordingly.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Resistance           | r             | 50            | -       | X       | -     |
| Port number          | num           |               | -       | X       | -     |
| triggerValue         | triggerValue  | 1             | -       | X       | -     |
| Delay time           | tdPrbs        |               | -       | X       | -     |
| Zero value           | val0          |               | -       | X       | -     |

| CDF Parameter Prompt        | CDF Parameter       | Default Value                                          | hspiceD | spectre | auCdl |
|-----------------------------|---------------------|--------------------------------------------------------|---------|---------|-------|
| One value                   | val1                |                                                        | -       | X       | -     |
| Bit period                  | perPrbs             | 1n                                                     | -       | X       | -     |
| Rise time                   | tr                  |                                                        | -       | X       | -     |
| Fall time                   | tf                  |                                                        | -       | X       | -     |
| Transition reference        | transitionreference |                                                        | -       | X       | -     |
| Lower threshold(percent)    | lowerthreshold      |                                                        | -       | X       | -     |
| Upper threshold(percent)    | upperthreshold      |                                                        | -       | X       | -     |
| Edge type                   | edgetype            |                                                        | -       | X       | -     |
| Trigger                     | trigger             | Internal                                               | -       | X       | -     |
| Threshold                   | triggerthreshold    |                                                        | -       | X       | -     |
| LFSR Mode                   | lfsrmode            | PN32                                                   | -       | X       | -     |
| Browse and specify bit file | prbsFileBox         | 0                                                      | -       | X       | -     |
| Bit data file               | prbsFile            |                                                        | -       | X       | -     |
| Seed                        | lfsrseed            | 1 3 5 7 9<br>11 13 15<br>17 19 21<br>23 25 27<br>29 31 | -       | X       | -     |
| Taps                        | lfsrtaps            |                                                        | -       | X       | -     |
| RJ(rms)                     | rjrms               |                                                        | -       | X       | -     |
| RJ(seed)                    | rjseed              |                                                        | -       | X       | -     |
| Number of periodic jitters  | nPeriodicJitter     | 0                                                      | -       | X       | -     |
| PJ1(amplitude)              | pjamp1              |                                                        | -       | X       | -     |
| PJ1(frequency)              | pjfreq1             |                                                        | -       | X       | -     |

| CDF Parameter Prompt    | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-------------------------|---------------|---------------|---------|---------|-------|
| PJ1(type)               | pjtype1       | sine          | -       | X       | -     |
| PJ2(amplitude)          | pjamp2        |               | -       | X       | -     |
| PJ2(frequency)          | pjfreq2       |               | -       | X       | -     |
| PJ2(type)               | pjtype2       | sine          | -       | X       | -     |
| PJ3(amplitude)          | pjamp3        |               | -       | X       | -     |
| PJ3(frequency)          | pjfreq3       |               | -       | X       | -     |
| PJ3(type)               | pjtype3       | sine          | -       | X       | -     |
| Pattern Parameter data  | data          |               | -       | X       | -     |
| Reference Value         | ref           |               | -       | X       | -     |
| Random delay time       | jitter        |               | -       | X       | -     |
| Seed                    | seed          |               | -       | X       | -     |
| Feedback Shift Register | taps          |               | -       | X       | -     |
| Source type             | srcType       | prbs          | -       | X       | -     |
| Multiplier              | m             |               | -       | X       | -     |
| Version                 | version       | 2             | -       | -       | -     |

## Example

```
PORT0 (out 0) port r=50 type=prbs period=1n registerlength=32
lfsrseed=[1 3 5 7 \
 9 11 13 15 17 19 21 23 25 27 29 31]
```

## Controlling Version Number for Legacy Designs

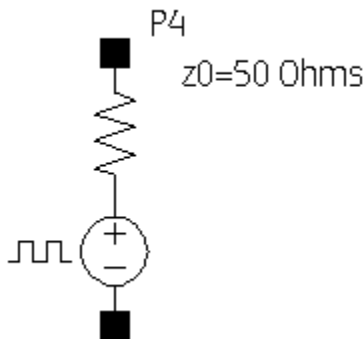
To control the version number for legacy designs, create and use the [dbAnalogLibPprbsVersion](#) Tcl preference before the analogLib library is loaded.

## Updating Port Number Automatically

You can create and use the [dbAnalogLibAutoUpdatePortNum](#) Tcl preference to control whether to automatically update the port number of port family devices.

ppulse

Port element with pulse transient function.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Source Type          | sourceType    | port          | -       | -       | -     |
| Source type          | srcType       | pulse         | -       | X       | -     |
| Port Number          | num           |               | X       | X       | -     |
| Impedance            | z0            | 50            | X       | -       | -     |
| Resistance           | r             | 50            | -       | X       | -     |
| Power Switch         | power         |               | X       | -       | -     |
| DC Value             | dc            |               | X       | -       | -     |
| DC Voltage           | vdc           |               | -       | X       | -     |
| Value 1              | v1            | 0             | X       | X       | -     |
| Value 2              | v2            | 0             | X       | X       | -     |
| Delay Time           | td            | 0             | X       | X       | -     |
| Rise Time            | tr            |               | X       | X       | -     |
| Fall Time            | tf            |               | X       | X       | -     |
| Pulse Width          | pw            |               | X       | X       | -     |
| Period               | per           |               | X       | X       | -     |

| CDF Parameter Prompt                     | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|-------------------|---------------|---------|---------|-------|
| # HB Tones                               | hbtones           | 0             | X       | -       | -     |
| HB Magnitude 1-7                         | hbmag1-7          |               | X       | -       | -     |
| HB Phase1-7                              | hbphase1-7        | 0             | X       | -       | -     |
| HB Harmonic1-7                           | hbharm1-7         | 1             | X       | -       | -     |
| HB Tone1-7                               | hbtone1-7         | 1-7           | X       | -       | -     |
| HB ModHarm1-7                            | hbmodharm1-7      |               | X       | -       | -     |
| HB ModTone1-7                            | hbmodtone1-7      |               | X       | -       | -     |
| Open DC Connection                       | dcopen            |               | X       | -       | -     |
| DC Resistance                            | rdc               |               | X       | -       | -     |
| AC Resistance                            | rac               |               | X       | -       | -     |
| HB Resistance                            | rhb               |               | X       | -       | -     |
| HBAC Resistance                          | rhbac             |               | X       | -       | -     |
| TRAN Resistance                          | rtran             |               | X       | -       | -     |
| HBAC Magnitude                           | hbacm             |               | X       | -       | -     |
| HBAC Phase                               | hbacp             |               | X       | -       | -     |
| SNAC Magnitude                           | snacm             |               | X       | -       | -     |
| SNAC Phase                               | snacp             |               | X       | -       | -     |
| Noise file name                          | noisefile         |               | -       | X       | -     |
| Noise file name is variable              | noise_file_as_var | 0             | -       | X       | -     |
| Number of noise/frequency pairs          | FNpairs           | 0             | -       | X       | -     |
| Frequency1-50                            | F1-50             |               | -       | X       | -     |
| Noise1-50                                | N1-50             |               | -       | X       | -     |
| Frequency name for fundamental frequency | fundname          |               | -       | X       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Multiplier           | m             |               | -       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| PAC magnitude        | pacm          |               | -       | X       | -     |
| PAC phase            | pacp          |               | -       | X       | -     |
| XF magnitude         | xfm           |               | -       | X       | -     |
| Temp coefficient 1   | tc1           |               | -       | X       | -     |
| Temp coefficient 2   | tc2           |               | -       | X       | -     |
| Nominal temperature  | tnom          |               | -       | X       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

Example

```
p4 n1 gnd! port=1 dc=3.3 pulse ( 0 2.5 0 50p 50p 2n 4n ) z0=50
```

See Also

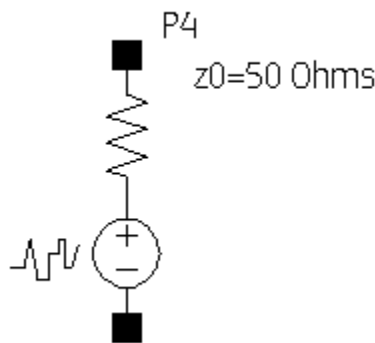
- [Port Element](#) in the *HSPICE® User Guide: Elements*

Updating Port Number Automatically

You can create and use the `dbAnalogLibAutoUpdatePortNum` Tcl preference to control whether to automatically update the port number of port family devices.

ppwl

Port element with PWL transient function.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Source Type          | sourceType    | port          | -       | -       | -     |
| Source Type          | srcType       | pwl           | -       | X       |       |
| Port Number          | num           |               | X       | X       | -     |
| Impedance            | z0            | 50            | X       | -       | -     |
| Resistance           | r             | 50            | -       | X       | -     |
| Power Switch         | power         |               | X       | -       | -     |
| DC Value             | dc            |               | X       | -       | -     |



| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Voltage           | vdc           |               | -       | X       | -     |
| Time Value Pairs     | tvpairs       | 2             | -       | X       | -     |
| Time1-50             | t1-50         |               | X       | X       | -     |
| Value1-50            | v1-50         | 0             | X       | X       | -     |
| Repeat               | rpt           |               | -       | -       | -     |
| Repeat Stop Time     | rstop         |               | X       | -       | -     |
| Repeat Stop Value    | stopvalue     |               | X       | -       | -     |
| Repeat Stop Slope    | stopslope     |               | X       | -       | -     |
| Delay Time           | td            | 0             | -       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | X       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |

| CDF Parameter Prompt                     | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|-------------------|---------------|---------|---------|-------|
| HBAC Phase                               | hbacp             |               | X       | -       | -     |
| SNAC Magnitude                           | snacm             |               | X       | -       | -     |
| SNAC Phase                               | snacp             |               | X       | -       | -     |
| Number of noise/frequency pairs          | FNpairs           | 0             | -       | X       | -     |
| Noise file name                          | noisefile         |               | -       | X       | -     |
| Noise file name is variable              | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                            | F1-50             |               | -       | X       | -     |
| Noise1-50                                | N1-50             |               | -       | X       | -     |
| Frequency name for fundamental frequency | fundname          |               | -       | X       | -     |
| Multiplier                               | m                 |               | -       | X       | -     |
| Noise temperature                        | noisetemp         |               | -       | X       | -     |
| Nominal temperature                      | tnom              |               | -       | X       | -     |
| PAC magnitude                            | pacm              |               | -       | X       | -     |
| PAC phase                                | pacp              |               | -       | X       | -     |
| PWL period                               | pwlperiod         |               | -       | X       | -     |
| Amplitude scale factor                   | scale             |               | -       | X       | -     |
| Transition width                         | twidht            |               | -       | X       | -     |
| Desired rms value                        | rms               |               | -       | X       | -     |
| Time scaling factor                      | stretch           |               | -       | X       | -     |
| Temp coefficient 1                       | tc1               |               | -       | X       | -     |
| Temp coefficient 2                       | tc2               |               | -       | X       | -     |
| XF magnitude                             | xfm               |               | -       | X       | -     |
| Offset voltage                           | vo                |               | -       | X       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Breakpoints          | allbrkpts     |               | -       | X       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

## Example

```
p4 n1 gnd! port=1 dc=0 pw1 ( 0 0.0 1n 2.5 td=0 ) z0=50
```

## See Also

- [Port Element](#) in the *HSPICE® User Guide: Elements*

## Controlling Version Number for Legacy Designs

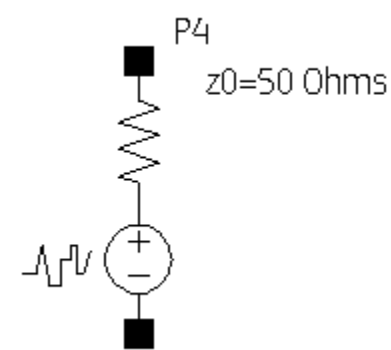
To control the version number for legacy designs, create and use the [dbAnalogLibPpw1Version](#) Tcl preference before the analogLib library is loaded.

## Updating Port Number Automatically

You can create and use the `dbAnalogLibAutoUpdatePortNum` Tcl preference to control whether to automatically update the port number of port family devices.

### ppwlf

Port element with PWL transient function specified using a file, containing time voltage value pairs.



### CDF Parameters

| CDF Parameter Prompt  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------|---------------|---------------|---------|---------|-------|
| Source Type           | sourceType    | port          | -       | -       | -     |
| Source Type           | srcType       | pwlf          | -       | X       | -     |
| Port Number           | num           |               | X       | X       | -     |
| Impedance             | z0            | 50            | X       | -       | -     |
| Resistance            | r             | 50            | -       | X       | -     |
| Power Switch          | power         |               | X       | -       | -     |
| DC Value              | dc            |               | X       | -       | -     |
| DC Voltage            | vdc           |               | -       | X       | -     |
| PWL File              | fileName      |               | X       | X       | -     |
| File name is variable | file_as_var   | 0             | X       | X       | -     |
| Repeat                | rpt           |               | X       | -       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Repeat Stop Time     | rstop         |               | X       | -       | -     |
| Repeat Stop Value    | stopvalue     |               | X       | -       | -     |
| Repeat Stop Slope    | stopslope     |               | X       | -       | -     |
| Delay Time           | td            | 0             | X       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | X       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |
| SNAC Magnitude       | snacm         |               | X       | -       | -     |
| SNAC Phase           | snacp         |               | X       | -       | -     |
| PAC magnitude        | pacm          |               | -       | X       | -     |
| PAC phase            | pacp          |               | -       | X       | -     |

| CDF Parameter Prompt                     | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|-------------------|---------------|---------|---------|-------|
| PWL period                               | pwlperiod         |               | -       | X       | -     |
| Amplitude scale factor                   | scale             |               | -       | X       | -     |
| Transition width                         | twidth            |               | -       | X       | -     |
| Desired rms value                        | rms               |               | -       | X       | -     |
| Time scaling factor                      | stretch           |               | -       | X       | -     |
| Temp coefficient 1                       | tc1               |               | -       | X       | -     |
| Temp coefficient 2                       | tc2               |               | -       | X       | -     |
| XF magnitude                             | xfm               |               | -       | X       | -     |
| Number of noise/frequency pairs          | FNpairs           | 0             | -       | X       | -     |
| Noise file name                          | noisefile         |               | -       | X       | -     |
| Noise file name is variable              | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                            | F1-50             |               | -       | X       | -     |
| Noise1-50                                | N1-50             |               | -       | X       | -     |
| Frequency name for fundamental frequency | fundname          |               | -       | X       | -     |
| Multiplier                               | m                 |               | -       | X       | -     |
| Nominal temperature                      | tnom              |               | -       | X       | -     |
| Offset voltage                           | vo                |               | -       | X       | -     |
| Breakpoints                              | allbrkpts         | no            | -       | X       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

**Example**

```
p4 n1 gnd! port=1 dc=0 pw1 pw1file='pw1_data.txt' td=0 z0=50
```

**See Also**

- [Port Element](#) in the *HSPICE® User Guide: Elements*

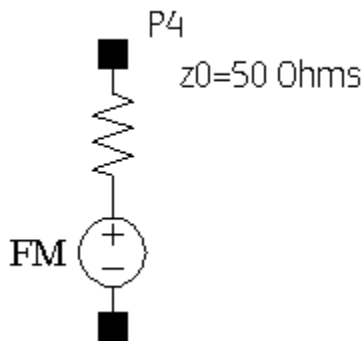
**Updating Port Number Automatically**

You can create and use the `dbAnalogLibAutoUpdatePortNum` Tcl preference to control whether to automatically update the port number of port family devices.

---

**psffm**

Port element with FM transient function.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Source Type          | sourceType    | port          | -       | -       | -     |
| Source Type          | srcType       | sffm          | -       | -       | -     |
| Port Number          | num           |               | X       | -       | -     |
| Impedance            | z0            | 50            | X       | -       | -     |
| Resistance           | r             | 50            | -       | -       | -     |
| Power Switch         | power         |               | X       | -       | -     |
| DC Value             | dc            |               | X       | -       | -     |
| DC Voltage           | vdc           |               | -       | -       | -     |
| Output Amplitude     | va            | 0             | X       | -       | -     |
| Offset Value         | vo            | 0             | X       | -       | -     |
| Carrier Frequency    | freq          | 1             | X       | -       | -     |
| Modulation Index     | mdi           | 0             | X       | -       | -     |
| Signal Frequency     | fs            |               | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | -       | -     |
| AC Phase             | acp           |               | X       | -       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude 1-7     | hbmag1-7      |               | X       | -       | -     |



| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | X       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |
| SNAC Phase           | snacp         |               | X       | -       | -     |

## Example

```
p4 n1 gnd! port=1 dc=0 sffm ( 0.35 500m 400Meg 5 500k ) z0=50
```

## See Also

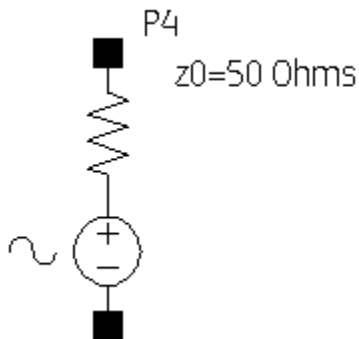
- [Port Element](#) in the *HSPICE® User Guide: Elements*

## Updating Port Number Automatically

You can create and use the [dbAnalogLibAutoUpdatePortNum](#) Tcl preference to control whether to automatically update the port number of port family devices.

## psin

Port element with sinusoidal transient function.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Source Type          | sourceType    | port          | -       | -       | -     |
| Source Type          | srcType       | sine          | -       | X       | -     |
| Port Number          | num           |               | X       | X       | -     |
| Impedance            | z0            | 50            | X       | -       | -     |
| Resistance           | r             | 50            | -       | X       | -     |
| Power Switch         | power         |               | X       | -       | -     |
| DC Value             | dc            |               | X       | -       | -     |
| DC Voltage           | vdc           |               | -       | X       | -     |
| Offset Value         | vo            | 0             | X       | X       | -     |
| Amplitude Value      | va            |               | X       | X       | -     |
| Frequency            | freq          |               | X       | X       | -     |
| Delay Time           | td            | 0             | X       | X       | -     |
| Damping Factor       | theta         | 0             | X       | X       | -     |
| Phase for sinusoid   | sinephase     | 0             | X       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |

| CDF Parameter Prompt      | CDF Parameter      | Default Value | hspiceD | spectre | auCdl |
|---------------------------|--------------------|---------------|---------|---------|-------|
| HB Magnitude1-7           | hbmag1-7           |               | X       | -       | -     |
| HB Phase1-7               | hbphase1-7         | 0             | X       | -       | -     |
| HB Harmonic1-7            | hbharm1-7          | 1             | X       | -       | -     |
| HB Tone1-7                | hbtone1-7          | 1-7           | X       | -       | -     |
| HB ModHarm1-7             | hbmodharm1-7       |               | X       | -       | -     |
| HB ModTone1-7             | hbmodtone1-7       |               | X       | -       | -     |
| Open DC Connection        | dcopen             |               | X       | -       | -     |
| DC Resistance             | rdc                |               | X       | -       | -     |
| AC Resistance             | rac                |               | X       | -       | -     |
| HB Resistance             | rhb                |               | X       | -       | -     |
| HBAC Resistance           | rhbac              |               | X       | -       | -     |
| TRAN Resistance           | rtran              |               | X       | -       | -     |
| HBAC Magnitude            | hbacm              |               | X       | -       | -     |
| HBAC Phase                | hbacp              |               | X       | -       | -     |
| SNAC Magnitude            | snacm              |               | X       | -       | -     |
| SNAC Phase                | snacp              |               | X       | -       | -     |
| Amp modulation frequency  | ammodfreq          |               | -       | X       | -     |
| Amp modulation index      | ammodindex         |               | -       | X       | -     |
| Amp modulation phase      | ammodphase         |               | -       | X       | -     |
| Number of FM Files        | filenums           | none          | -       | X       | -     |
| Name of FM File1          | fmmodfile1         |               | -       | X       | -     |
| FM file1 name is variable | fmmodfile1_as _var | 0             | -       | X       | -     |
| Name of FM file 2         | fmmodfile2         |               | -       | X       | -     |

| CDF Parameter Prompt                     | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|-------------------|---------------|---------|---------|-------|
| FM file2 name is variable                | fmmodfile2_as_var | 0             | -       | X       | -     |
| FM frequency                             | fmmodfreq         |               | -       | X       | -     |
| FM index                                 | fmmodindex        |               | -       | X       | -     |
| Frequency 2                              | freq2             |               | -       | X       | -     |
| Number of noise/frequency pairs          | FNpairs           | 0             | -       | X       | -     |
| Noise file name                          | noisefile         |               | -       | X       | -     |
| Noise file name is variable              | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                            | F1-50             |               | -       | X       | -     |
| Noise1-50                                | N1-50             |               | -       | X       | -     |
| Frequency name for fundamental frequency | fundname          |               | -       | X       | -     |
| Second frequency name                    | fundname2         |               | -       | X       | -     |
| Multiplier                               | m                 |               | -       | X       | -     |
| Noise temperature                        | noisetemp         |               | -       | X       | -     |
| Nominal temperature                      | tnom              |               | -       | X       | -     |
| PAC magnitude                            | pacm              |               | -       | X       | -     |
| PAC phase                                | pacp              |               | -       | X       | -     |
| Phase for sinusoid 2                     | sinephase2        |               | -       | X       | -     |
| Number of frequencies                    | freqvec           | 0             | -       | X       | -     |
| Sinusoid frequency 1-9                   | freqv1-9          |               | -       | X       | -     |
| Sinusoid phase 1-9                       | phasev1-9         |               | -       | X       | -     |
| Sinusoid max harmonic 1-9                | harmv1-9          |               | -       | X       | -     |

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Voltage level                | sinedc        |               | -       | X       | -     |
| Temp coefficient 1           | tc1           |               | -       | X       | -     |
| Temp coefficient 2           | tc2           |               | -       | X       | -     |
| XF magnitude                 | xfm           |               | -       | X       | -     |
| Amplitude (dBm)              | vaDBm         |               | -       | X       | -     |
| Amplitude 2                  | va2           |               | -       | X       | -     |
| Amplitude 2 (dBm)            | vaDBm2        |               | -       | X       | -     |
| PAC magnitude (dBm)          | pacDBm        |               | -       | X       | -     |
| Sinusoid amplitude 1-9 (DBm) | vaDBmv1-9     |               | -       | X       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.
- If parameter va is empty, writes out 0 in the Hspice-format netlisting result. For Spectre netlister users, create and use the [dbAnalogLibPsinVersion](#) Tcl preference to get legacy designs.

## Example

```
p4 n1 gnd! port=1 dc=0 sin ( 0.35 500m 400Meg 0 0 0 ) z0=50
```

## See Also

- [Port Element](#) in the *HSPICE® User Guide: Elements*

## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibPsinVersion](#) Tcl preference before the analogLib library is loaded.

## Updating Port Number Automatically

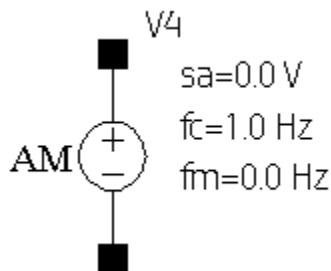
You can create and use the [dbAnalogLibAutoUpdatePortNum](#) Tcl preference to control whether to automatically update the port number of port family devices.

---

## vam

Single-frequency AM voltage source.

$$sourcevalue = sa \cdot \{oc + SIN[2 \cdot \pi \cdot fm \cdot (Time - td)]\} \cdot SIN[2 \cdot \pi \cdot fc \cdot (Time - td)]$$



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Voltage           | vdc           | 0             | X       | -       | -     |
| Source Type          | srcType       | am            | -       | -       | -     |
| Signal Amplitude     | sa            | 0             | X       | -       | -     |
| Carrier Frequency    | fc            | 1             | X       | -       | -     |
| Modulation Frequency | fm            | 0             | X       | -       | -     |
| Offset Constant      | oc            | 0             | X       | -       | -     |
| Delay Time           | td            | 0             | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | -       | -     |
| AC Phase             | acp           |               | X       | -       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7      | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | X       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| SNAC Magnitude       | snacm         |               | X       | -       | -     |
| SNAC Phase           | snacp         |               | X       | -       | -     |
| Impedance            | z0            |               | -       | -       | -     |

Example

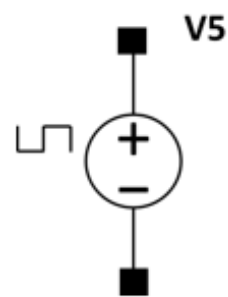
```
vp4 n1 gnd! dc=0 am ( 500m 0.0 200Meg 100Meg 0 )
```

See Also

- [Single-frequency AM Source](#) in the *HSPICE® User Guide: Elements*

vbit

Bit independent voltage source.



CDF Parameters

| CDF Parameter Prompt               | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------------|---------------|---------------|---------|---------|-------|
| Pattern Parameter data             | data          |               | X       | X       | -     |
| Pattern Parameter data is variable | data_as_var   |               | X       | X       | -     |
| Pattern Parameter rptstart         | rptstart      | 1             | X       | X       | -     |
| Pattern Parameter rpttimes         | rpttimes      | 0             | X       | X       | -     |



| CDF Parameter Prompt       | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|----------------------------|-----------------|---------------|---------|---------|-------|
| Trigger                    | trigger         |               | -       | X       | -     |
| RJ(rms)                    | rjrms           |               | X       | X       | -     |
| RJ(seed)                   | rjseed          |               | X       | X       | -     |
| Number of periodic jitters | nPeriodicJitter | 0             | X       | X       | -     |
| PJ%d(amplitude)            | pjamp1-3        |               | X       | X       | -     |
| PJ%d(frequency)            | pjfreq1-3       |               | X       | X       | -     |
| PJ%d(type)                 | pjtype1-3       | sine          | X       | X       | -     |
| One Value                  | val1            |               | X       | X       | -     |
| Zero Value                 | val0            |               | X       | X       | -     |
| Delay Time                 | td              |               | X       | X       | -     |
| Rise Time                  | tr              | 10f           | X       | X       | -     |
| Fall Time                  | tf              | 10f           | X       | X       | -     |
| Period                     | per             |               | X       | X       | -     |
| Source Type                | srcType         | bit           | -       | X       | -     |

## Example

In the example, the first b-string is 1011, which repeats once and then repeats from the second bit, which is 0. The pattern is repeated 5 times. The final b-string is 1011011011011011011.

```
v4 n1 n2 pat ( 1.2 0.25 0 1n 1n 5n b1011 r=5 rb=2 )
```

Where:

- High voltage level is 1.2 V
- Low voltage level is 0.25 V
- Time delay is 0 n
- Rise time is 1 n

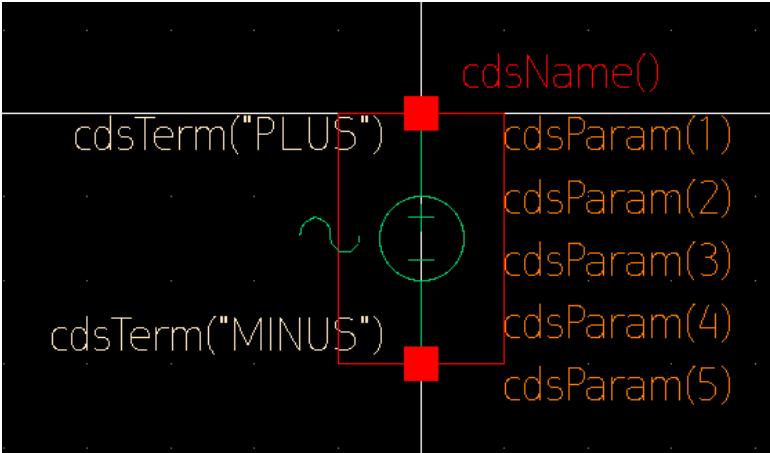
- Fall time is 1 n
- Sample time (period) is 5 n

See Also

- [Pattern Source](#) in the *HSPICE® User Guide: Elements*

VCOS

Voltage source.



CDF Parameters

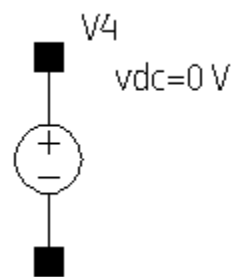
| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Source Type          | srcType       | cosine        | -       | -       | -     |
| DC Voltage           | vdc           | 0             | X       | -       | -     |
| Voltage Offset       | vo            | 0             | X       | -       | -     |
| Voltage Amplitude    | va            | 0             | X       | -       | -     |
| Frequency            | freq          |               | X       | -       | -     |
| Delay Time           | td            | 0             | X       | -       | -     |
| Damping Factor       | q             | 0             | X       | -       | -     |
| Phase for sinusoid   | sinephase     | 0             | X       | -       | -     |
| Period Jitter        | perjitter     |               | X       | -       | -     |

| CDF Parameter Prompt     | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------|---------------|---------------|---------|---------|-------|
| Seed                     | seed          |               | X       | -       | -     |
| AC Magnitude             | acm           |               | X       | -       | -     |
| AC Phase                 | acp           |               | X       | -       | -     |
| # HB Tones               | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7          | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7              | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7           | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7               | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7            | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7            | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection       | dcopen        |               | -       | -       | -     |
| DC Resistance            | rdc           |               | X       | -       | -     |
| AC Resistance            | rac           |               | X       | -       | -     |
| HB Resistance            | rhb           |               | X       | -       | -     |
| HBAC Resistance          | rhbac         |               | X       | -       | -     |
| TRAN Resistance          | rtran         |               | X       | -       | -     |
| HBAC Magnitude           | hbacm         |               | X       | -       | -     |
| HBAC Phase               | hbacp         |               | X       | -       | -     |
| SNAC Magnitude           | snacm         |               | X       | -       | -     |
| SNAC Phase               | snacp         |               | X       | -       | -     |
| Amp modulation frequency | ammodfreq     |               | -       | -       | -     |
| Amp modulation index     | ammodindex    |               | -       | -       | -     |
| Amp modulation phase     | ammodphase    |               | -       | -       | -     |
| Number of FM Files       | filenums      | none          | -       | -       | -     |

| CDF Parameter Prompt                     | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|---------------|---------------|---------|---------|-------|
| Name of FM File1                         | fmmodfile1    |               | -       | -       | -     |
| Name of FM File2                         | fmmodfile2    |               | -       | -       | -     |
| FM frequency                             | fmmodfreq     |               | -       | -       | -     |
| FM index                                 | fmmodindex    |               | -       | -       | -     |
| Frequency 2                              | freq2         |               | -       | -       | -     |
| Number of noise/frequency pairs          | FNpairs       | 0             | -       | -       | -     |
| Noise file name                          | noisefile     |               | -       | -       | -     |
| Frequency 1-10                           | F1-10         |               | -       | -       | -     |
| Noise 1-10                               | N1-10         |               | -       | -       | -     |
| Frequency name for fundamental frequency | fundname      |               | -       | -       | -     |
| Second frequency name                    | fundname2     |               | -       | -       | -     |
| Noise temperature                        | noisetemp     |               | -       | -       | -     |
| Nominal temperature                      | tnom          |               | -       | -       | -     |
| PAC magnitude                            | pacm          |               | -       | -       | -     |
| PAC phase                                | pacp          |               | -       | -       | -     |
| Phase for sinusoid 2                     | sinephase2    |               | -       | -       | -     |
| Temp coefficient 1                       | tc1           |               | -       | -       | -     |
| Temp coefficient 2                       | tc2           |               | -       | -       | -     |
| XF magnitude                             | xfm           |               | -       | -       | -     |
| Phase shift                              | phase         |               | -       | -       | -     |
| Amplitude 2                              | va2           |               | -       | -       | -     |
| Impedance                                | z0            |               | -       | -       | -     |

**vdc**

DC voltage source.



**CDF Parameters**

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Voltage           | vdc           | 0             | X       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7      | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | -       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |

| CDF Parameter Prompt            | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------------|---------------|---------------|---------|---------|-------|
| HBAC Magnitude                  | hbacm         |               | X       | -       | -     |
| HBAC Phase                      | hbacp         |               | X       | -       | -     |
| SNAC Magnitude                  | snacm         |               | X       | -       | -     |
| SNAC Phase                      | snacp         |               | X       | -       | -     |
| Source Type                     | srcType       | dc            | -       | X       | -     |
| Number of noise/frequency pairs | FNpairs       | 0             | -       | X       | -     |
| Noise file name                 | noisefile     |               | X       | X       | -     |
| Frequency1-50                   | F1-50         |               | -       | X       | -     |
| Noise1-50                       | N1-50         |               | -       | X       | -     |
| Nominal temperature             | tnom          |               | X       | X       | -     |
| PAC magnitude                   | pacm          |               | -       | X       | -     |
| PAC phase                       | pacp          |               | -       | X       | -     |
| Temp coefficient 1              | tc1           |               | X-      | X       | -     |
| Temp coefficient 2              | tc2           |               | X       | X       | -     |
| XF magnitude                    | xfm           |               | -       | X       | -     |
| Impedance                       | z0            |               | -       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output | spectre Output |
|-------------------|----------------|----------------|
| noise_file_as_var |                |                |

| CDF Parameter   | hspiceD Output           | spectre Output        |
|-----------------|--------------------------|-----------------------|
| false (default) | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true            | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, str(NOISEFILE) is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

Example

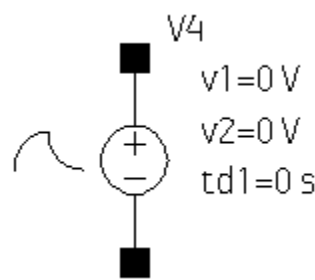
```
v4 n1 gnd! dc=2.5 ac=1
```

See Also

- [DC Sources](#) in the *HSPICE® User Guide: Elements*

vexp

Exponential voltage source.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Voltage           | vdc           | 0             | X       | X       | -     |
| Source Type          | srcType       | exp           | -       | X       | -     |
| Voltage1             | v1            | 0             | X       | X       | -     |

| CDF Parameter Prompt            | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------------|---------------|---------------|---------|---------|-------|
| Voltage2                        | v2            | 0             | X       | X       | -     |
| Rise Delay Time                 | td1           | 0             | X       | X       | -     |
| Fall Delay Time                 | td2           |               | X       | X       | -     |
| Rise Time Constant              | tau1          |               | X       | X       | -     |
| Fall Time Constant              | tau2          | 0             | X       | X       | -     |
| AC Magnitude                    | acm           |               | X       | X       | -     |
| AC Phase                        | acp           |               | X       | X       | -     |
| # HB Tones                      | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7                 | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7                     | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7                  | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7                      | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7                   | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7                   | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection              | dcopen        |               | -       | -       | -     |
| DC Resistance                   | rdc           |               | X       | -       | -     |
| AC Resistance                   | rac           |               | X       | -       | -     |
| HB Resistance                   | rhb           |               | X       | -       | -     |
| HBAC Resistance                 | rhbac         |               | X       | -       | -     |
| TRAN Resistance                 | rtran         |               | X       | -       | -     |
| HBAC Magnitude                  | hbacm         |               | X       | -       | -     |
| HBAC Phase                      | hbacp         |               | X       | -       | -     |
| SNAC Magnitude                  | snacm         |               | X       | -       | -     |
| SNAC Phase                      | snacp         |               | X       | -       | -     |
| Number of noise/frequency pairs | FNpairs       | 0             | -       | X       | -     |



| CDF Parameter Prompt        | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|-------------------|---------------|---------|---------|-------|
| Noise file name             | noisefile         | -             | -       | X       | -     |
| Noise file name is variable | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50               | F1-50             | -             | -       | X       | -     |
| Noise1-50                   | N1-50             | -             | -       | X       | -     |
| Nominal temperature         | tnom              | -             | X       | X       | -     |
| PAC magnitude               | pacm              | -             | -       | X       | -     |
| PAC phase                   | pacp              | -             | -       | X       | -     |
| Temp coefficient 1          | tc1               | -             | X       | X       | -     |
| Temp coefficient 2          | tc2               | -             | X       | X       | -     |
| XF magnitude                | xfm               | -             | -       | X       | -     |
| Delay time                  | td                | -             | -       | X       | -     |
| Impedance                   | z0                | -             | -       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noisefile parameter is set as *NOISEFILE*, the netlisting behavior for noisefile and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

Example

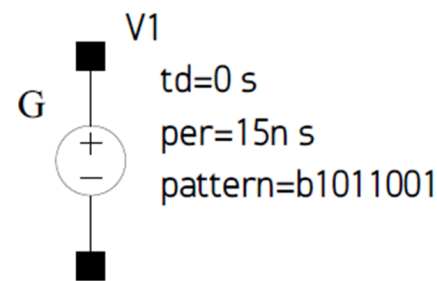
```
v4 n1 gnd! dc=2.5 ac=1 exp ( 0 5 5n 3n 0.5n 15n )
```

See Also

- [Exponential Source Function](#) in the *HSPICE® User Guide: Elements*

vgpulse

Gaussian voltage pulse.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Voltage           | vdc           | 0             | X       | -       | -     |
| Voltage1             | v1            |               | X       | -       | -     |
| Voltage2             | v2            |               | X       | -       | -     |
| Delay time           | td            |               | X       | -       | -     |
| Pulse Width          | Pw            |               | X       | -       | -     |
| Period               | per           |               | X       | -       | -     |

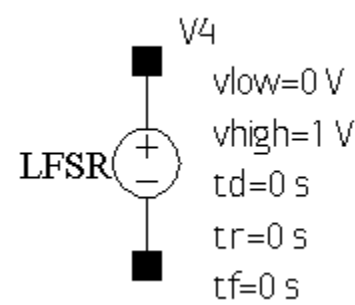
| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Pattern              | pattern       |               | X       | -       | -     |
| Repeat               | r             |               | X       | -       | -     |
| Repeat Starts        | rb            |               | X       | -       | -     |
| Pseudo Random Seq.   | bprbs         |               | X       | -       | -     |

Example

```
v9 a gnd! dc=0 gpulse (0 1 0 5n 15n b1011001 bprbs=12 r=100n)
```

vlfsr

Pseudo Random Bit Generator Source (PRBS) function as an independent voltage source. The PRBS source uses a Linear Feedback Shift Register (LFSR) to generate a pseudo random bit sequence. HSPICE supports eight-bit data byte conversion to a 10-bit transmission character-8B/10B encoding for the LFSR keyword.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Voltage           | vdc           | 0             | X       | -       | -     |
| Source Type          | srcType       | lfsr          | -       | -       | -     |
| Voltage Low          | vlow          | 0             | X       | -       | -     |
| Voltage High         | vhigh         | 1             | X       | -       | -     |
| Delay Time           | td            | 0             | X       | -       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Rise Time            | tr            | 0             | X       | -       | -     |
| Fall Time            | tf            | 0             | X       | -       | -     |
| Bit Rate             | rate          | 1M            | X       | -       | -     |
| Seed                 | seed          | 1             | X       | -       | -     |
| Taps                 | taps          | [1 2]         | X       | -       | -     |
| Output Resistance    | rout          |               | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | -       | -     |
| AC Phase             | acp           |               | X       | -       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7      | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | -       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |
| SNAC Magnitude       | snacm         |               | X       | -       | -     |
| SNAC Phase           | snacp         |               | X       | -       | -     |

| CDF Parameter Prompt | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|----------------------|-----------------|---------------|---------|---------|-------|
| PAM4 modulation      | pam4_modulation | none          | X       | -       | -     |
| PAM4 mapping         | pam4_mapping    |               | X       | -       | -     |
| Impedance            | z0              |               | -       | -       | -     |

## Example

The following example shows the pattern voltage source connected between nodes n1 and n2.

Example of LFSR, output is 1100011111001101:

```
v4 n1 n2 dc=0 lfsr ( 0 50u 0 1n 1n 10meg 3 [2,5] rout=10 )
```

Where:

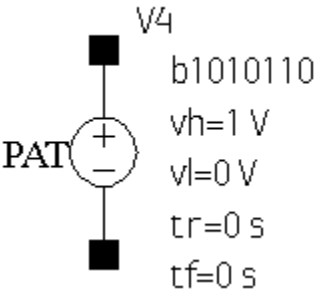
- The output low voltage is 0, and the output high voltage is 1 v.
- The delay time is 0 ms.
- The rise and fall times are each 1 ns.
- The bit rate is 10 meg bits/s.
- The seed is 3 (bits: 00011).
- The taps are at the output of the 2nd and 5th shift registers and are xor'd together as the input to the first shift register.
- The output resistance is 10 ohms.

## See Also

- [Pseudo Random Bit Generator Source](#) in the *HSPICE® User Guide: Elements*

## vpat

Pattern voltage source function. It uses four states, '1','0','m', and 'z', which represent the high, low, middle voltage, or current and high impedance state respectively. The series of these four states is a "b-string." HSPICE supports 8-bit data byte conversion to a 10-bit transmission character-8B/10B encoding, 64-bit data conversion to a 66-bit transmission character-64B/66B encoding, 128-bit data byte conversion to a 130-bit or 132-bit transmission character-128B/130B or 128B/132B encoding, for the PAT sources.



CDF Parameters

| CDF Parameter Prompt       | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|----------------------------|-----------------|---------------|---------|---------|-------|
| DC Voltage                 | vdc             | 0             | X       | -       | -     |
| Source Type                | srcType         | pat           | -       | -       | -     |
| Voltage High               | vh              | 1             | X       | -       | -     |
| Voltage Low                | vl              | 0             | X       | -       | -     |
| Delay Time                 | td              | 0             | X       | -       | -     |
| Rise Time                  | tr              | 0             | X       | -       | -     |
| Fall Time                  | tf              | 0             | X       | -       | -     |
| Sample Time                | tsample         | 1n            | X       | -       | -     |
| Data                       | data            | b1010<br>110  | X       | -       | -     |
| Repeat                     | r               |               | X       | -       | -     |
| Repeat Bit                 | rb              |               | X       | -       | -     |
| AC Magnitude               | acm             |               | X       | -       | -     |
| AC Phase                   | acp             |               | X       | -       | -     |
| RJ(rms)                    | rjrms           |               | X       | X       | -     |
| RJ(seed)                   | rjseed          |               | X       | X       | -     |
| Number of periodic jitters | nPeriodicJitter | 0             | X       | X       | -     |
| PJ%d(amplitude)            | pjump1-3        |               | X       | X       | -     |

| CDF Parameter Prompt | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|----------------------|-----------------|---------------|---------|---------|-------|
| PJ%d(frequency)      | pjfreq1-3       |               | X       | X       | -     |
| PJ%d(type)           | pjtype1-3       | sine          | X       | X       | -     |
| # HB Tones           | hbtones         | 0             | X       | -       | -     |
| HB Magnitude1-7      | hbmag1-7        |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7      | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7       | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7       | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7    |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7    |               | X       | -       | -     |
| Open DC Connection   | dcopen          |               | X       | -       | -     |
| DC Resistance        | rdc             |               | X       | -       | -     |
| AC Resistance        | rac             |               | X       | -       | -     |
| HB Resistance        | rhb             |               | X       | -       | -     |
| HBAC Resistance      | rhbac           |               | X       | -       | -     |
| TRAN Resistance      | rtran           |               | X       | -       | -     |
| HBAC Magnitude       | hbacm           |               | X       | -       | -     |
| HBAC Phase           | hbacp           |               | X       | -       | -     |
| SNAC Magnitude       | snacm           |               | X       | -       | -     |
| SNAC Phase           | snacp           |               | X       | -       | -     |
| PAM4 modulation      | pam4_modulation | none          | X       | -       | -     |
| PAM4 mapping         | pam4_mapping    |               | X       | -       | -     |
| Impedance            | z0              |               | -       | -       | -     |

## Example

```
v4 n1 n2 pat ( 1.2 0.25 0 1n 1n 5n b1011 r=5 rb=2 )
```

In this example:

- High voltage level is 1.2 V
- Low voltage level is 0.25 V
- Time delay is 0 n
- Rise time is 1 n
- Fall time is 1 n
- Sample time is 5 n

The first b-string is 1011, which repeats once and then repeats from the second bit, which is 0. The pattern is repeated 5 times.

### See Also

- [Pattern Source](#) in the *HSPICE® User Guide: Elements*

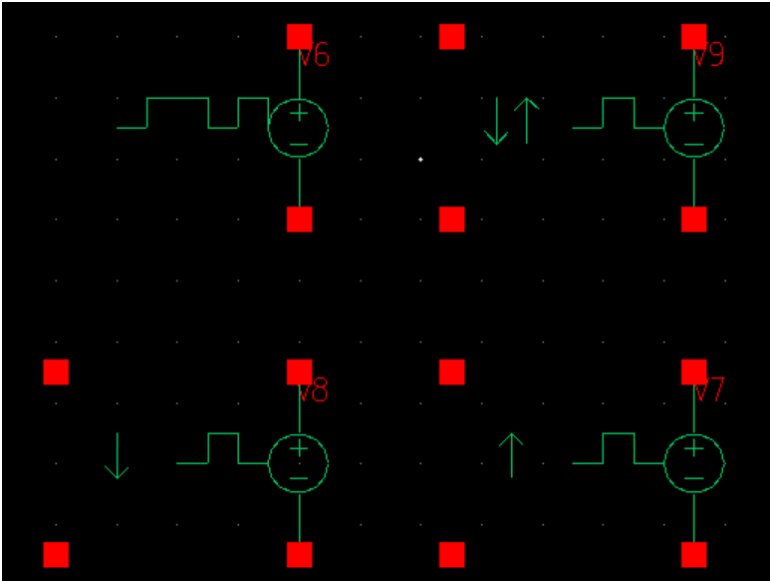
---

## vprbs

Voltage source with pseudo random bit generator source transient function source.

The vprbs device is a PCell controlled by the trigger parameter. The following figure shows four vprbs devices that have different trigger modes, which are Internal (on the upper left), External by both edges (on the upper right), External by falling edges (on the lower left), and External by rising edges (on the lower right) modes. The internal trigger mode has two pins (PLUS and MINUS) and other 3 external trigger modes have 4 pins (P1 M1 P2 M2). The netlisting procedure writes out their pins accordingly.





CDF Parameters

| CDF Parameter Prompt     | CDF Parameter       | Default Value | hspiceD | spectre | auCdl |
|--------------------------|---------------------|---------------|---------|---------|-------|
| triggerValue             | triggerValue        | 1             | -       | X       | -     |
| Delay time               | tdPrbs              |               | -       | X       | -     |
| Zero value               | val0                |               | -       | X       | -     |
| One value                | val1                |               | -       | X       | -     |
| Bit period               | perPrbs             | 1n            | -       | X       | -     |
| Rise Time                | tr                  |               | -       | X       | -     |
| Fall Time                | tf                  |               | -       | X       | -     |
| Transition reference     | transitionreference |               | -       | X       | -     |
| Lower threshold(percent) | lowerthreshold      |               | -       | X       | -     |
| Upper threshold(percent) | upperthreshold      |               | -       | X       | -     |
| Edge type                | edgetype            |               | -       | X       | -     |
| Trigger                  | trigger             | Internal      | -       | X       | -     |

| CDF Parameter Prompt        | CDF Parameter     | Default Value                                          | hspiceD | spectre | auCdl |
|-----------------------------|-------------------|--------------------------------------------------------|---------|---------|-------|
| Threshold                   | triggerthresh old |                                                        | -       | X       | -     |
| LFSR Mode                   | lfsrmode          | PN32                                                   | -       | X       | -     |
| Browse and specify bit file | prbsFileBox       | 0                                                      | -       | -       | -     |
| Bit data file               | prbsFile          |                                                        | -       | X       | -     |
| Seed                        | lfsrseed          | 1 3 5 7 9<br>11 13 15<br>17 19 21<br>23 25 27<br>29 31 | -       | X       | -     |
| Taps                        | lfsrtaps          |                                                        | -       | X       | -     |
| RJ(rms)                     | rjrms             |                                                        | -       | X       | -     |
| RJ(seed)                    | rjseed            |                                                        | -       | X       | -     |
| Number of periodic jitters  | nPeriodicJitter   | 0                                                      | -       | X       | -     |
| PJ%d(amplitude)             | pjamp1-3          |                                                        | -       | X       | -     |
| PJ%d(frequency)             | pjfreq1-3         |                                                        | -       | X       | -     |
| PJ%d(type)                  | pjtype1-3         | sine                                                   | -       | X       | -     |
| Pattern Parameter data      | data              |                                                        | -       | X       | -     |
| Reference Value             | ref               |                                                        | -       | X       | -     |
| Random delay time           | jitter            |                                                        | -       | X       | -     |
| Seed                        | seed              |                                                        | -       | X       | -     |
| Feedback Shift Register     | taps              |                                                        | -       | X       | -     |
| Source Type                 | srcType           | prbs                                                   | -       | X       | -     |
| Version                     | version           | 2                                                      | -       | -       | -     |

**Note:**

The nPeriodicJitter value is limited to the range of 0 and 3. If the input value is smaller than 0, it is reset to 0. If the input value is greater than 3, it is reset to 3. No message is displayed for resetting the limit value.

## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the [dbAnalogLibVprbsVersion](#) Tcl preference before the analogLib library is loaded.

## vpulse

This symbol specifies an independent source element and modes pulse generator.

### CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | vdc           | 0             | X       | X       | -     |
| Source Type          | srcType       | pulse         | X       | X       | -     |
| Voltage1             | v1            | 0             | X       | X       | -     |
| Voltage2             | v2            | 0             | X       | X       | -     |
| Delay Time           | td            | 0             | X       | X       | -     |
| Rise Time            | tr            |               | X       | X       | -     |
| Fall Time            | tf            |               | X       | X       | -     |
| Pulse Width          | pw            |               | X       | X       | -     |
| Period               | per           |               | X       | X       | -     |
| Period Jitter        | perjitter     |               | X       | -       | -     |
| Seed                 | seed          |               | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7      | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |

| CDF Parameter Prompt                     | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|-------------------|---------------|---------|---------|-------|
| HB Harmonic1-7                           | hbharm1-7         | 1             | X       | -       | -     |
| HB Tone1-7                               | hbtone1-7         | 1-7           | X       | -       | -     |
| HB ModHarm1-7                            | hbmodharm1-7      |               | X       | -       | -     |
| HB ModTone1-7                            | hbmodtone1-7      |               | X       | -       | -     |
| Open DC Connection                       | dcopen            |               | X       | -       | -     |
| DC Resistance                            | rdc               |               | X       | -       | -     |
| AC Resistance                            | rac               |               | X       | -       | -     |
| HB Resistance                            | rhb               |               | X       | -       | -     |
| HBAC Resistance                          | rhbac             |               | X       | -       | -     |
| TRAN Resistance                          | rtran             |               | X       | -       | -     |
| HBAC Magnitude                           | hbacm             |               | X       | -       | -     |
| HBAC Phase                               | hbacp             |               | X       | -       | -     |
| SNAC Magnitude                           | snacm             |               | X       | -       | -     |
| SNAC Phase                               | snacp             |               | X       | -       | -     |
| Number of noise/frequency pairs          | FNpairs           | 0             | -       | X       | -     |
| Noise file name                          | noisefile         |               | -       | X       | -     |
| Noise file name is variable              | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                            | F1-50             |               | -       | X       | -     |
| Noise1-50                                | N1-50             |               | -       | X       | -     |
| Frequency name for fundamental frequency | fundname          |               | -       | X       | -     |
| Nominal temperature                      | tnom              |               | X       | X       | -     |
| PAC magnitude                            | pacm              |               | -       | X       | -     |
| PAC phase                                | pacp              |               | -       | X       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Edge type            | edgetype      |               | X       | X       | -     |
| Temp coefficient 1   | tc1           |               | X       | X       | -     |
| Temp coefficient 2   | tc2           |               | X       | X       | -     |
| XF magnitude         | xfm           |               | -       | X       | -     |
| Impedance            | z0            |               | -       | -       | -     |

### Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

### Example

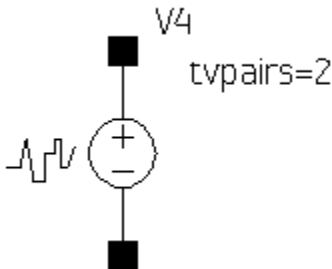
```
v1 plus minus dc=0 pulse ( 0 0 0 )
```

### See Also

- [Independent Source Elements](#) in the *HSPICE® User Guide: Elements*

vpwl

PWL independent voltage source.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Voltage           | vdc           | 0             | X       | X       | -     |
| Source Type          | srcType       | pwl           | -       | X       | -     |
| Time Value Pairs     | tvpairs       | 2             | X       | X       | -     |
| Display Pairs Count  | dpcount       | 5             | X       | -       | -     |
| Time1-50             | t1-50         |               | X       | X       | -     |
| Voltage1-50          | v1-50         | 0             | X       | X       | -     |
| Repeat               | r             |               | X       | -       | -     |
| Repeat Stop Time     | rstop         |               | X       | -       | -     |
| Repeat Stop Value    | stopvalue     |               | X       | -       | -     |
| Repeat Stop Slope    | stopslope     |               | X       | -       | -     |
| Delay Time           | td            | 0             | X       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7      | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |

| CDF Parameter Prompt                     | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|-------------------|---------------|---------|---------|-------|
| HB Harmonic1-7                           | hbharm1-7         | 1             | X       | -       | -     |
| HB Tone1-7                               | hbtone1-7         | 1-7           | X       | -       | -     |
| HB ModHarm1-7                            | hbmodharm1-7      |               | X       | -       | -     |
| HB ModTone1-7                            | hbmodtone1-7      |               | X       | -       | -     |
| Open DC Connection                       | dcopen            |               | -       | -       | -     |
| DC Resistance                            | rdc               |               | X       | -       | -     |
| AC Resistance                            | rac               |               | X       | -       | -     |
| HB Resistance                            | rhb               |               | X       | -       | -     |
| HBAC Resistance                          | rhbac             |               | X       | -       | -     |
| TRAN Resistance                          | rtran             |               | X       | -       | -     |
| HBAC Magnitude                           | hbacm             |               | X       | -       | -     |
| HBAC Phase                               | hbacp             |               | X       | -       | -     |
| SNAC Magnitude                           | snacm             |               | X       | -       | -     |
| SNAC Phase                               | snacp             |               | X       | -       | -     |
| Number of noise/frequency pairs          | FNpairs           | 0             | -       | X       | -     |
| Noise file name                          | noisefile         |               | -       | X       | -     |
| Noise file name is variable              | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                            | F1-50             |               | -       | X       | -     |
| Noise1-50                                | N1-50             |               | -       | X       | -     |
| Frequency name for fundamental frequency | fundname          |               | -       | X       | -     |
| Nominal temperature                      | tnom              |               | X       | X       | -     |
| PAC magnitude                            | pacm              |               | -       | X       | -     |
| PAC phase                                | pacp              |               | -       | X       | -     |

| CDF Parameter Prompt   | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------|---------------|---------------|---------|---------|-------|
| PWL period             | pwlperiod     |               | -       | X       | -     |
| Amplitude scale factor | scale         |               | -       | X       | -     |
| Edge type              | edgetype      | linear        | -       | X       | -     |
| Transition width       | twidth        |               | -       | X       | -     |
| Time scaling factor    | stretch       |               | -       | X       | -     |
| Temp coefficient 1     | tc1           |               | X       | X       | -     |
| Temp coefficient 2     | tc2           |               | X       | X       | -     |
| XF magnitude           | xfm           |               | -       | X       | -     |
| Delay time             | delay         |               | -       | X       | -     |
| Offset voltage         | vo            |               | -       | X       | -     |
| Impedance              | z0            |               | -       | -       | -     |
| PWL                    | pwl           |               | -       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |



- For Hspice-format netlisting, a pair of single quotes is added to noisefile parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NoiseFile)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

Example

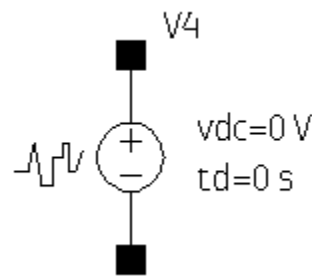
```
v4 n1 n2 dc=0 pwlf ( 0 0.0 1n 0.02 1.2n 0.5 2n 1.2 )
```

See Also

- [PWL Source](#) in the *HSPICE® User Guide: Elements*

vpwlf

PWL source, that accepts a text file containing time and voltage pairs. The data in the text file can be either a table or a list.



CDF Parameters

| CDF Parameter Prompt  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------|---------------|---------------|---------|---------|-------|
| Source Type           | srcType       | pwlf          | -       | X       | -     |
| PWL File              | fileName      |               | X       | X       | -     |
| File name is variable | file_as_var   | 0             | X       | X       | -     |
| DC Voltage            | vdc           | 0             | X       | X       | -     |
| Repeat                | r             |               | X       | -       | -     |

| CDF Parameter Prompt  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------|---------------|---------------|---------|---------|-------|
| Repeat Stop Time      | rstop         |               | X       | -       | -     |
| Repeat Stop Value     | stopvalue     |               | X       | -       | -     |
| Repeat Stop Slope     | stopslope     |               | X       | -       | -     |
| Delay Time            | td            | 0             | X       | X       | -     |
| AC Magnitude          | acm           |               | X       | X       | -     |
| AC Phase              | acp           |               | X       | X       | -     |
| # HB Tones            | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7       | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7           | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7        | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7            | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7         | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7         | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection    | dcopen        |               | -       | -       | -     |
| DC Resistance         | rdc           |               | X       | -       | -     |
| AC Resistance         | rac           |               | X       | -       | -     |
| HB Resistance         | rhb           |               | X       | -       | -     |
| HBAC Resistance       | rhbac         |               | X       | -       | -     |
| TRAN Resistance       | rtran         |               | X       | -       | -     |
| HBAC Magnitude        | hbacm         |               | X       | -       | -     |
| HBAC Phase            | hbacp         |               | X       | -       | -     |
| SNAC Magnitude        | snacm         |               | X       | -       | -     |
| SNAC Phase            | snacp         |               | X       | -       | -     |
| Desired rms value     | rms           |               | -       | X       | -     |
| Filter for pwl source | pwlfilter     |               | -       | X       | -     |

| CDF Parameter Prompt                     | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|-------------------|---------------|---------|---------|-------|
| Rolloff factor                           | rolloff           |               | -       | X       | -     |
| Bandwidth of the pwl signal              | pwlbandwidth      |               | -       | X       | -     |
| PAC magnitude                            | pacm              |               | -       | X       | -     |
| PAC phase                                | pacp              |               | -       | X       | -     |
| PWL period                               | pwlperiod         |               | -       | X       | -     |
| Amplitude scale factor                   | scale             |               | -       | X       | -     |
| Edge type                                | edgetype          | linear        | -       | X       | -     |
| Transition width                         | twidth            |               | -       | X       | -     |
| Time scaling factor                      | stretch           |               | -       | X       | -     |
| Temp coefficient 1                       | tc1               |               | X       | X       | -     |
| Temp coefficient 2                       | tc2               |               | X       | X       | -     |
| XF magnitude                             | xfm               |               | -       | X       | -     |
| Number of noise/frequency pairs          | FNpairs           | 0             | -       | X       | -     |
| Noise file name                          | noisefile         |               | -       | X       | -     |
| Noise file name is variable              | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                            | F1-50             |               | -       | X       | -     |
| Noise1-50                                | N1-50             |               | -       | X       | -     |
| Frequency name for fundamental frequency | fundname          |               | -       | X       | -     |
| Nominal temperature                      | tnom              |               | X       | X       | -     |
| Offset voltage                           | vo                |               | -       | X       | -     |
| Impedance                                | z0                |               | -       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

**Example**

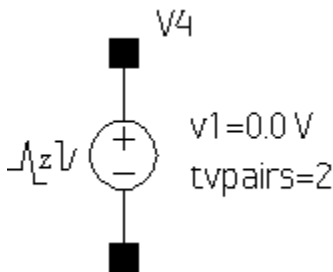
```
v4 n1 n2 PWL PWLFILE='Vmod.dat'
```

**See Also**

- [File-driven PWL Source](#) in the *HSPICE® User Guide: Elements*

**vpwlz**

The high Z state of the PWLZ source adds the capability to disconnect the PWL source for time periods marked with the keyword Z.



CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Voltage           | vdc           | 0             | X       | X       | -     |
| Source Type          | srcType       | pwlz          | -       | X       | -     |
| Time Value Pairs     | tvpairs       | 2             | X       | X       | -     |
| Time1-50             | t1-50         |               | X       | X       | -     |
| Voltage1-50          | v1-50         | 0             | X       | X       | -     |
| Z State1-50          | Z1-50         | 0             | X       | X       | -     |
| Repeat               | r             |               | X       | -       | -     |
| Delay Time           | td            | 0             | X       | X       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7      | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | -       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |

| CDF Parameter Prompt                     | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|-------------------|---------------|---------|---------|-------|
| AC Resistance                            | rac               |               | X       | -       | -     |
| HB Resistance                            | rhb               |               | X       | -       | -     |
| HBAC Resistance                          | rhbac             |               | X       | -       | -     |
| TRAN Resistance                          | rtran             |               | X       | -       | -     |
| HBAC Magnitude                           | hbacm             |               | X       | -       | -     |
| HBAC Phase                               | hbacp             |               | X       | -       | -     |
| SNAC Magnitude                           | snacm             |               | X       | -       | -     |
| SNAC Phase                               | snacp             |               | X       | -       | -     |
| Number of noise/frequency pairs          | FNpairs           | 0             | -       | X       | -     |
| Noise file name                          | noisefile         |               | -       | X       | -     |
| Noise file name is variable              | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                            | F1-50             |               | -       | X       | -     |
| Noise1-50                                | N1-50             |               | -       | X       | -     |
| Frequency name for fundamental frequency | fundname          |               | -       | X       | -     |
| Nominal temperature                      | tnom              |               | X       | X       | -     |
| PAC magnitude                            | pacm              |               | -       | X       | -     |
| PAC phase                                | pacp              |               | -       | X       | -     |
| PWL period                               | pwlperiod         |               | -       | X       | -     |
| Amplitude scale factor                   | scale             |               | -       | X       | -     |
| Edge type                                | edgetype          | linear        | -       | X       | -     |
| Transition width                         | twidth            |               | -       | X       | -     |
| Time scaling factor                      | stretch           |               | -       | X       | -     |
| Temp coefficient 1                       | tc1               |               | X       | X       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Temp coefficient 2   | tc2           |               | X       | X       | -     |
| XF magnitude         | xfm           |               | -       | X       | -     |
| Delay time           | delay         |               | -       | X       | -     |
| Offset voltage       | vo            |               | -       | X       | -     |
| Impedance            | z0            |               | -       | -       | -     |

#### Note:

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise file parameter is set as *NOISEFILE*, the netlisting behavior for noise file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noise file parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

#### Example

```
v4 n1 n2 dc=0 pwlz ( 0 0.0 1n z 1.2n 0.5 2n 1.2 )
```

#### See Also

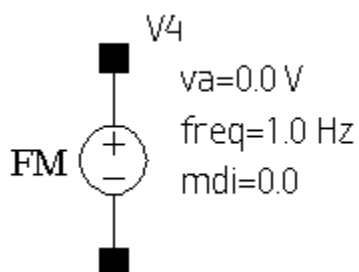
- [High Impedance State PWL \(PWLZ\)](#) in the *HSPICE® User Guide: Elements*

## vsffm

Single-frequency FM voltage source.

The following expression defines the waveform shape:

$$sourcevalue = vo + va \cdot SIN[2 \cdot \pi \cdot fc \cdot Time + mdi \cdot SIN(2 \cdot \pi \cdot fs \cdot Time)]$$



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Current           | idc           | 0             | -       | -       | -     |
| Source Type          | srcType       | sffm          | -       | -       | -     |
| Output Amplitude     | va            | 0             | X       | -       | -     |
| Offset Voltage       | vo            | 0             | X       | -       | -     |
| Carrier Frequency    | freq          | 1             | X       | -       | -     |
| Modulation Index     | mdi           | 0             | X       | -       | -     |
| Signal Frequency     | fs            |               | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | -       | -     |
| AC Phase             | acp           |               | X       | -       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7      | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |



| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| HB ModHarm1-7        | hbmodharm1-7  | X             | -       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7  | X             | -       | -       | -     |
| Open DC Connection   | dcopen        | -             | -       | -       | -     |
| DC Resistance        | rdc           | X             | -       | -       | -     |
| AC Resistance        | rac           | X             | -       | -       | -     |
| HB Resistance        | rhb           | X             | -       | -       | -     |
| HBAC Resistance      | rhbac         | X             | -       | -       | -     |
| TRAN Resistance      | rtran         | X             | -       | -       | -     |
| HBAC Magnitude       | hbacm         | X             | -       | -       | -     |
| HBAC Phase           | hbacp         | X             | -       | -       | -     |
| SNAC Magnitude       | snacm         | X             | -       | -       | -     |
| SNAC Phase           | snacp         | X             | -       | -       | -     |
| Impedance            | z0            | -             | -       | -       | -     |

## Example

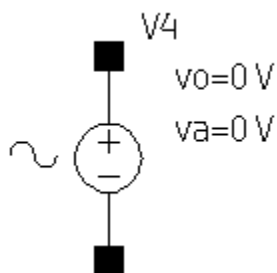
```
v4 n1 gnd! dc=0.5 sffm ( 0.0 1.1 20k 10 5k )
```

## See Also

- [Single-frequency FM Source](#) in the *HSPICE® User Guide: Elements*

## vsin

Sinusoidal voltage source that can be either a constant sinusoid or a damped, that produces a dying exponential with a sine wave.



## CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Source Type          | srcType       | sine          | -       | X       | -     |
| DC Voltage           | vdc           | 0             | X       | X       | -     |
| Voltage Offset       | vo            | 0             | X       | X       | -     |
| Voltage Amplitude    | va            | 0             | X       | X       | -     |
| Frequency            | freq          |               | X       | X       | -     |
| Delay Time           | td            | 0             | X       | X       | -     |
| Damping Factor       | theta         | 0             | X       | X       | -     |
| Phase for sinusoid   | sinephase     | 0             | X       | X       | -     |
| Period Jitter        | perjitter     |               | X       | -       | -     |
| Seed                 | seed          |               | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | X       | -     |
| AC Phase             | acp           |               | X       | X       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7      | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |

| CDF Parameter Prompt      | CDF Parameter           | Default Value | hspiceD | spectre | auCdl |
|---------------------------|-------------------------|---------------|---------|---------|-------|
| HB ModTone1-7             | hbmodtone1-7            |               | X       | -       | -     |
| Open DC Connection        | dcopen                  |               | -       | -       | -     |
| DC Resistance             | rdc                     |               | X       | -       | -     |
| AC Resistance             | rac                     |               | X       | -       | -     |
| HB Resistance             | rhb                     |               | X       | -       | -     |
| HBAC Resistance           | rhbac                   |               | X       | -       | -     |
| TRAN Resistance           | rtran                   |               | X       | -       | -     |
| HBAC Magnitude            | hbacm                   |               | X       | -       | -     |
| HBAC Phase                | hbacp                   |               | X       | -       | -     |
| SNAC Magnitude            | snacm                   |               | X       | -       | -     |
| SNAC Phase                | snacp                   |               | X       | -       | -     |
| Amp modulation frequency  | ammodfreq               |               | -       | X       | -     |
| Amp modulation index      | ammodindex              |               | -       | X       | -     |
| Amp modulation phase      | ammodphase              |               | -       | X       | -     |
| Number of FM Files        | filenums                | none          | -       | X       | -     |
| Name of FM File1          | fmmodfile1              |               | -       | X       | -     |
| FM file1 name is variable | fmmodfile1_as 0<br>_var |               | -       | X       | -     |
| Name of FM file 2         | fmmodfile2              |               | -       | X       | -     |
| FM file2 name is variable | fmmodfile2_as 0<br>_var |               | -       | X       | -     |
| FM frequency              | fmmodfreq               |               | -       | X       | -     |
| FM index                  | fmmodindex              |               | -       | X       | -     |
| Frequency 2               | freq2                   |               | -       | X       | -     |

| CDF Parameter Prompt                     | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|------------------------------------------|-------------------|---------------|---------|---------|-------|
| Number of noise/frequency pairs          | FNpairs           | 0             | -       | X       | -     |
| Noise file name                          | noisefile         |               | -       | X       | -     |
| Noise file name is variable              | noise_file_as_var | 0             | -       | X       | -     |
| Frequency1-50                            | F1-50             |               | -       | X       | -     |
| Noise1-50                                | N1-50             |               | -       | X       | -     |
| Frequency name for fundamental frequency | fundname          |               | -       | X       | -     |
| Second frequency name                    | fundname2         |               | -       | X       | -     |
| Noise temperature                        | noisetemp         |               | -       | X       | -     |
| Nominal temperature                      | tnom              |               | X       | X       | -     |
| PAC magnitude                            | pacm              |               | -       | X       | -     |
| PAC phase                                | pacp              |               | -       | X       | -     |
| Phase for sinusoid 2                     | sinephase2        |               | -       | X       | -     |
| Temp coefficient 1                       | tc1               |               | X       | X       | -     |
| Temp coefficient 2                       | tc2               |               | X       | X       | -     |
| XF magnitude                             | xfm               |               | -       | X       | -     |
| Phase shift                              | phase             |               | -       | X       | -     |
| Amplitude 2                              | va2               |               | -       | X       | -     |
| Impedance                                | z0                |               | -       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise\_file parameter is set as *NOISEFILE*, the netlisting behavior for noise\_file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noise\_file parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

**Example**

```
v4 n1 0 dc=0 sin ( 0 1v 100Meg 1n 1e+6 0 )
```

This damped sinusoidal source connects between nodes n1 and 0. In this waveform:

- Peak value is 1V
- Offset is 0 V
- Frequency is 100 MHz
- Time delay is 1 ns
- Damping factor is 1e+6
- Phase delay is 0 deg

**See Also**

- [Sinusoidal Source Function](#) in the *HSPICE® User Guide: Elements*

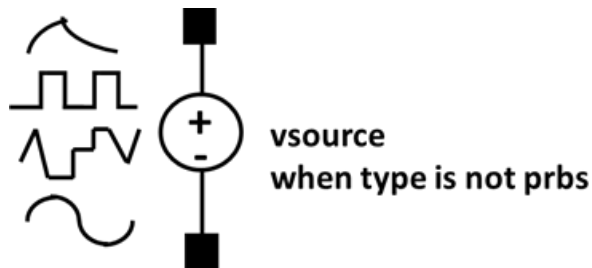
## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the `dbAnalogLibVsinVersion` Tcl preference before the analogLib library is loaded.

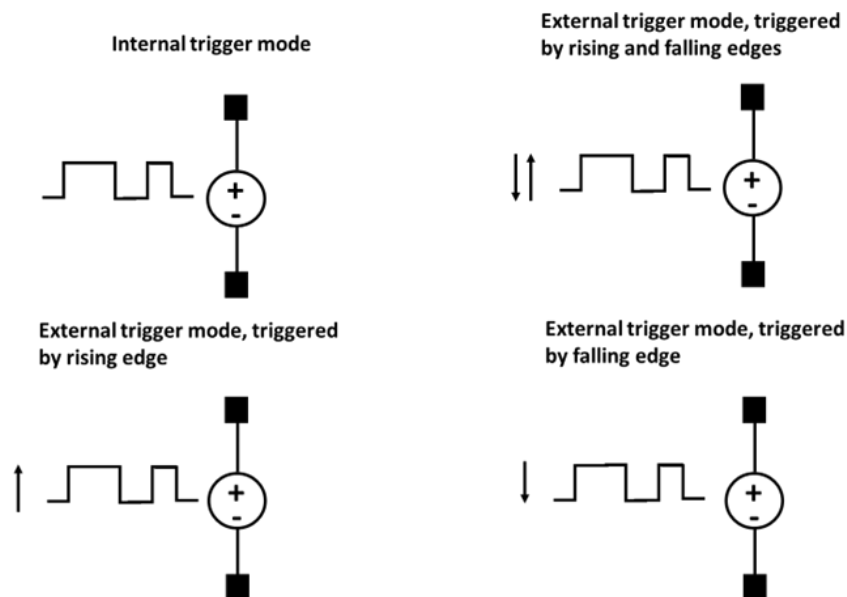
### vsources

Voltage source element with various parameters settings.

The vsources device is a PCell controlled by its source type. For source type other than prbs, it is a 2-pin device and its symbol looks as follows:



For prbs source type, its PCell behavior is affected by the trigger parameter. The following figure shows four vsources devices that have different trigger modes, which are Internal (on the upper-left), External by rising edges (on the upper-right), External by falling edges (on the lower-left), and External by both edges (on the lower-right) modes. The internal trigger mode has 2 pins (PLUS and MINUS) and other three external trigger modes have 4 pins (P1, M1, P2, and M2). The netlisting procedure writes out their pins accordingly.



## CDF Parameters

| CDF Parameter Prompt  | CDF Parameter  | Default Value | hspiceD | spectre | auCdl |
|-----------------------|----------------|---------------|---------|---------|-------|
| DC Voltage            | vdc            | 0             | X       | X       | -     |
| srcTypeValue          | srcTypeValue   | 1             | -       | -       | -     |
| Source Type           | srcType        | dc            | X       | X       | -     |
| Frequency name 1      | fundname       |               | -       | X       | -     |
| Waveform entry method | pwlEntryMethod | File          | X       | X       | -     |
| File name             | fileName       |               | X       | X       | -     |
| File name is variable | file_as_var    | 0             | X       | X       | -     |
| Time Value Pairs      | tvpairs        | 0             | X       | X       | -     |
| Time1-50              | t1-50          |               | X       | X       | -     |
| Z State1-50           | Z1-50          | 0             | X       | X       | -     |
| Voltage1-50           | v1-50          |               | X       | X       | -     |
| Zero value            | val0           |               | -       | X       | -     |
| One value             | val1           |               | -       | X       | -     |
| Rise Delay Time       | td1            | 0             | X       | X       | -     |
| Fall Delay Time       | td2            | 0             | X       | X       | -     |
| Rise Time Constant    | tau1           | 0             | X       | X       | -     |
| Fall Time Constant    | tau2           | 0             | X       | X       | -     |
| Exp period            | expperiod      |               | -       | X       | -     |
| Rise Time             | tr             | 0             | X       | X       | -     |
| Fall Time             | tf             | 0             | X       | X       | -     |
| Pulse Width           | pw             | 0             | X       | X       | -     |
| Period                | per            | 0             | X       | X       | -     |
| Repeat                | r              |               | X       | -       | -     |
| Repeat Stop Time      | rstop          |               | X       | -       | -     |

| CDF Parameter Prompt | CDF Parameter  | Default Value | hspiceD | spectre | auCdl |
|----------------------|----------------|---------------|---------|---------|-------|
| Repeat Stop Value    | stopvalue      |               | X       | -       | -     |
| Repeat Stop Slope    | stopslope      |               | X       | -       | -     |
| Voltage Offset       | vo             | 0             | X       | -       | -     |
| Frequency 1          | freq           |               | X       | X       | -     |
| Voltage Amplitude    | va             | 0             | X       | X       | -     |
| Phase for sinusoid   | sinephase      | 0             | X       | X       | -     |
| Voltage level        | sinedc         |               | -       | X       | -     |
| Delay Time           | td             | 0             | X       | X       | -     |
| Damping Factor       | theta          |               | X       | X       | -     |
| DC offset            | offset         |               | -       | X       | -     |
| Desired rms value    | rms            |               | -       | X       | -     |
| Time scaling factor  | stretch        |               | -       | X       | -     |
| PWL period           | pwlperiod      |               | -       | X       | -     |
| Edge type            | edgetype       |               | X       | X       | -     |
| Width of transition  | twidth         |               | -       | X       | -     |
| Breakpoints          | allbrkpts      | no            | -       | X       | -     |
| Period start time    | pwlperiodstart |               | -       | X       | -     |
| # HB Tones           | hbtones        | 0             | X       | -       | -     |
| HB Magnitude1-7      | hbmag1-7       |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7     | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7      | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7      | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7   |               | X       | -       | -     |
| HB ModTone1-7        | hbmodtone1-7   |               | X       | -       | -     |
| Power Switch         | power          | V             | X       | -       | -     |



| CDF Parameter Prompt       | CDF Parameter     | Default Value | hspiceD | spectre | auCdl |
|----------------------------|-------------------|---------------|---------|---------|-------|
| Impedance                  | z0                |               | -       | -       | -     |
| Open DC Connection         | dcopen            |               | X       | -       | -     |
| Display second sinusoid    | numofsines        | 0             | -       | X       | -     |
| Frequency name 2           | fundname2         |               | -       | X       | -     |
| Frequency 2                | freq2             |               | -       | X       | -     |
| Amplitude 2 (Vpk)          | va2               |               | -       | X       | -     |
| Phase for sinusoid 2       | sinephase2        |               | -       | X       | -     |
| Display multi sinusoid     | vecofsines        | 0             | -       | X       | -     |
| Number of frequencies      | freqvecs          | 0             | -       | X       | -     |
| Sinusoid frequency 1-9     | freqv1-9          |               | -       | X       | -     |
| Sinusoid Ampl 1-9 (Vpk)    | vav1-9            |               | -       | x       | -     |
| Sinusoid phase 1-9         | phasev1-9         |               | X       | X       | -     |
| Sinusoid max harmonic 1-9  | harmv1-9          |               | X       | X       | -     |
| Display modulation params  | modulation        | 0             | -       | X       | -     |
| Amp modulation frequency 1 | ammodfreq         |               | -       | X       | -     |
| Amp modulation index 1     | ammodindex        |               | -       | X       | -     |
| Amp modulation phase 1     | ammodphase        |               | -       | X       | -     |
| Number of FM files         | filenums          | none          | -       | X       | -     |
| Name of FM file 1          | fmmodfile1        |               | -       | X       | -     |
| FM file1 name is variable  | fmmodfile1_as_var | 0             | -       | X       | -     |

| CDF Parameter Prompt      | CDF Parameter       | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------------|---------------|---------|---------|-------|
| Name of FM file 2         | fmmodfile2          |               | -       | X       | -     |
| FM file2 name is variable | fmmodfile2_as_var   | 0             | -       | X       | -     |
| FM index 1                | fmmodindex          |               | -       | X       | -     |
| FM frequency 1            | fmmodfreq           |               | -       | X       | -     |
| IQ modulation I File      | imodfile            |               | -       | X       | -     |
| I File name is variable   | imodfile_as_var     | 0             | -       | X       | -     |
| IQ modulation Q File      | qmodfile            |               | -       | X       | -     |
| Q File name is variable   | qmodfile_as_var     | 0             | -       | X       | -     |
| Delay time                | tdPrbs              |               | -       | X       | -     |
| Bit period                | perPrbs             | 1n            | -       | X       | -     |
| Transition reference      | transitionreference |               | -       | X       | -     |
| Lower threshold(percent)  | lowerthreshold      |               | -       | X       | -     |
| Upper threshold(percent)  | upperthreshold      |               | -       | X       | -     |
| Edge type                 | edgetypePrbs        |               | -       | X       | -     |
| Rise delay                | td01                |               | -       | X       | -     |
| Fall delay                | td10                |               | -       | X       | -     |
| PAM4 modulation           | pam4_modulation     | none          | -       | X       | -     |
| PAM4 mapping              | pam4_mapping        |               | -       | X       | -     |
| triggerValue              | triggerValue        | 1             | -       | X       | -     |
| Trigger                   | trigger             | Internal      | -       | X       | -     |
| Threshold                 | triggerthreshold    |               | -       | X       | -     |

| CDF Parameter Prompt               | CDF Parameter   | Default Value                                          | hspiceD | spectre | auCdl |
|------------------------------------|-----------------|--------------------------------------------------------|---------|---------|-------|
| LFSR Mode                          | lfsrmode        | PN32                                                   | -       | X       | -     |
| Browse and specify bit file        | prbsFileBox     | 0                                                      |         |         |       |
| Bit data file                      | prbsFile        |                                                        | -       | X       | -     |
| Bit data file name is variable     | prbsFile_as_var | 0                                                      | -       | X       | -     |
| Seed                               | lfsrseed        | 1 3 5 7 9<br>11 13 15<br>17 19 21<br>23 25 27<br>29 31 | -       | X       | -     |
| Taps                               | lfsrtaps        |                                                        | -       | X       | -     |
| RJ(rms)                            | rjrms           |                                                        | -       | X       | -     |
| RJ(seed)                           | rjseed          |                                                        | -       | X       | -     |
| Number of periodic jitters         | nPeriodicJitter | 0                                                      | -       | X       | -     |
| PJ%d(amplitude)                    | pjamp1-3        |                                                        | -       | X       | -     |
| PJ%d(frequency)                    | pjfreq1-3       |                                                        | -       | X       | -     |
| PJ%d(type)                         | pjtype1-3       |                                                        | -       | X       | -     |
| Pattern Parameter data             | data            |                                                        | -       | X       | -     |
| Pattern Parameter data is variable | data_as_var     | 0                                                      | -       | X       | -     |
| Pattern Parameter rptstart         | rptstart        | 1                                                      | -       | X       | -     |
| Pattern Parameter rpttimes         | rpttimes        | 0                                                      | -       | X       | -     |
| Amplitude scale factor             | scale           |                                                        | -       | X       | -     |
| Filter for pwl source              | pwlfilter       |                                                        | -       | X       | -     |
| Rolloff factor                     | rolloff         |                                                        | -       | X       | -     |

| CDF Parameter Prompt              | CDF Parameter   | Default Value | hspiceD | spectre | auCdl |
|-----------------------------------|-----------------|---------------|---------|---------|-------|
| Bandwidth of the pwl signal       | pwlbandwidth    |               | -       | X       | -     |
| Display small signal params       | smallSig        | 0             | X       | X       | -     |
| HBAC Magnitude                    | hbacm           |               | X       | X       | -     |
| HBAC Phase                        | hbacp           |               | X       | X       | -     |
| SNAC Magnitude                    | snacm           |               | X       | -       | -     |
| SNAC Phase                        | snacp           |               | X       | -       | -     |
| AC Magnitude                      | acm             |               | X       | X       | -     |
| AC Phase                          | acp             |               | X       | X       | -     |
| PAC magnitude                     | pacm            |               | -       | X       | -     |
| PAC phase                         | pacp            |               | -       | X       | -     |
| XF magnitude                      | xfm             |               | -       | X       | -     |
| Display resistance parameters     | resistanceParam | 0             | X       | -       | -     |
| DC Resistance                     | rdc             |               | X       | -       | -     |
| AC Resistance                     | rac             |               | X       | -       | -     |
| HB Resistance                     | rhb             |               | X       | -       | -     |
| HBAC Resistance                   | rhbac           |               | X       | -       | -     |
| TRAN Resistance                   | rtran           |               | X       | -       | -     |
| Display temperature params        | tempParam       | 0             | -       | X       | -     |
| Nominal temperature               | tnom            |               | X       | X       | -     |
| Linear temperature coefficient    | tc1             |               | X       | X       | -     |
| Quadratic temperature coefficient | tc2             |               | X       | X       | -     |

| CDF Parameter Prompt                                         | CDF Parameter          | Default Value                     | hspiceD | spectre | auCdl |
|--------------------------------------------------------------|------------------------|-----------------------------------|---------|---------|-------|
| Display noise parameters                                     | noiseParam             | 0                                 | -       | X       | -     |
| Generate noise?                                              | isnoisyNoiseModel      |                                   | -       | X       | -     |
| Noise entry method                                           | noiseEntryMethod       | File                              | -       | X       | -     |
| Noise file name                                              | noisefile              |                                   | -       | X       | -     |
| Noise file name is variable                                  | noise_file_as_var      | 0                                 | X       | X       | -     |
| Noise type                                                   | noiseModelType         | Noise Voltage(V <sup>2</sup> /Hz) | -       | X       | -     |
| Interpolation method                                         | noiseModelInterMethod  |                                   | -       | X       | -     |
| Number of noise/freq pairs                                   | FNpairs                | 0                                 | -       | X       | -     |
| Frequency1-50                                                | F1-50                  |                                   | -       | X       | -     |
| Noise1-50                                                    | N1-50                  |                                   | -       | X       | -     |
| Noise temperature                                            | noisetemp              |                                   | -       | X       | -     |
| This dummy parameter exists only for using its default value | paramForDefault        | evalDefaultZeroOrEmpty("")        |         |         |       |
| High-Z impedance                                             | highz                  |                                   | -       | X       | -     |
| Min high-Z transition width                                  | min_z_transition_width |                                   | -       | X       | -     |
| Multiplier                                                   | m                      |                                   | -       | X       | -     |
| Version                                                      | version                | 2                                 | -       | -       | -     |

**Note:**

- When frequency/noise pairs have empty values or the number of frequency/noise pairs does not match with the FNpairs parameter, the Spectre netlister shows an error in the console.
- The default of noise\_file\_as\_var parameter is *false*. If the noise\_file parameter is set as *NOISEFILE*, the netlisting behavior for noise\_file and noise\_file\_as\_var parameters is as follows:

| CDF Parameter     | hspiceD Output           | spectre Output        |
|-------------------|--------------------------|-----------------------|
| noise_file_as_var |                          |                       |
| false (default)   | noisefile='NOISEFILE'    | noisefile="NOISEFILE" |
| true              | noisefile=str(NOISEFILE) | noisefile=NOISEFILE   |

- For Hspice-format netlisting, a pair of single quotes is added to noise\_file parameter when noise\_file\_as\_var is *false*. When noise\_file\_as\_var parameter is *true*, `str(NOISEFILE)` is used as the value format.
- For Spectre-format netlisting, a pair of double quotes is added when noise\_file\_as\_var parameter is *false*. No single/double quotes are added when noise\_file\_as\_var parameter is *true*.

## Example

```
v4 n1 n2 dc=1 power=0
v4 n1 n2 dc=0 exp ( 0 1 0 20n 0 30n )
v4 n1 n2 dc=0.35 pulse ( 0 0.75 1n 1n 1n 10n 20n )
```

## See Also

- [Independent Sources and Stimuli](#) in the *PrimeSim Continuum User Guide: Elements*

## Controlling Version Number for Legacy Designs

To control the version number for legacy designs, create and use the `dbAnalogLibVsourceVersion` Tcl preference before the analogLib library is loaded.

## Using vsource to Set up Harmonic Balance or Shooting PSS Simulations

vsource can be used as stimulus for one-, two-, and N-tone Harmonic Balance and Shooting Newton simulations. Refer to [port](#) for details, because the setup is similar and most parameters are shared.

**Note:**

vsources **Resistance parameters**, **#HB Tones parameters**, and **Power Switch** are kept for reasons of compatibility with PrimeSim HSPICE, but their ongoing use is not recommended.

## vvmrf

Vector Modulated RF voltage source. It allows digital modulation of an RF carrier using in-phase and quadrature components created from a binary data stream.

### CDF Parameters

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| DC Voltage           | vdc           | 0             | X       | -       | -     |
| Source Type          | srcType       | vmrf          | -       | -       | -     |
| Voltage Amplitude    | amp           |               | X       | -       | -     |
| Carrier Frequency    | freq          |               | X       | -       | -     |
| Carrier Phase        | phase         |               | X       | -       | -     |
| Modulation Method    | mod           | bpsk          | X       | -       | -     |
| Filter Method        | filter        | rect          | X       | -       | -     |
| Filter Parameter     | filcoef       |               | X       | -       | -     |
| Bit Rate             | rate          |               | X       | -       | -     |
| Bitstream            | bitstream     |               | X       | -       | -     |
| AC Magnitude         | acm           |               | X       | -       | -     |
| AC Phase             | acp           |               | X       | -       | -     |
| # HB Tones           | hbtones       | 0             | X       | -       | -     |
| HB Magnitude1-7      | hbmag1-7      |               | X       | -       | -     |
| HB Phase1-7          | hbphase1-7    | 0             | X       | -       | -     |
| HB Harmonic1-7       | hbharm1-7     | 1             | X       | -       | -     |
| HB Tone1-7           | hbtone1-7     | 1-7           | X       | -       | -     |
| HB ModHarm1-7        | hbmodharm1-7  |               | X       | -       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| HB ModTone1-7        | hbmodtone1-7  |               | X       | -       | -     |
| Open DC Connection   | dcopen        |               | -       | -       | -     |
| DC Resistance        | rdc           |               | X       | -       | -     |
| AC Resistance        | rac           |               | X       | -       | -     |
| HB Resistance        | rhb           |               | X       | -       | -     |
| HBAC Resistance      | rhbac         |               | X       | -       | -     |
| TRAN Resistance      | rtran         |               | X       | -       | -     |
| HBAC Magnitude       | hbacm         |               | X       | -       | -     |
| HBAC Phase           | hbacp         |               | X       | -       | -     |
| SNAC Magnitude       | snacm         |               | X       | -       | -     |
| SNAC Phase           | snacp         |               | X       | -       | -     |
| Multiplier           | m             |               | X       | -       | -     |
| Impedance            | z0            |               | X       | -       | -     |

## Example

```
v4 n1 n2 dc=0 vmrf amp=0.85 freq=10k phase=0 rate=100Meg
+ mod=bpsk filter=rect bitstream=01010010011100b
```

## See Also

- [Vector-modulated RF \(VMRF\) Source](#) in the *HSPICE® User Guide: Elements*

## Dependent Sources

The following topics describe:

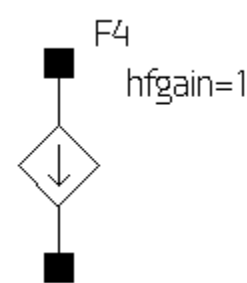
- [cccs](#)
- [ccvs](#)
- [pcccs](#)
- [pccvs](#)



- [pvccs](#)
- [pvcvs](#)
- [pvcvs2](#)
- [svccs](#)
- [svcvs](#)
- [vccs](#)
- [vcvs](#)
- [zvccs](#)
- [zvcvs](#)

CCCS

Current-controlled current source. Can be of *linear*, *pwl*, and *delay* types.



CDF Parameters

| CDF Parameter Prompt   | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------|---------------|---------------|---------|---------|-------|
| Name of Voltage Source | vref          |               | X       | X       | -     |
| Current Gain (Spectre) | fgain         | 1             | -       | X       | -     |
| Current Gain (HSPICE)  | hfgain        | 1             | X       | -       | -     |
| Type                   | csType        | linear        | X       | -       | -     |
| Controlling Pairs      | xypairs       | 2             | X       | -       | -     |

| CDF Parameter Prompt        | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|---------------|---------------|---------|---------|-------|
| Delta                       | delta         |               | -       | X       | -     |
| Control Current 1-50        | x1-50         |               | X       | -       | -     |
| Corresp Element1-50         | y1-50         |               | X       | -       | -     |
| Maximum Output Current      | maxi          |               | X       | -       | -     |
| Minimum Output Current      | mini          |               | X       | -       | -     |
| Maximum Output Current      | maxm          |               | -       | X       | -     |
| Minimum Output Current      | minm          |               | -       | X       | -     |
| Delay Time                  | htd           | 0             | X       | -       | -     |
| Scale Factor                | scale         |               | X       | X       | -     |
| Temperature Coefficient 1   | tc1           |               | X       | X       | -     |
| Temperature Coefficient 2   | tc2           |               | X       | X       | -     |
| Absolute Output Current     | absol         | 0             | -       | X       | -     |
| Absolute Output Current     | habs          | 0             | X       | -       | -     |
| Initial Output Current      | hic           |               | X       | -       | -     |
| Probe 1-20                  | p1-20         |               | -       | X       | -     |
| Port 1-20                   | port1-20      |               | -       | X       | -     |
| Probe count                 | probeCnt      | 4             | -       | X       | -     |
| Port                        | port          |               | -       | X       | -     |
| PWL vector source           | iVectSpec     | PwlVect       | -       | X       | -     |
| File containing PWL Vectors | pwlFile       |               | -       | X       | -     |
| Scale for input             | stretchpwl    |               | -       | X       | -     |

| CDF Parameter Prompt  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------|---------------|---------------|---------|---------|-------|
| Scale for output      | scalepwl      |               | -       | X       | -     |
| Smoothing factor      | smoothing     |               | -       | X       | -     |
| Type of source        | typesrc       | cccs          | -       | X       | -     |
| Type of transfer char | trfType       | Linear        | -       | X       | -     |
| Number of PWL Vectors | pwlCnt        | 4             | -       | X       | -     |
| PWL Input             | pwi1-20       |               | -       | X       | -     |
| PWL Output            | pwo1-20       |               | -       | X       | -     |
| Multiplier            | m             |               | -       | X       | -     |
| Time Delay            | td            |               | -       | X       | -     |

Example

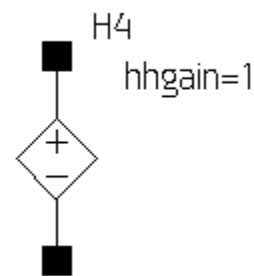
```
f4 n1 gnd! cccs vcntrl 0.45 abs=0
```

See Also

- [Current-dependent Current Sources—F-elements](#) in the *HSPICE® User Guide: Elements*

CCVS

Current-controlled voltage source. Can be of *linear*, *pwl*, and *delay* types.



## CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Name of Voltage Source    | vref          |               | X       | X       | -     |
| Transresistance (Spectre) | hgain         | 1             | -       | X       | -     |
| Transresistance (HSPICE)  | hhgain        | 1             | X       | -       | -     |
| Type                      | csType        | linear        | X       | -       | -     |
| Controlling Pairs         | xypairs       | 2             | X       | -       | -     |
| Delta                     | delta         |               | X       | X       | -     |
| Control Current 1-50      | x1-50         |               | X       | X       | -     |
| Corresp Element1-50       | y1-50         |               | X       | -       | -     |
| Maximum Output Voltage    | maxv          |               | X       | -       | -     |
| Minimum Output Voltage    | minv          |               | X       | -       | -     |
| Maximum Output Voltage    | maxm          |               | -       | X       | -     |
| Minimum Output Voltage    | minm          |               | -       | X       | -     |
| Delay Time                | htd           | 0             | X       | -       | -     |
| Scale Factor              | scale         |               | X       | X       | -     |
| Temperature Coefficient 1 | tc1           |               | X       | X       | -     |
| Temperature Coefficient 2 | tc2           |               | X       | X       | -     |
| Absolute Output Voltage   | absol         | 0             | -       | X       | -     |
| Absolute Output Voltage   | habs          | 0             | X       | -       | -     |
| Initial Output Voltage    | hic           |               | X       | -       | -     |

| CDF Parameter Prompt        | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|---------------|---------------|---------|---------|-------|
| Probe 1-20                  | p1-20         |               | -       | X       | -     |
| Port 1-20                   | port1-20      |               | -       | X       | -     |
| Probe count                 | probeCnt      | 4             | -       | X       | -     |
| Port                        | port          |               | -       | X       | -     |
| PWL vector source           | iVectSpec     | PwlVect       | -       | X       | -     |
| File containing PWL Vectors | pwlFile       |               | -       | X       | -     |
| Scale for input             | stretchpwl    |               | -       | X       | -     |
| Scale for output            | scalepwl      |               | -       | X       | -     |
| Smoothing factor            | smoothing     |               | -       | X       | -     |
| Type of Source              | typesrc       |               | -       | X       | -     |
| Type of transfer char       | trfType       | Linear        | -       | X       | -     |
| Number of PWL Vectors       | pwlCnt        | 4             | -       | X       | -     |
| PWL Input                   | pwi1-20       |               | -       | X       | -     |
| PWL Output                  | pwo1-20       |               | -       | X       | -     |
| Multiplier                  | m             |               | -       | X       | -     |

Example

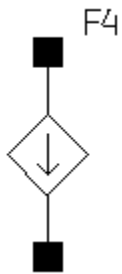
```
h4 n1 gnd! ccvs vcntrl 1 max='vmax' min='vmin' abs=0
```

See Also

- [Current-dependent Voltage Sources—H-elements](#) in the *HSPICE® User Guide: Elements*

pcccs

Polynomial current-controlled current source.



## CDF Parameters

| CDF Parameter Prompt                  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------------------|---------------|---------------|---------|---------|-------|
| Dimension (Number of Voltage Sources) | dim           | 1             | X       | X       | -     |
| Name of Voltage Source 1              | vref1         |               | X       | X       | -     |
| Name of Voltage Source 2              | vref2         |               | X       | X       | -     |
| Name of Voltage Source 3              | vref3         |               | X       | X       | -     |
| Type                                  | csType        | poly          | X       | -       | -     |
| Initial Controlling Current           | hic           |               | X       | -       | -     |
| Scale Factor                          | scale         |               | X       | -       | -     |
| Absolute output current               | absol         | off           | X       | X       | -     |
| Maximum number of coefficients        | polyCoef      | 4             | X       | X       | -     |
| Poly coefficient 0-20                 | c0-20         | 0             | X       | X       | -     |
| Coefficients to be specified          | coefSpec      | form          | -       | X       | -     |
| File containing coefficients          | filecoef      |               | -       | X       | -     |
| Gain                                  | gain          | 1             | -       | X       | -     |
| Smoothing factor                      | smoothing     |               | -       | X       | -     |

| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Temp coefficient 1   | tc1           |               | -       | X       | -     |
| Temp coefficient 2   | tc2           |               | -       | X       | -     |
| Probe 1-20           | p1-20         |               | -       | X       | -     |
| Port 1-20            | port1-20      |               | -       | X       | -     |
| Probe count          | probeCnt      | 4             | -       | X       | -     |
| Min output current   | minm          |               | X       | X       | -     |
| Max output current   | maxm          |               | X       | X       | -     |
| Multiplier           | m             |               | X       | X       | -     |

Example

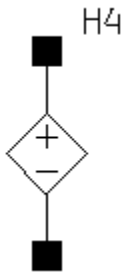
```
f4 n1 gnd! cccs poly(2) v1 v2 ic=1u 1 1 3 1 6
```

See Also

- [Polynomial \(POLY\)](#) in the *HSPICE® User Guide: Elements*
- [Polynomial Functions](#) in the *HSPICE® User Guide: Elements*

pccvs

Polynomial current-controlled voltage source.



## CDF Parameters

| CDF Parameter Prompt                  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------------------|---------------|---------------|---------|---------|-------|
| Dimension (Number of Voltage Sources) | dim           | 1             | X       | X       | -     |
| Name of Voltage Source 1              | vref1         |               | X       | X       | -     |
| Name of Voltage Source 2              | vref2         |               | X       | X       | -     |
| Name of Voltage Source 3              | vref3         |               | X       | X       | -     |
| Type                                  | csType        | poly          | X       | -       | -     |
| Initial Controlling Current           | hic           |               | X       | -       | -     |
| Scale Factor                          | scale         |               | X       | -       | -     |
| Absolute output voltage               | absol         | off           | X       | X       | -     |
| Maximum number of coefficients        | polyCoef      | 4             | X       | X       | -     |
| Poly coefficient 0-20                 | c0-20         | 0             | X       | X       | -     |
| Coefficients to be specified          | coefSpec      | form          | -       | X       | -     |
| File containing coefficients          | filecoef      |               | -       | X       | -     |
| Gain                                  | gain          | 1             | -       | X       | -     |
| Smoothing factor                      | smoothing     |               | -       | X       | -     |
| Temp coefficient 1                    | tc1           |               | X       | X       | -     |
| Temp coefficient 2                    | tc2           |               | X       | X       | -     |
| Probe 1-20                            | p1-20         |               | -       | X       | -     |
| Port 1-20                             | port1-20      |               | -       | X       | -     |
| Probe count                           | probeCnt      | 4             | -       | X       | -     |
| Min output voltage                    | minm          |               | X       | X       | -     |



| CDF Parameter Prompt | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------|---------------|---------------|---------|---------|-------|
| Max output voltage   | maxm          |               | X       | X       | -     |
| Multiplier           | m             |               | -       | X       | -     |

Example

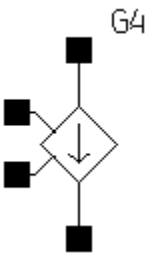
```
h4 n1 gnd! ccvs poly(2) v1 v2 1 1 3 1 6 ic=1u
```

See Also

- [Polynomial \(POLY\)](#) in the *HSPICE® User Guide: Elements*
- [Polynomial Functions](#) in the *HSPICE® User Guide: Elements*

pvccs

Polynomial voltage-controlled current source.



CDF Parameters

| CDF Parameter Prompt                  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------------------|---------------|---------------|---------|---------|-------|
| Dimension (Number of Voltage Sources) | dim           | 1             | X       | X       | -     |
| Type                                  | csType        | poly          | X       | -       | -     |
| Initial Controlling Voltage           | hic           |               | X       | -       | -     |
| Scale Factor                          | scale         |               | X       | -       | -     |
| Absolute output current               | absol         | off           | X       | X       | -     |

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Maximum number of coefficients | polyCoef      | 4             | X       | X       | -     |
| Poly coefficient 0-20          | c0-20         | 0             | X       | X       | -     |
| Coefficients to be specified   | coefSpec      | form          | -       | X       | -     |
| File containing coefficients   | filecoef      |               | -       | X       | -     |
| Gain                           | gain          | 1             | -       | X       | -     |
| Smoothing factor               | smoothing     |               | -       | X       | -     |
| Temp coefficient 1             | tc1           |               | X       | X       | -     |
| Temp coefficient 2             | tc2           |               | X       | X       | -     |
| Min output current             | minm          |               | X       | X       | -     |
| Max output current             | maxm          |               | X       | X       | -     |
| Multiplier                     | m             |               | X       | X       | -     |

Example

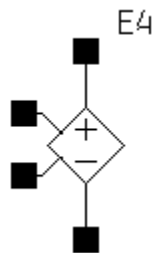
```
g5 n1 gnd! vccs poly(1) n2 gnd! ic=0.5 3 1 5
```

See Also

- [Polynomial \(POLY\)](#) in the *HSPICE® User Guide: Elements*

pvcvs

Polynomial voltage-controlled voltage source.



## CDF Parameters

| CDF Parameter Prompt                  | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------------------|---------------|---------------|---------|---------|-------|
| Dimension (Number of Voltage Sources) | dim           | 1             | X       | X       | -     |
| Type                                  | csType        | poly          | X       | -       | -     |
| Initial Controlling Voltage           | hic           |               | X       | -       | -     |
| Scale Factor                          | scale         |               | X       | -       | -     |
| Maximum number of coefficients        | polyCoef      | 4             | X       | X       | -     |
| Poly coefficient 0-20                 | c0-20         | 0             | X       | X       | -     |
| Coefficients to be specified          | coefSpec      | form          | -       | X       | -     |
| File containing coefficients          | filecoef      |               | -       | X       | -     |
| Gain                                  | gain          | 1             | -       | X       | -     |
| Smoothing factor                      | smoothing     |               | -       | X       | -     |
| Temp coefficient 1                    | tc1           |               | X       | X       | -     |
| Temp coefficient 2                    | tc2           |               | X       | X       | -     |
| Min output voltage                    | minm          |               | X       | X       | -     |
| Max output voltage                    | maxm          |               | X       | X       | -     |
| Multiplier                            | m             |               | -       | X       | -     |
| Absolute output voltage               | absol         | off           | X       | X       | -     |

## Example

```
e5 n1 gnd! vcvs poly(1) n2 gnd! 3 1 5 ic=0.5
```

## See Also

- [Polynomial \(POLY\)](#) in the *HSPICE® User Guide: Elements*

## pvcvs2

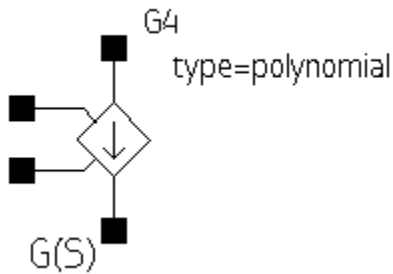
Polynomial voltage controlled voltage sources, which is controlled by two voltage sources.

### CDF Parameters

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Max Coefficient Number    | polyCoef      | -             | -       | X       | -     |
| Poly Coeff 0-20           | c0 – c20      | -             | -       | X       | -     |
| Initial condition         | ic            | -             | -       | X       | -     |
| Scale factor              | scale         | -             | -       | X       | -     |
| Absolute value            | abs           | -             | -       | X       | -     |
| Gain                      | gain          | -             | -       | X       | -     |
| Multiplier                | m             | -             | -       | X       | -     |
| Minimum Output Voltage    | minm          | -             | -       | X       | -     |
| Maximum Output Voltage    | maxm          | -             | -       | X       | -     |
| Absolute Output Voltage   | absol         | -             | -       | X       | -     |
| Smoothing Factor          | smoothing     | -             | -       | X       | -     |
| Temperature coefficient 1 | tc1           | -             | -       | X       | -     |
| Temperature coefficient 2 | tc2           | -             | -       | -       | -     |

## SVCCS

Voltage-controlled current source with Laplace transform and Pole-Zero function.



CDF Parameters

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Laplace Specification Type     | type          | polynomial    | X       | -       | -     |
| Order of Transfer Function     | order         | 1             | X       | -       | -     |
| Coefficient of Num. Const Term | k0            |               | X       | -       | -     |
| Coefficient of Num. Term 1     | k1            |               | X       | -       | -     |
| Coefficient of Num. Term 2     | k2            |               | X       | -       | -     |
| Coefficient of Num. Term 3     | k3            |               | X       | -       | -     |
| Coefficient of Num. Term 4     | k4            |               | X       | -       | -     |
| Coefficient of Num. Term 5     | k5            |               | X       | -       | -     |
| Coefficient of Num. Term 6     | k6            |               | X       | -       | -     |
| Coefficient of Num. Term 7     | k7            |               | X       | -       | -     |
| Coefficient of Num. Term 8     | k8            |               | X       | -       | -     |
| Coefficient of Num. Term 9     | k9            |               | X       | -       | -     |

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Coefficient of Num. Term 10    | k10           |               | X       | -       | -     |
| Coefficient of Den. Const Term | d0            |               | X       | -       | -     |
| Coefficient of Den. Term 1     | d1            |               | X       | -       | -     |
| Coefficient of Den. Term 2     | d2            |               | X       | -       | -     |
| Coefficient of Den. Term 3     | d3            |               | X       | -       | -     |
| Coefficient of Den. Term 4     | d4            |               | X       | -       | -     |
| Coefficient of Den. Term 5     | d5            |               | X       | -       | -     |
| Coefficient of Den. Term 6     | d6            |               | X       | -       | -     |
| Coefficient of Den. Term 7     | d7            |               | X       | -       | -     |
| Coefficient of Den. Term 8     | d8            |               | X       | -       | -     |
| Coefficient of Den. Term 9     | d9            |               | X       | -       | -     |
| Coefficient of Den. Term 10    | d10           |               | X       | -       | -     |
| Zero Multiplier Gain           | a             |               | X       | -       | -     |
| Zero Real Part 1               | az1           |               | X       | -       | -     |
| Zero Imaginary Part 1          | fz1           |               | X       | -       | -     |
| Zero Real Part 2               | az2           |               | X       | -       | -     |
| Zero Imaginary Part 2          | fz2           |               | X       | -       | -     |
| Zero Real Part 3               | az3           |               | X       | -       | -     |
| Zero Imaginary Part 3          | fz3           |               | X       | -       | -     |

| CDF Parameter Prompt   | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------|---------------|---------------|---------|---------|-------|
| Zero Real Part 4       | az4           |               | X       | -       | -     |
| Zero Imaginary Part 4  | fz4           |               | X       | -       | -     |
| Zero Real Part 5       | az5           |               | X       | -       | -     |
| Zero Imaginary Part 5  | fz5           |               | X       | -       | -     |
| Zero Real Part 6       | az6           |               | X       | -       | -     |
| Zero Imaginary Part 6  | fz6           |               | X       | -       | -     |
| Zero Real Part 7       | az7           |               | X       | -       | -     |
| Zero Imaginary Part 7  | fz7           |               | X       | -       | -     |
| Zero Real Part 8       | az8           |               | X       | -       | -     |
| Zero Imaginary Part 8  | fz8           |               | X       | -       | -     |
| Zero Real Part 9       | az9           |               | X       | -       | -     |
| Zero Imaginary Part 9  | fz9           |               | X       | -       | -     |
| Zero Real Part 10      | az10          |               | X       | -       | -     |
| Zero Imaginary Part 10 | fz10          |               | X       | -       | -     |
| Pole Multiplier Gain   | b             |               | X       | -       | -     |
| Pole Real Part 1       | ap1           |               | X       | -       | -     |
| Pole Imaginary Part 1  | fp1           |               | X       | -       | -     |
| Pole Real Part 2       | ap2           |               | X       | -       | -     |
| Pole Imaginary Part 2  | fp2           |               | X       | -       | -     |
| Pole Real Part 3       | ap3           |               | X       | -       | -     |
| Pole Imaginary Part 3  | fp3           |               | X       | -       | -     |
| Pole Real Part 4       | ap4           |               | X       | -       | -     |
| Pole Imaginary Part 4  | fp4           |               | X       | -       | -     |
| Pole Real Part 5       | ap5           |               | X       | -       | -     |
| Pole Imaginary Part 5  | fp5           |               | X       | -       | -     |

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Pole Real Part 6          | ap6           |               | X       | -       | -     |
| Pole Imaginary Part 6     | fp6           |               | X       | -       | -     |
| Pole Real Part 7          | ap7           |               | X       | -       | -     |
| Pole Imaginary Part 7     | fp7           |               | X       | -       | -     |
| Pole Real Part 8          | ap8           |               | X       | -       | -     |
| Pole Imaginary Part 8     | fp8           |               | X       | -       | -     |
| Pole Real Part 9          | ap9           |               | X       | -       | -     |
| Pole Imaginary Part 9     | fp9           |               | X       | -       | -     |
| Pole Real Part 10         | ap10          |               | X       | -       | -     |
| Pole Imaginary Part 10    | fp10          |               | X       | -       | -     |
| Scale Factor              | scale         |               | X       | -       | -     |
| Temperature Coefficient 1 | tc1           |               | X       | -       | -     |
| Temperature Coefficient 2 | tc2           |               | X       | -       | -     |
| Multiplier                | m             |               | X       | -       | -     |

## Example

```
glowpass out gnd! laplace in gnd! 1.0 / 1.0, 2.0, 2.0, 1.0
ghigh_pass out gnd! pole in gnd! 1 / 1 0.001,0.0
```

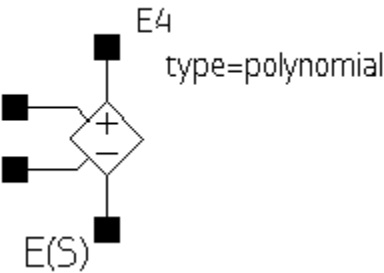
## See Also

- [Laplace Transform](#) in the *HSPICE® User Guide: Elements*
- [Pole-Zero Function](#) in the *HSPICE® User Guide: Elements*

## SVCVS

Voltage-controlled voltage source with Laplace transform and Pole-Zero function.





CDF Parameters

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Laplace Specification Type     | type          | polynomial    | X       | -       | -     |
| Order of Transfer Function     | order         | 1             | X       | -       | -     |
| Coefficient of Num. Const Term | k0            |               | X       | -       | -     |
| Coefficient of Num. Term 1     | k1            |               | X       | -       | -     |
| Coefficient of Num. Term 2     | k2            |               | X       | -       | -     |
| Coefficient of Num. Term 3     | k3            |               | X       | -       | -     |
| Coefficient of Num. Term 4     | k4            |               | X       | -       | -     |
| Coefficient of Num. Term 5     | k5            |               | X       | -       | -     |
| Coefficient of Num. Term 6     | k6            |               | X       | -       | -     |
| Coefficient of Num. Term 7     | k7            |               | X       | -       | -     |
| Coefficient of Num. Term 8     | k8            |               | X       | -       | -     |
| Coefficient of Num. Term 9     | k9            |               | X       | -       | -     |

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Coefficient of Num. Term 10    | k10           |               | X       | -       | -     |
| Coefficient of Den. Const Term | d0            |               | X       | -       | -     |
| Coefficient of Den. Term 1     | d1            |               | X       | -       | -     |
| Coefficient of Den. Term 2     | d2            |               | X       | -       | -     |
| Coefficient of Den. Term 3     | d3            |               | X       | -       | -     |
| Coefficient of Den. Term 4     | d4            |               | X       | -       | -     |
| Coefficient of Den. Term 5     | d5            |               | X       | -       | -     |
| Coefficient of Den. Term 6     | d6            |               | X       | -       | -     |
| Coefficient of Den. Term 7     | d7            |               | X       | -       | -     |
| Coefficient of Den. Term 8     | d8            |               | X       | -       | -     |
| Coefficient of Den. Term 9     | d9            |               | X       | -       | -     |
| Coefficient of Den. Term 10    | d10           |               | X       | -       | -     |
| Zero Multiplier Gain           | a             |               | X       | -       | -     |
| Zero Real Part 1               | az1           |               | X       | -       | -     |
| Zero Imaginary Part 1          | fz1           |               | X       | -       | -     |
| Zero Real Part 2               | az2           |               | X       | -       | -     |
| Zero Imaginary Part 2          | fz2           |               | X       | -       | -     |
| Zero Real Part 3               | az3           |               | X       | -       | -     |
| Zero Imaginary Part 3          | fz3           |               | X       | -       | -     |

| CDF Parameter Prompt   | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------|---------------|---------------|---------|---------|-------|
| Zero Real Part 4       | az4           |               | X       | -       | -     |
| Zero Imaginary Part 4  | fz4           |               | X       | -       | -     |
| Zero Real Part 5       | az5           |               | X       | -       | -     |
| Zero Imaginary Part 5  | fz5           |               | X       | -       | -     |
| Zero Real Part 6       | az6           |               | X       | -       | -     |
| Zero Imaginary Part 6  | fz6           |               | X       | -       | -     |
| Zero Real Part 7       | az7           |               | X       | -       | -     |
| Zero Imaginary Part 7  | fz7           |               | X       | -       | -     |
| Zero Real Part 8       | az8           |               | X       | -       | -     |
| Zero Imaginary Part 8  | fz8           |               | X       | -       | -     |
| Zero Real Part 9       | az9           |               | X       | -       | -     |
| Zero Imaginary Part 9  | fz9           |               | X       | -       | -     |
| Zero Real Part 10      | az10          |               | X       | -       | -     |
| Zero Imaginary Part 10 | fz10          |               | X       | -       | -     |
| Pole Multiplier Gain   | b             |               | X       | -       | -     |
| Pole Real Part 1       | ap1           |               | X       | -       | -     |
| Pole Imaginary Part 1  | fp1           |               | X       | -       | -     |
| Pole Real Part 2       | ap2           |               | X       | -       | -     |
| Pole Imaginary Part 2  | fp2           |               | X       | -       | -     |
| Pole Real Part 3       | ap3           |               | X       | -       | -     |
| Pole Imaginary Part 3  | fp3           |               | X       | -       | -     |
| Pole Real Part 4       | ap4           |               | X       | -       | -     |
| Pole Imaginary Part 4  | fp4           |               | X       | -       | -     |
| Pole Real Part 5       | ap5           |               | X       | -       | -     |
| Pole Imaginary Part 5  | fp5           |               | X       | -       | -     |

| CDF Parameter Prompt      | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|---------------------------|---------------|---------------|---------|---------|-------|
| Pole Real Part 6          | ap6           |               | X       | -       | -     |
| Pole Imaginary Part 6     | fp6           |               | X       | -       | -     |
| Pole Real Part 7          | ap7           |               | X       | -       | -     |
| Pole Imaginary Part 7     | fp7           |               | X       | -       | -     |
| Pole Real Part 8          | ap8           |               | X       | -       | -     |
| Pole Imaginary Part 8     | fp8           |               | X       | -       | -     |
| Pole Real Part 9          | ap9           |               | X       | -       | -     |
| Pole Imaginary Part 9     | fp9           |               | X       | -       | -     |
| Pole Real Part 10         | ap10          |               | X       | -       | -     |
| Pole Imaginary Part 10    | fp10          |               | X       | -       | -     |
| Scale Factor              | scale         |               | X       | -       | -     |
| Temperature Coefficient 1 | tc1           |               | X       | -       | -     |
| Temperature Coefficient 2 | tc2           |               | X       | -       | -     |

## Example

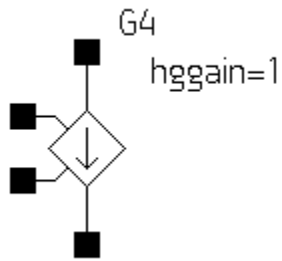
```
ehipass out 0 LAPLACE in 0 0.0,0.0,0.0,1.0 / 1.0,2.0,2.0,1.0
elow_pass out 0 POLE in 0 1.0 / 1.0, 1.0,0.0 0.5,0.1379
```

## See Also

- [Laplace Transform](#) in the *HSPICE® User Guide: Elements*
- [Pole-Zero Function](#) in the *HSPICE® User Guide: Elements*

## VCCS

Voltage-controlled current source with *linear*, *pwl*, and *delay* functions.



## CDF Parameters

| CDF Parameter Prompt       | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|----------------------------|---------------|---------------|---------|---------|-------|
| Transconductance (Spectre) | ggain         | 1             | -       | X       | -     |
| Transconductance (HSPICE)  | hggain        | 1             | X       | -       | -     |
| Type                       | csType        | linear        | X       | X       | -     |
| Controlling Pairs          | xypairs       | 2             | X       | -       | -     |
| Delta                      | Delta         |               | X       | X       | -     |
| Control Volt 1-50          | x1-50         |               | X       | -       | -     |
| Corresp Element1-50        | y1-50         |               | X       | -       | -     |
| Maximum Output Current     | maxi          |               | X       | -       | -     |
| Minimum Output Current     | mini          |               | X       | -       | -     |
| Delay Time                 | htd           | 0             | X       | -       | -     |
| Scale Factor               | scale         |               | X       | X       | -     |
| Temperature Coefficient 1  | tc1           |               | X       | X       | -     |
| Temperature Coefficient 2  | tc2           |               | X       | X       | -     |
| Absolute Output Current    | absol         | 0             | -       | X       | -     |
| Absolute Output Current    | habs          | 0             | X       | -       | -     |

| CDF Parameter Prompt         | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------------|---------------|---------------|---------|---------|-------|
| Initial Output Current       | hic           |               | X       | -       | -     |
| Multiplier                   | hm            |               | X       | -       | -     |
| Multiplier                   | m             |               | -       | X       | -     |
| PWL vector source            | iVectSpec     | PwlVect       | -       | X       | -     |
| File containing PWL Vectors  | pwlFile       |               | -       | X       | -     |
| Scale for input              | stretchpwl    |               | -       | X       | -     |
| Scale for output             | scalepwl      |               | -       | X       | -     |
| Smoothing factor             | smoothing     |               | -       | X       | -     |
| Type of source               | typesrc       | vccs          | X       | X       | -     |
| Capacitance Calculation Mode | ctype         | 0             | X       | -       | -     |
| Type of input of source      | inputtype     |               | -       | X       | -     |
| Type of transfer char        | trfType       | Linear        | -       | X       | -     |
| Number of PWL Vectors        | pwlCnt        | 4             | -       | X       | -     |
| PWL Input                    | pwi1-20       |               | -       | X       | -     |
| PWL Output                   | pwo1-20       |               | -       | X       | -     |
| Time Delay                   | td            | 0             | X       | -       | -     |
| Multiplier                   | m             |               | -       | X       | -     |

## Example

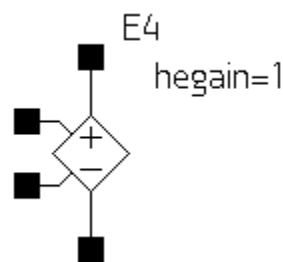
```
G4 out gnd! vccs cntrl gnd! 1 max=100u min=10u abs=0
```

## See Also

- [Voltage-dependent Current Sources—G-elements](#) in the *HSPICE® User Guide: Elements*

VCVS

Voltage-controlled voltage source with *linear*, *pwl*, and *delay* functions.



CDF Parameters

| CDF Parameter Prompt   | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|------------------------|---------------|---------------|---------|---------|-------|
| Voltage Gain (Spectre) | egain         | 1             | -       | X       | -     |
| Voltage Gain (HSPICE)  | hegain        | 1             | X       | -       | -     |
| Type                   | csType        | linear        | X       | X       | -     |
| Controlling Pairs      | xypairs       | 2             | X       | X       | -     |
| Delta                  | delta         |               | X       | X       | -     |
| Control Volt 1-50      | x1-50         |               | X       | X       | -     |
| Corresp Element1-50    | y1-50         |               | X       | X       | -     |
| Maximum Output Voltage | maxv          |               | X       | -       | -     |
| Minimum Output Voltage | minv          |               | X       | -       | -     |
| Maximum Output Voltage | maxm          |               | -       | X       | -     |
| Minimum Output Voltage | minm          |               | -       | X       | -     |
| Delay Time             | htd           | 0             | X       | X       | -     |
| Scale Factor           | scale         |               | X       | X       | -     |

| CDF Parameter Prompt        | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|-----------------------------|---------------|---------------|---------|---------|-------|
| Temperature Coefficient 1   | tc1           |               | X       | X       | -     |
| Temperature Coefficient 2   | tc2           |               | X       | X       | -     |
| Absolute Output Voltage     | absol         | 0             | -       | X       | -     |
| Absolute Output Voltage     | habs          | 0             | X       | -       | -     |
| Initial Output Voltage      | hic           |               | X       | -       | -     |
| PWL vector source           | iVectSpec     | PwlVect       | -       | X       | -     |
| File containing PWL Vectors | pwlFile       |               | -       | X       | -     |
| Scale for input             | stretchpwl    |               | -       | X       | -     |
| Scale for output            | scalepwl      |               | -       | X       | -     |
| Smoothing factor            | smoothing     |               | -       | X       | -     |
| Type of source              | typesrc       | vcvs          | -       | X       | -     |
| Multiplier                  | m             |               | -       | X       | -     |

## Example

```
E4 out gnd! vcvs cntrl gnd! 1 max=1 min=0.25 abs=0
```

## See Also

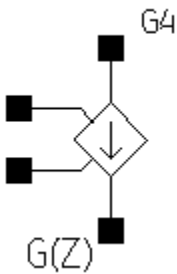
- [Voltage-Dependent Voltage Sources—E-elements](#) in the *HSPICE® User Guide: Elements*

---

## ZVCCS

Voltage-controlled current source with Z transform function.





CDF Parameters

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Order of Transfer Function     | order         | 1             | X       | -       | -     |
| Coefficient of Num. Const Term | k0            |               | X       | -       | -     |
| Coefficient of Num. Term 1-10  | k1-10         |               | X       | -       | -     |
| Coefficient of Den. Const Term | d0            |               | X       | -       | -     |
| Coefficient of Den. Term 1-10  | d1-10         |               | X       | -       | -     |
| Nyquist Critical Frequency     | maxf          |               | X       | -       | -     |
| Scale Factor                   | scale         |               | X       | -       | -     |
| Temperature Coefficient 1      | tc1           |               | X       | -       | -     |
| Temperature Coefficient 2      | tc2           |               | X       | -       | -     |
| Multiplier                     | m             |               | X       | -       | -     |

Example

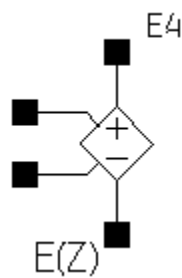
```
glow_pass 0 out ZTRANS in 0 0.0317 0.0951
+ 0.0951 0.0317 / 1.0 -1.459 0.9104 -0.1978
```

See Also

- [Z Transform Function Cell](#) in the *HSPICE® User Guide: Elements*

ZVCVS

Voltage-controlled voltage source with Z transform function.



CDF Parameters

| CDF Parameter Prompt           | CDF Parameter | Default Value | hspiceD | spectre | auCdl |
|--------------------------------|---------------|---------------|---------|---------|-------|
| Order of Transfer Function     | order         | 1             | X       | -       | -     |
| Coefficient of Num. Const Term | k0            |               | X       | -       | -     |
| Coefficient of Num. Term 1-10  | k1-10         |               | X       | -       | -     |
| Coefficient of Den. Const Term | d0            |               | X       | -       | -     |
| Coefficient of Den. Term 1-10  | d1-10         |               | X       | -       | -     |
| Nyquist Critical Frequency     | maxf          |               | X       | -       | -     |
| Scale Factor                   | scale         |               | X       | -       | -     |
| Temperature Coefficient 1      | tc1           |               | X       | -       | -     |
| Temperature Coefficient 2      | tc2           |               | X       | -       | -     |

## Example

```
ehigh_pass out 0 ZTRANS in 0 -0.0082 -0.1793  
+ 0.6579 -0.1793 -0.0082 / 1
```

## See Also

- [Z Transform Function Cell](#) in the *HSPICE® User Guide: Elements*

---

## Global Nets

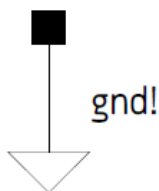
The following topics describe:

- [gnd](#)
- [gnd\\_connDef](#)
- [gndd!](#)
- [vcc](#)
- [vccd!](#)
- [vdd](#)
- [vdd\\_connDef](#)
- [vdda](#)
- [vddd](#)
- [vee!](#)
- [veed](#)
- [vssa!](#)
- [vssd!](#)

---

## **gnd**

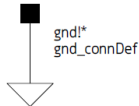
The symbol specifies a gnd! global ground net. The cell has no CDF parameters.



## gnd\_connDef

The symbol specifies a ground net with a connection definition and gnd! default net name. This symbol is similar to *gnd!* and has no CDF parameters.

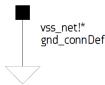
The cell has one connection assignment with the name “gnd”. With this connection assignment it is possible to create a new global power net and pass it to a higher hierarchy level of the design.



## Example

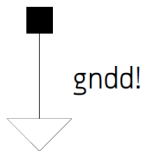
| Connection Assignments |          |
|------------------------|----------|
| name                   | net      |
| gnd                    | vss_net! |
| click to add           |          |

gnd = vss\_net!



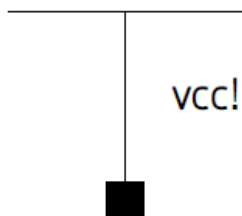
## gndd!

The symbol specifies a gndd! global ground net. The cell has no CDF parameters.



## vcc

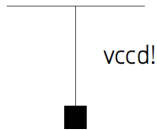
The symbol specifies vcc! power global net. The cell has no CDF parameters.



---

## vccd!

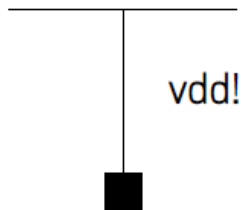
The symbol specifies a power connection with vccd! global net. The cell has no CDF parameters.



---

## vdd

The symbol specifies a power connection with vdd! global net. The cell has no CDF parameters.

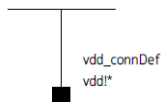


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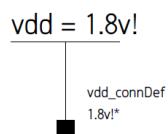
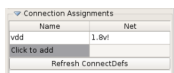
## vdd\_connDef

The symbol specifies a power net with a connection definition and the default net name vdd!. The cell has no CDF parameters.

The cell has one connection assignment with the name “vdd”. With this connection assignment it is possible to create a new global power net and pass it to a higher hierarchy level of the design.



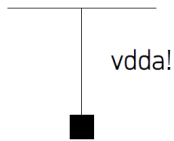
## Example



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## **vdda**

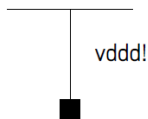
The symbol specifies a vdda! global power net. The cell has no CDF parameters.



---

## **vddd**

The symbol specifies a vddd! global power net. The cell has no CDF parameter.



---

## **vee!**

The symbol specifies a ground connection with vee! global power net. The cell has no CDF parameters.



---

## **veed**

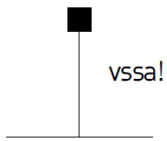
The symbol specifies a veed! global power net. The cell has no CDF parameters.



---

## **vssa!**

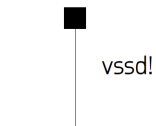
The symbol specifies a vssa! global ground net. It is similar to vss! and gnd! symbols and the cell has no CDF parameters.



---

## **vssd!**

The symbol specifies a vssd! global power net. The cell has no CDF parameters.



# 2

## analogLib Preferences

---

*Summarizes the supported Tcl preferences of analogLib.*

Tcl preferences of analogLib are used for version control and parameter settings. Each preference has:

- Preference purpose and name in the page's title
- A table lists:
  - The Custom Compiler build with a new version of related element.
  - The description of that version.
  - The preference value which points to that version.
- Sometimes the preference values are enabled or redefined in a newer build than the element version.
  - The build information related to preference is in the note under the table. The preference values are valid with that build.
- Check [dbAnalogLibSwitchVersion](#) for an example.

### Syntax

The syntax to use an analogLib preference:

```
db::createPref <preference_name> -value <preference_value>
```

### Example

```
db::createPref dbAnalogLibIsinVersion -value 1
```



## Guidelines

Guidelines to use analogLib preferences:

- If need a third-party compatible element: don't set version control preferences. The default version is third-party compatible.
- If there is any back-compatible issue in legacy schematics: try a version control preference. But whenever it is possible, it is recommended to migrate to the latest analogLib versions.
- Always set analogLib preferences right after launching Custom Compiler, to make sure the preferences work as expected.

The following sections describe the preferences in details.

- [setAnalogLibCompatibleVersion](#)
- [dbAnalogLibAutoUpdatePortNum](#)
- [dbAnalogLibCccsVersion](#)
- [dbAnalogLibCcvsvVersion](#)
- [dbAnalogLibIndVersion](#)
- [dbAnalogLibIndqVersion](#)
- [dbAnalogLibIprbsVersion](#)
- [dbAnalogLibIsinVersion](#)
- [dbAnalogLibIsourceVersion](#)
- [dbAnalogLibNcsVersion](#)
- [dbAnalogLibNportVersion](#)
- [dbAnalogLibNvsVersion](#)
- [dbAnalogLibNxPortVersion](#)
- [dbAnalogLibMTlineVersion](#)
- [dbAnalogLibPcapacitorVersion](#)
- [dbAnalogLibPdiodeVersion](#)
- [dbAnalogLibPinductorVersion](#)
- [dbAnalogLibPortVersion](#)
- [dbAnalogLibPprbsVersion](#)

- [dbAnalogLibPpwlVersion](#)
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- [dbAnalogLibVccsVersion](#)
- [dbAnalogLibVcresVersion](#)
- [dbAnalogLibVcvsVersion](#)
- [dbAnalogLibVprbsVersion](#)
- [dbAnalogLibVsinVersion](#)
- [dbAnalogLibVsourceVersion](#)
- [dbAnalogLibWindingVersion](#)
- [nlHSPICESkipEmptyDC](#)

---

## setAnalogLibCompatibleVersion

Sets all analogLib preferences to the default value in specified version.

### Syntax

```
db::setAnalogLibCompatibleVersion <release version number>
```

### Examples

```
db::setAnalogLibCompatibleVersion U-2023.03-SP1
```

### Note:

- Use this command line before opening any libraries or library manager.
- This preference sets all analogLib preferences to specified release version. After closing Custom Compiler, all analogLib preferences are written to the file: `<work_dir>/synopsys_custom/prefs.xml`

Next time when launching Custom Compiler, these analogLib preferences are loaded automatically.

- After using the preference `setAnalogLibCompatibleVersion`, the analogLib preferences are not changed with newer CC versions. If want to use the latest analogLib, manually delete all analogLib preferences in the file, `<work_dir>/synopsys_custom/prefs.xml.`,
- When using top level preference and individual preference together, individual preference wins.

For example:

```
db::setAnalogLibCompatibleVersion U-2023.03
db::setAnalogLibCompatibleVersion U-2023.03
```

In this case, the individual preference wins. The `lprbs` version is set to 2, all other elements are set to the default version in U-2023.02. However, the individual preference set in console won't be remembered next time you open Custom Compiler

The permanent solution is to open `<work_dir>/synopsys_custom/prefs.xml`, change the value of the `dbAnalogLibIprbsVersion` preference to 2 directly

## dbAnalogLibAutoUpdatePortNum

Disables auto-assignment of port number.

| First Build to Enable the Enhancements | Description                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before S-2021.09-SP2                   | <b>Automatically assign port number if the parameter “num” is not specified</b><br>This applies when creating an instance, performing Copy action or Property Editor display on a selected device that has an empty port number.<br>Port family devices include port, port_ref, mport, pdc, pexp, ppulse, ppwl, ppwlf, plfsr, pp sin, pam, and psffm. |
| N.A.                                   | Turn off the auto-assignment of the port number.                                                                                                                                                                                                                                                                                                      |

**Note:**

Preference values true and false are enabled in U-2023.03-SP2. The default value is true.

## dbAnalogLibCccsVersion

Controls cccs versions.

| First Build to Enable the Enhancements | Description |
|----------------------------------------|-------------|
|----------------------------------------|-------------|

Before V-2023.12

**Legacy cccs**

V-2023.12

**More enhancements for third-party compatibility**

Spectre-format: changed several CDF parameter definitions. Now it is compatible with third-party's spectre-format

Hspice-format: no change

**Note:**

Preference values 1-2 are redefined or enabled in V-2023.12. The default is 2.

## dbAnalogLibCcvsvVersion

Controls ccvs versions.

| First Build to Enable the Enhancements | Description |
|----------------------------------------|-------------|
|----------------------------------------|-------------|

Before V-2023.12

**Legacy ccvs**

V-2023.12

**More enhancements for third-party compatibility**

Spectre-format: changed several CDF parameter definitions. Now it is compatible with third-party's spectre-format

Hspice-format: no change

**Note:**

Preference values 1-2 are redefined or enabled in V-2023.12. The default is 2.

## dbAnalogLibIndVersion

Controls ind versions.

| First Build to Enable the Enhancements | Description |
|----------------------------------------|-------------|
|----------------------------------------|-------------|

Before V-2023.12-SP1

**Legacy ind**

V-2023.12-SP1

Use c1 ~ c20 instead of c0 ~c20 for poly coefficients in spectre-format display and netlis

**Note:**

Preference values 1-2 are redefined or enabled in V-2023.12-SP1. The default is 2.

## dbAnalogLibIndqVersion

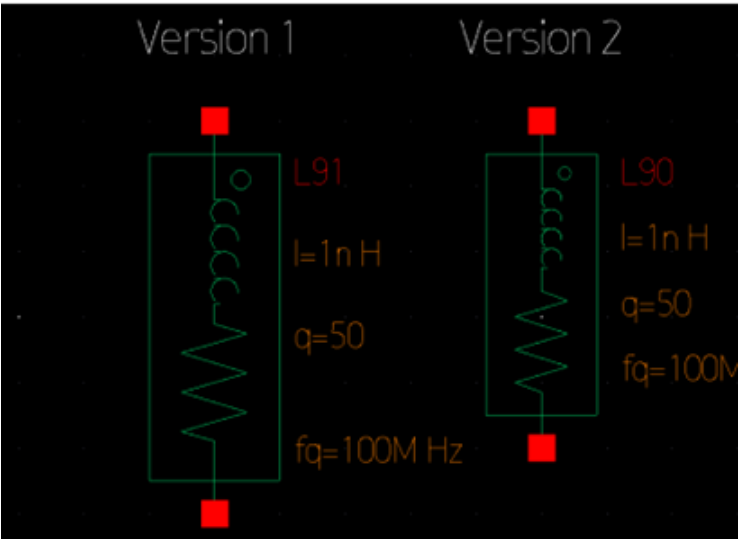
Controls `ind` versions.

| First Build to Enable the Enhancements | Description                                    |
|----------------------------------------|------------------------------------------------|
| Before V-2023.12-SP1                   | <b>Legacy pinductor</b>                        |
| V-2023.12-SP1                          | Use new symbol, for third-party compatibility. |

**Note:**

Preference values 1-2 are redefined or enabled in V-2023.12-SP1. The default value is 2.

The comparison of symbols in version 1 and version 2 (The height is changed):



## dbAnalogLiblprbsVersion

Controls `lprbs` versions.

| First Build to Enable the Enhancements | Description |
|----------------------------------------|-------------|
|----------------------------------------|-------------|

|               |                                                     |
|---------------|-----------------------------------------------------|
| U-2023.03-SP2 | <b>Add iprbs in analogLib</b><br>Symbol: is a PCell |
|---------------|-----------------------------------------------------|

**Note:**

Preference value 1 is enabled in U-2023.03-SP2. The default value is 1.

dbAnalogLibIsinVersion

Controls isin versions.

| First Build to Enable the Enhancements | Description |
|----------------------------------------|-------------|
|----------------------------------------|-------------|

|                  |                                                                                                                                                                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before U-2023.03 | <b>Legacy isin</b><br>CDF parameters: ia default is 0, io default is 0                                                                                                                                                                                                                            |
| U-2023.03        | <b>Remove default value of io</b><br>Spectre-format: netlist is changed, simulation result is not changed<br>Hspice-format: netlist is not changed<br><br><b>Remove default value of ia</b><br>Spectre-format: netlist and simulation result are changed<br>Hspice-format: netlist is not changed |

**Note:**

Preference values 1-2 are redefined and enabled in U-2023.03-SP2. The default value is 2.

dbAnalogLibIsourceVersion

Controls isource versions.

| First Build to Enable the Enhancements | Enhancements Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before U-2023.03                       | <b>Legacy isource</b><br>Symbol: is not a PCell<br>CDF parameters: tvpairs default is 2, ia default is 0<br>Netlisting behavior: <ul style="list-style-type: none"> <li>Noise related parameters (noisefile, F/N pairs) are only netlisted when they are displayed in the Property Editor (parameter noiseParam is selected)</li> </ul>                                                                                                                                                                                                                         |
| U-2023.03                              | <b>Remove default value of ia</b><br>Spectre-format: netlist and simulation result are changed<br>Hspice-format: netlist is not changed<br><br><b>Inherited from last version:</b> tvpair default is 2, netlisting behavior of noisefile and F/N pairs is not a PCell                                                                                                                                                                                                                                                                                           |
| U-2023.03-SP2                          | <b>Symbol is changed to PCell (for “type=prbs” &amp; “external trigger modes”)</b><br>No impact on legacy designs if not in this type and mode<br><br><b>Change tvpairs default value to 0</b><br>Both of Spectre-format and Hspice-format: netlist and simulation result are changed<br><br><b>Netlist noise-related parameters (noisefile, F/N pairs) when they are hidden in PE</b><br>Spectre-format: netlist and simulation result are changed<br>Hspice-format: netlist is not changed<br><br><b>Inherited from last version:</b> ia has no default value |

**Note:**

Preference values 1-3 are in U-2023.03-SP2. The default value is 3.

## dbAnalogLibNcsVersion

Controls ncs versions.

| First Build to Enable the Enhancements | Enhancements Description                                                                                                                                                                          |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before U-2023.03-SP2                   | <b>Legacy ncs</b><br>Property Editor displays noise, noisefile, and noisefile_as_var at the same time<br>Hspice netlister writes noise and noisefile at the same time if they both have non-empty |

| First Build to Enable the Enhancements | Description                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| U-2023.03-SP2                          | <b>Add new CDF parameter noiseEntryMethod to control the display and netlisting of and noise file</b><br>Choices are: file and expression<br>Default value is: file<br><br>Spectre-format: no impact. (ncs is for Hspice-format only)<br>Hspice-format: legacy schematic is impacted if the ncs element sets Noise Expression a<br>The noise expression is not netlisted because it is hidden by default noiseEntryMethod. |

**Note:**

Preference values 1-2 are enabled in U-2023.03-SP2. The default value is 2.

## dbAnalogLibNportVersion

Controls nport versions.

| First Build to Enable the Enhancements | Description                                                       |
|----------------------------------------|-------------------------------------------------------------------|
| Before S-2021.09                       | Legacy nport                                                      |
| Before S-2021.09                       | Redefine nport device to make it more compatible with third party |

**Note:**

Preference values 1-2 are enabled before S-2021.09. The default value is 2.

## dbAnalogLibNvsVersion

Controls nvs versions.

| First Build to Enable the Enhancements | Description                                                                                                                                                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before U-2023.03-SP2                   | <b>Legacy nvs</b><br>Property Editor displays noise, noise file, and noise file _as_ var at the same time<br>Hspice netlister writes noise and noise file at the same time if they both have non-empty |



| First Build to Enable the Enhancements | Description                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| U-2023.03-SP2                          | <b>Add new CDF parameter noiseEntryMethod to control the display and netlisting of and noise file</b><br>Choices are: file and expression<br>Default value is: file<br><br>Spectre-format: no impact. (nvs is for Hspice-format only)<br>Hspice-format: legacy schematic is impacted if the ncs element sets Noise Expression a<br>The noise expression is not netlisted because it is hidden by default noiseEntryMethod. |

**Note:**

Preference values 1-2 are enabled in U-2023.03-SP2. The default value is 2.

## dbAnalogLibNxPortVersion

Controls n1port, n2port, n3port, and n4port versions.

| First Build to Enable the Enhancements | Description                                                                                                                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before U-2023.03                       | <b>Legacy n1port</b><br>CDF parameter: interp default value is default                                                                                                                     |
| U-2023.03                              | <b>Change default value of interp to linear</b><br>Spectre-format: netlist and simulation result are changed<br>Hspice-format: netlist is not changed. (interp is for Spectre-format only) |

**Note:**

Preference values 1-2 are enabled in U-2023.03-SP1-1. The default value is 2.

## dbAnalogLibMTlineVersion

Controls mtline versions.

| First Build to Enable the Enhancements | Description                                                 |
|----------------------------------------|-------------------------------------------------------------|
| Before S-2021.09                       | Legacy mtline                                               |
| Before S-2021.09                       | Redefine mtline to make it more compatible with third party |

| First Build to Enable the Enhancements | Description                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| V-2023.12-SP1                          | Use CDF parameter <code>file</code> instead of <code>rlgcfile</code> in spectre-format display and netlisting.<br>spectre-format: Set this preference to value 2 to use <code>rlgcfile</code> in spectre-format display netlisting.<br>Hspice-format: Use <code>rlgcfile</code> only for Hspice-format. |

**Note:**  
Preference values 1-2 are enabled before S-2021.09.  
Preference value 3 is enabled in V-2023.12-SP1. The default is 3.

## dbAnalogLibPcapacitorVersion

Controls pcapacitor versions.

| First Build to Enable the Enhancements | Description                                                                                                                                        |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Before V-2023.12                       | <b>Legacy pcapacitor</b>                                                                                                                           |
| V-2023.12                              | <b>More enhancements for third-party compatibility</b><br>Spectre-format: use CDF parameter “trise” instead of “dtemp”<br>Hspice-format: no change |

**Note:**  
Preference values 1-2 are redefined or enabled in V-2023.12. The default is 2.

## dbAnalogLibPdiodeVersion

Controls pdiode versions.

| First Build to Enable the Enhancements | Description                                                                                                                                                                |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before V-2023.12                       | <b>Legacy pdiode</b>                                                                                                                                                       |
| V-2023.12                              | <b>More enhancements for third-party compatibility</b><br>Spectre-format: the CDF parameter definition and netlist are third-party compatible.<br>Hspice-format: no change |

**Note:**

Preference values 1-2 are redefined or enabled in V-2023.12. The default is 2.

---

## dbAnalogLibPinductorVersion

Controls `pinductor` versions.

| First Build to Enable the Enhancements | Description                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Before V-2023.12-SP1                   | <b>Legacy pinductor</b>                                                                                     |
| V-2023.12-SP1                          | Use CDF parameter <code>trise</code> instead of <code>dtemp</code> in spectre-format display and netlisting |

**Note:**

Preference values 1-2 are redefined or enabled in V-2023.12-SP1. The default value is 2.

---

## dbAnalogLibPortVersion

Controls port versions.

| First Build to Enable the Enhancements | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before U-2023.03-SP2                   | <b>Legacy port</b><br>Symbol: is not a PCell<br>CDF parameter: <code>tvpairs</code> default is 2<br>Netlisting behavior: <ul style="list-style-type: none"><li>Noise related parameters (<code>noisefile</code>, <code>F/N</code> pairs) are only netlisted when they are displayed in the Property Editor (parameter <code>noiseParam</code> is selected)</li><li>Modulation parameter (<code>type=sine</code>) <code>fmmodfiles</code> and <code>scale</code> are only netlisted when they are displayed in the Property Editor (parameter <code>modulation</code> is selected)</li></ul> |

| First Build to Enable the Enhancements | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| U-2023.03-SP2                          | <p><b>Symbol is changed to PCell (for “type=prbs” &amp; “external trigger modes”)</b><br/>No impact on legacy designs if not in this type and mode</p> <p><b>Change tvpairs default value to 0</b><br/>Both of spectre-format and hspice-format: netlist and simulation result are changed</p> <p><b>Netlist noise-related parameters (noisefile, F/N pairs) when they are hidden in the Property Editor by parameter noiseParam</b><br/>Spectre-format: netlist and simulation result are changed<br/>Hspice-format: netlist is not changed</p> <p><b>Netlist (type=sine) fmmodfiles and scale when they are hidden in the Property Editor by parameter modulation</b><br/>Spectre-format: netlist and simulation result are changed<br/>Hspice-format: netlist is not changed</p> |

Note:

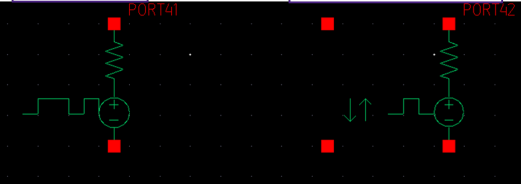
Preference values 1-2 are enabled in U-2023.03-SP2. The default value is 2.

dbAnalogLibPprbsVersion

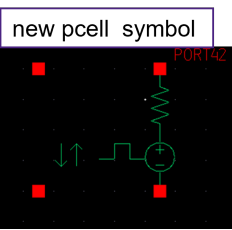
Controls pprbs versions.

| First Build to Enable the Enhancements | Description                                                                                                                                                                               |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before U-2023.03-SP2                   | <p><b>Legacy pprbs</b><br/>Symbol: is not a PCell<br/>When the trigger mode is external, because there is no pin to connect the external trigger signal, the netlist is incorrect</p>     |
| U-2023.03-SP2                          | <p><b>Change the symbol to PCell</b><br/>When the trigger mode is external, adds pins to connect the external signals<br/>No impact on legacy designs if the trigger mode is internal</p> |

old symbol



new pcell symbol



|                      |                           |
|----------------------|---------------------------|
| Transition reference |                           |
| Edge type            |                           |
| Trigger              | External both edges       |
| Threshold            |                           |
| LFSR Mode            | PN32                      |
| Seed                 | 1 3 5 7 9 11 13 15 17 ... |

**Note:**

Preference values 1-2 are enabled in U-2023.03-SP2. The default value is 2.

---

## dbAnalogLibPpwlVersion

Controls ppwl versions.

| First Build to Enable the Enhancements | Enhancements Description                                                                                                                                                                           |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before U-2023.03-SP2                   | <b>Legacy ppwl</b><br>CDF parameter: allbrkpts default is no                                                                                                                                       |
| U-2023.03-SP2                          | <b>Remove default value of allbrkpts</b><br>Spectre-format: netlist is changed. Simulation result may be changed depending on simulator's default values.<br>Hspice-format: netlist is not changed |

**Note:**

Preference values 1-2 are enabled in U-2023.03-SP2. The default value is 2.

---

## dbAnalogLibPsinVersion

Controls psin versions.

| First Build to Enable the Enhancements | Enhancements Description                                                                                                                |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Before U-2023.03-SP2                   | <b>Legacy psin</b><br>CDF parameter: va default is 0                                                                                    |
| U-2023.03-SP2                          | <b>Remove default value of va</b><br>Spectre-format: netlist and simulation result are changed<br>Hspice-format: netlist is not changed |

**Note:**

Preference values 1-2 are enabled in U-2023.03-SP2. The default value is 2.

## dbAnalogLibPulseExpV1V2Remap

Disables using val0 and val1 in vsource, isource, or port devices (type=pulse/exp) in Hspice-format.

| First Build to Enable the Enhancements | Description                                                                                                                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before U-2023.03                       | <b>Legacy vsource, isource, or port devices (type=pulse/exp) in Hspice-format</b><br>CDF parameter for high/low values: v1 and v2, prompts are “Voltage1” and “Voltage 2”                                    |
| U-2023.03                              | <b>Use CDF parameters val0 and val1 for high/low values, prompts are “Zero value” and “One value”</b><br>Hspice-format: netlist is not changed<br>Spectre-format: no change. This is only for Hspice-format. |

**Note:**

Preference values true and false are enabled in U-2023.03. The default value is true.

## dbAnalogLibSp1tswitchVersion

Controls sp1tswitch versions.

| First Build to Enable the Enhancements | Description                                                     |
|----------------------------------------|-----------------------------------------------------------------|
| Before S-2021.09                       | Legacy sp1tswitch                                               |
| Before S-2021.09                       | Redefine sp1tswitch to make it more compatible with third party |

**Note:**

Preference values 1-2 are enabled before S-2021.09. The default value is 2.

## dbAnalogLibSp1t2tswitchEmptyOffset

Write offset=0 in netlist by default.

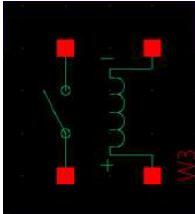
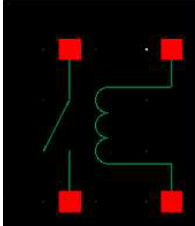
| First Build to Enable the Enhancements | Enhancements Description                                                                        |
|----------------------------------------|-------------------------------------------------------------------------------------------------|
| Before V-2023.12                       | <b>Write offset=0 by default in netlist</b><br>CDF parameter transfactor has the default 1.     |
| V-2023.12                              | <b>Do not write the offset=0 by default in netlist</b><br>The simulation result is not changed. |

**Note:**

Preference values true and false are redefined or enabled in V-2023.12. The default is true.

dbAnalogLibSwitchVersion

Controls switch versions.

| First Build to Enable the Enhancements | Enhancements Description                                                                                                                                                                      |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before T-2022.06                       | <b>Legacy switch</b><br>Symbol: version 1<br>                                                              |
|                                        | CDF parameters: vt1 default value is 0.1, vt2 default value is 0                                                                                                                              |
| T-2022.06                              | <b>Change the symbol to version 2</b><br>Impact both Hspice-format design and Spectre-format schematic<br> |
|                                        | <b>Inherited from last version:</b> vt1 default value is 0.1, vt2 default value is 0                                                                                                          |

| First Build to Enable the Enhancements | Enhancements Description                                                                                                                                                                                                                                                                                                     |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| U-2023.03                              | <b>Remove default values of vt1 and vt2</b><br>Spectre-format: netlist and simulation result are changed<br>Hspice-format: netlist is not changed                                                                                                                                                                            |
| V-2023.12-SP2                          | <b>Change the hspice-format netlist syntax</b><br>Spectre-format netlist: no change<br>hspice-format netlist: <ul style="list-style-type: none"> <li>• Previous syntax: <code>gw0 n+ n- vcr pwl(1) nc+ nc- vt2 r vt1 ropen</code></li> <li>• New syntax: <code>gw0 n+ n- vcr relay nc+ nc- vt1 vt2 ropen r</code></li> </ul> |

#### Note:

Preference values 1-3 are redefined or enabled in U-2023.03-SP1. Preference value 4 is enabled in V-2023.12-SP2. The default value is 4.

#### Examples

A legacy schematic was created with Custom Compiler T-2022.06-SP1. The switch in the design should have symbol version 2, vt1 default value is 0.1, vt2 default value is 0. Suppose the legacy schematic used these default values.

- If opening the legacy schematic with U-2023.03-SP1, set the preference value to 2 for back-compatibility.
- If opening the example schematic with version earlier than U-2023.03-SP1, the preference value 2 may not work as the table shown.
- If the switch does not use the default vt1 and vt2, no need to set the preference.

## dbAnalogLibTlineVersion

Controls tline versions.

| First Build to Enable the Enhancements | Enhancements Description                                                                                                                                                                                          |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before T-2022.06-SP2                   | <b>Legacy tline</b><br>CDF parameter "td" has default as 0<br>CDF parameter "rs" has default as 1K<br>CDF parameter "nl" has no default<br>Use CDF parameter "z0" and "l" in spectre-formant netlisting procedure |



| First Build to Enable the Enhancements | Description                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T-2022.06-SP2                          | <b>Remove td default value</b><br>Spectre-format: netlist is changed, simulation result is not changed<br>Hspice-format: no change                                                                                                                                                                                                                                                                   |
| V-2023.12                              | <b>More enhancements for third-party compatibility</b><br>Remove CDF parameter “rs” default<br>Set CDF parameter “nl” default to 0.25<br>Use CDF parameter “zo” instead of “z0” in spectre-format<br>Use CDF parameter “len” instead of “l” in spectre-format<br>Spectre-format: netlist is changed. Simulation result is changed<br>Hspice-format: netlist is changed. Simulation result is changed |

**Note:**

Preference values 1-2 are redefined or enabled in V-2023.12. The default is 3.

---

## dbAnalogLibVccsVersion

Controls vccs versions.

| First Build to Enable the Enhancements | Description                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before V-2023.12                       | <b>Legacy vccs</b>                                                                                                                                                                                      |
| V-2023.12                              | <b>More enhancements for third-party compatibility</b><br>Spectre-format: changed several CDF parameter definitions. Now it is compatible with third-party's spectre-format<br>Hspice-format: no change |

**Note:**

Preference values 1-2 are redefined or enabled in V-2023.12. The default is 2.

---

## dbAnalogLibVcresVersion

Controls writing hm=1 in spectre-format netlist by default.

---

| First Build to Enable the Enhancements | Description |
|----------------------------------------|-------------|
|----------------------------------------|-------------|

---

|                  |                                                                                                                                                                                                                                     |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before V-2023.12 | <b>Legacy vres</b><br>CDF parameter transfactor has the default 1.                                                                                                                                                                  |
| V-2023.12        | <b>Remove the default value of transfactor</b><br>Spectre-format: netlist does not have gm=1 by default. Simulation result changed<br>Hspice-format: netlist has gm=2 by default. No change<br>The simulation result is not changed |

---

**Note:**

Preference values 1-2 are redefined or enabled in V-2023.12. The default is 2.

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## dbAnalogLibVcvsVersion

Controls vcvs versions.

---

| First Build to Enable the Enhancements | Description |
|----------------------------------------|-------------|
|----------------------------------------|-------------|

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|                  |                                                                                                                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before T-2022.06 | <b>Legacy vcvs</b>                                                                                                                                                                                      |
| T-2022.06        | <b>Add CDF parameter trfType for spectre-format</b><br>Spectre-format: replace CDF parameter “csType” with “trfType”<br>Hspice-format: not changed                                                      |
| V-2023.12        | <b>More enhancements for third-party compatibility</b><br>Spectre-format: changed several CDF parameter definitions. Now it is compatible with third-party's spectre-format<br>Hspice-format: no change |

---

**Note:**

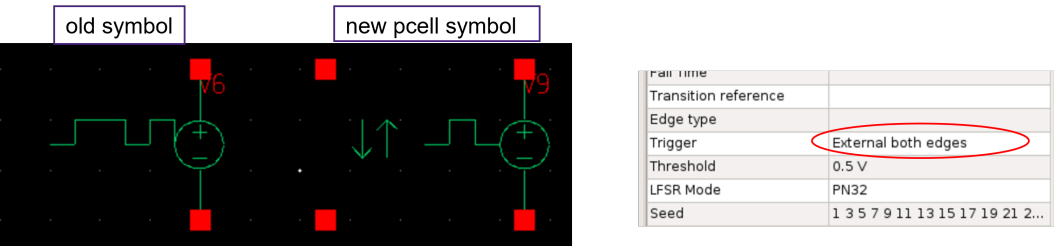
Preference values 1-3 are redefined or enabled in V-2023.12. The default is 3.

---

## dbAnalogLibVprbsVersion

Controls vprbs versions.

| First Build to Enable the Enhancements | Description                                                                                                                                                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before U-2023.03-SP1                   | <b>Legacy vprbs</b><br>Symbol: is not a PCell<br>When the trigger mode is external, because there is no pin to connect the external trigger signal, the netlist is incorrect     |
| U-2023.03-SP1                          | <b>Change the symbol to PCell</b><br>When the trigger mode is external, adds pins to connect the external signals<br>No impact on legacy designs if the trigger mode is internal |



**Note:**  
Preference values 1-2 are enabled in U-2023.03-SP1. The default value is 2.

## dbAnalogLibVsinVersion

Controls vsin versions.

| First Build to Enable the Enhancements | Description                                                                                                                                                                                               |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before S-2021.09-SP2-1                 | <b>Legacy vsin</b><br>CDF parameters: va default is 0, vo default is 0                                                                                                                                    |
| S-2021.09-SP2-1                        | <b>Remove default value of vo</b><br>Spectre-format: netlist is changed, simulation result is not changed<br>Hspice-format: netlist is not changed<br><b>Inherited from last version:</b> va default is 0 |
| U-2023.03                              | <b>Remove default value of va</b><br>Spectre-format: netlist and simulation result are changed<br>Hspice-format: netlist is not changed<br><b>Inherited from last version:</b> vo has no default value    |

**Note:**

Preference values 1-3 are redefined or enabled in U-2023.03-SP2. The default value is 3.

---

## dbAnalogLibVsourceVersion

Controls vsource versions.

| First Build to Enable the Enhancements | Description                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before U-2023.03                       | <b>Legacy vsource</b><br>Symbol: is not a PCell<br>CDF parameters: tvpairs default is 2, va default is 0                                                                                                                                                                                                                                                 |
| U-2023.03                              | <b>Remove default value of va</b><br>Spectre-format: netlist and simulation result are changed<br>Hspice-format: netlist is not changed<br><br><b>Inherited from last version:</b> tvpairs default is 2, symbol is not a PCell                                                                                                                           |
| U-2023.03-SP1                          | <b>Symbol is changed to PCell (for “type=prbs” &amp; “external trigger modes”)</b><br>No impact on legacy designs if not in this type and mode<br><br><b>Change tvpairs default value to 0</b><br>Both of spectre-format and hspice-format: netlist and simulation result are changed<br><br><b>Inherited from last version:</b> va has no default value |

**Note:**

Preference values 1-3 are redefined or enabled in U-2023.03-SP2. The default value is 3.

---

## dbAnalogLibWindingVersion

Version control for winding devices.

| First Build to Enable the Enhancements | Description                                                                                       |
|----------------------------------------|---------------------------------------------------------------------------------------------------|
| V-2023.12-SP1                          | Use parameter <code>r</code> and map it as <code>resis</code> in Spectre flow.                    |
| V-2023.12-SP2                          | Use parameter <code>resis</code> and remove the mapping of <code>r</code> to <code>resis</code> . |

**Note:**

Preference values 1-2 are redefined or enabled in V-2023.12-SP2. The default value is 2.

---

## nIHSPICESkipEmptyDC

Controls whether to write dc=0 by default in hspice-format netlist.

---

| First Build to Enable the Enhancements | Description |
|----------------------------------------|-------------|
|----------------------------------------|-------------|

|                  |                                                       |
|------------------|-------------------------------------------------------|
| Before V-2023.12 | Write dc=0 by default for independent sources.        |
| V-2023.12        | Do not write dc=0 by default for independent sources. |

---

**Note:**

Preference values true and false are redefined or enabled in V-2023.12. The default is true.

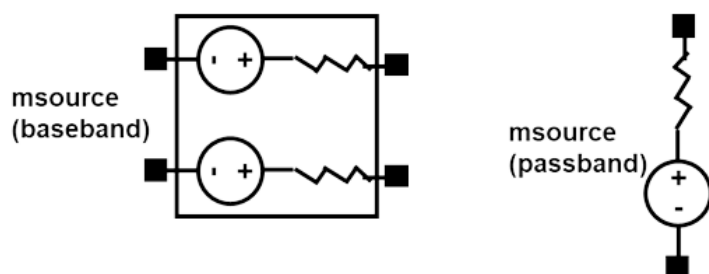
# 3

## rfLib

Support for third-party compatible RF sample library (rfLib) for migration of user's design created by third-party tools. Also, this library can be used by designers to create RF designs.

### msource

IQ modulated source with various parameters settings. The msource device is a PCell. Its PCell behavior is controlled by its signal type. When the signal type is `passband`, the symbol is a two-terminal device. When the signal type is `baseband`, the symbol is a four-terminal device.



### CDF Parameters

| CDF Parameter Prompt | CDF Parameter        | Default  | hspiceD | spectre | auCdl |
|----------------------|----------------------|----------|---------|---------|-------|
| Signal type          | wtype                | passband | X       | X       | —     |
| Output type          | output_type          | port     | X       | —       | —     |
| Source type          | wsource_type         | txInput  | —       | X       | —     |
| Carrier harmonic     | wsource_carrier_harm | 1        | —       | X       | —     |

| CDF Parameter Prompt                        | CDF Parameter     | Default | hspiceD | spectre | auCdl |
|---------------------------------------------|-------------------|---------|---------|---------|-------|
| LO source type                              | local_osc_type    | I&Q     | –       | X       | –     |
| LO/I source name                            | local_osc_name_i  |         | –       | X       | –     |
| LO/Q source name                            | local_osc_name_q  |         | –       | X       | –     |
| LO source name                              | local_osc_name    |         | –       | X       | –     |
| Average Power                               | dbm               | 6.99    | X       | X       | –     |
| Amplitude of signal                         | amp               |         | X       | –       | –     |
| Resistance                                  | r                 | 50      | X       | X       | –     |
| Number of frames                            | framenum          | 1       | –       | X       | –     |
| Oversample ratio                            | oversample        | 4       | –       | X       | –     |
| SNR (dB)                                    | snr               |         | –       | X       | –     |
| Channel frequency                           | MSource_Frequency | 5G      | X       | X       | –     |
| Data rate (bps)                             | MSource_DataRate  | 48M     | X       | X       | –     |
| Modulation type                             | MSource_ModType   | QAM16   | –       | X       | –     |
| Modulation type                             | mod               | QAM16   | X       | –       | –     |
| Modulation order                            | order             |         | X       | –       | –     |
| Number of symbols                           | MSource_SymbolNum | 512     | –       | X       | –     |
| Symbols per Impulse                         | filter_symbols    | 20      | X       | –       | –     |
| artParameterInToolDisplay('filter_symbols') | filter_samples    | 4       | X       | –       | –     |
| PN code                                     | MSource_DataType  | PN15    | X       | X       | –     |
| OFDM type                                   | MSource_OFDMType  | No      | X       | X       | –     |

| CDF Parameter Prompt      | CDF Parameter         | Default | hspiceD | spectre | auCdl |
|---------------------------|-----------------------|---------|---------|---------|-------|
| Pulse shaping filter type | MSource_FilterType    | None    | –       | X       | –     |
| Roll-off factor           | MSource_Alpha         | 0.2     | –       | X       | –     |
| Pulse shaping filter type | filter                | None    | X       | –       | –     |
| Roll-off factor           | filter_coef           | 0.2     | X       | –       | –     |
| IFFT/FFT size             | MSource_FFT Length    | 64      | X       | X       | –     |
| Number of Data carriers   | MSource_Data Carriers | 52      | –       | X       | –     |
| Cyclic prefix             | MSource_CP            | 1/4     | –       | X       | –     |
| Guard band                | ofdm_gband            | 6.5     | X       | –       | –     |
| Prefix length             | ofdm_prefix           | 16      | X       | –       | –     |
| Number of DC carriers     | MSource_DC Num        | 1       | –       | X       | –     |
| Number of carriers in CA  | MSource_CA Num        | 1       | –       | X       | –     |

## Examples

Passband netlist in spectre-format:

```
W0_pass (net1 net2) wsource wsource_type=txInput wsource_carrier_harm=1 \
    standard="MSource" dbm=6.99 r=50 framenum=10 oversample=4
    snr=3 \
    MSource_Frequency=5G MSource_DataRate=48M MSource_ModType=QAM16 \
    MSource_SymbolNum=512 MSource_DataType=PN15 \
    MSource_FilterType=None MSource_OFDMType=No
```

Baseband netlist in spectre-format:

```
W1_base (net4 net3 net6 net5) wsource wsource_type=txInput \
    wsource_carrier_harm=1 local_osc_name_i=test_I \
    local_osc_name_q=test_Q standard="MSource" dbm=6.99 r=50 \
    framenum=10 oversample=4 snr=3 MSource_Frequency=5G \
    MSource_DataRate=48M MSource_ModType=QAM16 MSource_SymbolNum=512
    \
    MSource_DataType=PN15 MSource_FilterType=None MSource_OFDMType=No
```



**Passband netlist in hspice-format:**

```
PW_pass_ofdm net3 0 iqsrc ofdm_fft=64 ofdm_prefix=16 ofdm_gband=6,5
freq=5e+09 mod=QAM16
+      bit_rate=4.8e+07 filter_symbols=20 data=WP15 axis=iq
filter_samples=4 z0=50
+      dbm=6.99
```

**Baseband netlist in hspice-format:**

```
PW5 net24 net23 iqsrc ofdm_fft=64 ofdm_prefix=16 ofdm_gband=6,5 mod=QAM16
+      bit_rate=4.8e+07 filter_symbols=20 data=WP15 axis=i
filter_samples=4 z0=50 dbm=6.99
PW5@1 net26 net25 iqsrc ofdm_fft=64 ofdm_prefix=16 ofdm_gband=6,5
mod=QAM16
+      bit_rate=4.8e+07 filter_symbols=20 data=WP15 axis=q
filter_samples=4 z0=50 dbm=6.99
```